19-G-0309/19-G-0310 - 2-10-20 - Brooklyn Union Gas

CASE 19-G-0309 - Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service.

CASE 19-G-0310 - Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a National Grid for Gas Service

EVIDENTIARY CONFERENCE

DATE: February 10, 2020

LOCATION: 19th Floor Hearing Room
Agency Building Three,
Empire State Plaza
Albany, New York

ALJ JAMES A. COSTELLO, DPS

ALJ MAUREEN F. LEARY, DPS

19-G-0309/19-G-0310 - 2-10-20 - Brooklyn Union Gas

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19-G-0309/19-G-0310 - 2-10-20 - Brooklyn Union Gas

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19-G-0309/19-G-0310 - 2-10-20 - Brooklyn Union Gas

A.L.J. COSTELLO: I call case 19-G-0309 proceeding on motion of the commission as to the rates, charges, rules and regulations of the Brooklyn Union Gas Company doing business as National Grid New York for Gas Service. And I also call case 19-G-0310 proceeding on motion of the commission as to the rates, charges, rules and regulations of KeySpan Gas East Corp. doing business as National Grid for Gas Service.

We're here pursuant to a notice of an evidentiary hearing that was issued by the secretary on January 13th, 2020. The purpose of the hearing is to accept -- primarily the purpose is to accept pre-filed testimony exhibits into the evidentiary record and to allow cross examination of witnesses.

My name is Jim Costello. I'm one of the Administrative Law Judges from the Department of Public Service. To my right is Maureen Leary, the other Administrative Law Judge who will be -- will be presiding in this case. So what we're going to do is start with take -- by taking appearances and we'll start with the company.

MR. GAVILONDO: Good morning, Your Honors. Carlos Gavilondo for the company. With me

19-G-0309/19-G-0310 - 2-10-20 - Brooklyn Union Gas at counsel table are Patric O'Brien, Phillip Decicco, Kenneth Maloney of Cullen and Dykman, and Jeremy Euto of the company. Thank you.

A.L.J. COSTELLO: Good morning. And - -

A.L.J. LEARY: Jeremy Euto, would you spell that last name?

MR. EUTO: Spelled E-U-T-O, Your Honor.

A.L.J. LEARY: Thank you.

A.L.J. COSTELLO: D.P.S. staff.

MR. GOODRICH: Good morning, Your Honors. This is Brandon Goodrich for the Department of Public Service and joining me is Nicholas Forst and Raquel Parks.

A.L.J. COSTELLO: Good morning. Okay. The Utility Intervention Unit.

MS. KASOW: This is Jill -- is this on? Yes.

A.L.J. LEARY: Speak up.

MS. KASOW: Jillian Kasow, from the New York Department of State, Utility Intervention Unit.

A.L.J. COSTELLO: Okay.

MS. O'HARE: Kathleen O'Hare from the Department of State, Utility Intervention Unit.

A.L.J. LEARY: Those microphones back there don't seem to be picking up a lot. I heard you just fine, Mr. Goodrich, but --

MR. GOODRICH: Is the light on?

A.L.J. LEARY: Is there volume? Is there a volume? No.

MS. O'HARE: Okay. I'll just try to speak louder.

A.L.J. LEARY: Thank you.

MS. O'HARE: Okay. Kathleen O'Hare from the New York State Department of State, Utility Intervention Unit.

A.L.J. COSTELLO: Okay. The Public Utility Law Project of New York.

MR. BERKLEY: Good morning, Your Honors. Richard Berkley for the Public Utility Law Project.

MS. WHEELOCK: And also Laurie Wheelock with the Public Utility Law Project.

A.L.J. LEARY: Again, cannot hear you.

MS. WHEELOCK: Laurie Wheelock with the Public Utility Law Project.

A.L.J. LEARY: Thank you.

A.L.J. COSTELLO: And I may call some parties that may not be here today. So I'm just if -- if I do that, just understand I'm going off my list. So the City of New York.

MR. LOUGHNEY: Yes, Your Honors, the -- representing the City of New York is the law firm of Couch White. I am Robert Loughney. Also appearing at different times will be Adam Conway, Justin Fung and perhaps Russell King. Thank you.

A.L.J. LEARY: When those gentlemen, Conway, what -- was it Fung?

MR. LOUGHNEY: Yes, Justin Fung, F-U-N-G.

A.L.J. LEARY: And the third?

MR. LOUGHNEY: Russell -- Russell King.

A.L.J. LEARY: Last name again?

MR. LOUGHNEY: King, K-I-N-G.

A.L.J. LEARY: Thanks. When they come into the room -- or are they here?

MR. LOUGHNEY: They're not here, no.

A.L.J. LEARY: Okay. Just --

MR. LOUGHNEY: Just Justin Fung.

A.L.J. LEARY: Okay. Just when -- when they come into the room, it's important that you have the court reporter understand who they are and so forth so that they can be identified when -- if they speak. Are they planning on doing cross or --?

MR. LOUGHNEY: Mr. Conway will be, yes.

A.L.J. LEARY: Okay. Thanks.

MR. LOUGHNEY: And Mr. Fung will be.

A.L.J. LEARY: Thank you.

MR. LOUGHNEY: Thanks.

A.L.J. COSTELLO: Environmental Defense Fund.

MS. MURPHY: Good morning. This is Erin Murphy on behalf of the Environmental Defense Fund, and I'm joined by Marty Rothfelder, who is with Rothfelder Stern representing E.D.F.

A.L.J. COSTELLO: E.D.F.?

MS. MURPHY: Yeah.

A.L.J. COSTELLO: Bob Wyman. Mr. Wyman just state your name. I know I've stated it but --

MR. WYMAN: Bob Wyman.

A.L.J. COSTELLO: Okay. Thank you. The Sane Energy Project.

MS. FRACZEK: Kim Fraczek from Sane Energy Project.

A.L.J. LEARY: Would you spell your last name, please?

MS. FRACZEK: F like Frank R-A-C like Charlie Z like zebra E-K.

A.L.J. LEARY: Thank you.

MS. ZIESCHE: Lee Ziesche, Sane Energy Project. Want me to spell my last name?

A.L.J. LEARY: I do.

MS. ZIESCHE: Z-I-E-S-C-H-E.

A.L.J. COSTELLO: Okay. Is there New York State Laborers' Organizing Fund? No? Estates New York Real Estate Services? Okay. Instead of going through the rest of the list, is there anyone here who's making an appearance who's -- who I haven't called yet? Yes?

MR. BLAKE: Yes, Your Honor. Kevin Blake, on behalf of Family Energy.

A.L.J. COSTELLO: Family Energy?

MR. BLAKE: With me is Jeff Donnelly, D-O-N-N-E-L-L-Y.

A.L.J. COSTELLO: Okay. Good?

A.L.J. LEARY: Yeah.

A.L.J. COSTELLO: Anyone else? Okay. Not hearing any. Just so you know, few housekeeping rules. When you get up to speak any time that you're starting or starting new, please state your name for the record so that the court reporter has a clear indication of who is speaking. Try not to speak over each other.

We have confidential testimony that was pre-filed as well as redacted testimony. Where there's confidential pre-filed testimony, we're going to make a separate transcript. We'll also be making a separate transcript of the redacted information that will go in with the unredacted informa -- I mean the non-redacted information and the testimony that's given here today.

For -- when we go on -- nothing is planned for today, but at some point, the Department of Public Service indicated they may be doing some cross examination that will involve confidential information. And at that time, we are going to clear the room of anyone who has not signed a protective order or who was not and otherwise entitled to be

here by statute, entitled to be subject to hearing confidential information.

A.L.J. LEARY: I don't have anybody that signed the protective order since our last call, I think, on the 5th. Has anyone -- does anyone intend to sign so that we should be on the lookout for that so you can stay in the room? Anyone? Okay. Thanks.

A.L.J. COSTELLO: Okay. We're not going to collect affidavits today, but people who are not being cross-examined unless you're going to get up on the stand and have your testimony admitted that way, we would ask that you provide affidavits for any witnesses who are not being cross-examined. We can do that as -- as the proceeding goes on. And then at some point when we have everything collected we'll put the testimony of people who are not being crossed – we'll put that into the record.

I want to just ask, does anyone at this point have any affidavits just so we can collect them? No?

MS. MURPHY: Environmental Defense Fund does.

A.L.J. COSTELLO: Okay.

THE REPORTER: What was your name?

A.L.J. LEARY: Please identify yourself before you speak.

MS. MURPHY: Sure, sorry, Erin Murphy with E.D.F.

MR. LOUGHNEY: Your Honor, I have a question.

A.L.J. LEARY: You need to speak into the microphone.

MR. LOUGHNEY: Yes, Robert Loughney, for the City. For the affidavits, can we have -- can we use scanned signatures, or do you need the originals?

A.L.J. LEARY: You can do a P.D.F.

A.L.J. COSTELLO: Yeah.

MR. LOUGHNEY: Okay. Thank you.

A.L.J. COSTELLO: And we'll -- we can -- we'll collect that at a break. I'll collect that from you. Okay?

MS. MURPHY: Okay. Sounds good.

A.L.J. LEARY: We need them electronically too.

A.L.J. COSTELLO: Yeah, we can ask them.

A.L.J. LEARY: We're going to need them electronically as well, and -- just F.Y.I.

A.L.J. COSTELLO: Okay. This -- early this morning I sent around a corrected exhibit list, that will need to be further corrected. But just so you know, we -- the last three exhibits listed on that need to be corrected to 596, 597 and 598. So that we will be starting with Exhibit 599. Okay. And just -- just so you know, exhibits will be identified during the testimony, but it will be admitted into the record at the end of the hearing.

Okay. Does anyone have any questions before we begin? Yes, sir.

MR. LOUGHNEY: It's Robert Loughney again. Your Honor, the -- there was cross examination for the low income panel and Lane Collin, who are witnesses for the city. The only party that had asked for cross examination was the company. And the company has waived cross-examination so I just wanted to make sure that there's no one -- that Your Honors don't have any questions for those witnesses because otherwise I would bring their testimony for --

A.L.J. LEARY: What day?

MR. LOUGHNEY: -- affidavit.

A.L.J. LEARY: What day is their testimony scheduled for?

MR. LOUGHNEY: Lane Collin was scheduled for the 13th. And the low income panel was also scheduled for the 13th.

A.L.J. COSTELLO: Yeah, I don't have any questions.

MR. LOUGHNEY: It was 30 minutes for Lane Collin and 30 minutes for the low income panel.

A.L.J. LEARY: Mr. Loughney, I just -- I'm going to be a Nazi, everybody in this room will know about the record. You need to speak into the microphone, even if you hold it, like an inch away. So I can hardly hear you. I don't know if the record is going to pick you up clearly or not. But I have to protect this record. So I'm telling everybody in this room, see how close you can hear me. That's how I want to hear you and so does the court reporter and Judge Costello. So please speak closely into the microphone, and so we can protect the record. Thanks. Go ahead.

MR. LOUGHNEY: Thank you, Your Honor. So Mr. Collin and the low income panel were both

19-G-0309/19-G-0310 - 2-10-20 - Brooklyn Union Gas scheduled for Wednesday the 13th. And there was a half an hour of cross examination for each of them by the company. And the company has waived that.

A.L.J. LEARY: We don't have questions.

A.L.J. COSTELLO: We have no questions, so.

A.L.J. LEARY: Does anybody else have questions for those witnesses? Okay. I think you're in good shape.

MR. LOUGHNEY: Okay. Thank you, Your Honor.

A.L.J. LEARY: We -- we've taken them off the list already.

A.L.J. COSTELLO: Yeah.

A.L.J. LEARY: You sent us an email, Mr. Loughney, I think.

A.L.J. COSTELLO: Yeah.

MR. LOUGHNEY: Yes.

A.L.J. LEARY: Thank you.

A.L.J. COSTELLO: Okay. So we're going to get started with the Future of Heat Panel.

MR. GAVILONDO: Thank you, Your Honor. Carlos Gavilondo for National Grid, the companies.

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I'd like to call to the stand the Future of Heat Panel which consists of five witnesses.

A.L.J. LEARY: We have enough chairs? Good.

A.L.J. COSTELLO: What I'm going to do is I'm going to ask you to stand and I'll start at the person on the end there. You can all stand with your hand, you don't have to raise your hands. I'm just asking one at a time to state your name and your business address for the record.

MS. SCHWARTZ: My name is Mackenzie Schwartz. My business address is 1 MetroTech Center in Brooklyn, New York.

A.L.J. COSTELLO: Okay. And I'm just going to go through and then what I'll do is I'll swear you all in at one time after I get your names and business addresses.

MS. TALLET: Good morning. My name is Lisa Tallet. My business address is 300 Erie Boulevard, Syracuse, New York.

MR. TRACZYK: Good morning. My name is Owen Brady Traczyk, last name spelled T-R-A-C-Z-Y-K and my business address is 1 MetroTech Center in Brooklyn, New York.

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MR. CHAHBAZPOUR: Good morning. Donald Chahbazpour also 1 MetroTech, Brooklyn, New York.

THE REPORTER: Spell your last name.

MR. CHAHBAZPOUR: C-H-A-H-B-A-Z-P-O-U-R.

MR. HAMLIN: Arthur W. Hamlin. My business address is 300 Erie Boulevard West, Syracuse, New York.

A.L.J. COSTELLO: Now, I just ask you to stand and raise your right hand? And just respond one at a time so that they can get your responses down. Do you swear or affirm that the testimony you will provide is the truth?

MS. SCHWARTZ: Yes.

WITNESS; MACKENZIE SCHWARTZ; Sworn

MS. TALLET: Yes.

WITNESS; LISA TALLET; Sworn

MR. TRACZYK: Yes.

WITNESS; OWEN BRADY TRACZYK; Sworn

MR. CHAHBAZPOUR: Yes.

WITNESS; DONALD CHAHBAZPOUR; Sworn

MR. HAMLIN: Yes.

WITNESS; ARTHUR W. HAMLIN; Sworn

A.L.J. COSTELLO: Okay. Thank you. You may be seated. Counsel?

MR. GAVILONDO: Thank you, Your Honor.

DIRECT EXAMINATION

BY MR. GAVILONDO:

Q. Good morning, panel. We're here today for the -- the gas rate case hearing for KEDNY-KEDLI. And I'd like you to focus on pre-filed testimony that was submitted by the panel. And I'd like to ask whether or not you have before you a document entitled direct testimony of the Future of Heat Panel dated April 2019, consisting of 120 pages, a cover page and a table of contents?

A.L.J. COSTELLO: You have to -- you have to say it orally.

BY MR. GAVILONDO: (Cont'g.)

Q. Speak affirmatively, no shaking.

A. (Panel) Yes.

Q. Thank you. And I'd also like to ask whether or not you have a document before you entitled, corrections and updates testimony of the Future of Heat Panel dated July 3, 2019, consisting of 7 pages and a cover page?

A. Yes.

Q. Thank you. And do you have before you a document entitled rebuttal testimony of the Future of Heat Panel dated September 18th, 2019, consisting of 66 pages, a cover page and a table of contents?

A. Yes.

Q. Thank you. And finally, because this has been a long proceeding, I'd like to ask whether or not you have a document before you entitled, supplemental testimony of the Future of Heat Panel dated January 29th, 2020 consisting of 12 pages, a cover page and a table of contents?

A. Yes.

Q. Okay. You will have to speak up so the court reporter can hear you. Okay. Thank you. So the documents that you've just identified, were they prepared by you or under your direction and supervision?

A. Yes.

Q. And if I asked you the questions contained in those documents today would your answers be the same?

A. Yes.

Q. Okay. Do you adopt those documents as your testimony in this proceeding?

A. Yes.

MR. GAVILONDO: Your Honors, I ask that the direct testimony, the corrections and updates testimony, the rebuttal testimony, and the supplemental testimony pre-filed by the Future of Heat Panel be transcribed into the record as if given orally.

A.L.J. COSTELLO: Okay. The motion is granted. And at this point in the record, these documents should be inserted. It's the KEDNY-KEDLI Future of Heat Panel direct testimony. KEDNY-KEDLI Future of Heat Panel corrections and updates testimony. KEDNY-KEDLI Future of Heat Panel rebuttal testimony. And the KEDNY-KEDLI Future of Heat Panel supplemental testimony.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY AND KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Direct Testimony

of the

Future of Heat Panel

Dated: April 2019

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I. Introduction and Qualifications

Q. Please introduce the members of the Future of Heat Panel.

A. The Panel consists of Donald Chahbazpour, Lisa M. Tallet, Arthur W. Hamlin, and Owen S. Brady.

Q. Mr. Chahbazpour, please state your name and business address.

A. My name is Donald Chahbazpour. My business address is One MetroTech Center, Brooklyn, New York 11201.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid USA Service Company, Inc. ("National Grid Service Company"), a subsidiary of National Grid USA ("National Grid"), and currently hold the position of Director of Gas Utility of the Future. My responsibilities include leading efforts to reduce methane and carbon emissions through policy, strategy, and technology for National Grid's operating companies, including The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI") (collectively, the "Companies"). I am also responsible for engaging stakeholders to raise awareness regarding the potential of renewable natural gas ("RNG").

Q. Please describe your educational background and business experience.

A. I received a Bachelor of Science in Mechanical Engineering from New Jersey Institute of Technology in 1998 and a Master of Public Administration from Columbia University's School of International and Public Affairs in 2000. Prior to working at National Grid, I worked for a startup energy technology company. I joined National Grid in 2004 and have held various positions of increasing responsibility in strategic planning, energy procurement, mergers and acquisitions, gas operations, and regulatory and customer strategy.

Q. Ms. Tallet, please state your name and business address.

A. My name is Lisa M. Tallet. My business address is 300 Erie Boulevard West, Syracuse, New York 13202.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid Service Company and currently hold the position of Director of New York Customer Energy Management. My responsibilities include the policy, design, and strategy for the energy efficiency programs that support the New York Public Service Commission's (the "Commission") clean-energy targets set forth in the

December 13, 2018 *Order Adopting Accelerated Energy Efficiency Targets* in Case 18-M-0084 (the "December 2018 EE Order").

Q. Please describe your educational background and business experience.

A. I received a Bachelor of Business Administration from St. Bonaventure University in 1985. I worked for New York State Electric and Gas Corporation ("NYSEG") and Rochester Gas and Electric Corporation ("RG&E") from 1991 to 2007, where I held positions of increasing responsibility working on gas marketing, customer service, customer advocacy, and low-income programs. I joined National Grid in September 2007 and have held positions of increasing responsibility in customer financial services, energy efficiency implementation, policy, and strategy and evaluation. I assumed my current role in August 2018.

Q. Mr. Hamlin, please state your name and business address.

A. My name is Arthur W. Hamlin. My business address is 300 Erie Boulevard West, Syracuse, New York 13202.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid Service Company and currently hold the position of Manager, Economic Development, with responsibility for managing National Grid's portfolio of economic development programs, including programs that have provided enduring benefits to customers and communities in the Companies' respective service territories.

Q. Please describe your educational and professional background.

A. I received a Bachelor of Science from the University of Michigan in 1981, majoring in environmental policy and management. In 1982, I received a Master of Science in resource economics, also from the University of Michigan. I was employed for five years as an economist for the U.S. Department of Labor in Washington, D.C., and was subsequently hired by Niagara Mohawk Power Corporation d/b/a National Grid ("NMPC") in 1989. Since that time, I have held a variety of positions with National Grid, including leadership roles in the economic development, corporate citizenship, and market research functions.

Q. Please briefly describe your current areas of responsibility for National Grid.

A. I am responsible for managing National Grid's economic development activities and programs, which feature initiatives that support sustainable economic growth in communities throughout the Companies' service territories.

Q. Mr. Brady, please state your name and business address.

A. My name is Owen S. Brady. My business address is One MetroTech Center, Brooklyn, New York 11201.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid Service Company and currently hold the position of Product Management Specialist in the Emerging Products group. My responsibilities include developing the business models, technical design, and strategy for new product offerings that will meet customers' changing energy needs. Additionally, I currently manage the Companies' Gas Demand Response Demonstration Project, which the Commission adopted in Cases 16-G-0058 and 16-G-0059 (the "2016 KEDNY and KEDLI Rate Cases").

Q. Please describe your education background and business experience.

A. I received a Bachelor of Science in Mechanical Engineering from the University of Vermont in 2010. Thereafter, I worked for Vermont Gas Systems from 2011 until 2017, where I held positions of increasing responsibility in areas of strategic planning, policy development, and customer-account management. In 2017, I was hired by National Grid as a member of the New Energy Solutions group, where I was responsible for managing demonstration projects and overseeing investment in research and development. In August 2018, I assumed my current role. I am also enrolled in an Executive MBA program through Columbia University and the London Business School, with an expected graduation date of February 2020.

II. Purpose of Testimony

Q. What is the purpose of the Panel's testimony?

A. As the energy provider to nearly two million customers in downstate New York, the Companies are committed to advancing clean energy solutions that further National Grid's commitment to environmental stewardship. Today, demand for natural gas remains strong, as customers seek a cost-effective, reliable heating source that generates fewer emissions than alternatives such as heavy oil. In this way, natural

gas continues to play a critical role in driving economic opportunity in New York. Yet, with the challenges presented by climate change, the State, the Commission, and the Companies recognize that more is needed to meaningfully change the current climate trajectory.

Governor Cuomo committed to aggressively pursuing clean energy policies aimed at reducing greenhouse gas emissions from the energy sector 40 percent (from 1990 levels) by 2030 and the longer-term goal of decreasing total carbon emissions 80 percent by 2050, as well as an aggressive new renewable energy goal that 100 percent of electricity consumed in New York be carbon neutral by 2040. For its part, National Grid similarly launched its "Northeast 80x50 Pathway" (the "80x50 Pathway") complementing New York State's efforts. The Companies' gas networks will play an integral role in meeting these ambitious goals and delivering the low-carbon economy of the future.

The purpose of this testimony is to set forth the Companies' vision for the future of the heating sector; a vision aimed at achieving those shared goals through a comprehensive plan built on the Companies' core obligations to provide safe, reliable, and affordable gas service to customers in New York. As more fully discussed below, the Companies

are sponsoring a suite of proposals directed at: (i) reducing methane emissions from the gas distribution system; (ii) achieving greater energy efficiency; (iii) promoting demand response and other non-pipes alternatives ("NPAs"); (iv) encouraging the development of sustainable heating options; (v) promoting economic development through investment in energy infrastructure; (vi) reducing carbon emissions on the gas network; and (vii) supporting further research, development, and demonstration ("RD&D") to advance RNG technologies.

Q. Please describe the Companies' vision for the future of the heating sector.

A. National Grid's vision for a sustainable gas future in New York is founded on a core commitment to exceed the expectations of its customers and communities while making possible the energy systems of tomorrow. Delivering on this vision requires a proactive, customer-centric, mission-driven approach – an approach that combines the Companies' ongoing obligation to provide safe, affordable, and reliable gas service with a renewed focus on advancing clean energy goals. To do this, the Companies developed a four-pronged strategy that establishes the goals, tools, and incentives for driving meaningful change in the heating sector:

Reducing methane emissions from the gas distribution system 60 percent by 2035: Building on the 80x50 Pathway, National Grid proposes an aggressive goal of reducing total network methane emissions 60 percent by 2035; continuing its leadership role in national initiatives aimed at reducing emissions; identifying, prioritizing, and repairing large-system leaks; and implementing work procedures to further reduce emissions going forward.

Empowering and Enabling Customers to Sustainably Meet Their Heating Needs: The Companies have developed a suite of programs, products, and demonstration projects aimed at empowering and enabling customers to take control of their energy usage through: incorporation of accelerated energy efficiency goals consistent with budgets and targets set forth in the Commission's December 2018 EE Order; access to RNG supplies through a Green Gas Tariff offering; an expanded Gas Demand Response Demonstration Project that complements the Companies' robust interruptible ("IT") and temperature control ("TC") service offerings; the development of a Sustainable Heat Initiative that includes deployment of geothermal heat pump technology, support for clean oil-to-gas conversions, and additional heating-related technology for low-to-moderate income ("LMI") customers; as well as enhanced economic

development programs capable of driving environmentally beneficial outcomes.

Reducing Carbon Emissions on the Gas Network: The integration of RNG and other sustainable gas supplies will reduce the carbon footprint of the Companies' gas networks. To facilitate carbon reductions through the integration of RNG supplies, the Companies propose to: deliver the Newtown Creek RNG project – one of the first projects in the U.S. to directly inject RNG into a local distribution system using biogas generated from a wastewater treatment plant and food waste; coordinate with the Department of Energy's National Renewable Energy Laboratory ("NREL") and potentially partner with New York City ("NYC") to develop a power-to-gas ("P2G") demonstration project that will convert excess renewable electricity to RNG and show how the gas system can serve as a long-term storage solution for renewable energy supplies; study, in coordination with Stony Brook University's Institute of Gas Innovation and Technology ("I-GIT"), the opportunities and challenges with blending hydrogen into the gas distribution system; eliminate barriers to RNG interconnections, and facilitate and incentivize new RNG projects through Company ownership of portions of the

interconnection equipment; and continuing productive end-use RD&D activities.

Performance-Based Incentives and Revenue Sharing: To align the Companies' incentives with a sustainable vision for the future of the heating sector and overarching energy policy goals, the Companies propose three Earnings Adjustment Mechanisms ("EAMs"), two Platform Service Revenue ("PSR") opportunities, and a NPA Incentive Mechanism.

Q. Please describe the anticipated benefits of the Companies' Future of Heat strategy.

A. The Companies' multi-faceted approach empowers customers to make energy choices that further clean energy goals, while also establishing the Companies as key drivers of the energy transition through development and deployment of RNG and a suite of NPA technologies. Collectively, the Companies believe the solutions presented here will drive a positive change in how customers meet their energy requirements and have a beneficial impact on the environment with projected reductions in carbon dioxide ("CO_2") emissions of approximately 73,000 metric tons (equivalent to taking 15,000 cars off the road for one year).

Q. What are the costs of the Companies' Future of Heat strategy?

A. Altogether, the Companies' propose to invest a total of $89.788 million over four years in incremental Future of Heat initiatives to reduce carbon emissions on the gas network and empower customers in support of a cleaner, more sustainable energy future. These amounts were provided to the Revenue Requirements Panel to develop the revenue requirements for KEDNY and KEDLI in the Rate Year (the 12 months ending March 31, 2021) and Data Years (Data Year 1 is the 12 months ending March 31, 2022; Data Year 2 is the 12 months ending March 31, 2023; and Data Year 3 is the 12 months ending March 31, 2024) (collectively, the "Data Years). The projected non-labor operations and maintenance ("O&M") costs and related full-time equivalent ("FTE") employees for the Future of Heat initiatives are set forth in Exhibit ___ (FOH-13). The projected capital costs for the Future of Heat initiatives are set forth in Exhibit ___ (GIOP-1) and Exhibit ___ (GIOP-7).

Q. Does the Panel sponsor any exhibits as part of its testimony?

A. Yes. The Panel sponsors the following exhibits that were prepared and compiled under our direction and supervision:

(i) Exhibit ___(FOH-1) are data sheets with summaries of the Companies' significant customer products, programs, and demonstration project proposals;

(ii) Exhibit ___ (FOH-2) are Benefit-Cost Analyses ("BCAs");

(iii) Exhibit ___ (FOH-3) is the Gas Demand Response REV Demonstration Project Quarterly Report for the Fourth Quarter of 2018, ending December 31, 2018;

(iv) Exhibit ___ (FOH-4) is the Geothermal Demonstration Project Quarterly Report for the Fourth Quarter of 2018, ending December 31, 2018;

(v) Exhibit ___ (FOH-5) is a Newtown Creek Project data sheet;

(vi) Exhibit ___ (FOH-6) is the Power-to-Gas ("P2G") Demonstration Project ("P2G Project") data sheet;

(vii) Exhibit ___ (FOH-7) provides detailed program descriptions including proposed modifications to KEDNY and KEDLI's current economic development grant programs;

(viii) Exhibit ___ (FOH-8) is the cost forecast of KEDNY and KEDLI's proposed economic development grant programs;

(ix) Exhibit ___ (FOH-9) is an example of the E-Commerce Platform PSR calculation;

(x) Exhibit ___ (FOH-10) is a summary of the proposed EAMs;

(xi) Exhibit ___ (FOH-11) describes the NPA Incentive Mechanism;

(xii) Exhibit ___ (FOH-12) describes the EmPower Replacement Earnings Incentive Mechanism; and

(xiii) Exhibit ___ (FOH-13) is a summary of labor and non-labor O&M expenses for the Future of Heat initiatives.

III. Reducing Methane Emissions

Q. What is methane and why are the Companies focused on reducing emissions?

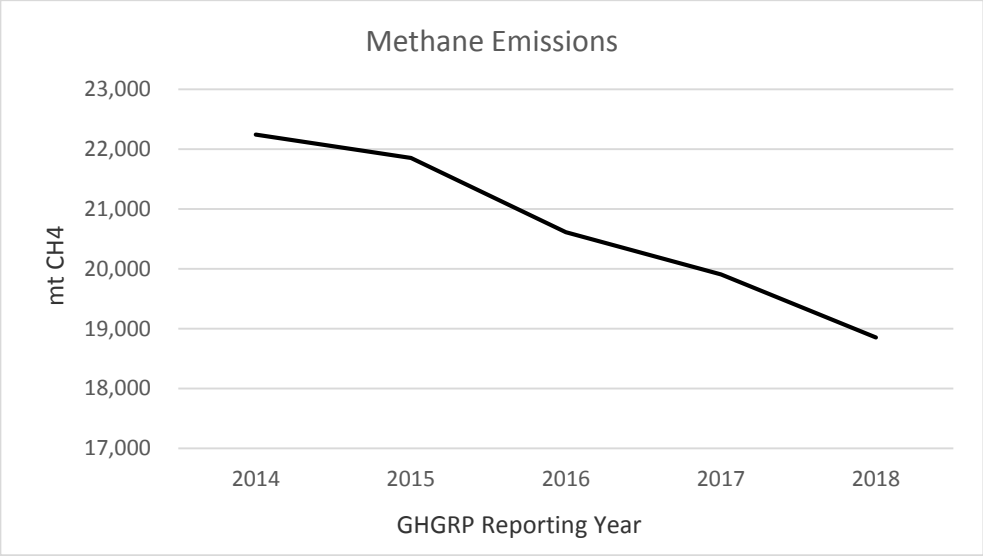
A. Methane (CH_4), the primary component of natural gas, has been identified by the U.S. Environmental Protection Agency ("EPA") as a greenhouse gas. The Companies, as distributors of natural gas, are committed to taking actions that reduce methane emissions from the gas network, ensuring that gas deliveries are sustainable and consistent with the Companies' clean energy goals.

Q. What steps have the Companies taken to address methane emissions?

A. The most cost-effective means of reducing methane emissions is to eliminate system leaks through the replacement of aging infrastructure. As shown in Figure 1 below, between 2014 and 2018, the Companies

replaced approximately 733 miles of leak-prone pipe ("LPP"), preventing approximately 3,388 metric tons of methane emissions each year.[1]

Figure 1: Methane Emission Reductions



In addition to infrastructure upgrades, the Companies have implemented work practices and procedures to mitigate methane emissions released during construction and maintenance activities. As discussed more below, in May 2016, the Companies established a blowdown protocol that limits methane emissions when purging gas from a pipeline – which can otherwise significantly contribute to total system emissions.

Q. How do the Companies plan to build on these efforts?

[1] Emission reduction estimates are based on the calculation methodology set forth in the Environmental Protection Agency's ("EPA") regulations (40 C.F.R. § 98, Subpart W).

A. The Companies are building on their initial success by deploying a multi-faceted strategy aimed at achieving meaningful methane reductions, including a proposal to increase the pace and scale of LPP replacement in NYC, identifying and repairing the highest emitting leaks, deploying a new mobile methane detection program, and continued utilization of blowdown reduction procedures.

A. Commitment to Methane Reduction Policies

Q. Please describe the Companies' efforts to lead the policy discussion on reducing methane emissions.

A. To drive a positive step change in natural gas production and distribution, the Companies have taken a series of policy actions aimed at reducing methane emissions. First, the Companies are establishing an ambitious target to reduce methane emissions from the gas distribution system 60 percent (from 1990 levels) by 2035. Second, as part of the EPA's Methane Challenge Program's ONE Future Emissions Intensity Commitment Option, the Companies are working to reduce the methane leakage rate from the production, processing, transmission, and distribution of natural gas to one percent or less by 2025. Collectively, these policy initiatives are driving transparency and accountability for

reducing methane emissions across all segments of the natural gas value chain.

B. Leak-Prone Pipe Replacement and Leak Reduction Programs

Q. Please describe the Companies' LPP replacement efforts.

A. The direct testimony of KEDNY and KEDLI's Gas Infrastructure and Operations Panels ("GIOP") highlight the Companies' continuing commitment to replacing older, high-risk LPP, which disproportionately contributes to leaks on the Companies' systems. The EPA, as part of its Methane Challenge Program, identified this strategy – upgrading natural gas infrastructure – as a best management practice for reducing methane emissions; noting that it also mitigates operational risks, increases efficiency, and advances environmental goals.

To that end, the Companies propose to continue their aggressive proactive LPP replacement efforts, increasing the replacement miles to 225 per year for a total of 930 miles over four years. Altogether, the Companies anticipate these efforts, once complete, will result in greenhouse gas reductions of approximately 3,627 metric tons of methane each year.

Q. What are the Companies' commitments to reducing system leaks?

A. Over the last several years, the Companies have eliminated thousands of leaks from their backlogs of non-hazardous leaks, which has driven further reductions in overall methane emissions. Each year, KEDNY and KEDLI eliminate more than 150 and 750 system leaks, respectively, through a combination of main/service replacements and proactive leak repairs. The Companies are additionally incentivized to repair up to 250 incremental Type 3 leaks that have been ranked as high emitters - and these incentive targets were achieved in 2017 and 2018. In this case, the Companies are proposing to maintain aggressive leak reductions targets that would reduce the Companies' total system leaks by at least 900 leaks each year, as well as retain the current incentive to repair the highest emitting system leaks. These leak reduction targets are discussed in more detail by the Companies' Gas Safety Panel.

C. Mobile Methane Detection Program

Q. What is mobile methane detection?

A. Mobile methane detection uses sensors (*e.g.,* cavity ring-down spectroscopy) attached to a vehicle equipped with mapping technology. This technology has the potential to quickly and efficiently identify large-volume natural gas leaks. With this information, utilities can target

infrastructure repairs and upgrades in high-priority areas with significant leaks.

Q. Please describe the Companies' mobile methane detection efforts.

A. The Companies, in partnership with Google, the Environmental Defense Fund, and Colorado State University, have been evaluating the potential use of an enhanced high-emitter methane detection vehicle, also referred to as the "Google Vehicle," to help with leak identification and prioritization. As set forth in the direct testimony of KEDNY and KEDLI's Gas Safety Panel, the Companies are proposing a program to expand the use of this technology in high-leak concentration areas. The technology could prove useful in monitoring, mapping, and helping the Companies to quickly and cost-effectively evaluate leak size and volume. With that information, the Companies can, in turn, prioritize repair and replacement activities in areas that can achieve the greatest methane emission reductions. Importantly, this effort is focused on identifying environmentally significant leaks, and as such it will augment, not replace, the Companies traditional leak surveying methods of identifying leaks.

D. Blowdown Procedures

Q. What are blowdown procedures and how do they help reduce methane emissions?

A. Natural gas distributors use blowdown mitigation procedures to bleed down and evacuate natural gas from an isolated pipeline or vessel and then re-inject the gas into the distribution system instead of letting it escape into the atmosphere. With the isolated pipeline free of gas, the pipeline owner or operator can then safely commence construction and repair activities. The EPA's Methane Challenge Program, through its Natural Gas STAR Program, identified the injection of blowdown gas into low-pressure mains as a best management practice for reducing methane emissions.

Q. Please describe the Companies' blowdown procedures.

A. In May 2016, National Grid adopted procedures for "Drawdown Compressor to Bleed Down Evacuate and Re-Inject Natural Gas from an Isolated Pipeline." The Companies use the procedures to minimize methane emissions when performing work in furtherance of their Methane Challenge Program commitments and clean energy priorities. In 2018, the Companies estimate the blowdown procedures reduced methane emissions by 156.442 million cubic feet (mcf).

IV. Customer Empowerment and Enablement

Q. What role do customers play in the Companies' vision for the future of heat?

A. Customers are at the heart of the Companies' commitment to adapting the gas system to meet new demands with a cleaner product and drive future benefits, while continuing to deliver safe, affordable, and reliable service. In practice, this means seamlessly enhancing the customer experience by: empowering them to take control of their energy usage through robust energy efficiency offerings; advancing new products and services that allow customers to actively participate in achieving clean energy goals; and enabling new customers to sustainably meet their heating needs.

A. Energy Efficiency

Q. Please describe the Companies' current energy efficiency programs.

A. The Companies' energy efficiency programs consist of a portfolio of programs designed to achieve energy savings across various market sectors (*e.g.,* residential, small business, and commercial and industrial ("C&I")). A description of the portfolio for program years 2019 and 2020 is set forth in the Companies' Updated 2019-2020 ETIPs, filed with the Commission on February 19, 2019 in Cases 15-M-0252 and 18-M-0084.

Over the past ten years, KEDLI's energy efficiency programs processed over 407,000 applications and helped customers save approximately 16.5 million therms. Over the same period, KEDNY's energy efficiency programs processed more than 490,000 applications and helped customers save approximately 23.1 million therms.

Q. How do the Companies recover energy efficiency program costs?

A. Energy efficiency program costs are currently recovered through the System Benefits Charge ("SBC") outside of base rates, except for labor and associated overhead costs, which were transitioned to base rates in the 2016 KEDNY and KEDLI Rate Cases. As set forth in the Companies' 2019 ETIP filings, the current annual budgets, excluding low-income budgets, for KEDNY and KEDLI's energy efficiency programs are as follows:

Table 1 - Energy Efficiency Program Costs ($000s)

Year	KEDNY	KEDLI
2019	$15,118	$8,137
2020	$16,877	$8,866

Q. How are energy efficiency program budgets determined?

A. Utility-specific targets and budgets for calendar years ("CY") 2019 and 2020 were adopted by the Commission in the December 2018 EE Order. Program targets and budgets for the years 2021-2025 are currently being reviewed in Case 18-M-0084. The Commission established "presumptive targets and budgets" in the December 2018 EE Order as a starting point for the development of utilities' specific portfolio proposals. Until the Commission directs otherwise, the Companies are using the adopted budget amounts for CY 2020, along with the presumptive targets and budgets for CYs 2021-2025 set forth in Appendix C of the order, to determine the proposed budget amounts for the Rate Year and Data Years.

Q. Are the Companies proposing any changes to the manner in which energy efficiency program costs are recovered?

A. Yes, the Companies are proposing three changes, two of which impact the revenue requirements in the Rate Years and Data Years. First, the Companies propose to move evaluation, measurement, and verification ("EM&V") administrative costs from the ETIP portfolio budget into base rates. Second, the Companies propose to maintain the current ETIP portfolio budget levels, inclusive of the amount removed for EM&V. This approach is consistent with the Commission's Order in Cases 17-E-0238

and 17-G-0239 (the "2017 NMPC Rate Case"), and it will enable the Companies to fund additional customer programs. Third, the Companies propose to move all energy efficiency costs from the SBC into base rates. Moving energy efficiency costs to base rates is also consistent with the approach adopted by the Commission in the 2017 NMPC Rate Case.

Q. What are EM&V activities?

A. These are activities to evaluate, measure, verify, track, and report energy efficiency savings. The current annual budget for KEDLI's EM&V activities was set at five percent of the annual ETIP portfolio budget, $0.408 million in 2019 and $0.445 million in 2020, as set forth in KEDLI's ETIP. The current annual budget for KEDNY's EM&V activities was also set at five percent of the annual ETIP portfolio budget, $0.785 million in 2019 and $0.876 million in 2020, as set forth in KEDNY's ETIP.

Q. Please explain why the Companies are proposing to move the costs associated with EM&V activities from the SBC surcharge to base rates, beginning in the Rate Year.

A. The Companies' proposal aligns with the Commission's direction in the REV Track One Order to begin transitioning energy efficiency costs into

base rates. The costs proposed to be moved are administrative costs associated with EM&V activities. These costs are either not directly related to delivery of the energy efficiency portfolio contained in the ETIPs or are scoped to integrate initiatives beyond energy efficiency only. The proposal is also consistent with the treatment recently approved by the Commission in the 2017 NMPC Rate Case Order. Tables 2a and 2b show the annual costs the Companies propose to shift from the ETIPs to base rates in the Rate Year and Data Years.

Table 2a – KEDLI Costs Shifted from ETIP to Base Rates

Costs Shifted from the ETIP	CY 2020 Budget Amounts	FY 2021 Rate Year	FY 2022 Data Year 1	FY 2023 Data Year 2	FY 2024 Data Year 3
EM&V	$444,977	$473,898	$513,426	$563,054	$629,575

Table 2b – KEDNY Costs Shifted from ETIP to Base Rates

Costs Shifted from the ETIP	CY 2020 Budget Amounts	FY 2021 Rate Year	FY 2022 Data Year 1	FY 2023 Data Year 2	FY 2024 Data Year 3
EM&V	$876,207	$924,353	$1,038,023	$1,169,814	$1,395,742

The adjustment to move budgeted EM&V costs to base rates is shown in Exhibit ___ (RRP-3), Schedule 27, page 6.

Q. Please explain the Companies' proposal to maintain the ETIPs at their current 2019-2020 budgets.

A. In the December 2018 EE Order, the Commission established increased utility energy efficiency budgets and targets to help achieve the State's carbon reduction goals. The Companies believe the level of energy efficiency funding adopted in the December 2018 EE Order, which is currently reflected in the Companies ETIP, should be maintained.

The Companies are proposing to maintain the current annual ETIP budgets through 2020. Furthermore, with the previously described movement of EM&V costs to base rates, the Companies propose to add additional programs and initiatives to the ETIP portfolios that would backfill the ETIP budgets and keep overall ETIPs at the same amount. The new incremental programs and initiatives will enable achievement of additional energy savings that are reflected in the higher savings targets approved in the December 2018 EE Order. For purposes of the revenue requirement in this filing, the Companies modeled the energy efficiency revenues and expenses and proposed budgets to align with the December 2018 EE

Order. KEDNY's ETIP budget moving to base rates is $18.102 million for the Rate Year, $20.464 million for Data Year 1, $23.128 million for Data Year 2, and $27.551 million for Data Year 3. KEDLI's ETIP budget moving to base rates is $9.478 million for the Rate Year, $10.274 million for Data Year 1, $11.269 million for Data Year 2, and $12.592 million for Data Year 3.

Q. Please explain the Companies' proposal to transition energy efficiency program costs from the SBC surcharge to base rates.

A. Consistent with the Companies' understanding of Staff and the Commission's expectations, the Companies propose to transition the entire ETIP program costs to base rates beginning in the Rate Year. The Companies further propose to reconcile, over the term of the rate plan, actual ETIP costs to the amount included in rates. Any under expenditure in a given year will be carried forward and reconciled at the end of Data Year 3. If there is an underspend at that time, the Company will defer the difference for future refund to customers.

Q. Are the Companies forecasting the need for additional resources to manage the energy efficiency portfolio?

A. Yes. In the December 2018 EE Order, the Commission proposed increases to the existing approved portfolios. Given the size of the Companies current programs and anticipated future program growth, the Companies forecast the need for seven incremental FTEs to support portfolio design development and implementation. The Companies propose to include four program manager roles focused on product development and delivery to the customers through third-party vendors and direct sales. In addition to these roles, the Companies propose to add one FTE supporting the sales activities for delivery of C&I products and two FTEs providing technical sales support.

Q. What are the proposed FTE costs?

A. As shown in Exhibit__ (RRP-3) Schedule 27, forecast O&M costs for the FTEs are as follows:

Table 3 - FTE Costs
($000s)

	KEDNY	KEDLI
Rate Year	$534	$552
Data Year 1	$545	$564
Data Year 2	$561	$581
Data Year 3	$570	$589

Q. Does KEDLI plan to continue the EmPower Replacement Program (known as "HEAT")?

A. Yes. In the 2016 KEDNY and KEDLI Rate Cases, the Commission approved KEDLI's low-income energy efficiency HEAT Program, which is currently funded through KEDLI's existing Low-Income Discount Program deferral balance. In CY 2017, the program was capped at $1.9 million, and in CYs 2018 and 2019 the cap was adjusted for inflation. KEDLI's administrative costs for the program were limited to no more than 15 percent of the annual budget amount. The HEAT Program also included an incentive mechanism, which, as set forth in the performance-based incentive section below, KEDLI proposes to continue.

KEDLI proposes to continue the program as designed but include the HEAT program costs in base rates. In addition, KEDLI proposes to increase the budget from $1.9 million to $2.5 million annually. KEDLI will use the additional funding to expand the offering to "moderate-income" customers, defined as those between 60 percent and 80 percent of state or area median income, whichever is higher. This program will assist in filling a gap that exists on Long Island for moderate income gas customers without electric air conditioning in the absence of a New York

State Energy Research and Development Authority ("NYSERDA") sponsored program.

B. New Products and Services

Q. Please describe the Companies' proposals for new customer products and services.

A. The Companies are committed to empowering and enabling customers to take more control over their energy usage, reduce consumption, and proactively embrace products and services that align with the State's clean energy goals. To that end, the Companies are proposing the following products and services as a means of providing customers with new options they can use to optimize energy usage and reduce their environmental impact:

(i) Green Gas Tariff;

(ii) Utility Energy Services Contracts ("UESC");

(iii) Fuel-Switching Calculator; and

(iv) Natural Gas Vehicle ("NGV") Rates.

Each product and service is supported by a detailed data sheet attached here to as Exhibit ___(FOH-1), summarizing the Companies' proposals, as well as the benefits of the proposed investments.

1. *Green Gas Tariff Offering*

Q. Please describe the Companies' proposed Green Gas Tariff offering.

A. As set forth in Exhibit ___ (FOH-1), Schedule 1, the Companies propose a Green Gas Tariff offering beginning in Data Year 1 that will enable customers to voluntarily purchase RNG to meet all or a portion of their energy needs. The offering will include four tiers, allowing customers to select a level of green gas procurement that works for their budget and their environmental aspirations.

Q. What Green Gas Tariff tiers will the Companies offer?

A. The Companies recognize that different customers prefer different pricing structures, residential customers tend to prefer price certainty, while non-residential customers tend to prefer costs that scale as a percentage of use. To accommodate these differing customer preferences, the Companies propose a tiered structure as follows

Tier	Residential Flat Rate	Non-Residential	
		% of Monthly Consumption	Flat Rate
Low	$5/month	5%	$25/month
Low-Middle	$20/month	10%	$50/month
High-Middle	$25/month	25%	$100/month
High	$50/month	100%	$200/month
			$500/month

Residential customers would be required to commit to the Green Gas Tariff program for twelve months, while non-residential customers must commit to a twenty-four-month term for either the flat rate or percentage of monthly consumption option.

Q. How do the Companies propose to address any potential over collection or under collection RNG gas costs for the Green Gas Tariff offering?

A. As further discussed in the direct testimony of the Companies' respective Rate Design Panels, the Companies propose a RNG-specific cost of gas to maintain an accurate cost of RNG apart from the cost for traditional gas. To the extent the Companies monetize the environmental attributes of excess gas, such amounts would reduce RNG costs for participating Green Gas Tariff customers.

Q. What is the cost of the Green Gas Tariff offering?

A. As set forth in Exhibit ___ (FOH-13), the Companies propose two FTEs to implement the new offering. The FTEs will be responsible for: contracting with RNG suppliers; tracking quantities of green gas purchased and sold monthly; managing the balance between purchases and sales; acquiring environmental attributes (to the extent sales exceed purchases); selling environmental attributes (to the extent purchases exceed sales); tracking green gas costs and revenues; making all associated transaction system entries; and performing reporting requirements. For KEDNY, the labor-related costs include $0.157 million in Data Year 1, $0.162 million in Data Year 2, and $0.164 million in Data Year 3. For KEDLI, the costs include $0.163 million in Data Year 1, $0.167 million in Data Year 2, and $0.170 million in Data Year 3.

Q. What are the benefits associated with the proposed Green Gas Tariff offering?

A. Offering a Green Gas Tariff to customers means they will have the opportunity to purchase a renewable fuel without the need to modify their existing equipment. Because RNG is made up of the same constituents as fossil-based natural gas, it can be introduced into the gas

distribution network safely and utilized by existing natural gas-fired equipment. This allows customers to reduce the environmental impact of their heating, without the need to replace appliances. Adoption by customers will also help advance the RNG market by providing predictable demand and certainty to RNG developers looking to invest in RNG projects. Additionally, customers who operate NGVs can purchase RNG for their fleet, helping decarbonize transportation (the sector with the largest volume of carbon emissions in the State). Finally, the Green Gas Tariff directly supports clean energy policy objectives, including: achieving a 40 percent reduction in greenhouse gas emissions by 2030; National Grid's 80x50 Pathway; cleaner transportation; protecting New York's natural resources by diverting waste streams and reducing organics in the water system; and building a more resilient energy system.

2. *Utility Energy Services Contracts ("UESC") Program*

Q. What is the UESC program?

A. The UESC program is an energy services contracting program, where the Companies and customers, typically government agencies and other large commercial entities, enter limited-source contracts for energy management services, such as energy and water improvements. With a

UESC, the Companies can help customers analyze their current energy usage, identify opportunities for improvement, and work with pre-selected contractors to achieve efficiencies, reduce costs, and adopt renewable energy systems, provided the efficiencies include some measure of gas savings. The customers, in turn, pay for the contracted services for the improvements delivered under the UESC. As further set forth in Exhibit ___ (FOH-1), Schedule 2, the Companies and their New York affiliates have participated in UESCs since 2011, delivering more than $18 million in value.

Q. Are the Companies proposing any changes to the UESC program?

A. Yes. The Companies propose an incentive to help customers offset the cost of energy feasibility studies, as well as the inclusion of a marketing budget to engage new customers. The Companies also propose a PSR for fees received from entities who enter a UESC with either of the Companies to implement the feasibility study findings. The PSR proposal is set forth in the Performance-Based Incentives and Revenue Sharing Section below.

Q. Please describe the proposed energy feasibility study incentive.

A. Feasibility studies are often the first step in the UESC process, providing customers with a range of improvements that could help better manage their energy usage and the potential cost savings associated with implementing the study's findings. The studies, however, can be expensive, with upfront costs deterring customers from initiating the UESC process. To overcome this hurdle, and only in so far as such costs are not covered by another program or entity like NYSERSDA, the Companies propose an incentive equivalent to 50 percent of any feasibility study that exceeds $0.010 million.

Q. Please describe the Companies' proposed increase in program marketing?

A. The Companies propose to create a marketing budget for the program and add two FTEs. The marketing increase includes outreach (*e.g.*, customer leads), sales, and procurement. The FTEs will include a marketing manager to drive additional customer participation in the program and a project engineer to assist customers through the energy feasibility study and implementation process.

Q. Please summarize the costs included in the revenue requirements for the UESC program?

A. The revenue requirement, as shown in Exhibit ___ (FOH-13), includes incremental O&M for marketing, feasibility study incentives, and FTEs. For KEDNY the revenue requirement includes $0.257 million in the Rate Year, $0.455 million in Data Year 1, $0.596 million in Data Year 2, and $0.688 million in Data Year 3. For KEDLI, the revenue requirement includes $0.261 million in the Rate Year, $0.459 million in Data Year 1, $0.600 million in Data Year 2 and $0.692 million in Data Year 3.

Q. What are the benefits of the UESC program?

A. UESCs allow government agencies and large commercial customers to benefit from the Companies' energy expertise, creating a streamlined approach to contracting using a limited-source contract vehicle that meets public procurement process requirements. With the energy savings achieved through the UESC program, customers will further benefit by saving money and achieving clean energy goals and mandates. For the Companies, UESCs provide an opportunity to enhance customer relationships while driving energy policy objectives. By becoming a partner in better energy management, the Companies can improve

customer load profiles, advance customer and State clean energy goals, and help to alleviate system constraints.

3. *Fuel-Switching Calculator*

Q. Please describe the Companies' proposed fuel-switching calculator.

A. The Companies propose to develop a web-based fuel-switching calculator similar to one developed by Central Hudson Gas and Electric Company. Using current costs, desired heating technology (*e.g.,* natural gas, ground or air sourced heat pumps), and existing equipment, the calculator can provide customers an estimated annual cost, payback period, and savings for alternative energy options as compared to their current load profile. Additionally, the calculator will highlight low-carbon fuel offerings, such as RNG available through the Companies' Green Gas Tariff offering, to allow users further clean-energy comparisons. A more detailed description of the proposed calculator is included in Exhibit ___ (FOH-1), Schedule 3.

Q. What is the cost of the fuel-switching calculator?

A. As shown in Exhibit ___ (FOH-13), the fuel-switching calculator proposal includes incremental O&M expense for the development and operation of the calculator of $0.125 million for KEDNY and KEDLI in

the Rate Year, and $0.018 million for the Companies in each of the Data Years.

Q. What are the benefits associated with the proposed fuel-switching calculator?

A. The fuel-switching calculator will empower customers to make more informed energy choices, providing them with information to assess the financial impacts of alternative energy options. The Companies believe that through such energy insights, customers will discover how they can use low-carbon solutions (*e.g.*, RNG, geothermal) at a reasonable cost. This, in turn, may animate the market for low-carbon products and services, leading to increased adoption and lowering emissions in support of the State's clean energy goals and the REV objectives.

4. *Natural Gas Vehicle ("NGV") Rates*

Q. Please describe the Companies' proposed adjustment to NGV Rates.

A. The Companies propose modernizing the NGV rates currently contained in their respective tariffs to ensure the rates are appropriate under current market conditions and designed to incentivize the use of RNG in the transportation sector – the sector that is the largest source of greenhouse gas emissions in the State. The specific changes include updates to

conventional rates, as set forth in the respective direct testimonies of the Rate Design Panels and establishing firm and non-firm (full or partial interruptible) service. Additional details regarding the Companies' proposal are set forth in Exhibit ___ (FOH-1), Schedule 4.

Q. What rates do the Companies propose?

A. The proposed rates for KEDNY and KEDLI are supported by the Rate Design Panels. The Companies further propose to monitor the market and, if the market conditions warrant, the Companies will petition the Commission to adjust the rates.

Q. What are the benefits of updating NGV rates?

A. Updating NGV rates will support increased use of NGVs in the heavy-duty transportation sector, lowering emissions as compared to diesel vehicles. The Companies believe this adjustment will animate the market for NGVs, leading to their increased use as a lower-carbon transportation resource. Displacing diesel plays an important role in achieving the State's clean energy goals through the decarbonization of the transportation sector. Additionally, by introducing an interruptible service offering, the Companies add to their suite of NPAs, potentially

reducing system peaks, alleviating constraints, and delaying traditional infrastructure investments.

C. Demand Response

Q. What role does demand response play in the Companies' future of heat strategy?

A. Demand response has the potential to play an important role in the Companies' future of heat strategy. Accordingly, the Companies are promoting demand response in two ways. First, the Companies' IT/TC service is one of the largest gas demand response programs in the country. Among other things, IT/TC service enables the Companies to reduce gas demand during periods of peak usage by offering a reduced delivery rate to customers who agree to interrupt service when demand peaks. The program benefits all customers by allowing the Companies to utilize the gas system more efficiently while avoiding the need for costly capital upgrades to meet a higher peak demand. Following a collaborative with Staff and interested parties, the Companies proposed to implement certain enhancements to the IT/TC offering. In the February 7, 2019 *Order Approving Tariff Revisions and Directing Further Tariff Filings* in Cases 16-G-0058 and 16-G-0059 ("IT/TC Order"), the Commission adopted these changes, which will blend the

existing IT and TC services into a single non-firm service classification with two pricing tiers determined by a customer's fuel switching capabilities. The testimony of KEDNY and KEDLI's Rate Design Panels discuss the proposed enhancements to IT/TC services.

Second, to complement the IT/TC program, the Companies' propose to expand the existing Demand Response Demonstration Project for firm customers who commit to minimum reductions. Expanding the project will allow the Companies to better determine whether they can achieve system benefits through peak demand reductions. Taken together, the non-firm and firm demand response programs have potential as components of the Companies' suite of NPA planning tools.

Q. Please describe the Gas Demand Response Program adopted by the Commission in the 2016 KEDNY and KEDLI Rate Cases.

A. Under the Gas Demand Response Program, the Companies provide a market-based credit to commercial firm customers who reduce their gas usage by a pre-determined amount when called upon by either company to do so. The initial purpose of the Demand Response Demonstration Project was to assess the effectiveness of voluntary peak reductions in

terms of reducing intraday demand and whether market-based credits will drive customer behavior to reduce consumption.

Q. What results have the Companies seen thus far?

A. In the project's first year, the Companies collectively called five events (two were exclusive to KEDLI customers), with 100 percent of the 16 project enrollees participating. During events in which both KEDNY and KEDLI customers participated, they collectively contributed a reduction of 192 dekatherm ("Dth") per hour of nameplate capacity (135 Dth in KEDNY, 57 Dth in KEDLI) during the peak hour. Such a reduction alleviates the burden on the system, and, with sufficient assurances, could be considered a system-management tool in the future. As part of the demonstration, the Companies provided $0.303 million in incentive payments to customers.

For the current year, only one customer has withdrawn from the project for operational reasons and a large property management company, Estates NY Real Estate Services LLC ("Estates"), has shown considerable interest in participating, submitting multiple applications for different facilities. The Companies were able to incorporate one of the facilities into the project – creating further opportunities for collaboration

and learning as discussed by the Commission in the IT/TC Order. During the second year, the Companies collectively called five events (two were exclusive to KEDLI customers). Customer participation across all events was 94 percent. In the events with 100 percent participation, customers provided 241 Dth per hour of nameplate capacity (184 Dth in KEDNY and 57 Dth in KEDLI) during the peak hour. Further information on the Demand Response Demonstration Project is included in Exhibit ___(FOH-3).

Q. Please describe the Companies' proposed expansion of the Demand Response Demonstration Project.

A. The Companies propose to build off the initial success of the Demand Response Demonstration Project, expanding it in scale and continuing to market to firm gas delivery customers that meet minimum annual demand values and agree to a minimum reduction amount (Dth/event hour). The Companies will notify enrolled customers in advance of an event. Such customers will then receive an incentive for reducing their gas demand by the pre-determined, minimum reduction value for the entire duration of a peak event.

Reductions in excess of the committed minimum will be incentivized at a prorated amount. The reduction will be determined based on a baseline that will be developed for each participating customer. To achieve the level of certainty needed for firm customer demand response to provide meaningful system benefits, the Companies propose to include a penalty that exceeds the incentive for customers who fail to respond or fail to meet the minimum committed reduction. Furthermore, customers who fail to respond or meet their committed reduction for more than a predetermined number of events may be removed from the project and prohibited from participating in future years. Further detail regarding the proposed expansion of the Demand Response Demonstration Project is included in Exhibit ___ (FOH-1), Schedule 5.

Q. What are the benefits associated with the expanded Gas Demand Response Project?

A. Gas demand response is a potentially valuable tool in the Companies' NPA toolbox. The project has already illustrated the capability to meaningfully decrease system pressure during peak periods. Customers have also proven willing to reduce their gas usage in response to financial incentives. Structural changes to the program, including the use of a baseline to determine reductions and increasing the scale of the

program to potentially interest the aggregator community, may make this program a viable way to reduce, delay, or eliminate infrastructure investment, especially when used in conjunction with other NPA technologies. In addition to supporting system pressure, gas demand response has the potential to reduce the quantity of supply that may be required on a peak day, which can provide cost savings. In addition, throughout the demonstration project, customers have been given access to their usage data, providing them insights that they previously have not had about their gas use. Such insights may encourage greater participation in the Companies' energy efficiency programs and help to achieve the REV objective of empowering customers to make more informed energy choices.

Q. What are the minimum values the Companies propose for customer eligibility and reduction targets?

A. For KEDNY, eligible customers must consume at least 6,000 Dth per year and commit to reducing demand over the duration of a peak event by at least two Dth per hour (*i.e.*, six Dth over the three-hour demand response event). For KEDLI, eligible customers must consume at least 4,000 Dth per year and commit to reducing demand over the duration of

an event by at least two Dth per hour (*i.e.*, six Dth over the three-hour demand response event).

Customers will specify their committed reduction amount during the application process. The committed reduction must be greater than 10 percent of either the customer's usage during a peak period or the sum of the nameplate requirements of the customer, whichever is less. This can be determined based on either data from the Companies or data from customer measurement systems. Customers must achieve the minimum reduction in each of the three hours of the event. Reducing peak usage more than the minimum may occur in any of the three hours.

Q. What are the costs of the expanded Gas Demand Response Demonstration Project?

A. As shown in Exhibit ___ (GIOP-7), the costs of the project for KEDNY include a capital investment of $0.236 million in FY 2020 and $0.059 million in each of the Rate Year and Data Years 1 and 2. For KEDLI, the costs of the project include a capital investment of $0.107 million in FY 2020 and $0.027 million in each of the Rate Year and Data Years 1 and 2.

In addition, the project includes incremental O&M expense to cover the cost of the incentives, demand response software fees, and FTEs to administer the project. As shown in Exhibit ___ (FOH-13), Schedule 1, the revenue requirement includes incremental O&M costs for KEDNY of $1.320 million in the Rate Year, $1.638 million in Data Year 1, $1.962 million in Data Year 2, and $2.326 million in Data Year 3. For KEDLI, the revenue requirement includes incremental O&M costs of $0.628 million in the Rate Year, $0.772 million in Data Year 1, $0.921 million in Data Year 2, and $1.112 million in Data Year 3, as shown in Exhibit ___ (FOH-13), Schedule 2. The costs include the addition of one FTE in the Rate Year and Data Year 1 split between the Companies and the addition of two FTEs in Data Year 2 and Data Year 3 split between the Companies. The FTEs will manage the project, by signing up customers and aggregators, evaluating data, coordinating with the gas operations group, and responding to regulatory inquiries and filing requirements.

Q. Will the Companies continue to report the findings from the Demand Response Demonstration Project?

A. Yes, the Companies will continue to submit quarterly report updates regarding the expanded Gas Demand Response Demonstration Project.

D. Sustainable Heat Initiative

Q. Please describe the Companies' Sustainable Heat Initiative.

A. The Companies propose a Sustainable Heat Initiative to ensure customers have an opportunity to participate in the clean-energy future through access to lower-carbon heating technology, while at the same time supporting the efficient operation of the natural gas network. The initiative includes three components. First, the Companies propose to draw on the lessons learned from KEDLI's current geothermal demonstration project, by expanding the offering to interested customers, giving priority to those located outside the footprint of the current and planned gas networks. This effort will help customers determine whether geothermal technology may be a suitable NPA for meeting heating requirements and facilitate customer access and adoption of geothermal systems.

Second, customers for whom geothermal is not the preferred option would have the ability to participate in the Companies' Clean Conversion Program – a scaled-down version of the former Neighborhood Expansion Program – intended to address anticipated customer demand for access to gas heating, as opposed to higher emitting (and costlier) heating options such as oil. Third, for those customers facing economic barriers to

adopting lower-carbon heating technology, the Companies propose expanding the Low-to-Moderate Income Gas Conversion Program ("LMI Gas Conversion Program") to provide incentives that will help customers access cleaner, more efficient heating options.

Q. How will the Sustainable Heat Initiative benefit customers?

A. To meet the 80x50 goal and transition to a cleaner energy future, the Companies believe aggressive action is required by both the Companies and their customers. In some cases, however, customers have few options for meeting their heating requirements based on practical challenges (*e.g.*, proximity to clean energy sources) and economic barriers (*e.g.*, high upfront costs). The Companies propose this initiative as a comprehensive approach to meeting customer demand in a clean and sustainable way. It combines a robust NPA offering (*i.e.,* geothermal) with gas conversions and incentives, as a means of giving customers additional clean-energy choices. Taken together, the Companies believe the initiative will achieve meaningful reductions in greenhouse gas emissions.

1. *Expanded Geothermal Demonstration Project*

Q. Please describe KEDLI's current Geothermal Demonstration Project.

A. The Geothermal Demonstration Project for KEDLI was adopted in the 2016 KEDNY and KEDLI Rate Cases to test shared-loop ground-source heat pump ("GSHP") systems as a cost-effective heating and cooling alternative to natural gas infrastructure in the Glenwood Village community of Riverhead, New York. The community is not currently served by KEDLI's gas network. The geothermal technology used in the project offers a way to reduce the carbon intensity of the heating and cooling sector while providing homeowners with potentially significant energy cost savings.

Q. What results has KEDLI seen thus far from the project?

A. In the first year of the project, KEDLI connected a total of ten homes with shared-loop GSHP systems, representing a total system heating capacity of 30 tons. The systems performed well during a period of extended cold weather in January 2018, with customers not reporting any loss in comfort. Likewise, during prolonged periods of warm weather in the summer, the systems did not encounter any interruptions or service outages.

The systems used in the demonstration project have also achieved high coefficients of performance ("COP") in the range of 2.2 to 3.5. High COPs, such as those realized in the Glenwood Village community, are a hallmark of GSHP, which leverages stable ground temperatures. With continued customer education and coordinated energy efficiency, the project results suggest GSHPs have the potential to achieve COPs up to 4.

Finally, customers identified comfort-related benefits, including improved air quality, quiet equipment operation, simplicity, and more even distribution of hot and cold air. Further information from the first year of the Geothermal Demonstration Project is included in Exhibit ___(FOH-4).

Q. Why are the Companies seeking to expand the Geothermal Demonstration Project?

A. Based on the project's initial success, the Companies believe geothermal technology has the potential to play an important role in the suite of NPAs called upon to sustainably meet customer heating requirements. The Companies estimate that approximately 200,000 residential buildings are more than 200 feet from the Companies' gas distribution

system and are therefore prospects for an alternative heating solution. By facilitating the installation of geothermal installations in areas of new construction (*i.e.*, areas where the existing gas distribution system is unlikely to expand), as well as converting existing systems, the Companies can provide further customer, system, and environmental benefits.

The Companies also believe that expanding the Geothermal Demonstration Project, in partnership with geothermal installers and other stakeholders, is key to animating the geothermal market, driving further innovation, economies of scale, and reduced costs. The reduction of costs is critically important with the impending expiration of the federal investment tax credit, which currently supports the installation of GHSP systems.

The Companies estimate each geothermal customer could realize an average annual energy cost savings of between $1,000 and $1,500 and nearly 6.75 metric tons of CO_2 emission reductions from the displacement of approximately 800 gallons of heating oil. Finally, geothermal also supports several REV objectives, including greenhouse gas emission reductions in furtherance of the 80x50 goal, and aligns with

the Commission's goal of reducing energy use by five TBtu through the adoption of heat pumps.

Q. How do the Companies propose to expand the Geothermal Demonstration Project?

A. The Companies propose to own and install the ground-loop portion of GSHP system in parts of KEDNY and KEDLI's service territories. In return, geothermal customers would pay the respective company a fee for design work and company-owned equipment installed. The expanded project will prioritize LMI and C&I customers. The Companies expect the ground loop utility-ownership model will prove particularly useful, helping customers to spread costs over time and improving adoption rates.

The goal of the expanded Geothermal Demonstration Project is to help scale the market for this technology, reducing costs as economies of scale are achieved. The Companies would own the ground-loop portion of the GSHP systems and would partner with qualified installers, similar to the Companies' Value-Plus Installers list, that could complete the above-ground/in-home work for customers. In this way, the Companies would assist in the development of the geothermal market. Furthermore, the

Companies would seek to collaborate with industry stakeholders to develop and adopt best practices for installing GSHP ground loops, with specific consideration to how best to operate GSHP systems in an area that has a high saturation rate of natural gas infrastructure. The expanded project is supported by a detailed data sheet included as Exhibit ___(FOH-1), Schedule 6.

Q. What is the cost of the expanded Geothermal Demonstration Project?

A. As shown in Exhibit ___ (FOH-13), the expanded Geothermal Demonstration Project includes incremental O&M costs for the ground loops, marketing, and FTEs who will manage the portfolio of installations. For KEDNY, the projected O&M costs are $0.216 million in the Rate Year, $0.326 million in Data Year 1, $0.500 million in Data Year 2, and $0.63 million in Data Year 3. For KEDLI, the revenue requirement includes incremental O&M costs of $0.980 million in the Rate Year, $1.896 million in Data Year 1, $3.293 million in Data Year 2, and $4.287 million in Data Year 3, as reflected in Exhibit ___ (FOH-13), Schedule 2. The costs include the addition of one FTE in the Rate Year split between KEDNY and KEDLI, increasing to an additional two FTEs in Data Year 2 and Data Year 3 split between KEDNY and KEDLI. The

FTEs will manage the geothermal installations, coordinating RFPs, contacting project coordinators, and ensuring the increased electric load will not cause adverse impacts for the electric utility serving areas where the Companies have installed geothermal systems.

Q. Do the Companies propose to reconcile the revenue requirement amount with the level of actual spending on the project?

A. Yes, to the extent either of the Companies do not spend the rate allowance over the course of the rate plan, the underspend will be deferred for the benefit of customers.

Q. How do the Companies propose to treat any fees received from customers participating in the project?

A. The Companies intend to charge a fixed monthly fee to participating customers (approximately $440 per year), which will offset the cost of the engineering and design work, as well as the portion of the GSHP system owned by either KEDNY or KEDLI. The Companies believe this will be an important component of understanding customers' willingness to pay for this technology. The following forecast revenues for the expanded Geothermal Demonstration Project are included in the revenue

requirement, and offset project costs, as reflected in Exhibit ___ (FOH-13):

Forecast of Expanded Geothermal Demonstration Project Revenues

	Project Participants	KEDNY	KEDLI
Rate Year	75	$3,300	$29,700
Data Year 1	150	$9,900	$89,100
Data Year 2	275	$22,000	$198,000
Data Year 3	400	$39,600	$356,400
Total	**900**	**$74,800**	**$673,200**

Q. Will the Companies continue to report the findings from the expanded Geothermal Demonstration Project?

A. Yes, the Companies will continue to submit quarterly report updates regarding the expanded project. If the project proves successful during the rate period, the Companies will petition the Commission for approval to scale the project and defer additional costs for recovery in the Companies' next rate case.

2. *Clean Conversion Program*

Q. Please discuss KEDLI's Clean Conversions Program.

A. As KEDLI transitions to a clean energy future, it must also fulfill its obligation to serve customers seeking to connect to the gas network. Often, such connections are preferred to the more carbon intensive,

costlier alternatives available. For this reason, KEDLI proposes a Clean Conversion Program, which will replace the existing Neighborhood Expansion Program. Under the Clean Conversion Program KEDLI will educate customers about their energy options using, among other things, the web-based fuel-switching calculator proposed in this case, as well as the RNG benefits available through the Green Gas Tariff offering. For customers whom geothermal or other NPAs are not the preferred option, KEDLI will offer them an opportunity to make an environmentally beneficial decision to convert their heating requirements from oil to natural gas through the Clean Conversion Program. Under the program, KEDLI will extend gas service to such customers under the same density and minimum customer connections requirements established in Case 14-G-0214.

Q. What are the benefits of the Clean Conversion Program approach?

A. The Clean Conversion Program benefits customers by giving them greater control over their energy options. In addition, by converting oil to natural gas, the Clean Conversion Program will enable KEDLI to take additional steps toward meeting the 80x50 target while also displacing significant quantities of heating oil.

Q. What is the cost of the Clean Conversion Program?

A KEDLI proposes $26 million in capital costs for the Clean Conversion Program in the Rate Year, $23 million in Data Year 1, $19 million in Data Year 2, and $16 million in Data Year 3. The costs are supported by the direct testimony of KEDLI's GIOP, as shown in Exhibit ___ (GIOP-1), Schedule 1.

3. *LMI Gas Conversion Program*

Q. Please explain the Companies' LMI Gas Conversion Program.

A. The Companies propose to replace the existing low-income gas conversion rebate program with the LMI Gas Conversion Program. The new program adds an important component to the Companies' Sustainable Heat Initiative by ensuring that LMI customers are not foreclosed, due to high upfront costs, from taking additional control over their energy usage. The program offerings will include products, such as high-efficiency heating and hot water systems, as well as smart thermostats, that will enable customers to save costs and help the environment.

Q. How will the Companies identify customers for purposes of participating in the LMI Gas Conversion Program?

A. The Companies offering will be available to low-income customers who are eligible to participate in the Energy Affordability Program, the criteria for which is further discussed in the direct testimony of the Shared Services Panel. For moderate income customers, the Companies will expand LMI Gas Conversion Program participation to those whose income is 60 percent to 80 percent of the State or area median income, whichever is higher.

Q. Why do the Companies believe the LMI Gas Conversion Program would be beneficial for customers?

A. The program would benefit LMI customers who will realize lower energy bills from a less carbon intensive heating source. Overall, this would improve their energy affordability and their carbon footprint. Installing high efficiency heating and hot water equipment would also allow the Companies to connect customers with other energy efficiency and social services to further decrease their energy burden. Finally, to the extent the LMI Gas Conversion Program retrofits are combined with the Residential Methane Detector Program supported by the Gas Safety Panel, customers will also enjoy an added safety benefit. The BCA for

the LMI Gas Conversion Program is set forth in Exhibit ___ (FOH-2), Schedules 5 and 6.

Q. What are the projected costs for the proposed LMI Gas Conversion Program?

A. As shown in Exhibit ___ (FOH-13), Schedule 1, KEDNY's projected costs are $1.6 million for the Rate Year and the Data Years. For KEDLI, the projected costs are $1.8 million for the Rate Year and the Data Years, as shown in Exhibit ___ (FOH-13), Schedule 2.

E. Economic Development

1. *Grant Programs*

Q. Please describe the current economic development grant programs in the KEDNY and KEDLI service territories.

A. KEDNY and KEDLI currently offer identical portfolios of eight economic development grant programs in each of their respective service territories. Implemented in the 2016 KEDNY and KEDLI Rate Cases, these programs provide financial grants and incentives to customers to meet the following economic development objectives:

1. Urban Revitalization - the Cinderella program supports the redevelopment of vacant commercial and mixed-use buildings and revitalization of distressed commercial districts;
2. Infrastructure Assistance – the Capital Investment Incentive program offsets the cost of upgrading gas infrastructure for growing C&I customers;
3. Brownfield Redevelopment – supports remediation and redevelopment of brownfield sites and abandoned buildings in the service territories;
4. Industrial Building Redevelopment – funds infrastructure to redevelop large, vacant or underutilized industrial properties into multi-tenant commercial space;
5. Sustainable Gas and Economic Development – provides funding for demonstration projects involving sustainable gas technologies that generate regional economic development benefits;
6. Manufacturing Productivity – funds productivity improvement and efficiencies for small- and medium-sized manufactures that use gas in their processes;
7. Cleantech Incubation – provides financial support for clean energy and other high-tech businesses; and

8. Cooperative Business Recruitment – provides grants to promote new business investment and jobs in the Companies' service territories.

Q. What is the current level of funding for these grant programs?

A. The current level of funding is $2 million per year for each of the Companies.

Q. How much grant funding has been awarded to customers thus far during the current rate plan, and how much funding has been committed through approved applications?

A. Through February 28, 2019, a total of $0.193 million has been awarded to customers, $0.068 million for KEDNY and $0.125 million for KEDLI. An additional $1.990 million has been committed through approved applications, $1.550 million for KEDNY and $0.440 million for KEDLI.

Q. Please describe the current treatment of economic development grant expenditures compared to those allowed in base rates.

A. For both KEDNY and KEDLI, actual grant expenditures are reconciled against the amounts included in rates, subject to a downward only reconciliation. Any under expenditures in a given year are carried forward

and reconciled at the end of KEDNY and KEDLI's current three-year rate plan, which runs from January 1, 2017 through December 31, 2019 ("KEDNY and KEDLI Rate Plan").

Q. What is the forecast of economic development grant expenditures through the end of the KEDNY and KEDLI Rate Plan?

A. KEDNY and KEDLI project an accumulated under expenditure of $4.7 million and $5.2 million, respectively.

Q. What factors are responsible for the underspend?

A. The Companies' portfolio of grant programs was modified and expanded significantly in CY 2017. Seven new programs were implemented for both KEDNY and KEDLI, and the eighth program was modified significantly. While the Companies anticipated a gradual increase in grant awards for the new programs due to their dramatically different scope and scale, the ramp-up in activity has been slower than anticipated, particularly during CY 2017, the first year of program implementation. It has taken longer than expected to build customer and stakeholder awareness of the new programs, many applications have taken longer than expected to be developed and submitted by the applicant, and many of the approved applications have involved projects with relatively long development and

construction timelines. Lastly, the eligibility requirements for some of the new programs have proven more restrictive than anticipated, resulting in a lower than expected number of qualifying applicants.

Q. Are the Companies proposing any modifications to its existing portfolio of grant programs?

A. Yes. The Companies are proposing to create a new grant program that would encourage customers and developers to invest in projects involving alternatives to traditional gas delivery service and efficient, emerging natural gas technologies. Eligible technologies would include, but not be limited to, geothermal, solar thermal, thermal storage, combined heat and power ("CHP"), fuel cells, biomass, microgrids, anaerobic digestion, and thermal cooling/chilling. Eligible projects would also be required to demonstrate regional economic impacts through the expansion or attraction of businesses in the Companies' service territories. Applicants will also be encouraged to use self-generated or purchased RNG to meet some or all their gas supply requirements. The proposed program is intended to support the Companies' larger Future of Heat initiatives, strengthen the market for RNG supply, and generate regional economic development benefits in the form of new or retained jobs and new capital investment.

Additionally, KEDNY and KEDLI are proposing modifications to several of their existing Economic Development grant programs to make them more responsive to customer needs and more compatible with the clean energy economy. For example, the Companies propose to expand eligibility for the Cinderella Program to include partially occupied buildings under specific and limited circumstances. The Companies further propose to modify the eligibility requirements for the Capital Investment Incentive program to allow additional customer classifications and types of gas infrastructure investment to be eligible for funding. Finally, KEDNY and KEDLI propose to more closely align the existing programs with the broader Future of Heat proposal by adding funding and eligibility guidelines that would encourage the use of RNG and strengthen the connection between energy efficiency and economic development. A detailed description of each program, including the proposed modifications, is set forth in Exhibit ___ (FOH-7).

Q. Will these proposed changes increase the level of customer participation in the Economic Development grant programs?

A. Yes. The Companies expect the changes to drive increased levels of activity as measured by applications submitted, applications approved, and grant funding reimbursed to customers. A forecast of annual grant

expenditures compared to the proposed levels of funding over the four-year rate term is set forth in Exhibit ___ (FOH-8).

Q. Are the Companies proposing to adjust the level of funding for economic development grant programs?

A. Yes. As set forth in Exhibit ___ (FOH-13), KEDNY and KEDLI propose to reduce the level of incremental grant program funding to $0.500 million for each company in the Rate Year, and to increase that amount by $0.500 million in each of the Data Years, with the revenue requirement including $1.0 million for each company in Data Year 1, $1.5 million for each company in Data Year 2, and $2.0 million for each company in Data Year 3.

Q. Given the proposed reduction in incremental funding and the anticipated increase in grant program activity, how do KEDNY and KEDLI propose to maintain adequate funding for the programs?

A. The Companies believe adequate funding is available from the initial carryover balances ($4.7 million for KEDNY and $5.2 million for KEDLI), the proposed incremental funding discussed above, and the continuation of the existing deferral mechanism for reconciling and recovering grant program expenditures. Exhibit ___ (FOH-8) further sets

forth the proposed grant program funding in the Rate Year and the three Data Years, along with projected program spending in each year.

Q. Please describe the deferral mechanism.

A. Under the deferral mechanism, actual grant expenditures would continue to be reconciled against the amounts included in rates, subject to a downward only reconciliation. Any under expenditures would be carried forward and reconciled at the end of Data Year 3. In the event of any anticipated over expenditures at the end of Data Year 3, the Companies would petition the Commission for deferral treatment of such amounts.

2. *Discount Programs*

Q. Please describe the Companies' current economic development discount programs.

A. KEDNY offers a Business Incentive Rate ("BIR") under which eligible new and expanding customers receive a distribution rate discount of 50 percent on eligible incremental gas usage for a twelve-year period, after which time the discount is phased out over an additional three-year period. Among the criteria that may qualify a customer for BIR discounts is receipt of benefits through an Industrial Development Agency ("IDA"). KEDNY also offers an Excelsior Jobs Program, under which certified

participants in the New York State Excelsior Jobs program receive a marginal cost-based delivery rate (currently $0.104/therm) on eligible incremental gas usage for up to ten years. Lastly, KEDNY offers an Area Development Rate which provides distribution rate discounts of 35 percent for a five-year period to customers that locate or expand in specific development areas of the KEDNY service territory.

KEDLI offers a BIR under which eligible new and expanding customers receive a distribution rate discount of 35 percent for a seven-year period, after which time the discount is phased out over an additional three-year period. KEDLI also offers an Excelsior Jobs Program, under which certified participants in the New York State Excelsior Jobs program receive a marginal cost-based delivery rate (currently $0.184/therm) on eligible incremental gas usage for up to ten years.

Q. Please describe the recent and current levels of customer participation in KEDNY and KEDLI's economic development discount programs.

A. Customer participation in the discount programs has generally been flat in both KEDNY and KEDLI's service territories, as measured by the number of customers participating and total discounts provided. The KEDNY

Area Development Rate has become virtually inactive, with only two participants remaining in the program.

Q. Why has the KEDNY Area Development Rate become inactive?

A. The development areas targeted by the KEDNY Area Development Rate were identified more than 25 years ago, when those areas of the KEDNY's service territory were considered economically depressed. Since that time, the targeted areas have changed dramatically in terms of their level of economic activity and their need for development incentives such as the Area Development Rate.

Q. Are the Companies proposing any changes to their existing portfolio of economic development discount programs?

A. Yes. KEDNY is proposing to modify its Area Development Rate to remove the current targeted areas from program eligibility and replace them with areas that better reflect the current economic development priorities and objectives of NYC and New York State. The proposed targeted areas are: (i) Industrial Business Zones ("IBZs") that were established by NYC to protect existing manufacturing districts and encourage industrial growth by offering expanded services to industrial and manufacturing businesses within the designated IBZs, 15 of which are

located in KEDNY's service territory; (ii) State Brownfield Opportunity Areas ("BOAs"), which are designated by the New York Department of State as areas where municipalities and community-based organizations are eligible for financial and technical assistance to address the presence of brownfield sites in the BOA, eleven of which are located in the KEDNY and KEDLI service territories; and (iii) federal Opportunity Zones recently nominated by the State pursuant to the U.S. Department of Treasury's Opportunity Zones Program that are designed to spur investment and create jobs in distressed communities through tax benefits for investments located in the zones, which include census tracts in the Companies' service territories.

KEDLI is proposing to create a similar Area Development Rate, which would also rely on the same targeted areas that have already been designated as economic development priorities through federal and state development programs.

Lastly, KEDNY and KEDLI propose that eligibility for the BIRs be broadened to include qualifying tenant businesses locating or expanding in properties receiving benefits from an IDA. Currently, the BIR is available

only to the IDA beneficiaries themselves, and not to growing customers who are tenants within the designated properties.

Q. How are the Companies proposing to recover discounts provided to customers?

A. The Companies propose to implement a deferral mechanism whereby actual discounts are reconciled against targets (forecasts) established in rates. Each year of the Rate Plan, actual economic development discounts would be reconciled to the amount reflected in rates for refund to or recovery from customers. An example of the proposed mechanism is set forth in Exhibit ___ (RRP-9), Schedule 1.

Q. How would gas customers benefit from the Companies' proposed changes to its economic development grant and discount programs?

A. The proposed revisions to the Companies' economic development grant and discount programs will make them more responsive to customer needs and will generate benefits in the form of new and retained jobs, new capital investment, and higher regional earnings in the communities the Companies serve. The proposed modifications will also better align the KEDNY and KEDLI programs with New York State economic development policy, while also supporting the Companies' broader

efforts to transition its service territories economy to a cleaner energy future.

V. Reducing Carbon Emissions on the Gas Network

Q. Please explain why the reduction of carbon emissions is an important part of the Future of Heat strategy?

A. The Companies believe a holistic approach is necessary to drive meaningful change toward a low-carbon future. This approach includes the aforementioned focus on network efficiency (*e.g.*, reducing system leaks) and demand-side energy management (*e.g.*, energy efficiency), as well as supply-side products capable of delivering carbon reductions on the gas and electric networks. As set forth below, the Companies intend to accomplish this through the production and distribution of RNG, as well as continued RD&D to drive further improvements in technology and lower costs for customers.

By including gas supply in their efforts to develop the low-carbon energy system of the future, the Companies can provide customers access to clean-energy choices that do not require deep retrofits or lifestyle changes. Moreover, adding the supply-side component allows the Companies and local communities to beneficially use biogas created by

wastewater treatment and food waste, mitigating some of the environmental impacts of these waste streams. Finally, with the development of P2G, the Companies can combine complementary electric and gas technologies to enhance the environmental benefits for both energy sectors through the production of lower-carbon RNG and by using the gas network to store excess renewable energy.

A. Renewable Natural Gas ("RNG")

Q. What is RNG?

A. RNG is a term generally used to describe pipeline compatible gaseous fuel derived from biomass or other renewable sources that has lower lifecycle CO_{2e} emissions than geological natural gas. RNG feedstocks include manure, food waste, wastewater treatment plants, or other biomass sources, often using an anaerobic digester. With recent advancements to lower the cost of gasification technology, feedstocks with lower moisture content can also be used to produce RNG (*e.g.*, municipal solid waste or agricultural residues). Furthermore, with new technological innovations, production of RNG is moving beyond biomass to include renewable electricity, often referred to as power-to-gas or P2G. This concept includes either adding hydrogen to the existing gas system (*i.e.*, hydrogen blending) or producing synthetic methane by

combining hydrogen and carbon dioxide. Collectively, RNG offers new ways to decarbonize the gas network by reducing the carbon footprint of the fuel supply in a manner similar to the way solar and wind technology reduce the carbon footprint of electricity.

Q. Please describe how the Companies have used RNG.

A. The Companies have over 30 years of experience integrating RNG into the gas distribution network, starting with the Staten Island Landfill project. The Staten Island project – the oldest operating RNG facility in the U.S. – continues to contribute RNG to National Grid's distribution network. In addition, since 2009, the Companies have partnered with NYC to deliver RNG from the City's largest wastewater treatment plant. In doing so, the Companies and NYC seek to animate the RNG market and highlight opportunities for expanded use of this valuable energy resource. In 2010, National Grid published a white paper outlining the potential for RNG to make a significant contribution to the reduction of greenhouse gas emissions. The following year National Grid advised the American Gas Foundation on a national study, assessing the potential from RNG from biomass. More recently, the Companies collaborated on a New York Standard Interconnection Guideline to establish a clear

process for both project developers and utilities to successfully connect RNG projects.

Q. Please discuss the Companies' RNG proposals.

A. KEDNY is continuing to progress the Newtown Creek Project ("Newtown Project"), and KEDLI proposes a new P2G Demonstration Project ("P2G Project"). The Newtown Project is supported by the data sheet summary contained in Exhibit ___ (FOH-5), and the P2G Project is supported by the data sheet summary contained in Exhibit ___(FOH-6). The Companies also jointly propose a hydrogen blending study in partnership with I-GIT, and, as set forth in the GIOP testimony, they are seeking to facilitate RNG interconnections by reducing upfront costs and leveraging their technical expertise and economies of scale to own and install portions of RNG systems that are core utility functions and in doing so lower the barrier of entry for RNG developers.

1. *Newtown Creek Project*

Q. Please describe the Newtown Project and its current rate treatment.

A. The Newtown Project, located at the New York City Newtown Creek Wastewater Treatment Plant, seeks to reduce greenhouse gas emissions by promoting RNG as a long-term supply source. KEDNY will capture the biogas generated from the wastewater plant, which consists of approximately 60 percent methane and 40 percent CO_2, filter it through a purification facility and upgrade it to pipeline quality natural gas. The RNG is then odorized and injected into the gas system for direct use by customers.

Because the in-service date and capital costs were not known at the time rates were set in the 2016 KEDNY and KEDLI Rate Cases, the costs for the Newtown Creek project are not currently included in rates. However, the Joint Proposal authorizes KEDNY to defer the return of and on the Newtown Project once it was placed in service. In addition, the Joint Proposal provides that the revenue requirement in a future rate case would be subject to a potential $1.6 million annual exclusion (prorated from the in-service date) for 20 years and reflect a 40-year depreciable life. However, the potential exclusion would be reviewed for possible modification, with any property tax abatement or revenues from the sale

of gas or environmental credits produced by the Newtown Creek project used as potential offsets to the exclusion.

Q. What are the benefits associated with the Newtown Project, and how do those benefits align with the State's clean energy goals?

A. The Newtown Project will expand the concept of renewable energy to more fully develop RNG as a viable alternative to conventional natural gas resources. In doing so, the RNG will enhance the diversity of the gas supply, provide a new method for managing local waste resources, and reduce greenhouse gas emissions in furtherance of the State's clean energy policy goals and the REV objectives. In total, KEDNY anticipates the Newtown Project will produce approximately 277,500 Dth of pipeline quality RNG, reducing CO_2 emissions by approximately 16,000 tons annually. In a separate partnership, NYC is working with Waste Management to incorporate processed food waste into the wastewater sludge at Newtown Creek. The combined projects have the potential to reduce annual CO_2 emissions by 90,000 metric tons.

Q. What is the status of the Newtown Project?

A. By June 2018, KEDNY secured all necessary permits and subsequently began project construction in July. Those efforts are currently on

schedule, and KEDNY expects to bring the project online in November 2019. In addition to the construction activities, KEDNY issued a request for proposal ("RFP") in 2018, seeking a firm to monetize the environmental attributes KEDNY expects to generate from the injection of RNG into the local distribution system. KEDNY selected Element Markets to complete this task. KEDNY and Element Markets are currently collaborating to register the project under the federal Renewable Fuel Standard ("RFS") Program as a generator of Renewable Identification Numbers ("RINs").

Q. Is KEDNY proposing to include the costs of the Newtown Project in rates?

A. Yes. As shown in Exhibit ___ (RRP-7), Schedule 1, KEDNY expects the Newtown Project will be in service by the end of CY 2019 with a total capital cost of approximately $32 million. This includes $18.911 million in construction work in progress, as of December 31, 2018, as well as $3.929 million in the remainder of FY 2019, $9.099 million incurred in FY 2020, and $0.1 million in the Rate Year. The capital costs include design work, construction, and equipment. As further set forth in the direct testimonies of the Shared Services Panel and the Revenue Requirements Panel, the revenue requirement also includes the property

tax net of the abatement, as well as a 20-year depreciation life, respectively.

In addition, as shown in Exhibit ___ (FOH-13), Schedule 1, the revenue requirement includes O&M costs of $0.668 million in the Rate Year inflated through Data Year 3. The O&M costs include the amounts necessary to run the RNG upgrading facility, the primary component of which is the electricity costs, as well as amounts to purchase replacement parts and perform annual maintenance

Q. Does KEDNY expect to generate revenue from the Newtown Project?

A. Yes. KEDNY anticipates revenue both from the sale of RNG and from monetizing the project's environmental attributes.

Q. How does KEDNY propose to treat the project revenues?

A. As set forth in the Rate Design Panel testimony, KEDNY developed a revenue mechanism to offset project costs through the sale of RNG produced from the project, as well as any revenues realized from monetizing the associated environmental attributes. Specifically, Exhibit ___ (RDP-2) includes a revenue estimate of $1.9 million per year,

comprised of $1.0 million from the gas sales and $0.9 million from the sale of environmental attributes.

KEDNY proposes to true-up the actual revenues, such that any variances will be deferred for future refund to (or recovery from) KEDNY customers. Any revenues realized above the level necessary to fully reimburse customers for the project costs will be shared evenly between customers and NYC beginning in the fifth year after the project becomes operational. Revenue sharing will then be assessed every year thereafter for the remainder of the project. KEDNY will track these costs on an annual basis and compare to the project's cumulative revenue requirement.

Q. How did KEDNY calculate the estimated value of the environmental attributes included in the Rate Design Panel's revenue offset mechanism?

A. KEDNY calculated the value of the environmental attributes by evaluating the primary markets for RNG credits: (i) the federal RFS; and (ii) California's Low Carbon Fuel Standard ("LCFS"). In 2019 the value of RNG from food waste (classified as D5 under the RFS) is roughly $4/Dth, and the value of RNG from wastewater (classified as D3) is roughly $26/Dth. These values, however, vary due to market changes, and

KEDNY expects significant variability in the market values over time. The LCFS currently provides a financial incentive of approximately $5.70/Dth, which can be additive to the RFS value if the RNG is ultimately sold to a transportation customer in California. Based on this review, KEDNY conservatively estimates it will receive $0.900 million per year in revenue from marketing the attributes. The actual revenue generated, however, is highly dependent on the volume of biogas produced, what proportion of the volume is designated as D3 or D5, and the price volatility of the RFS and LCFS markets.

Q. Does KEDNY believe that circumstances warrant elimination of the $1.6 million exclusion in the Joint Proposal?

A. Yes. At the time the exclusion was established, there was some uncertainty with respect to the potential benefits to customers from the Newtown Project. As discussed above, the project is one of the first in the U.S. to directly inject RNG into a local distribution system, helping decarbonize the gas supply and meet New York State's clean energy goals. Between the property tax exemption KEDNY obtained from NYC (discussed in the Shared Services Panel's testimony) and the forecast revenues set forth above, the project will deliver monetary and societal

benefits in excess of the $1.6 million exclusion. Therefore, the Companies believe the exclusion is not warranted.

Q. How has the Newtown Project supported RNG development?

A. KEDNY's partnership with NYC on the Newtown Project was one of the first of its kind in the U.S. Although the partnership began with a focus on helping NYC meet its environmental goals, it is also helping to demonstrate how energy providers and other stakeholders can collaborate to incorporate RNG into the gas supply portfolio to achieve clean energy goals, enhance reliability, and meet growing energy demands. The Companies' have already started using lessons learned from the development of the project to educate other utilities, RNG project developers, and internal teams on how to plan, develop, and incorporate RNG resources into the natural gas distribution system.

2. *P2G Demonstration Project ("P2G Project")*

Q. Please describe what is meant by the term "power-to-gas."

A. P2G refers to the technical and economic potential of converting excess renewable electricity to hydrogen or synthetic methane (*i.e.*, RNG) and utilizing the existing natural gas network to deliver the gas produced using these renewable resources. Indeed, P2G can serve as a form of

large-scale, long-duration energy storage when used to convert excess renewable electricity that would otherwise be curtailed – a situation that is expected to occur with increasing interconnection of renewable resources including large-scale offshore wind – is converted into RNG. P2G can also provide low- or zero-carbon RNG, depending on the feedstocks used for production. The technology holds considerable promise for addressing clean-energy goals, as it has the potential to support deep decarbonization of the transportation and heating sector, which are two sectors of the economy that have proven challenging to decarbonize.

Q. Please describe the P2G Project.

A. KEDLI proposes to develop a P2G design that combines existing hydrogen production technology (*i.e.*, an electrolyzer) and cutting-edge methanation technology (*i.e.*, a bioreactor) to produce pipeline-quality RNG capable of meeting gas system requirements. In doing so, KEDLI seeks to partner with federal and local government (*i.e.*, NREL and NYC, respectively), as well as industry collaborators, such as Electochaea GmbH, an innovative technology provider that develops bioreactors, to design and engineer the P2G Project.

Q. What are the benefits associated with the proposed P2G Project, and how do those benefits align with the Commission's REV policy objectives and the State's clean energy goals?

A. Combining electolyzers and bioreactors to produce pipeline quality RNG from renewable energy will help KEDLI lower carbon emissions, reduce gas constraints, and provides a key outlet for excess renewable energy. This last piece, the potential to store renewable energy, has taken on growing importance with the accelerated offshore wind targets and on-shore wind and solar generation goals announced by Governor Cuomo in the 2019 State-of-the-State address. The last piece is key, as use of renewable electricity to produce methane will prove an increasingly valuable method of energy storage with the proliferation of distributed generation and large-scale renewable energy projects. In effect, the gas distribution system will store excess renewable electricity that would otherwise be curtailed. Moreover, the RNG produced through the P2G process will help offset sectors, such as heavy transportation and industry, which have proven difficult to decarbonize.

Q. What is the P2G Project schedule?

A. KEDLI will begin scoping, design, and engineering of the P2G Project in 2019, including the execution of agreements with project partners and

site owners. KEDLI further anticipates full project design will be completed in October 2023. With design at 100 percent, KEDLI will submit a proposal for recovery of the full project costs.

Q. What is the projected cost of the P2G Project?

A. As shown in Exhibit ___ (FOH-13), Schedule 2, the KEDLI revenue requirement for the P2G Project includes O&M costs of $0.650 million in the Rate Year, $1.825 million in Data Year 1, $0.725 million in Data Year 2, and $0.325 million in Data Year 3. The O&M amounts reflect KEDLI's forecast for scoping, designing, and engineering the P2G Project. If the project scoping, design, and engineering is complete prior to KEDLI's next rate case, the company will petition the Commission for approval to move forward with project construction, deferring the revenue requirement associated with any costs that exceed the $3.525 million proposed here, for recovery in KEDLI's next rate case. Any such costs that receive deferral treatment would be excluded from KEDLI's net plant tracker.

3. *Hydrogen Blending*

Q. What is hydrogen blending?

A. Hydrogen blending involves adding hydrogen into the gas system to augment natural gas supplies. The hydrogen is either produced from water electrolysis using excess renewable electricity or from natural gas, a process referred to as steam methane reforming.

Q. Why do the Companies believe hydrogen blending is an important part of the strategy for the low-carbon future of the heating sector?

A. Hydrogen blending holds promise for the future of the heating sector for three reasons. First, combusting hydrogen can have zero carbon emissions if produced from renewable electricity. Second, using hydrogen blending as a low-carbon supply-side resource may prove to be an efficient way to rapidly lower the carbon intensity of end-uses fueled by natural gas without requiring the replacement of appliances or performing extensive building retrofits. Third, blending hydrogen with natural gas, like P2G, could serve as a method for storing excess energy generated by anticipated increases in renewable generation from solar and wind; effectively reducing carbon emissions while bolstering gas supplies.

Q. Please describe the Companies' hydrogen blending project.

A. The Companies propose a hydrogen blending study in partnership with I-GIT. The purpose of the study is to further analyze the impact of blending hydrogen into the natural gas system, to determine the percentage of hydrogen that can be safely blended into the Companies' gas distribution systems without negatively impacting the natural gas network or customer appliances. The Companies and I-GIT will evaluate the technology and risks associated with hydrogen blending, establish safety protocols, analyze the chemical changes to pipeline materials created by different mixtures of hydrogen, as well as the operational and engineering issues created by hydrogen blending. The Companies believe the hydrogen blending study is a necessary precursor to the development of a hydrogen blending demonstration project.

Q. What is the cost of the hydrogen blending study?

A. The Companies expect the hydrogen blending study to cost $0.446 million, split evenly between KEDNY and KEDLI to cover I-GIT's research work. To offset a portion of that amount, the Companies and I-GIT submitted a concept paper to NYSERDA in response to Program Opportunity Notice 3249. NYSERDA accepted the concept paper, and the Companies, along with I-GIT, submitted a full proposal for

NYSERDA's consideration in April. If awarded the grant, it could cover approximately $0.346 million of the study costs. Of the $0.100 million that would remain, $0.070 would be incremental to the Historic Test Year, and would be split evenly between the Companies, as shown in Exhibit ___ (FOH-13).

4. *RNG Interconnections*

Q. Please describe the Companies' RNG interconnection proposal.

A. As set forth in KEDNY and KEDLI's GIOP testimonies, the Companies propose to leverage their technical expertise and economies of scale to facilitate RNG interconnections by installing and owning parts of distributed RNG facilities. Like most renewable energy development, RNG projects require large upfront capital investments, which can prove an impediment to adoption. A portion of the upfront capital needed for RNG projects is due to the Companies' equipment and engineering requirements, which were established to ensure safe interconnection of RNG projects. To make it easier for customers to install low-carbon RNG resources, the Companies propose engineering, installing, and owning pieces of RNG interconnection equipment (*e.g.*, meters, odorizors, and spectrometers) at project sites.

The Companies are also focused on shortening the interconnection approval period to make RNG development a more efficient process for project developers. To that end, the Companies propose to create a new Future of Heat Engineering department that will include additional FTEs to review and process interconnection applications. These additional FTEs are discussed more fully below. The Companies are currently in discussions with six project developers looking to interconnect such distributed RNG facilities to serve KEDNY and KEDLI customers with feedstocks including municipal solid waste, food waste, and wastewater.

5. *Future of Heat Engineering Group*

Q. Do the Companies propose any FTEs to implement the gas decarbonization strategy?

A. Yes. As mentioned above, the Companies propose to establish a new Future of Heat Engineering Group under the Asset Management portion of the Gas Business Unit. The new group would consist of five incremental FTEs – a director and four engineers – with two-thirds of the directors' time and 100 percent of the engineers' time evenly split among KEDNY and KEDLI.

Q. Please describe the roles and responsibilities of the Future of Heat Engineering Group.

A. As the Companies shift their efforts to meeting heating needs through a decarbonized gas supply and to meet aggressive clean energy goals using NPAs and RNG projects, the Companies anticipate a need for additional FTEs to manage NPA market solicitations, timely review project proposals, and coordinate with project partners and other interested stakeholders. The new Future of Heat Engineering Group will focus on understanding and evaluating RNG, hydrogen blending, and NPA solicitations to determine whether they are cost-effective, viable solutions, and, in some instances, whether they are capable of displacing traditional capital projects.

Q. What are the costs of the new group?

A. As shown in Exhibit ___ (FOH-13), the costs of the new group include incremental O&M of $0.423 million for KEDNY and $0.313 million for KEDLI in the Rate Year. In the Data Years, KEDNY would incur $0.540 million in Data Year 1, $0.660 million in Data Year 2, and $0.776 million in Data Year 3. For KEDLI, the revenue requirement includes $0.374 million in Data Year 1, $0.438 million in Data Year 2, and $0.497 million in Data Year 3.

B. Research, Development, and Demonstration ("RD&D")

Q. Do the Companies' participate in any end-use RD&D efforts?

A. Yes. The Companies are active participants in two end-use RD&D programs: (i) the Gas Technology Institute's ("GTI") Utilization Technology Development ("UTD") program; and (ii) I-GIT, which is a unit of the Advanced Energy Research and Technology Center at Stony Brook University.

Q. Please describe the Companies participation in the UTD program.

A. The UTD program is a utility-funded collaborative with 20 members across North America. It supports research, analysis, and deployment of advanced technologies that are increasingly efficient and environmentally beneficial, such as thermal heat pumps that cost-effectively reduce peak and greenhouse gas emissions. Member utilities may allocate their contributions to specific projects, such as thermal heat pump research, that support their objectives.

National Grid, for example, is the product champion for the Thermolift, which is a client of the Clean Energy Business Incubator Program at Stony Brook University. The Thermolift research focuses on developing a thermal heat pump capable of using RNG or solar to meet heating

needs. Thus far, the Thermolift heat pump has exceeded its testing goals, and drawn support from the NREL and NYSERDA. Alternatively, UTD participants may make a block contribution, known as the Sustaining Members program, to a research portfolio determined by GTI. Collectively, the UTD research is supporting long-term projects that enable the use of RNG, hydrogen, and hydrogen blends. The technology investigated by the UTD collaborative includes items not eligible for support from the Millennium Fund, or which are otherwise unsuitable for funding by manufacturers.

Q. Are there any costs associated with the Companies participation in the UTD program?

A. Yes, as reflected in the Historic Test Year, the Companies each pay $0.175 million in annual costs to participate in the UTD program. The Companies propose to continue their membership in the UTD program at that level.

Q. Please describe the Companies participation in I-GIT.

A. The Companies, Stony Brook University, and other academic and industry leaders launched I-GIT in 2018 as a comprehensive initiative focused on finding clean, affordable energy solutions capable of meeting growing clean energy demands. I-GIT research goes beyond scientific

research, focusing on business and policy objectives related to the future of the heating sector. This work includes using I-GIT's innovative research, analysis, and education platform to accelerate deployment of advanced energy technologies and infrastructure that will provide community residents and businesses with value-added gas services.

Q. What work has I-GIT accomplished in its first year of operation?

A. Since its launch in 2018, I-GIT supported development RNG projects with dairy farmers, the utilization of RNG at State facilities, and the integration of hydrogen into gas distribution systems. In addition, I-GIT reviewed the Companies gas REV demonstration projects.

Q. Are there any costs associated with the Companies participation in I-GIT?

A. Yes. As shown in Exhibit ___ (FOH-13), the KEDNY and KEDLI revenue requirements include $0.050 million in annual costs for the I-GIT program.

Q. Do the Companies propose any non-end use operational RD&D?

A. Yes, the Companies' request for non-end use operational RD&D is supported by KEDNY and KEDLI's GIOP testimonies.

VI. Performance-Based Incentives and Revenue Sharing

A. EAMs

Q. Please summarize the Companies' EAM proposals.

A. The Companies propose the following EAMs, which are designed to support the Commission's REV objectives and the State's clean energy goals:

(i) System Efficiency;

(ii) Energy Efficiency; and

(iii) Carbon Reduction.

Q. Please describe how the Companies developed their EAM proposal.

A. The Companies EAM proposals are intended to encourage achievement of priorities like system efficiency, while also identifying unique ways to engage customers. The proposals focus on ways to achieve the State's clean energy goals, the Commission's REV objectives, and the Companies' commitment to sustainably and affordably meeting customers' heating requirements. To that end, the Companies identified a broad array of initiatives and EAM metrics that align with the Companies' strategic focus. The Companies then set targets for those metrics at levels that require incremental effort and robust customer adoption to achieve. Based on the costs and benefits of achieving the targets, the Companies

assigned basis points to the proposals. The resulting EAM metrics and basis points collectively establish meaningful incentives for the Companies. Summary tables of the proposed targets and associated basis points are included in Exhibit __ (FOH-10).

Q. How does KEDNY propose to allocate basis points across the three EAMs?

A. KEDNY proposes allocating the basis points as follows:

KEDNY EAM Maximum Incentive Basis Points (bps)				
EAMs and Associated Metrics	**2020**	**2021**	**2022**	**2023**
System Efficiency	25	25	25	25
Peak Reduction	5	5	5	5
Supply Diversification	20	20	20	20
Energy Efficiency	50	30	30	30
Incremental Energy Efficiency	40	20	20	20
Affordability*	10	10	10	10
Carbon Reduction	3	8	8	8
CO_{2e} Reduction – Full Service	1	3	3	3
CO_{2e} Reduction – Transportation	2	5	5	5
Total	78	63	63	63

*Basis points for the Affordability metric are estimated, as the metric is calculated based on the level of incremental achievement.

Q. How does KEDLI propose to allocate basis points across the three EAMs?

A. KEDLI proposes allocating the basis points as follows:

KEDLI EAM Maximum Incentive Basis Points (bps)

EAMs and Associated Metrics	2020	2021	2022	2023
System Efficiency	23	23	23	23
Peak Reduction	3	3	3	3
Supply Diversification	20	20	20	20
Energy Efficiency	50	30	30	30
Incremental Energy Efficiency	40	20	20	20
Affordability*	10	10	10	10
Carbon Reduction	4	9	10	11
CO_{2e} Reduction – Full Service	2	4	5	6
CO_{2e} Reduction – Transportation	2	5	5	5
Total	77	62	63	64

**Basis points for the Affordability metric are estimated, as the metric is calculated based on the level of incremental achievement.

Q. Do the Companies propose the EAMs become effective in CY 2020?

A. Yes. The Companies believe the EAMs should be measured on a calendar year basis, beginning in CY 2020, similar to how the Customer Service Quality and Gas Safety Performance metrics are measured. CY treatment would also align with NMPC's EAMs.

1. System Efficiency

Q. Please describe the Companies' proposed System Efficiency EAM.

A. The proposed System Efficiency EAM is composed of two metrics: (i) Peak Reduction; and (ii) Supply Diversification. The intent of the Peak Reduction metric is to encourage gas demand reduction during peak events, which will enable the Companies' to focus commodity and resources on constrained portions of the network. The Supply

Diversification metric will promote incremental development of local pipeline quality gas (*i.e.*, RNG). Increasing RNG interconnections diversifies the Companies' supply portfolio to include local sources, resulting in a more resilient system.

a. Peak Reduction

Q. Please explain the Peak Reduction metric.

A. The Companies' proposed Peak Reduction metric will be calculated by measuring reduction of gas usage during peak events and averaging the reduction across all peak events called during the winter season as part of the Companies' expanded Demand Response Demonstration Project. Reductions will be measured at participating customer sites, benchmarked against customers' weather-normalized baselines, and aggregated.

Q. What does KEDNY propose for its Peak Reduction targets and associated basis points?

A. KEDNY proposes the following net peak reduction targets and associated basis points for achieving the targets:

KEDNY Annual Peak Reduction Targets (average Dth/hour/event) and Basis Points (bps)					
	2020	**2021**	**2022**	**2023**	**Basis Points**
Minimum	510	638	766	893	1
Target	561	702	842	983	3
Maximum	612	766	919	1,072	5

Q. What does KEDLI propose for its Peak Reduction targets and associated basis points?

A. KEDLI proposes the following net peak reduction targets and associated basis points for achieving the targets:

KEDLI Annual Peak Reduction Targets (average Dth/hour/ event) and Basis Points (bps)					
	2020	**2021**	**2022**	**2023**	**Basis Points**
Minimum	232	290	348	406	1
Target	255	319	383	447	2
Maximum	278	348	418	487	3

Q. How did the Companies establish the targets?

A. The Companies used forecasts for the expanded Demand Response Demonstration Project as a starting point for the Peak Reduction targets, setting the minimum targets equivalent to the forecasts. Midpoint and maximum targets reflect incremental effort required to exceed the design of the expanded demonstration, which may be achieved by increasing customer enrollment and lowering program costs per Dth of reduced gas

consumption. The midpoint and maximum targets assume additional growth beyond the base levels of approximately 10 percent and 20 percent, respectively.

b. Supply Diversification.

Q. Please describe the proposed Supply Diversification metric.

A. The proposed outcome-based Supply Diversification metric will measure the sum of incremental RNG introduced into the natural gas network in the Companies' service territories each year. To achieve the targets, the Companies expect to work in collaboration with third parties to increase RNG development through initiatives such as the New York Standard Interconnection Guideline for RNG projects and the Companies' RNG interconnection proposal, which is supported by KEDNY and KEDLI's GIOP testimonies.

Q. What does KEDNY propose for its Supply Diversification targets and associated basis points?

A. KEDNY proposes the following annual Supply Diversification targets and associated basis points for achieving the targets:

KEDNY Supply Diversification Targets (incremental Dth/yr.) and Basis Points (bps)					
	2020	**2021**	**2022**	**2023**	**Basis Points**
Minimum	108,047	125,934	142,003	156,097	4
Target	270,118	314,836	355,007	390,244	10
Maximum	540,237	629,672	710,015	780,487	20

Q. What does KEDLI propose for its Supply Diversification targets and associated basis points?

A. KEDLI proposes the following annual Supply Diversification targets and associated basis points for achieving the targets:

KEDLI Supply Diversification Targets (incremental Dth/yr.) and Basis Points (bps)					
	2020	**2021**	**2022**	**2023**	**Basis Points**
Minimum	71,606	83,856	93,617	101,619	4
Target	179,015	209,640	234,042	254,048	10
Maximum	358,030	419,280	468,083	508,096	20

Q. How did the Companies calculate their Supply Diversification targets?

A. Using attributes of the six RNG projects in KEDNY and KEDLI's interconnection queue, the Companies estimated the net benefit per Dth of RNG connected to the natural gas distribution system. Taking this estimated benefit, the Companies then calculated the volume of Dths needed to achieve one basis point of value each year. The Companies believe that connecting RNG results in overall societal benefits and

propose setting the minimum target at one basis point. To establish midpoint and maximum targets, the Companies began by evaluating trends in RNG interconnections from the 1980s to the present and projecting future growth in RNG interconnections between 2020 and 2023 without further action by the Companies. The Companies then developed an estimated trendline for accelerated RNG interconnections to establish the midpoint and maximum targets. Based on the foregoing, the Companies propose a midpoint target of ten basis points, equivalent to a 300 percent improvement in interconnection volumes. The proposed maximum target is capped at 20 basis points or a 500 percent improvement in interconnection volumes.

Q. What are the benefits associated with the metrics that comprise the System Efficiency EAM?

A. The Peak Reduction and Supply Diversification metrics align with the Commission's finding that improving overall system efficiency is one of the most important objectives of REV. To that end, the initiatives and strategies the Companies anticipate using to reduce peak and diversify supply will likely result in avoided or deferred capital investment, active customer participation in dynamic load management, reduced energy constraints, and lower carbon emissions. Moreover, the Supply

Diversification metric incentivizes the Companies to increase RNG production volumes through market animation, creating a cleaner and more diverse energy supply, providing a local source of renewable energy that is consistent and reliable, and improving waste management.

2. Energy Efficiency

Q. Please describe the Companies' Energy Efficiency EAM.

A. The proposed Energy Efficiency EAM is composed of two metrics: (i) Incremental Energy Efficiency; and (ii) Affordability. In the Commission's REV Track Two Order, the Commission suggested that utilities consider applying the electric EAM approach to establish gas efficiency targets, with comparable EAMs to be considered in gas rate cases. Consistent with the Commission's direction, the Companies propose a program-based Energy Efficiency metric measuring incremental million British thermal units ("mmBtu") reductions, as well as an outcome-oriented Affordability metric measuring percentage reduction in average low-income customer bills.

a. Incremental Energy Efficiency

Q. Please describe how KEDNY and KEDLI will measure the Incremental Energy Efficiency metric.

A. The Companies propose a program-based energy-efficiency metric measuring incremental mmBtu reductions achieved through the programs described in the Companies' respective ETIP filings. As the Commission noted in the December 2018 EE Order, "[b]ecause increases in utility targets require a lead time to allow vendors and service providers to ramp up capacity, utilities will be granted flexibility in achievement of these targets and expenditure of funds through the full 2019 - 2020 period." With the consideration that utilities should be granted flexibility to align programs with increased targets, the Companies propose that the Incremental Energy Efficiency metric begin in CY 2020 with the first year EAM targets set to account for achievements in CYs 2019 and 2020 collectively. Each year after will consider only achievements against the EAM target in the respective year.

Q. What does KEDNY propose for its Incremental Energy Efficiency targets and associated basis points?

A. KEDNY proposes the following annual Incremental Energy Efficiency targets and associated basis points for achieving the targets:

KEDNY Annual Incremental Energy Efficiency Reduction Targets (mmBtu) and Basis Points (bps)				
	2019 / 2020	**2021**	**2022**	**2023**
Minimum	755,266 (0)	482,466 (0)	546,466 (0)	676,466 (0)
Target	1,009,930 (20)	618,953 (10)	688,542 (10)	815,165 (10)
Maximum	1,264,594 (40)	755,439 (20)	830,617 (20)	953,864 (20)

Q. What does KEDLI propose for its Incremental Energy Efficiency targets and associated basis points?

A. KEDLI proposes the following annual Incremental Energy Efficiency targets and associated basis points for achieving the targets:

KEDLI Annual Incremental Energy Efficiency Reduction Targets (mmBtu) and Basis Points (bps)				
	2019 / 2020	**2021**	**2022**	**2023**
Minimum	419,023 (0)	252,139 (0)	285,139 (0)	327,139 (0)
Target	544,169 (20)	314,842 (10)	347,943 (10)	380,603 (10)
Maximum	669,315 (40)	377,545 (20)	410,747 (20)	434,068 (20)

Q. Please explain how the Companies calculated the proposed targets.

A. The Companies believe the minimum threshold for this metric should be equal to the targets established in the December 2018 EE Order. The targets established in that order reflect a level of utility-specific cost savings to meet the statewide energy savings goal. The Companies therefore propose such targets as the minimum for this EAM metric and assigned them zero basis points. The maximum target was developed by assuming cost efficiencies equal to each company's respective lowest

average historic actual run rate from the last four years, as adjusted for anomalies and to remove behavioral program results. The mid-point target for the metric is set at the mid-point between the minimum and maximum targets.

b. Affordability

Q. Please describe how the Companies will measure the Affordability metric.

A. The Companies propose an outcome-oriented energy efficiency metric measuring the annual percentage reduction in average low-income customer bills. This metric incentivizes the Companies for programmatic efforts to reduce low-income customers' energy usage and for animating the market for third parties to likewise provide low-income benefits within the Companies' service territories.

Q. Please explain how the Companies will calculate achievements towards the Affordability metric.

A. The outcome-based Affordability metric measures reductions in the average monthly customer bill for low-income customers. The Companies will calculate the average monthly customer bill on an annual basis and

compare the year-over-year performance to the weather-normalized baseline from the previous year.

Q. What do the Companies propose for their respective Affordability targets and associated basis points?

A. The Companies propose that one basis point be awarded for each 0.2 percent reduction in the average low-income customer bill achieved year-over-year within their respective service territories.

Q. What are the benefits associated with the Gas Energy Efficiency EAM?

A. By encouraging increased deployment of gas energy-efficiency through specific measures and other delivery channels described in the Companies' ETIP, the Companies anticipate enhanced customer control of their energy use, market animation, reduced carbon emissions, and increased affordability. Taken together, these benefits support the REV objectives and the State's clean energy goals.

Q. Did the Companies analyze the benefits and costs of the Energy Efficiency EAMs?

A. Yes. For the Energy Efficiency EAM, the Companies estimated benefits based on the implied $/Dth numbers from its most recent ETIP filing. Annual costs include those in the Companies' ETIP filings, as well as the incremental cost-shift. Annual net benefits at the target levels are positive as shown in the Companies' ETIP filing.

3. Carbon Reduction

Q. Please describe the Companies' Carbon Reduction EAM.

A. The proposed Carbon Reduction EAM is composed of two metrics: (i) metric tons of CO_{2e} reduced from all full-service customers; and (ii) metric tons of CO_{2e} reduced from transportation customers. This metric will focus on the adoption rate of three specific offerings: oil-to-geothermal conversions; sales of RNG via the Green Gas Tariff; and use of compressed natural gas ("CNG") and RNG in NGVs to offset the use of gasoline and diesel.

a. CO_{2e} Reduced From Full-Service Customers

Q. Please describe the full-service Customer CO_{2e} Reduction metric.

A. Customers that are becoming KEDNY or KEDLI customers will typically be switching their fuel source, usually from delivered fuels to natural gas. This represents a reduction in CO_{2e} for those customers. The Companies have an opportunity to influence those customers when they are making a choice about their fuel switching so that their environmental impact will be further reduced using RNG or geothermal systems. Additionally, the Companies can market to existing customers to encourage them to explore ways that they can reduce their environmental impact, either with RNG or via replacing or supplementing their gas system with a geothermal system. The combined impact of all these customers will be aggregated and will provide a metric to measure how customers are tracking in terms of their environmental impact. By encouraging customers to consider oil-to-geothermal or using RNG rather than traditional fossil-based natural gas, KEDNY and KEDLI will be helping to reduce CO_{2e} emissions.

To calculate the carbon reduction, the volume of natural gas and RNG sold to the appropriate rate classes will be calculated and summed at the end of the calendar year. This sum will be multiplied by the lifecycle CO_{2e} factor for the combustion of fossil-derived natural gas. The resulting number will be the baseline factor for the number of customers and the amount of throughput that would have been expected in the absence of

environmentally beneficial offerings. Next, the CO_{2e} emissions for natural gas and RNG will be calculated separately using the lifecycle CO_{2e} emissions factors for each feedstock. These two will be summed and the resulting value will be subtracted from the first sum calculated above. The delta between these two will represent the carbon reduction achieved via RNG.

To calculate the carbon reduction from geothermal systems, the number of geothermal systems active and owned by each company will be calculated based on the type of customer class they are serving. The number of systems will be multiplied by an average consumption of natural gas for each of these customer classes. The resulting theoretical consumption will be multiplied by the lifecycle CO_{2e} emissions factor for natural gas to determine the amount of avoided carbon emissions via geothermal systems.

The two calculations will be added together to determine the total carbon reduction from all full-service customers.

Q. How did the Companies calculate basis points for the CO_{2e} metric?

A. KEDNY determined basis point values by developing a 100-point scale for achieving the carbon reduction target by 2023. To the extent carbon

reductions are achieved at a level above the minimum target, the Companies propose they earn 0.1 basis points for every 232 metric tons of CO_{2e} reduction.

Q. What does KEDNY propose for its full-service Customer $CO2_e$ Reduction metric rate targets?

A. KEDNY proposes the following targets:

KEDNY Annual Full-Service Customer CO_{2e} Reduction Targets				
	2020	**2021**	**2022**	**2023**
Minimum	1,647	3,432	3,632	3,893
Target	2,471	5,149	5,447	5,839
Maximum	3,295	6,865	7,263	7,785

Q What does KEDLI propose for its full-service Customer CO_{2e} Reduction metric rate targets?

A. KEDLI proposes the following targets:

KEDLI Annual Full-Service Customer CO_{2e} Reduction Targets				
	2020	**2021**	**2022**	**2023**
Minimum	1,792	4,301	5,563	7,369
Target	2,688	6,452	8,344	11,053
Maximum	3,584	8,603	11,126	14,737

Q. Please describe the transportation $CO2_e$ Reduction metric.

A. The Companies currently serve CNG fueling stations, which supply fuel for NGVs. By switching from gasoline and diesel to CNG, fleets can

reduce their environmental impact. However, this transition comes with a cost in the form of higher vehicle costs. With continued use of diesel as the business-as-usual scenario, the Companies propose calculating the reduction in CO_{2e} by comparing CNG use to diesel. The Companies believe this calculation is reasonable regardless of whether customers are full-service or work with an energy service company ("ESCO"), because the Companies can determine whether customers are using natural gas for their vehicles. Moreover, if a station signs up to use RNG for a portion of its load, the Companies will also incorporate those savings into the CO_{2e} calculation.

Q. What does KEDNY propose for its Transportation Customer CO_{2e} Reduction metric rate targets and associated basis points?

A. KEDNY proposes the following targets and associated basis points:

KEDNY Annual Transportation Customer CO_{2e} Reduction Targets and Basis Points					
	2020	**2021**	**2022**	**2023**	**Basis Points**
Minimum	2,891	5,840	5,899	5,958	3
Target	4,337	8,760	8,848	8,936	4
Maximum	5,782	11,680	11,797	11,915	5

Q. What does KEDLI propose for its Transportation Customer CO_{2e} Reduction metric targets and associated basis points?

A. KEDLI proposes the following targets and associated basis points:

KEDLI Annual Transportation Customer CO_{2e} Reduction Targets and Basis Points					
	2020	**2021**	**2022**	**2023**	**Basis Points**
Minimum	2,891	5,840	5,899	5,958	3
Target	4,337	8,760	8,848	8,936	4
Maximum	5,782	11,680	11,797	11,915	5

Q. Please explain how the Companies calculated the targets.

A. The Companies established the targets based on industry data regarding the adoption rate of renewable offerings that come with a cost premium. The minimum threshold was set at 23,176 metric tons CO_{2e} for CY 2020 given the high cost of renewable energy relative to non-renewable fuels. The target was set 50 percent higher at 34,746 metric tons and the maximum is 100 percent higher at 46,352 metric tons.

Q. What are the benefits associated with the Carbon Reduction EAM?

A. The Companies proposed Carbon Reduction EAM promote market activities that will reduce carbon emissions. By adopting the three offerings outlined in this metric, consumers will go beyond the standard level of CO_{2e} reduction that is achieved by switching to natural gas. This

will help to advance the State's and the Companies' goals of reducing statewide greenhouse gas emissions by 80 percent by 2050. Additionally, these offerings will give customers choices, enabling them to participate in a lower-carbon future in a way that fits with their preferences. The Carbon Reduction EAM will also support the use of renewable fuels in the transportation sector and it will animate the market for clean-energy innovation.

4. *Benefits and Costs*

Q. Did the Companies analyze the portfolio of benefits and costs associated with its EAMs?

A. Yes. The Companies evaluated the portfolio of benefits and costs associated with the proposed EAMs. The System Efficiency and Carbon Reduction BCAs are set forth in Exhibit ___ (FOH-2), and the Energy Efficiency BCA is included in the Companies' ETIP filing.

Q. What did the Companies' analysis of the benefits and costs conclude?

A. The Companies' analysis concluded that the portfolio of programs and products supporting the EAMs provide significant qualitative and quantitative benefits, including increased customer choice and avoided environmental impacts, in furtherance of State clean energy goals and the

REV objectives. Moreover, the Companies expect the benefits of the respective initiatives to increase as the Companies scale the products and services.

B. Evaluation and Reporting

Q. How do the Companies intend to report EAM results and recover incentives?

A. On March 31 of each year from 2020 to 2024, the Companies will make an annual compliance filing with the Commission in this docket. The annual filing will include a report on the Companies' prior CY performance relative to each EAM target, showing the savings and benefits achieved, as well as calculations for the incentives earned. For metrics where the Companies' performance falls between the minimum and the mid-point target or the mid-point target and the maximum, the incentive payouts will be prorated. The Companies will also provide an explanation of any targets not achieved, if applicable.

Q. How do the Companies propose to recover any incentive that is earned?

A. Any incentive that is achieved would be recovered through an EAM surcharge, as explained in the testimony of KEDNY and KEDLI's Gas Rate Design Panels.

Q. Are the Companies proposing a mid-period review to evaluate the EAMs?

A. Yes. By March 31, 2022, the Companies will evaluate the EAMs and, in their discretion, file a mid-period review if necessary to suggest regulatory changes and any EAM modifications the Companies believe are appropriate. Any proposed modifications to the EAMs would require Commission approval.

Q. Are there circumstances when the Companies would seek to modify the EAM proposal before the mid-period review?

A. Yes. Many of the proposed EAM metrics and associated targets are directly related to products and services supported by this Panel. To the extent the Commission does not approve or otherwise curtails those initiatives, the EAMs proposed may no longer provide meaningful incentives for the Companies, and the associated metrics may therefore require modification.

C. Platform Service Revenues ("PSRs")

Q. Are the Companies proposing any PSRs?

A Yes. The Companies propose two PSRs for revenues from: (i) the E-Commerce Platform; and (ii) the UESC program.

Q. Please describe the E-Commerce Platform PSR.

A. Consistent with the REV Track Two Order and the 2017 NMPC Rate Case Order, the Companies propose a PSR sharing mechanism for fees collected by the Companies from vendors who participate in the E-Commerce Platform. During the Rate Year and each Data Year, the Companies propose to retain 20 percent of the fees, deferring the remaining 80 percent for future credit to customers. An example of this mechanism is set forth in Exhibit ___(FOH-9).

Q. Please describe the UESC program PSR.

A. During the Rate Year and each of the Data Years, the Companies propose to retain one-third of the fees received from UESC participants, deferring the remaining two-thirds for future credit to customers.

Q. Why do the Companies believe the UESC sharing mechanism is reasonable?

A. For the UESC program, unlike the E-Commerce Platform, the Companies are undertaking additional marketing and feasibility management responsibilities without a guarantee that those efforts will result in a contract. Because the PSR would only apply if the initial marketing and feasibility study activities result in customers entering a

UESC with the Companies and moving forward with the improvements identified in the feasibility study, the Companies believe a higher sharing percentage is reasonable. It establishes a meaningful incentive for the Companies to continue to grow the UESC program, while also rewarding customers for the contributions toward marketing and the feasibility studies.

D. Non-Pipeline Alternatives ("NPA") Incentive Mechanism

Q. Do the Companies proposing an NPA Incentive Mechanism?

A. Yes. In support of the NPA proposal discussed in KEDNY and KEDLI's GIOPs' testimonies, and consistent with the "Report of Niagara Mohawk Power Corporation d/b/a National Grid Concerning the Non-Pipeline Alternatives Incentive Mechanism Collaborative" filed December 21, 2018 in Case 17-G-0239, the Companies propose a mechanism for each company to share 30 percent of the difference between the costs of a traditional investment and the proposed cost of an NPA, adjusted for other net benefits. This sharing mechanism follows the basic structure of the non-wires alternative ("NWA") mechanism approved by the Commission in the 2017 NMPC Rate Case. The proposed NPA incentive mechanism is set forth in Exhibit ___ (FOH-11).

E. KEDLI EmPower Replacement Program Incentive Mechanism

Q. Is KEDLI proposing to continue the EmPower Replacement Program Incentive Mechanism?

A. Yes. In the 2016 KEDNY and KEDLI Rate Cases, the Commission approved an earnings incentive mechanism for the EmPower Replacement Program, incentivizing KEDLI to achieve cost savings and promote program participation.

Q. How is the incentive mechanism structured?

A. The incentive amount is determined using a percentage of the approved program budget, excluding labor costs, and scaled so that each tier of the program design has a return that reflects the value of participation in that tier. KEDLI is not proposing any modifications to the existing structure of the earnings incentive mechanism.

Q. How will the incentive mechanism be calculated?

A. KEDLI proposes the earnings incentive mechanism be awarded annually after all committed projects have been completed and no encumbered funds remain. For each tier, KEDLI recommends applying the same percentages established in the 2016 Rate Case to the annual budget value for calculating the earnings incentive per household served. The

percentages are 0.007 percent for Tier Three and 0.014 percent for Tier Four. The earnings incentive amount per household served is then multiplied by participation in each tier, resulting in a total incentive award to the company. Exhibit ___ (FOH-12) includes an example of the earnings incentive mechanism.

VII. Conclusion

Q. Does that conclude the Panel's testimony.

A. Yes.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Corrections and Updates Testimony

of

The Future of Heat Panel

Dated: July 3, 2019

Q. Please identify the members of the Future of Heat Panel.

A. The Panel consists of Donald Chahbazpour, Lisa M. Tallet, Arthur W. Hamlin, and Owen S. Brady.

Q. Is this the same Future of Heat Panel that previously submitted prepared direct testimony in these proceedings?

A. Yes. The terms defined in the Panel's direct testimony have the same meanings here.

Q. What is the purpose of the Panel's corrections and updates testimony?

A. The purpose of the Panel's corrections and updates testimony is to make seven corrections and/or updates to its direct testimony. First, the Panel corrects the proposed Incremental Energy Efficiency EAM to include targets based on gross savings as opposed to net savings. Second, the Panel updates the forecast for increases in the economic development program discounts. Third, the Panel corrects typographical errors regarding economic development grant funding. Fourth, the Panel updates the Green Gas Tariff's residential participation tier to include an option that more closely aligns with the 100 percent option available for non-residential customers. Fifth, the Panel corrects the labor costs for the

Green Gas Tariff in the Rate Year. Sixth, the Panel updates the labor overhead costs, which are supported by the Revenue Requirements Panels, for the FTEs supporting the Future of Heat initiatives. Seventh, the Panel updates the Companies' ETIP budgets moving to base rates.

Q. Does the Panel sponsor any exhibits to its corrections and updates testimony?

A. Yes. The Panel sponsors the following exhibits that were prepared by one or more members of the Panel or under their supervision and direction:

Exhibit ___ (FOH-10CU)	Includes the updated Incremental Energy Efficiency EAM targets for KEDNY and KEDLI.
Exhibit ___ (FOH-13CU)	Sets forth the updated labor-related overhead costs and the Green Gas Tariff FTE expense for the Rate Year.
Exhibit ___ (FOH-14CU)	Sets forth the forecast revenue requirement impact associated with increased participation in the economic development discount programs.

In addition Appendix A includes relevant information request ("IR") responses supporting the corrections and updates included in this testimony.

Q. Please describe the Companies' correction to the Incremental Energy Efficiency EAM targets.

A. As set forth in the Companies' response to IR No. DPS-387, the Panel incorrectly used net savings, as opposed to gross savings, when establishing the Incremental Energy Efficiency EAM Targets for KEDNY and KEDLI.

Q. What are the updated Incremental Energy Efficiency EAM targets for KEDNY?

A. KEDNY proposes to replace the table located on page 105, line 1 of the Panel's direct testimony with the following updated annual Incremental Energy Efficiency targets. The proposed corrections are also reflected in Exhibit ___ (FOH-10CU).

KEDNY Annual Incremental Energy Efficiency Reduction Targets (mmBtu) and Basis Points (bps)				
	2019 / 2020	**2021**	**2022**	**2023**
Minimum	811,814 (0)	510,740 (0)	574,740 (0)	704,740(0)
Target	1,037,379 (20)	635,602 (10)	706,976 (10)	851,955 (10)
Maximum	1,262,944 (40)	760,463 (20)	839,212 (20)	999,169 (20)

Q. What are the updated Incremental Energy Efficiency EAM targets for KEDLI?

A. KEDLI proposes to replace the table located on page 105, line 5 of the Panel's direct testimony with the following updated annual Incremental Energy Efficiency targets. The proposed corrections are also reflected in Exhibit ___ (FOH-10CU).

KEDLI Annual Incremental Energy Efficiency Reduction Targets (mmBtu) and Basis Points (bps)				
	2019 / 2020	**2021**	**2022**	**2023**
Minimum	452,387 (0)	268,821 (0)	301,821 (0)	343,821 (0)
Target	556,586 (20)	322,047 (10)	355,919 (10)	399,029 (10)
Maximum	660,785 (40)	375,274 (20)	410,018 (20)	454,238 (20)

Q. Please describe the Companies' correction to the Economic Development discount programs.

A. In direct testimony, the Panel proposed changes to the economic development discount programs to make them more responsive to customer needs and generate customer and community benefits. As a result, the Companies anticipate increased participation in the programs; however, the revenue requirement impact of the anticipated increase was inadvertently omitted from the Panel's direct testimony. In total, the Companies estimate the proposed changes would increase annual

discounts in Data Years 2 and 3 by approximately $0.300 million for KEDNY and $0.150 million for KEDLI. The impact is reflected in Exhibit ___ (FOH-14CU).

Q. Please describe the correction regarding economic development grant funding.

A. As set forth in the Companies' responses to IR No. DPS-485, KEDLI identified typographical errors in the Panel's direct testimony regarding the amount of grant funding awarded and committed to customers. The grant amounts on page 63 of 120 of the Panel's direct testimony should be corrected to read as follows:

> Through February 28, 2019, a total of $0.198 million has been awarded to customers, $0.068 million for KEDNY and $0.130 million for KEDLI. An additional $2.313 million has been committed through approved applications, $1.551 million for KEDNY and $0.762 million for KEDLI.

Q. Please describe the Companies' proposed update to the residential tiers for Green Gas Tariff participants.

A. In its direct testimony, the Panel identified four flat-rate tiers for residential customers who participate in the Green Gas Tariff program. As

set forth in the Companies' response to IR No. DPS-365, the Companies propose to update the residential tiers to include a flat-rate option that more closely tracks the 100 percent option for non-residential customers, as shown in the following table:

Tier	Residential Flat Rate
Low	$5/month
Low-Middle	$25/month
High-Middle	$50/month
High	$100/month

Q. Please describe the Companies' correction to the labor costs for the Green Gas Tariff offering.

A. In the Panel's direct testimony, the Companies proposed to add two FTEs to implement the new Green Gas Tariff offering. The Panel, however, mistakenly omitted the associated Rate Year costs of $0.127 million for KEDNY and $0.125 million for KEDLI. Exhibit ___ (FOH-13CU), as well as the Revenue Requirements Panels corrections and updates testimony, include the labor-related Rate Year FTE costs for the Green Gas Tariff offering.

Q. Are there any additional changes required to labor costs?

A. Yes. As set forth in the Revenue Requirements Panels corrections and updates testimony, the Companies updated the labor-related overhead costs for all FTEs. Exhibit ___ (FOH-13CU) sets forth the associated update for the FTEs proposed by the Panel. The calculation of overhead costs is supported by the Revenue Requirements Panels.

Q. Please describe the update made to the ETIP budgets moving to base rates.

A. The Companies' updated the ETIP budget amounts moving to base rates due to the change in the revenue forecasts. The ETIP budget amounts on page 27 of the Panel's direct testimony should be corrected to read as follows:

> KEDNY's ETIP budget moving to base rates is $18.636 million for the Rate Year, $20.940 million for Data Year 1, $23.528 million for Data Year 2, and $27.949 million for Data Year 3. KEDLI's ETIP budget moving to base rates is $9.478 million for the Rate Year, $10.275 million for Data Year 1, $11.270 million for Data Year 2, and $12.592 million for Data Year 3.

Q. Does this conclude your corrections and updates testimony?

A. Yes.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY AND KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Rebuttal Testimony

of the

Future of Heat Panel

Case 19-G-0309
Case 19-G-0310

September 18, 2019

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I. Introduction

Q. Please introduce the members of the Future of Heat Panel.

A. The Future of Heat Panel ("Panel") consists of Donald Chahbazpour, Lisa M. Tallet, Arthur W. Hamlin, and Owen S. Brady.

Q. Is this the same Panel that submitted direct and corrections and updates testimony previously in this proceeding?

A. Yes. The terms defined in the Panel's direct and corrections and updates testimony have the same definitions here.

II. Purpose of Testimony

Q. What is the purpose of the Panel's rebuttal testimony?

A. The purpose of the Panel's rebuttal testimony is to respond to certain recommendations presented in the testimonies of the following Department of Public Service Staff ("Staff") witness panel: Staff Policy Panel, Staff Efficiency and Sustainability Panel, Staff Consumer Services Panel, and Staff Earnings Adjustment Mechanism Panel. The Panel also addresses certain recommendations presented in the testimonies of the following intervenor parties: The City of New York, Environmental Defense Fund, Utility Intervention Unit, William D. Yates, CPA for

Public Utility Law Project of New York, Inc. Estates NY Real Estate Services LLC, and Robert Mark Wyman.

Q. Does the Panel sponsor any exhibits as part of its rebuttal testimony?

A. Yes. The Panel sponsors the following exhibits that were prepared and compiled under our direction and supervision:

(i) Exhibit ___ (FOH-1R), Companies' Response to Information Request ("IR") DPS-268;

(ii) Exhibit ___ (FOH-2R), Geothermal Gas REV Demonstration Project – Q3 2018 Report;

(iii) Exhibit ___ (FOH-3R), American Gas Association Presentation — Developments in Renewable Natural Gas; and

(iv) Exhibit ___ (FOH-4R), Staff's Response to IR NG-13.

Q. Have there been any legislative or administrative changes since the filing of the direct testimony that impact the Panel's proposals?

A. Yes. There are two significant changes that impact the Panel's proposals. First, as mentioned in the Companies' Supplemental Testimony filed on June 11, 2019, the New York State Department of Environmental Conservation ("DEC") and the New Jersey Department of Environmental Protection denied without prejudice Transco's

applications for the necessary permits to construct the Northeast Supply Enhancement ("NESE") project, rendering the Companies' unable to fulfill applications for new firm service connections or requests for additional firm load from existing customers until the current natural gas supply constraints are addressed. Second, New York adopted the Climate Leadership and Community Protection Act of 2019 ("CLCPA"), codifying greenhouse gas ("GHG") emission reduction and renewable power generation requirements, while also creating a Climate Action Council to drive policy goals for achieving the clean energy requirements. The CLCPA takes effect January 1, 2020.

Q. How do the legislative and administrative actions impact the Panel's proposals?

A. The prospects of continued gas supply constraints due to the potential unavailability of NESE and the passage of the CLCPA underscore the importance of innovative proposals like those set forth in the Panel's direct testimony to meeting customers' heating needs. Projects such as the Newtown Creek renewable gas project with New York City, the Companies' power-to-gas ("P2G") demonstration project to develop new renewable energy storage opportunities, and the proposed expanded geothermal and demand response demonstrations, all are designed to

promote the transition to a decarbonized energy future and develop the solutions necessary to respond to the directives of the CLCPA.

III. Response to Staff's Recommendations

A. Reducing Methane Emissions

Q. Please address Staff's recommendations regarding the Companies' proposed methane reduction initiatives.

A. The Staff Pipeline Safety Panel opposes several of the Companies' initiatives aimed at reducing methane emissions from the gas distribution system, including eliminating the current leak prone pipe ("LPP") removal incentive, as well as the current incentive for repairing high-emitting Type 3 leaks. Removing LPP and repairing high emitting leaks are particularly effective means of reducing methane emissions from the gas system. Given the safety, reliability, and environmental benefits associated with eliminating system leaks, the Companies should be incented to drive continued improvements in this area through challenging but fair incentives that align with the Companies' ambitious target to reduce methane emissions from the gas distribution system 60 percent (from 1990 levels) by 2035. Staff's opposition to these incentives is counter to the Commission's stated policies for eliminating

LPP and lowering emissions. The Companies' Gas Safety Panel further addresses these incentive mechanisms.

B. Customer Empowerment and Enablement

1. Energy Efficiency

Q. Please address Staff's recommendation regarding the level of ETIP budget and the allocation of ETIP cost recovery.

A. The Staff Efficiency and Sustainability Panel ("SESP") recommends (pp. 20-22) using the Companies' proposed energy efficiency budgets as placeholders until the Commission sets budgets in the on-going statewide proceeding in Case 18-M-0084; and if Commission does not act in Case 18-M-0084 before rates are set for KEDNY and KEDLI in this case, the SESP proposes a deferral mechanism to reconcile the difference between amounts determined here and amounts set in the other case.

The Companies disagree with the SESP's recommendation. Establishing the budgets in this case will provide the Companies with the certainty needed to develop and implement their energy efficiency programs over the next several years. Therefore, the SESP's recommendation should be rejected.

Q. What is the position of the Companies on Staff's recommendation that the evaluation, measurement, and verification ("EM&V") costs remain combined in the energy efficiency budget and not moved to base rates as the Companies had proposed?

A. The Companies oppose Staff's recommendation (SESP, pp. 25-26) that EM&V costs remain within the ETIP budget and not be moved into base rates. The scope of EM&V activities is expanding to integrate initiatives beyond just energy efficiency and therefore should be separated from the energy efficiency portfolio budget.

The Companies would use the incremental budget amounts freed up by moving the EM&V funds out of the energy efficiency portfolio to evaluate and develop new customer offerings. Funding is needed to promote product development in line with new technologies, learnings, and public policy priorities. Moving EM&V costs into base rates also would align with how those costs are treated for Niagara Mohawk.

Q. Please address Staff's recommendation for a downward-only reconciliation for energy efficiency spend at year end.

A. The Companies disagree with Staff's recommendation for a downward-only reconciliation of energy efficiency spend in the Rate Year. The

proposed funding in this case is meant to help the Companies achieve statewide energy efficiency targets through the year 2025. The Companies believe that the funding should be retained for application towards these energy efficiency initiatives through 2025, regardless of the length of the rate period approved in this case.

Q. Please address Staff's recommendation (SESP pp. 26-30) that unspent energy efficiency funds, including the associated carrying charges, as well as carrying charges on NYSERDA Clean Energy Fund collections, be amortized over five years to offset energy efficiency collections.

A. The Companies oppose Staff's recommendation. In principle, the Companies are not against using a portion of the unspent/uncommitted energy efficiency funds, including associated carrying charges, to offset energy efficiency program costs. The Companies are concerned, however, that use of funds as the Staff proposes could negatively impact the Companies' cash flows and affect credit metrics.

Energy efficiency budgets are increasing significantly in the coming years and the current unspent/uncommitted energy efficiency funds will be needed to mitigate customers' costs associated with expanding energy

efficiency programs. The Companies believe the appropriate balance between the level of unspent/uncommitted energy efficiency funds, including associated carrying charges, to use now and the level to use in the future to offset the need to collect these costs in rates is that set forth in the Companies' response to DPS-268 and Attachment 5 to that response, a copy of which is included as Exhibit __ (FOH-1R). Specifically, the unspent/uncommitted EEPS funds should be used to fund the incremental budget and targets for 2019 and 2020, which would offset the need to collect these costs in rates for those years.

C. New Products and Services

1. Green Gas Tariff Offering

Q. Please address the Staff's position on the Companies' proposed Green Gas Tariff ("GGT").

A. The Staff Gas Supply Panel supports the proposed GGT, but would reduce the number of full time equivalent ("FTE") positions needed to implement the program from two to one. The Staff Gas Supply Panel also recommends that the Companies revise the proposed reconciliation mechanism such that GGT costs would be reconciled as part of the existing Gas Adjustment Clause.

Q. Do the Companies agree with Staff's recommendations?

A. In part. The Companies agree that GGT costs and revenues should be reconciled as part of the existing Gas Adjustment Clause. The Companies disagree, however, with the proposal to reduce the number of FTEs supporting the program. The two additional FTEs are needed to administer the GGT program. Their work will include contracting with RNG suppliers, monthly tracking of green gas purchases and sales quantities, balancing between purchases and sales, tracking costs and revenues, making all associated transaction system entries and complying with reporting requirements. Adequate resources are needed to successfully implement and maintain the GGT. The Companies' Energy Procurement team will be responsible for administering the program, and the required work is incremental to the current responsibilities of that team.

2. Utility Energy Services Contracts ("UESC") Program

Q. What is the Companies' position regarding Staff's recommendation on the UESC contracts program?

A. The SESP (pp. 36-38) proposed two modifications to the Companies' proposal: (i) reducing the Companies' share of net revenues generated from UESC program from 33.3 to 20 percent; and (ii) eliminating the two

incremental FTEs supported in the Companies' case. As described later in our testimony in the section on platform service revenues, the Companies disagree with Staff's proposal to reduce the Companies' share of platform service revenues. Likewise, the Companies do not accept Staff's proposal to eliminate the two FTE positions. The Commission has directed utilities in the state to identify opportunities to generate platform service revenues through the provision of value-added services to customers and stakeholders. The UESC program presents an opportunity for the Companies to implement this Commission directive. To have a reasonable chance to succeed, however, the program requires a dedicated set of resources. Requiring that the Companies redeploy resources from their energy efficiency programs—which are also charged with significantly increasing the penetration of those programs—would result in inadequate resources dedicated to both of these important areas. Accordingly, the Companies do not agree with Staff's recommendation to eliminate the two incremental FTEs to support the UESC program.

Q. Please address the Staff's proposal that the Companies should be required to develop specific energy efficiency targets and budgets for UESC program and incorporate the UESC program into their overall

energy efficiency portfolios, to be funded from authorized energy efficiency budgets.

A. The Companies do not agree that the UESC program should be incorporated into the energy efficiency portfolios and likewise do not agree that specific budgets and targets should be established. The UESC program is designed as an energy services contracting program rather than as a direct customer incentive program. Energy savings delivered through UESC programs are tracked and measured through the existing utility energy efficiency programs as customers participating in the UESC program have eligible measures installed. Therefore, any energy savings targets established for UESC would be duplicative of targets already established under the existing commercial and industrial energy efficiency programs.

Further, the scope of the feasibility studies conducted under the UESC program is not limited to gas savings, but also includes electric energy savings as well as water savings, operations and maintenance cost savings, and support integration of other distributed energy resources. Accordingly, the Companies do not support incorporating the UESC program into the existing gas-only energy efficiency portfolios. Rather, this program supports broader energy opportunities in line with expanded

customer tools and resources and therefore requires its own set of resources and should not be rolled into the Companies' overall energy efficiency portfolios.

3. Fuel-Switching Calculator

Q. Please address Staff's recommendations regarding the Companies' proposed fuel switching calculator.

A. The Companies' do not agree with the SESP's recommendation (pp. 38-40) that approval of the Fuel-Switching Calculator be subject to the costs being allocated at the outset to other National Grid companies, or that such allocation be based on National Grid's general allocator (8.29 percent KEDLI; 13.31 percent KEDNY).

First, the purpose of the calculator would be to enable customers to make informed energy choices by using location-specific assumptions – *e.g.*, incentives available and rates specific to KEDNY and KEDLI – to compare the cost and emissions estimates for different energy alternatives. Initially, the fuel-switching calculator would be used solely by KEDNY and KEDLI for their customers. Expanding the calculator for use by other National Grid companies will add to the costs of the fuel switching calculator (a fact Staff does not reflect in its proposed adjustment). To the extent the calculator is modified for use by other National Grid companies

in the future, the costs of those modifications will be reflected, and the cost allocations would be revised in accordance with National Grid's established allocation process for National Grid Service Company costs. Accordingly, the Companies do not agree with the recommendation that approval of the calculator be conditioned on the initial costs being initially charged to other National Grid companies.

In addition, when other National Grid companies begin using the calculator, National Grid Service Company will apply the appropriate allocator based on the benefitting companies at that time. The general allocator proposed by Staff, which allocates costs among all National Grid companies (including electric-only companies and National Grid Service Company), would not be the appropriate allocator for the costs of the fuel-switching calculator.

D. Demand Response

Q. Please address Staff's recommendations regarding the Companies' demand response programs.

A. The SESP recommends approval of the Companies' demand response program; however, the SESP recommends changes in the proposed rate treatment of non-labor O&M costs. The SESP also opposes recovery of the incremental FTE costs for this program.

Q. Do the Companies agree with Staff's proposal (SESP pp. 48-49) that non-labor O&M costs for the demand response program be recovered via a delivery surcharge line item, rather than in base rates?

A. No. The Companies have forecast the non-labor O&M costs associated with the proposed demand response demonstration ("DRD") project and reflected those costs in rates for current recovery. Moving those costs to a surcharge that would be recovered with a time lag that may be significant would adversely impact the Companies' cash flows. If the SESP is proposing the Companies recover the forecast non-labor O&M costs through the Delivery Rate Adjustment, however, the Companies would have no objection, as discussed in the rebuttal testimony of the Rate Design Panel.

Q. Do the Companies agree with the Panel's recommendation that no additional FTEs be approved for the DRD project?

A. No. As is described on pages 44 and 45 of the Panel's direct testimony, the Companies propose to scale the existing demonstration project , which will require additional support. The FTE referenced in the SESP testimony was included in the Joint Proposal approved by the Commission in Cases 16-G-0058 and 16-G-0059 and is assigned to

support the Companies' entire portfolio of Gas REV projects. The Companies' Gas REV portfolio includes the DRD project, MicroCHP Demonstration, Flood Zone Packages, and Geothermal Demonstration. The Companies believe that dedicated FTEs will be required to scale the program from providing 241 dth/hr of demand reduction to the proposed program target of 1,400 dth/hr as described in the Panel's direct testimony. The recommended FTEs will responsible for growing and managing the program. Their responsibilities would include enrolling customers and aggregators, evaluating program data, coordinating with the Companies' Gas Operations group, and complying with reporting requirements.

E. Sustainable Heat Initiative

1. Expanded Geothermal Demonstration Project

Q. Please address Staff's recommendation (SESP pp. 50-52) that the Companies' proposal to expand the Geothermal Demonstration Project be rejected because the existing Geothermal Demonstration Project has not yet proven to be cost-effective and the Companies have not demonstrated that utility ownership of geothermal systems would not adversely affect third-party market participants.

A. The Companies strongly disagree with the SESP's recommendation that the Commission reject the proposed expansion of the Geothermal Demonstration Project. The 2016 Gas REV Demonstration Project ("Riverhead Demonstration"), which ends December 2019, has allowed KEDLI to provide non-traditional heating and cooling services to participating customers since November 2017. Since then, the Riverhead Demonstration has shown this technology works and customers can achieve savings. As stated in the Panel's direct testimony (p. 35), it is estimated each geothermal customer could realize annual energy savings between $1,000 and $1,500. The expanded geothermal demonstration program proposes to develop and test a geothermal product that will be a cost-effective alternative to a customer's existing heating solution. The Riverhead Demonstration also confirmed the underground portion is a significant market barrier towards wider adoption of geothermal. Exhibit __ (FOH-2R) is a copy of KEDLI's Q3 2018 demonstration project status report, wherein it shows actual energy savings of 31 percent to 58 percent, which is $229.10 to $684.95 in per-customer savings from just January 2018 to May 2018 (much of the 2018 Heating Season).

The proposed expanded Geothermal Demonstration Proposal is designed to test that the business model of utility ownership of underground loops,

the most expensive part of the system, can accelerate greater market adoption. The proposal is for an expanded demonstration project, not a full ownership program, to determine if eliminating the high-cost barrier to entry for all customers but especially for low-to-moderate income, commercial, and industrial customers, can help the market grow to advance the state's energy and carbon goals.

Q. Are there any gas constraints that could be alleviated by a non-pipe alternative ("NPA") such as the proposed expanded geothermal demonstration program?

A. Yes. As a result of the denial of the NESE project, the Companies are not fulfilling applications for new firm service connections or requests for additional firm load from existing customers because of the existing gas supply constraints that the NESE project was to address. If these supply constraints are not addressed, ground-source heat pumps ("GSHPs") are a vital NPA that can be deployed by the Companies to serve new customers seeking to convert from other fossil fuels.

Q. Are there other customer benefits that can be provided by an expanded geothermal demonstration program?

A. Yes. An expanded program that increases the number of GSHP installations will provide customer benefits by reducing peak demand, energy intensity, and carbon emissions and increasing energy efficiency and distributed energy penetration.

Q. Do the Companies agree with Staff's position that utility ownership of geothermal systems could adversely affect third-party market participants that may be able to provide additional affordable alternatives?

A. No. As set forth above, the expanded demonstration project is not proposing full ownership of the entire heating system. The expanded demonstration program seeks to test whether the Companies' supportive role of funding and owning the geothermal ground loop can help the market grow and advance the state's energy and carbon goals by providing more choices in the market for New York residents in support of the clean energy transition. An increase in customers will support further growth in the third-party sector by providing more volume and predictability in GSHP installations.

It is critical for the GSHP market to grow significantly to meet New York's goals of a 40 percent reduction of GHG emissions by 2030 and an

80 percent reduction by 2050. The Commission's "Order Adopting Accelerated Energy Efficiency Targets" in Case 18-M-0084 (issued December 13, 2018) found that 83,000 total heat pumps must be installed by 2025 to meet the established five TBTU target. Gas utilities can play a positive role in meeting these targets because of the relationship they currently have with natural gas customers. Utility participation and ownership of geothermal loops has the potential to increase the numbers of geothermal installations, reducing costs and producing economies of scale that will benefit customers and third-party market participants.

Q. Do third-party market participants support testing the concept of utility ownership of geothermal loops?

A. Yes. The New York Geothermal Energy Organization ("NY-GEO"), a non-profit trade association representing geothermal heat pump installers, manufacturers, distributers, drillers and other geothermal stakeholders, in comments filed July 1, 2019 in Case 18-M-0084 stated: "NY-GEO supports the KEDLI, KEDNY rate case proposal to further test a utility loop ownership model for its LMI customers."

In testimony in this case, Jared Rodriguez for Estates NY Real Estate Services LLC notes (p. 9) that "[g]eothermal technology offers highly

energy efficient production of domestic hot water and space heating. The technology, particularly shared geothermal loop energy systems, will likely play a key role in reinforcing the Company's ability to meet decarbonization goals via heating electrification and reducing peak gas capacity."

Q. Do the Companies agree that a third-party solicitation as the SESP mentions (p. 52) is necessary to support approval of the expanded Geothermal Demonstration Program?

A. No. The Companies are proposing to expand the pilot program to obtain additional knowledge on if and how best to proceed with geothermal development in the future. The expanded pilot would test whether utility participation could increase consumer confidence in the technology, improve economic aspects of the project, or create additional value for customers through a complementary offering involving both utility-ownership and vendor participation. The Companies would explore these considerations by working with existing geothermal companies to test whether utility ownership of the geothermal ground loop can help the market to grow. We do not believe a third-party solicitation as suggested by the SESP would advance the pilot objectives.

Q. Why are new FTEs needed for the expanded Geothermal program proposed by the Companies?

A. The FTEs are necessary to manage the geothermal installations, coordinate RFPs, contact project coordinators, and ensure the increased electric load will not cause adverse impacts for the electric utility servicing areas where the Companies have installed geothermal systems. Accordingly, the Companies' disagree with Staff's recommendation to eliminate the one incremental FTE in the Rate Year and two additional FTEs in Data Years 2 and 3 to support the expanded Geothermal Demonstration Project.

F. Economic Development

1. Discount Programs

Q. Please address Staff's recommendation regarding the Companies' proposed changes to their existing portfolio of economic development discount programs.

A. The SCSP recommends (pp. 63-64) that KEDLI include non-firm, non-residential customer classes to the list of service classes eligible for the existing Business Incentive Rate ("BIR") and proposed Area Development Rate. The Companies agree that existing supply constraints warrant examining other options for serving customer needs and agree that expanding the KEDLI programs to non-firm service classes is reasonable. The SCSP's recommended expansion, however, would trigger incremental

billing requirements and costs. KEDLI's support for Staff's recommendation is subject to the ability to recover the incremental costs of the expanded programs, including if necessary through the proposed discount program deferral mechanism.

Q. Please address Staff's recommendation regarding the Companies' proposed discount programs deferral mechanism.

A. The SCSP recommends (pp. 61-62) that the Commission reject the Companies' proposed deferral mechanism to reconcile actual discounts against targets established in rates. The Companies disagree with Staff's recommendation. Although discounts associated with the KEDNY and KEDLI discount programs have been relatively stable in recent years, the Companies' proposed changes to these programs, including the proposed new Area Development Rate for KEDLI, Staff's recommended inclusion of non-firm rate classes in the KEDLI discount programs, as well as other new program offerings recommended by Staff, are expected to increase the level of activity and uncertainty associated with future customer discounts under these programs. These factors support the Companies' proposed deferral mechanism. Further, because the proposal is for a two-way reconciliation, customers would be protected from paying more than the programs actually cost.

2. Additional Staff Recommendations

Q. Does Staff make any other recommendations regarding the Companies' economic development programs?

A. Yes. The SCSP recommends (pp. 64-66) the Companies:

(i) Create a new grant program for prospective or current customers seeking additional natural gas supply to become a non-firm customer.

(ii) Modify existing programs pointed toward new customers or customers seeking a service upgrade for additional supply to become or convert to being non-firm customers.

The SCSP proposes that each Company not receive any additional rate recovery beyond the $0.5 million proposed Rate Year allowance.

Q. How do the Companies respond to these recommendations?

A. The Companies generally agree that including non-firm customers in the portfolio of economic development grant programs is reasonable. Of Staff's two proposed options, the second option – whereby the existing programs and the proposed new programs would be modified to allow non-firm customers to become eligible and to allow the infrastructure related costs of converting from firm to non-firm service to qualify for additional funding – would likely be the more effective approach.

Implementing either option, however, would represent an expansion of the programs and would increase the level of customer participation and grant expenditures. Therefore, the Companies disagree with Staff's proposal to implement these changes without identifying a corresponding source of funding, as discussed below.

Q. Does the Panel have any other observations regarding Staff's recommendations on the Companies' economic development programs?

A. Yes. As shown in Exhibit___ (FOH-8), the Companies' initial filing anticipated Rate Year grant program spending of $2.25 million and $2.75 million for KEDNY and KEDLI, respectively. These spending levels are exclusive of Staff's proposed changes to include non-firm customers in the programs. The Companies proposal to set their respective Rate Year rate allowances to $0.5 million was specifically premised on the use of funds from the existing economic development carryover balances to make up the difference between the rate allowance and program spending. The SCSP acknowledges this in its testimony (p. 61). The SCSP, however, is otherwise silent as to the Companies' use of the carryover balance amounts to fund projected economic development grant program spending. Absent additional funding or the ability to fund incremental

grant expenditures from the existing deferral balances, the Companies would have a Rate Year shortfall of $1.75 million and $2.25 million for KEDNY and KEDLI, respectively. These shortfalls would be even greater if the program expansions proposed by the SCSP and discussed above are adopted.

The Companies interpret the SCSP's silence on the proposed use of the unspent balances in the Rate Year to fund the costs of the programs in excess of the proposed $0.5 million rate allowances as Staff's acceptance of the Companies' proposal on this matter. To the extent Staff's position is that the unspent economic development program balances should not be available for such use, the Companies disagree with Staff.

Q. Please describe Staff's recommendations regarding reporting on the Companies' economic development programs.

A. Staff recommends the Companies file annual reports with the Secretary to the Commission 30 days following the end of each calendar year, beginning with the Rate Year. The proposed reports would provide detail regarding applications received and approved, grant money committed, grant funds disbursed and the amount of reserve funds, and

would include a summary of the success of the proposed new Future of Heat Economic Development program, if approved by the Commission. The Companies accept Staff's proposed reporting requirements, subject to the time for filing the report being 60 days following the end of the year. The additional time is needed to assemble the requisite data and assure the quality of the information reported. In addition, the Companies recommend that the annual filing provide an opportunity for the Companies to propose program modifications for the following year.

G. Reducing Carbon Emissions on the Gas Network

1. Renewable Natural Gas ("RNG")

a) Newtown Creek Project

Q. Please address Staff's recommendation (SPP pp. 39-57) with respect to the Newtown Creek Project.

A. The Panel disagrees with Staff's proposed adjustment regarding the Newtown Creek Project. Staff's agreed that KEDNY should proceed with the project as part of the Company's last rate cases (Case 16-G-0059). The adjustment Staff proposes here would discourage the Companies, and other utilities, from pursuing innovative solutions to address GHG emissions. As stated previously, climate change is an issue that requires action now, and decisions that discourage innovative approaches in favor

of the status quo can be detrimental. Further, even assuming the Company agreed to an adjustment as Staff proposes, Staff erred in how it calculated and applied the value of the adjustment.

The basis for Staff's downward adjustment and capping cost recovery based on revenues generated is that the cost of gas produced by the project is higher than the Company's weighted average cost of gas. The project was not intended to produce RNG at the cost of traditional natural gas. Rather, the project was intended to help meet New York City's environmental goals while also demonstrating a new source of renewable energy, and how energy providers and other stakeholders can collaborate to incorporate RNG into the supply portfolio to meet sustainability goals, enhance reliability and utilize an existing energy network to meet the energy demands of New York City.

The fact that the unit cost of RNG is greater today than conventionally produced gas is not surprising and is reminiscent of similar comparisons between wind and solar electric renewables to fossil-fired generation sources a decade ago. Just as the cost disparity between renewable and fossil-fired electric generation has shifted with time, investments need to be made now to encourage the scaling of RNG to achieve cost savings. It

is imperative that the economy move to a much greater share of clean energy sources. Indeed, the CLCPA mentioned earlier requires that the state move to 70 percent renewable generation by 2030 and that New York State achieve an 85 percent reduction in GHG emissions by 2050. Fulfilling this mandate will require innovation and the Company believes it will require decarbonization of the gas network through the development of RNG. It is the Panel's position that, for RNG to become more prevalent and for the costs of RNG to approach parity with conventionally produced resources as renewable and conventional electricity sources have, there must be regulatory support for investments such as the Newtown Creek project.

The benefits of the Newtown Creek project extend far beyond the average cost of the RNG produced. For example, Exhibit __(FOH-3R), is a copy of 2017 presentation delivered by the American Gas Association's Chief Regulatory Counsel at a national EPA Gas STAR conference that highlights the Newtown Creek project as a case study for utilities' RNG interconnections, and for the potential of public-private partnerships to reduce metropolitan GHG emissions. To focus on the unit cost of the gas produced and injected into the system ignores those other important

benefits. To make progress on the ambitious clean energy targets, the industry must innovate.

It should also be noted that this project was developed in collaboration with the City of New York at one of the most complex wastewater treatment plants in the country creating a set of unique challenges that escalated the cost of this project over time. These challenges include obtaining the approval of Public Design Commission to ensure that the purification facility was aligned with the Newtown Creek masterplan and its overall architecture. The project site also had to be moved after detailed engineering assessment highlighted a risk that was unforeseen in the preliminary engineering analysis.

Q. On what other basis does the Panel disagree with Staff's adjustment to the Newtown Creek Project?

A. Even if KEDNY agreed with the basis for Staff's adjustment, Staff erred in how it calculated and applied that adjustment for two reasons. First, Staff intended to increase the project's capital costs based on the current estimate of $37.898 million. However, Staff included the entire $37.898 million estimate amount as a fiscal year 2020 plant addition on Exhibit __ (SPP-3), but did not take into consideration the Company's fiscal year

2021 capital addition of $0.051 million in Exhibit __ (SPP-3). As a result, Staff overstated the estimated project cost in Exhibit __ (SPP-3) by $0.051 million.

Second, Staff's intent was to impute a revenue adjustment on the revenue requirement of the Newtown Creek project, which is the rationale behind their calculation including the Company's current project estimate of $37.898 million. However, Staff failed to increase their revenue requirement in Exhibit __ (SSP-3) to use the same $37.898 million project estimate. Staff confirmed this oversight in response to IR NG-13, a copy of which is included as Exhibit __ (FOH-4R). As a result, the Company is revising the Newtown Creek project estimate in its capital forecast shown in Exhibit __ (GIOP-1R) to include the current project estimate of $37.898 million, consistent with Staff's revised Exhibit __ (SGIOP-4).

b) P2G Demonstration Project ("P2G Project")

Q. Please address Staff's recommendation regarding the P2G Project.

A. The SESP recommends the proposal be rejected because it is not identified as one of the "envisioned solutions … for excess renewable electricity" in the Commission's December 12, 2018 Order Establishing

Energy Storage Goal and Deployment Policy in Case 18-E-0130 ("Storage Order"). Staff's basis for rejecting the proposal is contrary to the spirit of innovation necessary for the state to achieve its emissions reductions objectives and contrary to the intent of the Storage Order.

The Storage Order set a target of 1500 MW of energy storage deployment by 2025 and an aspirational deployment goal of 3000 MW energy storage statewide by 2030. It did not preclude technology types and neither does Public Service Law Section 74, which compelled the Commission to develop targets and policies. The law expressly identifies a range of storage types for various purposes, including the use of chemical reactions to store energy in usable forms by defining a "qualified energy storage system" as one which is "commercially available technology that is capable of absorbing energy, storing it for a period of time, and thereafter dispatching the energy using *mechanical, chemical, or thermal processes* to store energy that was generated at one time for *use at a later time.*" PSL §74 (emphasis added).

Even if one reads PSL §74 and the Storage Order narrowly to only target storage related to the electric system, the P2G demonstration meets this criteria. Adding net carbon neutral gas to the gas distribution system can promote a cleaner electric system by reducing gas system constraints and

thus allowing RNG to displace the combustion of dirtier fuels. Further, by providing an additional end use market for carbon-free electric generation, P2G may stimulate the development of additional renewable generation resources.

Further, as the Commission has articulated, the Energy Storage Roadmap "is technology-neutral and acknowledges that a range of energy storage solutions will be deployed to best meet customer and system needs." (Storage Order p. 5).

Finally, as the Panel has mentioned previously, the CLCPA is the most progressive decarbonization legislation in the United States and to achieve the state's decarbonization mandates, the Companies must enable innovation and facilitate the addition of new technologies and business models. If commercialized at scale, P2G has the potential to significantly decarbonize the gas network. This project will provide insight into how to integrate such resources into the gas network and streamline the interconnection process for this new gas source.

Accordingly, the Panel disagrees with Staff's rationale for opposing the P2G Demonstration Project.

c) Hydrogen Blending Study

Q. Please address Staff's recommendation regarding the Companies' proposed Hydrogen Blending project.

A. Similar to the P2G Demonstration project, the SESP opposes (pp. 59-60) the proposed Hydrogen Blending study because hydrogen blending is not a specific solution identified in the Commission's Storage Order. The Companies disagree with Staff's position.

The Companies' primary interest in hydrogen is as an additional potential pathway toward decarbonization of the gas network. Hydrogen production is not a function of biomass and waste feedstock availability, and thus hydrogen blending could play a significant role in expanding the potential of low carbon gas and achieving the state's deep decarbonization goals. In terms of cost and customer experience, hydrogen blending may be the least disruptive way to decarbonize the gas network. Through the hydrogen blending study, the Companies seek to understand what actions will be required to prepare gas system infrastructure for hydrogen.

The proposed hydrogen blending study would provide necessary information for the development of a hydrogen readiness plan, focusing on safety, reliability, and operability of the Companies' specific network.

d) RNG Interconnections and Future of Heat Engineering Group.

Q. Please address Staff's recommendation regarding the Companies' proposed RNG Interconnections program.

A. Staff states (SESP pp. 56-57) that it supports the RNG Interconnections program, but opposes adding incremental FTEs to implement it. Staff also opposes the proposed KEDNY capital expenditures related to the program, and opposes the proposed O&M for the KEDNY projects under the program. The Companies appreciate Staff's conceptual support for the proposed program; however, we disagree with Staff's opposition to funding the incremental labor needed to implement the program, as well as the adjustments to remove KEDNY's associated Rate Year operating and capital expenditures under the program.

The Companies' RNG interconnection queue has increased significantly over the past few years, growing from zero to six proposals within KEDNY and KEDLI. Additional resources are needed to support the development of this local and renewable supply of gas. The Companies'

proposal is designed to alleviate two barriers to RNG interconnection, the extensive time it takes for a developer to receive interconnection approval and the high capital cost of the interconnection.

The time it takes to complete the interconnection approval process is a barrier to the development of RNG projects. The additional FTEs requested for the program would be responsible for processing RNG interconnection proposals, which would reduce the time to evaluate requests. The Companies currently rely on existing gas engineering personnel to support the evaluation of interconnection proposals and conduct feasibility analyses and interconnection design. These existing FTEs are dedicated to supporting the Companies' traditional gas engineering activities, and RNG interconnection processing is additive to their existing workload. As a result, interconnection processing has been delayed. RNG developers rely on receiving timely interconnection approval to solidify project financing. Thus, delays in interconnection processing inhibits RNG development. For these reasons, and as is stated in the Companies' GIOP rebuttal testimony, the incremental FTEs included in the FOH Engineering Group are necessary even if the capital RNG Interconnection Program were to be deferred by a year.

The cost of RNG interconnections has been identified as a key barrier to commercializing RNG. Much of the cost of interconnection is due to equipment required by gas utilities, including meters, odorization systems and gas quality analyzers. Under the current process, project developers are responsible for the cost of this equipment, while the equipment itself is maintained and operated by the utility. Project developers in the Companies' interconnection queue have indicated that this upfront cost is prohibitive to the development of their proposed projects. The Companies' proposed RNG Interconnection program would unburden project developers from bearing the cost of equipment that is core to utility functions - *i.e.*, equipment required to verify gas quality and quantity before it is received into the gas network.

Q. Do the Companies' agree with Staff's position that FTEs can be reallocated from other portions of the gas business?

A. No. First, Staff (SESP p. 57) suggests that the RNG program is insufficiently mature to warrant dedicated FTE resources, and also proposes the potential reallocation of FTEs associated with the Clean Conversion Program. As described above, the RNG program is growing and accelerating the pace of RNG implementation is an important element in decarbonizing the energy system. Further, contrary to Staff's

suggestion, the previous elimination of the Clean Conversion Program also eliminates any funding associated with the labor resources needed to deliver that program. It would be a double count to propose that the Clean Conversion Program not be funded, and that the labor resources associated with that unfunded program be used to deliver the RNG Interconnections program. In addition, even if incremental resources were available within the Companies, it is unlikely such resources would be immediately qualified to perform the work needed of the Future of Heat Engineering Group.

As the Companies work to meet heating needs through a decarbonized gas supply and achieve aggressive clean energy goals using NPAs and RNG projects, additional resources will be needed to manage NPA market solicitations, review project proposals on a timely basis, and coordinate with project partners and other interested stakeholders. The new Future of Heat Engineering Group will need to have the skillset to understand and evaluate RNG, hydrogen blending, and NPA solicitations to determine whether they are cost-effective, viable solutions, and, in some instances, whether they are capable of displacing traditional capital projects.

H. Performance-Based Incentives and Revenue Sharing

1. EAMs

Q. Does the Panel agree with the Staff EAM Panel's recommendations regarding the Companies' proposed package of EAMs?

A. No. The SEAMP makes four overall recommendations (pp. 13-17) regarding the Companies' EAM proposals:

(i) reject KEDNY's proposal to earn up to 78 basis points and KEDLI's proposal to earn up to 77 basis points in the Rate Year at maximum achievement levels and replace those with a program allowing each company to earn up to 10 basis points maximum, plus earnings from a Share the Savings energy efficiency metric;

(ii) impose more aggressive performance measures on the Companies;

(iii) express the eligible EAM amounts in terms of absolute dollars rather than as basis points; and

(4) in the event a multi-year rate plan is adopted, establish EAM metrics, targets and financial incentive levels for the full term of the rate plan.

Staff also recommends (SEAMP p. 19) that EAMs that may be established in this case be "reconsidered as part of a collaborative among the Companies, Staff, and other stakeholders" if the Commission authorizes new programs or significantly increased budgets as part of a statewide

proceeding. However, Staff proposes that the number and value of basis points available should remain as set in this case.

Q, What is the Panel's position regarding the SEAMP's proposal to replace the Companies' proposed package of EAMs with Staff's two proposed EAMs?

A. The Panel generally disagrees with Staff's proposal. Staff's EAM proposal is most effectively illustrated in Exhibit __ (SEAMP-2), which lists six proposed EAM metrics each for KEDNY and KEDLI, with potential aggregate maximum values in the Rate Year of 78 and 77 basis points, respectively, compared to Staff's proposals of two EAMs each for KEDNY and KEDLI, with maximum values in the Rate Year of 25 basis points. Staff's proposed EAM maximum values are significantly below what the Companies proposed and would not produce the incentives necessary to encourage desired outcomes.

In the Track Two Order (Case 14-M-0101, Order Adopting a Ratemaking and Utility Revenue Model Policy Framework (May 19, 2016)), the Commission stated that incentive mechanisms should provide utilities with stretch targets that are reasonably and realistically achievable with significant utility efforts and with financially meaningful incentives. Thus,

the Commission selected 100 basis points as the reasonable and appropriate ceiling for EAMs. The potentially achievable maximum incentive levels proposed by Staff are only one-quarter this level and less that one-third what the Companies proposed. That level of incentive would not serve as the mechanism the Commission envisioned.

Further, although there may be merit to a system peak reduction metric, the Companies question whether the capabilities are in place to effectively implement and administer such a metric by the Rate Year. Likewise, while a Share the Savings metric may have benefits, the Companies do not support adopting such a metric to the exclusion of all other EAMs.

The Companies' positions regarding the SEAMP's specific application of its overall recommendations are presented below.

a) System Efficiency

(1) Peak Reduction

Q. Please address Staff's recommendation regarding the Peak Reduction metric of the System Efficiency EAM.

A. The Companies agree an EAM metric designed to encourage the reduction of gas peak demand would be beneficial, and support the Commission's preference for an outcome-based EAM. However, current capabilities are

insufficient to adequately measure peak reduction on a system-level and there are limitations on the ability to adequately measure peak reduction at a system-level or isolate it from other large macro-economic variables. Similar concerns regarding Staff's proposed outcome-based gas peak reduction EAM were raised in the recent Consolidated Edison gas case (Case 19-G-0066) (Customer Energy Solutions Update and Rebuttal Testimony, p. 219).

The Companies are open to working with Staff and other stakeholders to try to develop a system-level peak reduction methodology; however, until a more robust peak reduction metric can be developed and implemented, the Commission should approve the peak-reduction metric proposed by the Companies' in direct testimony (as corrected in the Panel's Corrections & Updates filing). A programmatic-based metric would push the Companies to continue to drive peak reduction, while the capabilities needed to allow for system-wide peak reduction measurements are developed.

(2) Supply Diversification

Q. Please address Staff's recommendation regarding the Supply Diversification metric of the System Efficiency EAM.

A. Staff (SEAMP p. 28) opposes the Supply Diversification metric on the basis that the Commission's support for fuel and resource diversity was intended in the context of diversifying the electric grid away from reliance on gas, and the Track Two Order did not identify diversification of fuel sources within a gas utility's supply portfolio as an EAM-worthy metric.

The Companies believe that the Supply Diversification EAM metric is consistent with the intent of the Track Two Order, which was to align utility shareholders' financial interests with customers' interests. As with Staff's recommended Peak Reduction EAM, which was not initially included in the Track Two Order for gas EAMs, the Supply Diversification EAM would incentivize the Companies to prioritize actions that are not currently aligned with the Companies' business model. Although RNG can help the Companies address supply constraints, they currently have no incentive to prioritize RNG interconnection over the interconnection of traditional natural gas supply infrastructure or non-traditional natural gas supply infrastructure – *e.g.*, CNG or LNG. The Companies currently generate no profit from interconnecting third-party RNG projects. The Companies should be incentivized to prioritize RNG interconnections, because of the unique benefits of RNG, including emissions reductions and improved local waste management.

Q. Do the Companies' agree with Staff's position (SEAMP pp. 29-30) that $1 of value created by RNG interconnection results in $4 in incentives?

A. The Future of Heat testimony cited by Staff is incorrect. Although the testimony (FOH direct p. 102) states the Companies "propose setting the minimum target at one basis point," the two tables setting forth the minimum, target and maximum levels (FOH direct p. 102) clearly show the minimum target for both KEDNY and KEDLI to be four basis points. The minimum target was set at the volume of RNG required to achieve four basis points worth of value and the incentive was also proposed to be four basis points.

b) Energy Efficiency

(1) Incremental Energy Efficiency

Q. Please address Staff's recommendation (SEAMP pp. 30-41) regarding the Incremental Energy Efficiency metric of the Energy Efficiency EAM.

A. The SEAMP explains (pp. 32-39) that the Incremental Energy Efficiency EAM is not based on verified gross savings and therefore is not responsive to Commission precedent and further recommends replacing the proposed Incremental Energy Efficiency EAM (as well as most of the Companies' other proposed EAMs) with Staff's proposed Share the Savings EAM.

Although the Companies acknowledge that Staff's proposed Share the Savings EAM model has aspects of a potentially meaningful incentive, the Staff's proposal to replace virtually all of the Companies' proposed EAMs with a metric projected to provide 15 basis points of maximum value is unacceptable to the Companies.

With respect to verified gross savings, the Companies agree that the methodology outlined in the recent guidance document issued by Staff should be applied against the Incremental Energy Efficiency EAM target as well as to other programmatic energy efficiency-related EAMs.

Q. Please address the SEAMP's recommendation (pp. 39-40) that the Companies forecast verified gross savings achievement on June 1 of the year following annual programmatic activity and begin to collect their EAM incentive payment based on that forecast in September.

A. The Companies accept the SEAMP's recommendation. By June 1 of each year, the Companies will forecast verified gross savings achievement for programmatic activity that took place during the prior calendar year, beginning June 1, 2020 for activity in calendar year 2019. In September of each year, the Companies will begin to collect any EAM incentive payment earned based on that year's forecast. The Companies also agree

to reconcile the forecast and adjust their collections as necessary at time of the forecast of their next EAM incentive payment.

Q. Do the Companies agree with the SEAMP's position (p. 40) that the Companies conduct an annual reconciliation of the prior year's forecasted EAM incentives when setting the incentive payment collection levels to be effective in September of the year following program activity?

A. Yes. The Companies agree with this recommendation.

Q. Are there any other SEAMP positions related to the Incremental Energy Efficiency metric of the Energy Efficiency EAM that you would like to address?

A. Yes. The Companies agree with the SEAMP's position (pp. 40-41) that, if the Companies have concerns with the recognition of EAM revenues, they can recognize a portion of forecasted incentive level until a full reconciliation occurs based on completion of the verified gross savings process, which would limit the risk of future downward adjustments.

(2) Affordability

Q. Please address Staff's recommendation regarding the Affordability metric of the Energy Efficiency EAM.

A. The Companies disagree with Staff (SEAMP pp. 41-42, 47) that the Affordability metric should be rejected. First, the Companies believe that the outcome-oriented design of the Affordability metric means that it encompasses not only programmatic efforts of the utility to directly influence low-income customer energy usage, but also that it reflects utility partnerships and platforms meant to encourage markets to reduce low-income customer bills. As affordability of energy bills remains a top priority for both the Companies and the State, this metric should be adopted to promote deeper engagement by the market in this population and to measure the ultimate impacts of those activities.

The Companies accept Staff's position that the targets proposed in the direct testimony should be higher. Therefore, the Companies propose to adjust the targets for the Affordability metric from 0.2 percent to 1.0 percent reduction in the low-income customer bill. That is, the Companies propose to earn one basis point for each one percent reduction in the average low-income customer bill. Unlike other proposed EAM metrics, the Companies have not proposed to earn prorated incentives for this metric.

c) Carbon Reduction

(1) CO_{2e} Reduced from Full-Service Customers

Q. Please address Staff's recommendations regarding the full-service Customer CO_{2e} Reduction metric and the transportation CO_{2e} Reduction metric.

A. Staff opposes (SEAMP pp. 47-52) the CO_{2e} Reduction metrics on the grounds that GHG reduction metrics should focus on decarbonization of end-uses, and the proposed metrics are wholly programmatic

The Companies do not agree. The CO_{2e} Reduction EAM metric aligns with the intent of the Track Two Order, which was to align utility shareholders' financial interests with customers' interests. The CO_{2e} Reduction EAM would incentivize the Companies to prioritize actions that are not currently aligned with their business model.

Reducing GHG emissions is clearly identified in Track Two and enshrined in state policy as a priority. Decarbonization of end uses can be more readily implemented, tracked, and measured, such as solutions like heat pumps or electric vehicles. Some end uses, however, are more challenging to decarbonize – i.e. existing building stock in dense urban environments. The Companies believe in this instance it may be more

cost effective and less disruptive to decarbonize the energy supply. The Companies' proposed EAM is inclusive of achieving emissions reductions in whatever manner makes most sense for customers – *e.g.*, converting to a geothermal heat pump or continuing to utilize their existing equipment and lowering their carbon footprint by consuming RNG.

2. Value of a Basis Point

Q. What is the Companies' position regarding Staff's recommendation to express the value of EAMs in terms of absolute dollars rather than as basis points once the capital structure and rate bases are determined in this case (SEAMP p. 15)?

A. The Companies agree with Staff's position in principle and will update the calculated value of a basis point for KEDNY and KEDLI at the end of these proceedings.

3. Platform Service Revenues ("PSRs")

Q. Please address Staff's position regarding PSRs for revenues from the E-Commerce Platform.

A. The Companies do not agree with Staff's recommendation (SEAMP pp. 54-57) that they should not receive a PSR for the E-Commerce Platform. Staff asserts the Companies did not satisfy the criteria in the Track Two Order to warrant justifying a PSR. However, in setting forth the criteria

for whether a PSR should be approved, the Commission specifically noted that the "criteria are intended to support balanced judgment, as opposed to a set of binary determinations that must all be made in the affirmative." Track Two Order, n. 57. Further, the Commission specifically recognized customer origination services as the type of service that would be suitable for a PSR, explicitly stating that "utilities may be best situated to provide origination in a cost-effective manner that facilitates market penetration, based on analysis of customer data and/*or connecting interested customers with DER providers via a marketplace portal*." *Id.* (italics added). The Commission recently approved a PSR for Central Hudson (Cases 17-E-0459 and 17-G-0460) relating to its E-Commerce Platform. There is no rational basis for denying a PSR for the Companies' proposed E-Commerce platform, and it should be approved.

Q. Please address Staff's recommendation (SEAMP pp. 54-60) regarding PSRs for the UESC Program.

A. The Companies disagree with Staff's recommendation to reduce the Companies' respective share of PSRs from the proposed amount of 33 percent to 20 percent. Under the UESC program, the Companies are providing an enhanced service to customers by analyzing their current

energy usage, identifying opportunities for improvement, and working with pre-selected contractors to achieve efficiencies, reduce costs, and adopt renewable energy systems. While benefits are achieved through the UESC program for gas distribution customers to compensate for the customer funds utilized to administer the program, the enhanced value achieved towards the State's energy goals through the Companies' holistic approach should be compensated above the typical 20 percent share.

4. Non-Pipeline Alternatives ("NPA") Incentive Mechanism

Q. Please address Staff's position (SEAMP pp. 75-77) regarding the Companies' proposed NPA Incentive Mechanism.

A. The Companies agree with Staff's proposal that a specific NPA Incentive Mechanism that would apply to all NPA projects not be established at this time. Rather, the Companies agree that the design of NPA Incentive Mechanisms should be presented at the time NPA proposals are presented with their proposed implementation plans.

5. KEDLI EmPower Replacement Program Incentive Mechanism

Q. Please address Staff's recommendations regarding KEDLI's EmPower Replacement Program Incentive Mechanism.

A. Staff agrees (SEAMP pp. 61-65) KEDLI should continue to be eligible for incentives to maximize participation in the program, however, Staff recommends redesigning the incentive mechanism to set it as a flat dollar amount per household, regardless of program budget and restructuring the Tier 4 participation incentive. Specifically, Staff proposes a Tier 3 incentive of $125/household and Tier 4 of $62.50/household ($187.50/household in both Tiers 3 and 4).

KEDLI agrees with the structure concept of the HEAT program incentive being based on a flat rate per household participating in each Tier 3 and Tier 4, and further agrees that the incentive amount for Tier 4 should be reduced comparative to Tier 3. However, the Company proposes that the incentive amounts be set at $240 for each household participating in Tier 3 and $120 for each household participating in Tier 4. These levels more closely match the earnings levels of the current earnings incentive mechanism while recognizing the energy savings value primarily achieved by customer participation in Tier 3.

The Companies also support incorporating the KEDLI HEAT Program into its energy efficiency portfolio with the $2.5 million budget being additional to the existing energy efficiency budget and operating the program in accordance with the rules and requirements associated with the current energy efficiency portfolio and the statewide LMI customer framework being developed. The KEDLI HEAT Program will follow the benefit-cost analysis ("BCA") requirements established in the December 2018 Order Adopting Accelerated Energy Efficiency Targets for LMI programs. Additionally, the KEDLI HEAT Program will report its energy savings in line with the process for reporting energy savings for the Companies' existing energy efficiency portfolio, but will report on and determine success by the number of unique households served, in line with the proposed incentive mechanism for the program.

IV. Response to Other Parties

A. City of New York

1. Energy Efficiency

Q. Please address the recommendation by the City of New York ("City") that the Companies increase their respective incremental

energy efficiency target savings based on lower per-unit costs for energy efficiency savings achieved in last two years.

A. The Companies disagree with this recommendation by the City's Energy Efficiency Panel ("NYCEEP"). Estimating the costs to achieve savings targets going forward based on a backward-looking analysis of average costs for 2017 and 2018 does not necessarily reflect the Companies' expectations of costs in the Rate Year and beyond, especially with the expectation that future energy efficiency investments will need to go "deeper" than in the past. The City itself acknowledges (NYCEEP p.8), increasing target savings without a corresponding increase in spending allowance could encourage a focus towards lower cost-quick payback measures, and away from higher cost solutions that might provide greater long-term benefit.

Q. Please address the City's recommendation that the Commission incentivize comprehensive packages of energy efficiency measures.

A. The City recommends (NYCEEP pp. 10-17) that energy efficiency program offerings should be modified in several ways, including incentivizing customers to pursue comprehensive energy efficiency packages (*e.g.*, stacking incentives), designing programs to be delivered in stages or over a longer period of time-rather than all at once, and/or

offer higher incentives when measures are combined, and implementing changes to enable or improve access to private funding sources.

The Companies are not opposed to evaluating other potentially effective energy efficiency strategies, and believe the City identifies a number of approaches worth considering. However, these rates case are not the appropriate forum for doing so. Rather, a more appropriate forum to consider such changes would be in the Commission's broader energy efficiency proceedings. Some of the recommendations proposed by the City may require changes to the existing Technical Resource Manual or Gross Savings Verification Guidance, and application of benefit costs analyses that to date have not be performed. The Companies are open to examining these issues in the context of a broader proceeding or collaborative; however, not as part of these rate cases.

Q. Please address the City's recommendation regarding increased energy efficiency reporting.

A. There are existing comprehensive reporting requirements applicable to the Companies' energy efficiency programs. Many of the reporting parameters have been established by Staff or the Commission and are set forth in guidance documents the Companies follow. In addition, the

Commission's comprehensive New Energy New York proceeding will likely establish new energy efficiency reporting requirements. Although the Companies are open to collaborating with the City (including the Retrofit Accelerator) to determine how to improve information sharing and communications consistent with applicable guidelines, we do not agree that new reporting requirements should be implemented in the context of these rate cases.

Q. Please address the City's recommendations regarding changes the Companies should make to their proposals in the event NESE is not approved.

A. The Companies address the City's recommendations in the rebuttal testimony of Elizabeth D. Arangio and the rebuttal testimony of the Companies' Gas Infrastructure and Operations Panel.

2. Demand Response

Q. Please address the City's recommendations regarding expanding the Companies' demand response programs.

A. The City appears to support (NYCEEP pp. 24-25) the Companies' proposed Demand Response Demonstration ("DRD") project expansion, and suggests that the Companies consider further expanding the project to reach a broader pool of potential participants. The Companies agree

that the DRD project has the potential to provide significant benefits and deliver valuable learnings. The Companies are open to exploring the potential further expansion of the DRD project once additional experience has been achieved; however, at this point expanding the reach of the project beyond what is currently proposed appears premature.

3. Sustainable Heat Initiative

Q. Please address the City's recommendation (NYCPP pp. 26-31) that the Companies conduct a feasibility study to assess potential district energy systems.

A. The Companies are proposing an expansion of the KEDLI Riverhead Demonstration project to assess potential utility ownership of geothermal underground loop systems. The City's proposed study could further the understandings of the feasibility and potential benefits of similar systems in institutional settings such as colleges. District-type systems can be complex, and completion of technical and feasibility studies would be necessary before a pilot is considered or pursued. At this point, the Companies have not estimated the costs to perform such studies, nor budgeted for the associated costs of the studies in their revenue requirements.

4. Performance-Based Incentives and Revenue Sharing (EAMs)

Q. Please address the City's recommendations regarding the Companies' proposed EAMs.

A. The City supports (NYCEEP pp. 30-42) the concept of EAMs, but opposes specific EAMs proposed by the Companies. The Companies agree in principle that EAMs should be designed to provide increasing levels of incentive for increasing levels of performance; however, we disagree with the City that the EAMs proposed by the Companies are inadequate or otherwise inappropriate.

Regarding the System Efficiency (Peak Reduction) metric, the Companies propose an increasing scale of incentives, where the minimum incentive requires the Companies to achieve their target performance. If they do not achieve the targets, the EAM is not triggered. The potential to earn above the minimum requires incremental effort and performance. As described above in relation to the Staff's position on this metric, the Companies support the concept of developing a more robust peak reduction metric design; however, until such a metric is developed and can be implemented, the Commission should approve the Companies' proposed peak-reduction metric.

Regarding the City's position (NYCEEP pp. 41-42) on the incremental energy efficiency metric, the City reiterates its argument summarized above that the targets should be restated based on the Companies' cost experience in 2017 and 2018. For the reasons the Panel stated above regarding the City's argument, the Companies do not agree with this position.

With respect to the City's position (NYCEEP pp. 33-36) on the proposed Affordability metric, the Companies accept the proposal to increase the performance target as described previously in the Panel's rebuttal testimony where we respond to the Staff's position on the Affordability EAM metric. However, subject to this modification to the proposed metric, the Companies do not agree with the position that this metric should be rejected.

Regarding the Carbon Reduction EAM, the Companies disagree with the City's recommendation (NYCEEP pp. 37-39) that the metrics be rejected. The City's filing in this case acknowledges the importance of GHG reductions and the development of projects to promote RNG, including for use in vehicles. The Companies' Carbon Reduction EAM metrics are

focused on achieving both of these important objectives. The City acknowledges (NYCEEP p. 39) that it has no specific metric to offer in place of the Companies' Carbon Reduction metrics. In the absence of more effective incentives to encourage these objectives, the Commission should approve the Companies' proposed Carbon Reduction EAM metrics.

5. Other Topics

Q. Please address the City's recommendations regarding expansion of eligibility for non-firm service.

A. The City makes several recommendations regarding the expansion of, and criteria relating to, non-firm service, which are addressed in the Rebuttal Testimony of the Companies' Rate Design Panel.

Q. Please address the City's recommendations (NYCEEP pp. 26-30) regarding expansion of electrification of the Companies' fleet.

A. The Companies acknowledge the importance of electrifying transportation in order to achieve the state's emissions objectives. However, the current commercially available electric vehicle ("EV") market is focused primarily on passenger vehicles, whereas the vast majority of the Companies' fleet is comprised of other vehicle types. The Companies know the EV market will continue to evolve and they will continue to

pursue appropriate opportunities to introduce EVs into their fleets; however, in the absence of commercially available EV options to meet many of the Companies' vehicle needs, the Companies do not agree with the City's proposal.

B. The Environmental Defense Fund ("EDF")

Q. Please address EDF's recommendations in this proceeding.

A. EDF sets forth recommendations that address, among other things: (1) development of comprehensive utility baseline GHG inventories and a broad-based GHG EAM (Ucar); (2) gas supply planning (Lander); and (3) leak detection and repair (Von Fischer). The Panel addresses item (1), above.

Q. Please describe EDF's recommendation regarding developing utility GHG inventories.

A. EDF recommends (Ucar pp. 9-12) that gas utilities work with state and local entities and electric utilities to gather economy-wide information to establish a current, holistic GHG emissions status and baseline attributable to the utility's geographical footprint. The Companies agree that understanding the nature and extent of GHG sources is vital to developing effective strategies to address emissions. We also recognize,

however, that GHG sources exist in all sectors of the economy, and utilities do not necessarily possess the resources necessary to develop such economy-wide GHG inventories. Although the Companies are open to participating in collaborative efforts to arrive at appropriate GHG inventories against which reductions can be assessed, we do not agree that utilities should be responsible for developing economy-wide GHG inventories as EDF proposes.

Regarding EDF's potential incentive for broad GHG emissions reductions, the Companies agree that incentives to promote such reductions are desirable; and in fact the Companies proposed two Carbon Reduction metrics as part of its proposed package of EAMs in this case.

C. Utility Intervention Unit ("UIU")

Q. Please address UIU's recommendation regarding the Companies' proposed Affordability EAM.

A. UIU recommends (Collar pp. 23-26) that the Companies' proposed Affordability metric be rejected. The Companies accept the proposal to increase the performance target as described previously in the Panel's rebuttal testimony where we respond to the Staff's position on the Affordability EAM metric. However, subject to this modification to the

proposed metric, the Companies do not agree with the position that this metric should be rejected.

D. Public Utility Law Project ("PULP")

Q. Please address PULP's recommendations regarding matters presented in the Future of Heat Panel's proposals.

A. PULP addresses (Yates pp. 40-43) two issues: (1) support for the new LMI Gas Conversion Program; and (2) opposition to the Affordability metric of the Energy Efficiency EAM.

Regarding the LMI Clean Conversion Program, the Companies have determined they will not pursue those programs in light of the current moratoria on new connections.

Regarding the Energy Efficiency EAM Affordability metric, the Companies accept the proposal to increase the performance target as described previously in the Panel's rebuttal testimony where we respond to the Staff's position on the Affordability EAM metric. However, subject to this modification to the proposed metric, the Companies do not agree with the position that this metric should be rejected.

E. Estates NY Real Estate Services LLS ("Estates/NYRES")

Q. Please address Estates/NYRES's recommendations regarding the Future of Heat Panel's proposals.

A. Estates/NYRES (Rodriguez pp. 7-9) recommends expanding the Companies' demand response proposal, including the possible combination of targeted electrification technologies to support gas demand management. The Companies are open to exploring the potential further expansion of the DRD project once additional experience has been achieved; however, at this point expanding the reach of the project beyond what is currently proposed appears premature.

Estates/NYRES (Rodriguez pp. 5, 11-12) also supports the Companies' proposals to promote increased RNG production and utilization, as well as hydrogen blending, as opportunities to decarbonize the energy system while avoiding waste associated with the abandonment of existing natural gas infrastructure; and to also explore using the gas network as a potential large-scale seasonal energy storage and delivery system.

Estates/NYRES also supports (Rodriguez pp. 9-11) expanded use of geothermal technologies such as the Geothermal Demonstration project,

and strongly encourages the Commission to explore a potential for a regulatory framework for the creation of thermal companies.

Estates/NYRES also urges caution in how the Companies proceed with decarbonization so as to avoid creating significant stranded costs that would be paid for by a shrinking pool of customers, most likely low-to-moderate income customers with fewer potential alternatives.

F. Sane Energy Project ("Sane")

Q. Please address Sane's recommendations.

A. Sane presents several positions and proposals in its testimony (Fraczek & Ziesche). The Company rejects Sane's proposal that the Companies not invest to address leak prone pipe ("LPP"). As Witness Von Fischer of EDF describes, addressing leaks from LPP is vital in reducing methane leaks and addressing GHG emissions. Sane also opposes the Companies' Clean Conversion Program; however, as the Panel explained above, the Companies are not pursuing the program in light of the current moratoria on new connections.

Regarding Sane's recommendations on LNG and CNG investments, the Companies strongly disagree. The proposed LNG and CNG investments in the Companies' capital plans are for projects to maintain the safety and

reliability of these critical facilities that are often needed to meet peak demands. Failure to adequately maintain the safety and integrity of these facilities could jeopardize the ability of the Companies to meet the energy needs of New Yorkers when they are most critical.

With regard to radon testing, Sane points to study findings of increased radon levels in buildings located near active hydraulic fracturing sites to support its recommendation that the Companies provide radon monitoring in all gas supplied to customers in New York City and Long Island. The Companies disagree. There is no basis for concluding that radon gas that may accumulate in buildings and basements proximate to fracturing operations in Pennsylvania is in the gas delivered to customers in New York.

Finally, the Companies respectfully oppose Sane's recommendation that it receive intervenor funding.

G. Robert Mark Wyman ("Bob Wyman")

Q. Please address the recommendations of Bob Wyman.

A. Bob Wyman recommends a number of changes in how utilities operate in order to eliminate use of gas. Bob Wyman's proposed approach, however, would likely cause substantial unintended disruption in the Companies'

ability to operate their networks to serve the energy needs of customers, not to mention the potential for significant financial impacts. Although he may hold sincere views on what should be done to curb fossil fuel use and reduce GHG emissions, Bob Wyman's recommendations, if adopted, would result in significant economic and operational disruption and would be infeasible in the near term. Accordingly, the Companies oppose the recommendations in Mr. Wyman's testimony.

V. Conclusion

Q. Does that conclude the Panel's rebuttal testimony.

A. Yes.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY AND KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Supplemental Testimony

of the

Future of Heat Panel

Dated: January 29, 2020

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I. Introduction and Qualifications

Q. Please identify the members of the Future of Heat Panel.

A. The Panel consists of Donald Chahbazpour, Lisa M. Tallet, Arthur W. Hamlin, Owen S. Brady, and McKenzie S. Schwartz.

Q. Is this the same Panel that previously submitted testimony on behalf of The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and the KeySpan Gas East Corporation d/b/a National Grid ("KEDLI") (collectively, the Companies) in this proceeding?

A. Yes, but with one change to the members of the Panel. McKenzie Schwartz is being added to the Panel. The terms defined in the Panel's previously filed testimony have the same meanings here.

Q. Ms. Schwartz, as a new member of the Panel, please state your name and business address.

A. My name is McKenzie S. Schwartz. My business address is One MetroTech Center, Brooklyn, New York 11201.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid Service Company and hold the position of Future of Heat Analyst in the Regulatory and Customer Strategy group. My responsibilities include developing the regulatory strategy and proposals for new products and programs focused on providing affordable, sustainable, and low-carbon heat.

Q. Please describe your education background and business experience.

A. I received a Bachelor of Science in Sociology from the University of Utah in 2015. After, I served as an AmeriCorps VISTA member at Summit Land Conservancy, where I continued to work after finishing the AmeriCorps program. In 2017, I completed a Master of Public Administration in Environmental Science and Policy from Columbia University's School of International and Public Affairs. Thereafter, I was hired by National Grid as an Associate Analyst in the Climate Change Compliance group, where I was responsible for filing mandated and voluntary emissions reports, preparing resiliency and climate change adaptation plans, and supporting climate change-related projects. In October 2018, I assumed my current role with the Regulatory and Customer Strategy group.

II. Purpose of Testimony

Q. What is the purpose of the Panel's supplemental testimony?

A. The purpose of the Panel's supplemental testimony is to discuss the impact of the Commission's "Order Authorizing Utility Energy Efficiency and Building Electrification Portfolios Through 2025" issued January 16, 2020 in Case 18-M-0084 ("NE:NY Order") on the Companies' Future of Heat proposals in these rate proceedings, consistent with the Administrative Law Judges' January 21, 2020 "Ruling Requiring Additional Testimony Related to Energy Efficiency Initiative Order."

Q. Does the Panel's Supplemental Testimony include any exhibits?

A. No.

III. NE:NY Order

Q. Please provide a summary of the NE:NY Order.

A. The NE:NY Order marks the next step in the State's energy transformation, driving significant market-enabling investments aimed at reducing energy consumption and achieving ambitious clean energy

goals. It establishes progressive energy efficiency budgets and targets for calendar years 2021 through 2025, directs resources to helping disadvantaged communities and low- and moderate-income ("LMI") customers achieve energy savings, identifies strategies to address gas supply-constrained areas, incentivizes the achievement of superior energy efficiency outcomes, and provides a framework to implement, review, and refine the overall approach to energy efficiency. The Companies share the State's vision for the clean energy future. To that end, this supplemental testimony incorporates specific elements of the shared vision, as set forth in the NE:NY Order.

Q. What elements of the NE:NY Order impact the Panel's Future of Heat proposals?

A. Most notably, the Companies have updated the revenue requirements to reflect the new energy efficiency budgets and targets set forth in the NE:NY Order. Furthermore, to deliver the benefits in the NE:NY Order and meet the reporting and compliance requirements contained therein, the Companies seek eight additional full-time equivalents ("FTEs"). The hiring of the FTEs will be staggered throughout the Rate Year and Data Year 1, as discussed below. The Companies also look to build on the

earnings adjustment mechanism ("EAM") framework to further incentivize achieving superior energy efficiency outcomes. Finally, the Companies highlight ways in which the forward-looking Future of Heat programs and services align with the "all cost-effective measures" approach to delivering clean energy solutions set forth in the NE:NY Order and embodied in the Climate Leadership and Community Protection Act ("CLCPA"). Each of these changes is described below more fully.

A. Budgets and Targets

Q. Please summarize the Companies' energy efficiency budget proposal.

A. The Companies' original rate filing used the presumptive targets and budgets in the Commission's December 13, 2018 "Order Adopting Accelerated Energy Efficiency Targets" in Case 18-M-0084 ("Accelerated Efficiency Order") to set the proposed Rate Year energy efficiency rate allowance. With the NE:NY Order, the Companies have incorporated the updated energy efficiency budgets for the Rate Year, which represents an incremental increase of $1.232 million for KEDLI and $0.070 million for KEDNY, as compared to the presumptive budgets

from the Accelerated Efficiency Order. Please see Exhibit __ (RRP-1TS) for further detail on the revenue requirement adjustments.

Q. How does the NE:NY Order impact the Companies' proposal to treat EM&V costs separately from the energy efficiency budget?

A. The NE:NY Order clarifies that the authorized energy efficiency budgets include EM&V costs. Accordingly, the Companies have adjusted their respective revenue requirement to remove EM&V costs from base rates and instead incorporate those costs in their energy efficiency program portfolio budgets.

Q. Will the Companies require additional FTEs to support the NE:NY Order?

A. Yes. The NE:NY Order ramps up the energy efficiency targets by 114 percent and 22 percent for KEDLI and KEDNY, respectively, for 2021 through 2025. As set forth in Exhibit __ (RRP-2TS), the Companies request eight FTEs in addition to the seven FTEs originally requested in the Companies' initial filing. The new positions are needed to enable the Companies to meet the new targets, optimize the energy efficiency

portfolios to achieve maximum customer benefits, and ensure compliance with regulatory requirements.

Q. Please describe the roles each of the eight additional FTEs would play in delivering the NE:NY Order initiatives.

A. Specifically, two FTEs would develop and build the business cases for additional products, defined as new programs, measures, or delivery streams. One FTE is requested to provide technical support to the energy efficiency offerings by reviewing technical/trade press for technology trends and providing technical input for aspects of new or existing pilots.

Two FTEs would have roles as product owners whose responsibilities include reviewing the performance of existing products, improving the commercial viability of products, and identifying challenges with existing products. The two product owners would work closely with product developers to launch solutions within the respective energy efficiency portfolios.

The Companies also seek one LMI project manager and one LMI marketing analyst to support the NE:NY Order's expanded focus on LMI

customer programs. The LMI project manager would work to identify LMI customers, collaborate with the New York State Energy Research and Development Authority ("NYSERDA"), other utilities, and stakeholders to better serve disadvantaged communities, and coordinate with the LMI marketing analyst to target energy efficiency-related outreach and marketing materials to LMI customers. The LMI marketing analyst would be responsible for marketing and outreach to LMI customers based on new initiatives aligned with NE:NY Order.

Finally, the Companies request one FTE, an analyst, to support the enhanced EM&V workload required by the NE:NY Order, which emphasizes close coordination with NYSERDA and Staff in performing evaluations. Likewise, the Verified Gross Savings guidance includes additional requirements that will increase the workload of the current EM&V resources.

Q. When would the Companies anticipate hiring the eight FTEs?

A. The Companies would seek to fill the three FTE positions dedicated to developing and building the business case and providing technical support at the beginning of the Rate Year. The FTEs would be split

equally between KEDNY and KEDLI. The two product owner FTEs would be needed once product developers have identified energy efficiency products, and thus would not be hired until the second half of the Rate Year. The allocation of the FTEs is shown in Exhibit ___ (RRP-2TS).

The Companies believe the two LMI positions (project manager and marketing analyst) would be needed early in calendar year 2021. Given the service territory demographics and increased targets in KEDLI, the Companies anticipate the FTEs would focus more time on KEDLI than KEDNY. Finally, the Companies anticipate hiring the EM&V analyst at the start of Data Year 1. The analyst's time would be split equally between KEDNY and KEDLI.

B. Incentive Mechanisms

Q. Please address the impacts of the revised energy efficiency targets in the NE:NY Order on the Companies' EAM proposal.

A. The NE:NY Order defers to rate proceedings for the development and refinement of EAM proposals while maintaining elements of statewide uniformity with the incentives. To that end, the Companies will use the

energy efficiency targets established in the NE:NY Order as the basis for the minimum threshold for their respective energy efficiency EAMs.

Q. Please describe the Companies' proposed changes to KEDLI's EmPower Replacement Program Incentive Mechanism.

A. As set forth in the Third Supplemental Testimony of the Revenue Requirements Panel, KEDLI supports incorporating the HEAT Program into its energy efficiency portfolio. Given the incremental funding authorized for LMI energy efficiency programs in the NE:NY Order, KEDLI no longer believes it is necessary to request additional funding for the program outside of the energy efficiency budget. Instead, the Companies will fund this program within the authorized energy efficiency portfolio budgets and the Companies will make an adjustment to the revenue requirement to remove the incremental $2.5 million. The Companies still propose that the Earnings Incentive Mechanism be applied to this program as discussed in the Panel's rebuttal testimony.

C. New Products and Services

Q. How does the NE:NY Order impact the Companies' proposals for new products and services?

A. The NE:NY Order emphasized the development of public-facing resources capable of providing energy efficiency information, as well as a focus on gas peak reduction in supply-constrained areas, while also highlighting the importance of heat pumps, even calling out the need for Consolidated Edison Company of New York to dedicate funding to heat pump deployment in KEDNY's service territory.

The Companies developed a suite of proposals in the Future of Heat filing that align with the NE:NY Order's clean energy focus, as well as the broader CLCPA clean energy goals. Specifically, the Companies' proposed fuel-switching calculator is a customer-facing technology intended to reduce barriers to information, facilitate the further development of energy efficiency markets, and accelerate the adoption of energy efficiency technologies. Likewise, the Companies' expanded geothermal and demand reduction demonstration projects address the precise issues raised in the NE:NY Order: supply constraints and building electrification through heat pump deployment. The Companies continue to believe that these demonstration projects can play an important role in achieving the State's clean energy goals and developing

lessons learned to continuously improve efforts to achieve the ambitious targets.

VII. Conclusion

Q. Does that conclude the Panel's supplemental testimony.

A. Yes.

MR. GAVILONDO: Thank you, Your Honor.

A.L.J. COSTELLO: Okay.

BY MR. GAVILONDO: (Cont'g.)

Q. Now, panel -- make sure this is on. With respect to your direct testimony, did the panel sponsor 13 exhibits designated as I think they've been pre-marked.

MR. GAVILONDO: Would-- would you like me to refer to the pre-marked designations, Your Honor?

A.L.J. COSTELLO: That's fine.

MR. GAVILONDO: Okay.

A.L.J. COSTELLO: Sure.

BY MR. GAVILONDO: (Cont'g.)

Q. 13 exhibits that have been pre-marked for identification as Exhibits 39 through 51.

I'll restate the question if you didn't understand it. The direct testimony of the Future of Heat Panel consisted of -- I'm sorry, sponsored 13 exhibits designated pre-marked for identification as Exhibits 39 through 51. Were those exhibits prepared by you or under your direct supervision?

A. (Panel) Yes.

Q. Thank you. And the corrections and updates testimony of the Future of Heat Panel included in Exhibit and 3 -- I'm sorry, an appendix and 3 exhibits identified as Appendix A, and 3 exhibit numbers, those had been pre-marked for identification as Exhibits 229 through 232. Were those exhibits and the appendix prepared by you or under your supervision and direction?

A. Yes.

Q. Thank you. And for your rebuttal testimony, did the pa -- did the panel sponsor 4 exhibits that have been pre-marked for identification 326 through 329. And were those exhibits prepared by you or under your supervision and direction?

A. Yes.

MR. GAVILONDO: Your Honor, I -- I have no further questions for this panel and I'd like to turn to them for cross.

A.L.J. COSTELLO: Okay. Had -- are there any objections with respect to the exhibits at this time? Okay.

MR. FORST: No, Your Honor.

A.L.J. COSTELLO: Okay, hearing none, we'll start with cross examine -- examination by the Department of Public Service. Oh, can I --one second, please, let me just ask. There was a question that Judge Leary had with respect to a witness.

A.L.J. LEARY: We have a witness here that is not listed on the testimony, at least the --

MR. GAVILONDO: Yes, Your Honor.

A.L.J. LEARY: So I have Chaba --

THE WITNESS: (Chahbazpour) Chahbazpour.

A.L.J. LEARY: Chahbazpour, Tallet, Hamlin and Brady. So who is the extra person?

MR. GAVILONDO: Your Honor, Mackenzie Schwartz was a witness in the supplemental testimony that was submitted on January 29th, 2020. Ms.

Schwartz has been instrumental to the development of the --

A.L.J. LEARY: Just the supplemental testimony is the testimony in which she was involved?

MR. GAVILONDO: That is correct.

A.L.J. LEARY: Okay. That's perfect.

MR. GAVILONDO: Okay. Thank you.

A.L.J. LEARY: Thank you.

A.L.J. COSTELLO: Mr. Goodrich, I'm sorry for the interruption.

CROSS EXAMINATION

BY MR. GOODRICH:

Q. Hello, my name is Brandon Goodrich and I apologize I seem to have come down with something just yesterday. So this will be a fun week. I have some questions first about the energy efficiency full-time employee equivalents. And first, what do you understand the term full-time equivalent employee or full-time employee equivalent or F.T.E.s?

A.L.J. LEARY: Can you hear him?

A.L.J. COSTELLO: Please use the mic.

MR. GOODRICH: I am.

A.L.J. LEARY: No, you need to be this close to the microphone.

MR. GOODRICH: Okay.

A.L.J. LEARY: For the people in the back to hear you.

MR. GOODRICH: Okay.

A.L.J. LEARY: Sorry.

MR. GOODRICH: I'm trying.

THE WITNESS: (Tallet) So for the ener -- can you hear me? For the energy efficiency, full-time equivalent, that could be resources that would be dedicated to the energy efficiency programs. So by full-time implying a full-time employee.

BY MR. GOODRICH: (Cont'g.)

Q. So if you have -- if an employee is only spending part of their time for -- on energy efficiency programs, how -- how is that person counted? Like, how is that person counted within the -- the meaning of full-time equivalent?

A. Sorry, I'm not sure I understand the question.

Q. So if you have a person who is working -- well, first off, what is your -- what is -- what is full-time for -- for National Grid?

A. It would be 40 hours.

Q. 40 hours a week. So if you have a person who -- who is a full-time employee of National Grid, they work 40 hours for -- for KEDNY or whichever company it is. But they only work 20 hours on energy efficiency, and they work 20 hours on some other projects. What -- how does that translate into a full-time equivalent?

A. Within the energy efficiency programs, those F.T.E.s are dedicated to the energy efficiency efforts. It would charge their time accordingly based on whatever company they were doing work for. So one F.T.E. could do energy efficiency activities for KEDNY and for KEDLI and might charge their time 50-50.

Q. Okay. Do you have any employees that are working on energy efficiency programs for part of their time and other programs entirely for part of their time?

A. No.

Q. So how many full-time equivalents does KEDNY have working for the energy efficiency programs presently?

A. I don't know that answer.

Q. And do you know that answer as of May 28th, the end of May 20 -- 2019?

A. I don't recall.

Q. And how many F.T.E.s does KEDLI have working on energy efficiency programs presently?

A. I don't recall the exact number.

A.L.J. LEARY: Excuse me, I just want to make sure that any of the questions that are directed to the panel that may be answered by someone else should be answered by someone else. Just as a reminder. So the last series of questions, do any other witnesses have an answer or know the answer to that question?

THE WITNESS: (Panel) No.

A.L.J. LEARY: Thank you.

BY MR. GOODRICH: (Cont'g.)

Q. And do you know how many F.T.E.s KEDLI has had working on energy efficiency programs as of May 28, 2019?

A. (Tallet) I don't recall.

Q. So I have the response to IR DPS-266 that I'd like to pass out.

A.L.J. COSTELLO: Mr. Goodrich, is this a -- is this a new exhibit or is this pre-marked?

MR. GOODRICH: This would be a new exhibit. This is not in any of the pre-filed exhibits.

A.L.J. COSTELLO: Okay. Thank you.

A.L.J. LEARY: So we're going to -- subject to objection, we're going to assign it proposed Exhibit 599.

MR. GOODRICH: 599, yeah. So if the panel would just review it and let me know when you're ready to proceed.

A.L.J. COSTELLO: Mr. Goodrich, also would you just identify it for the court reporter, just give an identification of the document?

MR. GOODRICH: Sure. This, marked for identification purposes as Exhibit 599, is a response provided by National Grid to interrogatory D.P.S.-266. It is 11 pages.

A.L.J. COSTELLO: Thank you.

A.L.J. LEARY: Do we have a date? May 28th, 2019 -- I'm sorry.

MR. GOODRICH: Yes, May 28th, 2019 is the date the response was provided.

A.L.J. LEARY: Thank you.

BY MR. GOODRICH: (Cont'g.)

Q. Are you ready?

A. (Tallet) Yes.

Q. So was this prepared by you or under your supervision?

A. It was.

Q. And is this a true and accurate copy of the response to I.R. DPS-266?

A. Yes.

MR. GOODRICH: So Your Honors, this has been marked and you said that we're going to wait until the end to -- to admit them?

A.L.J. COSTELLO: That's correct. I mean, you can have testimony on it and whatnot. If -- we'll see if there are any objections, but we'll move everything in at the end of the -- at the end of the hearing, we'll move them into the record.

MR. GOODRICH: Okay. Are you going to take objections now? I'm just trying to make sure.

A.L.J. COSTELLO: Yeah, we can -- we can hear objections and then -- is there any objections to this document?

MR. GAVILONDO: No objection, Your Honor.

A.L.J. COSTELLO: No? Hearing none, you may proceed.

BY MR. GOODRICH: (Cont'g.)

Q. So Question 1 here asks for an organizational chart that includes the current number of full-time employees or equivalents working for the energy efficiency programs for KEDLI and KEDNY, is that correct?

A. (Tallet) Yes.

Q. Please turn to page 5 of the exhibit. It's the one that says Brooklyn Union Gas, KEDNY, attachment 1, page 1 of 2. Please state the -- the number of KEDNY F.T.E.s identified in this chart.

MR. GAVILONDO: Your Honor -- Your Honors, I'm going to ask if -- I'm going to object because if I understand the question, it's asking the witness to add up the number of lines here. If

that's the question, I'm going to object. I -- I'm not sure that's appropriate for cross examination.

A.L.J. COSTELLO: Can we just stipulate to the number?

MR. GOODRICH: Actually, we can't, Your Honor, because it's unclear how many F.T.E.s the KEDNY has working on its energy efficiency programs. The panel testified that they do not know the number so I'm trying to figure out what that number is.

A.L.J. COSTELLO: Okay. Objection is overruled, you may proceed. And just so the court reporter has it F.T.E.s, when we're referring to F.T.E.s it's full-time employees or full-time equivalents.

MR. GOODRICH: Thank you.

And we can go -- we'll go off the record just while, let us know when you're --

(Off the record; 9:34 a.m. to 9:36 a.m.)

THE WITNESS: (Tallet) The total number is 163.

BY MR. GOODRICH: (Cont'g.)

Q. So from that you're stating that -- is it correct that you're stating that KEDNY has

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163 full-time equivalents working on its energy efficiency programs?

A. That's correct.

Q. Please turn to page 6, next page. From looking at this document, how many full-time equivalents does KEDLI currently have, or as of May 28th, 2019 have working on energy efficiency programs?

A. 163.

Q. So between the two companies, is -- does that mean that there are -- hold on, let me do my own math. 326 employees full-time equivalents work. Let me -- let me rephrase for clarity. So does this mean that for KEDNY and KEDLI together there are 326 full-time equivalents working on energy efficiency programs?

A. That's correct.

Q. Turning back to page 1 of the exhibit. Question 5. Question 5 asked separately for each of 2017, 2018 and 2019. Provide the total number of hours worked and indicate the number of overtime hours that were associated with energy efficiency employees. Provide a breakdown of the hours identifying the hours allocated to regulatory

compliance and the hours allocated to program implementation. Is that correct?

A. Yes.

Q. Now, turn -- please turn to the last page of the exhibit. This is attachment 3 which was the -- what the company stated was respond -- was the response to Question 5, is that correct?

A. Yes.

Q. So what period does this show? What -- what time period is this showing the hours for?

A. It appears to show the hours for the calendar years 2017, '18 and '19 as of the date of the response.

Q. And does this include KEDNY only, KEDLI only or both companies?

A. It would cover both companies.

Q. Does this include any other National Grid affiliates?

A. It does not.

Q. So from the total hours worked here, can you tell me how many full-time equivalents KEDNY and KEDLI had for 2017, 2018 and 2019?

A. Can you repeat the question?

Q. From -- from the information here, can you tell me how many full-time equivalents KEDNY and KEDLI had working on energy efficiency programs for, you said calendar years 2017, 2018 and 2019?

A. No.

Q. So the number you provided me before, was that together KEDNY and KEDLI have 326 full-time equivalents working on energy efficiency programs. Is that correct?

A. Yes.

Q. And in your initial testimony -- in the Future of Heat's initial or direct testimony, you requested 7 additional full-time equivalents for KEDNY and KEDLI together, correct?

A. Yes.

Q. So that would bring the total, if those were hired to 300 -- 333 full-time equivalents?

A. Yes.

Q. And in your supplemental testimony, you requested an additional 7 F.T.E.s for KEDNY and KEDLI during the rate year, is that correct?

A. Yes.

Q. So that would bring the total to 340 F.T.E.s working on energy efficiency programs at KEDNY and KEDLI together?

A. Yes.

Q. I have another I.R. for you to look at and this is -- this is looking at unspent energy efficiency funds. And the first I.R. is 1020 -- 1082.

A.L.J. LEARY: We're going to assign this 600. Hearing Exhibit 600.

MR. GAVILONDO: Your Honors, it's Carlos Gavilondo for the company. His -- I'm not sure if this I.R. has already been pre-marked for identification or if it's newly introduced.

MR. GOODRICH: I'm sorry, every – unless -- unless I say otherwise everything was -- is new for -- for the hearing.

A.L.J. LEARY: Thank you, Mr. Goodrich.

A.L.J. COSTELLO: And Mr. Goodrich just identify for the record and it's going to be as Judge Leary said, marked as Exhibit 600.

MR. GOODRICH: So this is a two-page document that is the company's response to

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A.L.J. LEARY: Two pages in length.

BY MR. GOODRICH: (Cont'g.)

Q. Have you completed reviewing?

A. (Tallet) Yes.

Q. Was this prepared -- response prepared by you or under your supervision?

A. Yes.

Q. Is this a true and accurate copy of the response to I.R. D.P.S. 1082?

A. Yes.

MR. GOODRICH: I don't have any questions on this I.R., I just wanted to get it into the record.

A.L.J. COSTELLO: Okay. Any objections at this point?

MR. GAVILONDO: No objection, Your Honor.

A.L.J. COSTELLO: Okay. Hearing --

MR. GOODRICH: I have another I.R., this is the response to D.P.S. 1085. Actually, this is the document from the company asking for an extension to provide the response to D.P.S. 1085.

A.L.J. COSTELLO: Okay. We're going to mark this as Exhibit 601?

MR. GOODRICH: It doesn't necessarily need to be marked.

A.L.J. COSTELLO: No. Okay. All right let's do it.

A.L.J. LEARY: Let's just make the record clear. This is D.P.S. 1085 the due date was February 6th, 2020. This is not -- this is new?

MR. GOODRICH: Correct. And this is a -- well, I was going to ask the --

A.L.J. LEARY: Yeah, please.

MR. GOODRICH: -- witness.

A.L.J. LEARY: Go ahead.

BY MR. GOODRICH: (Cont'g.)

Q. So do you -- do you recall, were you provided with a I.R. D.P.S. 1085 from staff?

A. (Tallet) Yes.

Q. And have you provided a response to that question yet?

A. The response is due February 13th per the extension the company requested.

Q. So you have not provided a response yet?

A. To date, no.

MR. GOODRICH: And Your Honors, I really am raising this just as this is an I.R. that we do want to review the response to and have the ability to get it into the record and I wanted to ask how Your Honors want us to proceed with that.

A.L.J. LEARY: Well, I guess this whole panel is going to have to come back if that's the case. So if the company wanted a February 13th day -- date obviously we need to provide the right to self-cross-examination, so the panel comes back or --

MR. GAVILONDO: Your Honor --

A.L.J. LEARY: -- if you have any questions.

MR. GAVILONDO: Your Honor, if I might? I'm not sure the date that this I.R. was actually served upon the company. And I'm not sure --

A.L.J. COSTELLO: Who? Just identify yourself, I'm sorry. Identify yourself again for the record.

MR. GAVILONDO: I'm sorry. Yes, thank you, Your Honor. Carlos Gavilondo for National Grid. I'm not sure the date that this I.R. was actually

served upon the company. If counsel -- D.P.S. counsel could remind us --

A.L.J. LEARY: Why is that relevant?

MR. GAVILONDO: Typically, we have 10 days to respond under the --

A.L.J. LEARY: Understood.

MR. GAVILONDO: -- commission rules.

A.L.J. LEARY: Understood.

MR. GAVILONDO: And I would just like to know when this was submitted to the company.

A.L.J. LEARY: Well, you can talk offline about that. What I am concerned about is giving staff the opportunity to cross examine at least one of, or if not 5 of these people who, so it's --

MR. GOODRICH: I'm not necessarily asking for the opportunity to cross examine. I would like staff to be able to review the response and not -- not automatically put it into the record if it is not useful. However, if staff deems it to be useful, I would like to be able to get the response into the record. That I -- at this time, I do not expect that I have any -- any substantive questions about the I -

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And so it's just a question for Your Honors of how we may be able to do that if it's holding an exhibit open. If it's -- if the company would need to provide it with an affidavit or --

A.L.J. LEARY: Why don't we give it hear -- proposed hearing Exhibit 601? You too can get together and stipulate --

MR. GOODRICH: Okay.

A.L.J. LEARY: -- that it comes in. If you have questions about the responses, Mr. Goodrich, I'm just trying to put counsel for the company on notice that I know it's 10 days, but if there are questions, somebody's going to have to come back. I'm just trying to protect the record in that regard. Not that Mr. Goodrich may need to do that, but I just want to leave that possibility open.

So what it may be moved, let's -- let's move forward --let's give it --

MR. GAVILONDO: We'll do. Thank you, your honor.

A.L.J. LEARY: -- 601.

BY MR. GOODRICH: (Cont'g.)

Q. So I'm turning now to the hydrogen blending study that the company proposed. Now, on pages 88 and 89 of your initial testimony, you state that the companies have applied for a NYSERDA grant of approximately 300 -- $346,000 for the proposed hydrogen blending study. Is that correct?

A. (Schwartz) Yes.

Q. Has NYSERDA announced the determination on National Grid's grant application?

A. Can I see the mic, please? Sorry. No, we've not made a public announcement on that.

Q. When do you -- you said we, I'm sorry, has NYSERDA -- do you know if NYSERDA has announced that?

A. NYSERDA has not made a public announcement that we know of.

Q. When do you anticipate that NYSERDA will announce its determination?

A. We anticipate to know in the next few months, their determination.

Q. So that may be before or after this case concludes?

A. It will likely be before this case concludes.

Q. I would ask that in -- in -- that the companies commit to -- to providing an update, should that occur during briefing stages, I guess.

MR. GAVILONDO: Should we reserve an exhibit number, Your Honors?

MR. GOODRICH: I don't know if anything is going to --

A.L.J. LEARY: No, but I think what's important here is that if there are these issues that remain outstanding that we deal with them later. Let's not speculate on what may or may not happen. But is there -- I mean, discovery is not closed, you can serve a discovery request if you don't already have one on this information, the update. You want an update?

A.L.J. COSTELLO: But my understanding, though, is that they just want to get a commitment from the company to provide an update when -- if that happens while the rate case is pending, the company is willing to do that.

MR. GAVILONDO: Yes, Your Honor.

A.L.J. COSTELLO: Okay. And we can deal with that in briefing. Is everyone agree to that?

MR. GOODRICH: Yes.

A.L.J. COSTELLO: Okay. Thank you.

MR. GAVILONDO: Yes.

A.L.J. COSTELLO: It -- I -- my understanding is there won't be an exhibit with respect to this. It's just a statement by the company or representation by the company of what has happened.

MR. GOODRICH: That -- that would be my expectation presuming, I mean, they -- they laid out in their testimony what they applied for and it's a question of whether NYSERDA granted it or not.

A.L.J. COSTELLO: Okay. And just -- just so -- so that we -- going forward, we know. If there is anything that for any reason we need to get exhibits then we can deal with that later on. But it's something that we can -- you can provide to us and we can add to the exhibit list before, you know, we've -- we may finalize it and submit it on D.M.M. So as long as we're all working together, we can both straighten all that out as the time goes on.

MR. GOODRICH: Thank you. And my -- my intention was just to make sure the record reflected that these were issues that needed to be addressed.

A.L.J. COSTELLO: Okay. Understood.

BY MR. GOODRICH: (Cont'g.)

Q. Turning to the company's E.A.M. for CO2, carbon dioxide equivalent emissions reductions. Beginning on page 111, line 12 of your direct testimony you describe a proposed incentive for National Grid based on customer conversions from diesel to compressed natural gas for vehicles. Is that correct?

A. (Traczyk) Yes.

Q. So on page 112, line 7 to 8 it states the companies can determine whether customers are using natural gas for their vehicles. Is that correct?

A. Yes.

Q. This means that customers are using natural gas delivered by KEDNY or KEDLI, correct?

A. Correct. Or it could be in sum total for sales to C.N.G. stations. So in the event

that there was a non-full service customer that was providing natural gas to a C.N.G. fueling station, presumably since we are delivering the fuel, we would know what the delivered total would be, even if we did not procure the commodity for them.

Q. So my -- my question was, this was that customers are using natural gas delivered by KEDNY or KEDLI, correct?

A. So yes, it -- forgive me if I -- I answered incorrectly. In that case, yes, we will be able to determine the sum total of all natural gas that is delivered to C.N.G. fueling stations within our territory.

Q. My question wasn't -- if you can calculate that that's in your testimony. My question was this means that the gas is being delivered by KEDNY and KEDLI, correct?

A. Yes.

Q. Do these vehicle operators or fueling station operators pay delivery rates to KEDNY and KEDLI for this gas?

A. Under our tariffed rates for C.N.G. sales, yes.

Q. And customers paying delivery rates to KEDNY and KEDLI for gas delivered for fuel for C.N.G. vehicles creates revenues for KEDNY and KEDLI, correct?

A. That is correct, yes.

A.L.J. COSTELLO: Let me just interrupt one second. I'm sorry. C.N.G., just for the record, just --

THE WITNESS: (Traczyk) Forgive me Your Honor, compressed natural gas.

A.L.J. COSTELLO: Okay. Thank you.

BY MR. GOODRICH: (Cont'g.)

Q. Moving now to another E.A.M. on the empower replacement program.

A.L.J. LEARY: Again, I'm sorry, E.A.M.?

MR. GOODRICH: I'm sorry, earnings adjustment mechanism.

A.L.J. LEARY: Thank you.

BY MR. GOODRICH: (Cont'g.)

Q. So in your supplemental testimony filed on January 29th, 2020, on pages -- on page 10, lines 8 through 14 your response -- you proposed to fund KEDLI's current empower replacement program, or

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for ease just empower, out of the low and moderate income portion of KEDLI's overall energy efficiency budget. Is that correct?

A. (Tallet) I believe the intention was to fund that out of the energy efficiency portfolio as a whole.

Q. Okay. What percentage of buildings addressed under the empower program are multifamily buildings?

A. The empower heat replacement program pertains to Long Island. I don't recall the exact number of multifamily buildings for Long Island.

Q. Does KEDLI expect the number of households served under Tier 4 of the empower program to remain the same as anticipated prior to this latest proposed change or do you expect a -- a modification of the number of -- number of households served under Tier 4?

A. With expanding the -- the funding, the funding originally going, funded at 1.9 million to 2.5 million would likely afford us to serve more customers in general. The intention would

be to drive as many as we could towards the higher tiers.

Q. To be clear, though, that the -- increasing the funding from 1.9 million to 2.5 million was included in the company's earlier proposals in this case, correct? Not -- not in the supplemental testimony?

A. In the supplemental -- in the supplemental, we intend to just bring that into the energy efficiency portfolio. Within that portfolio, the company has flexibility to assign funding to individual programs.

Q. So you may spend more than 2.5 million?

A. We could if it was treated as any other energy efficiency program.

Q. So as you proposed that empower would be part of your overall energy efficiency program and funded out of your energy efficiency budget, do you propose that KEDLI's efforts on the empower program would count toward the M.M.B.T.U. and I'm going to ask you to say what that means, because I'm going to screw it up if I try? The M.M.B.T.U. savings incentive that you proposed on pages 101 to

106 of your initial testimony, if it were adopted by the Commission. I think it's Million British Thermal Units, but I'm not hundred percent positive.

A. (Traczyk) That is correct.

Q. Okay.

A. (Tallet) And we would propose that any savings relative to that would be counted towards the statewide targets.

Q. That -- that wasn't my question. My question was you proposed in your initial testimony an M.M.B.T.U. savings incentive. Where in -- in E.A.M., do you propose that KEDLI's efforts in the empower program would count toward that M.M.B.T.U. incentive?

A. I believe that any effort would count towards that.

Q. And similarly, while staff didn't support the M.M.B.T.U. savings incentive, we proposed -- staff proposed they share the savings incentive. If that were adopted by the Commission, would you propose that your efforts on the empower program count toward that incentive?

A. It could.

Q. Turning now to power of gas -- not -- not power of gas, power to gas program on page 85, line 6 of your initial testimony. You state that your power to gas proposal would, "provide a key outlet for excess renewable energy." Is that correct?

A. (Chahbazpour) Correct, yes.

Q. Could you define excess renewable energy?

A. In this case, what we're referencing is when most likely it's renewable sources of energy so whether it's wind or solar. And there is a condition of either, you know -- ex -- excess supply. So that could be that power it's basically being curtailed.

Q. And is that excess renewable energy available at all times?

A. Probably, I mean, it depends, it could be. Right now, the places where you actually see this happening in U.S. at scale is it's California, it's happening in California. It's coming from solar mostly today, if you can imagine it could be, you know, very sunny, it's a sunny day, but

the temperature is not high, so that's occurring in California.

The other place that's occurring today in the United States, it's Colorado. There's onshore wind and there are days where the price of power in Colorado it's zero or negative. Essentially, those wind turbines are spinning, and they have nowhere to go. So there is not a lot of that happening in the State of New York, but in terms of either offshore wind or large-scale solar.

Q. So this is a proposal for KEDNY and KEDLI so we're talking about Brooklyn and the Long Island.

A. Uh-huh.

Q. Do you see -- is there excess renewable energy -- electricity energy available at all times on Long Island?

A. Today?

Q. Today.

A. I am not sure. I think -- the an -- I think no. I'm speculating.

Q. Do you expect that in the next year?

A. In the next year, no, probably not.

Q. So the power to gas project would use a bioreactor to create methane, correct?

A. Uh-huh.

Q. I need a verbal answer.

A. Yes. Yes.

Q. And please describe what a bioreactor is.

A. Sure. So I guess let me just give you some context, where we came into this. First the concept, we went to NREL, that's National Renewable Energy Laboratory labs at Colorado. It's one of D.O.E. labs. They are funding --

A.L.J. LEARY: One of the what labs?

THE WITNESS: The Department of Energy -- NREL, it's National Renewable Energy Laboratory. It's -- it's right outside Denver. They are currently building and have actually constructed the first demonstration project in the U.S. in the space of power to gas in the anticipation that we expect a lot of renewable energy sources to come online in the U.S.

They have partnered with a German technology company that has built a bioreactor. And what it essentially does, it takes hydrogen and it takes CO2 or it comes from various streams and the bioreactor is able to take the hydrogen and the CO2 that's fed into that bioreactor and what comes out is methane.

BY MR. GOODRICH: (Cont'g.)

Q. Using what? It's a bio -- what is a bioreactor?

A. Yes. So there are enzymes known as electrochaea. These are -- if you Google it it's a bacteria that's been around for billions of years. And essentially if you -- they're not really bacteria, bacteria is what anaerobic digester produce as biogas. These are single -- single cell organisms and they produce methane. So if you basically feed them co2 and hydrogen they produce methane. Q. So what happens to these single celled organisms if the bioreactor is shutdown?

A. So the way they've discovered these, like I said, they've been around for billions of years. And what they have found out is in -- through the research and the patent that this company

has gotten is they're very resilient. They have figured out a way how to basically keep them happy. So they have found a temperature where that if you have -- they have optimized the maximum production of methane. If you shut it down, which means if you lower the temperature where they stopped producing.

So they have found, again, the optimum range. The worst case scenario that, you know, we found out is what if the bioreactor spills, right. If it spills, they would eventually -- they will essentially die in the area where there's oxygen. But they are not toxic or anything and they're self-replicating single cell organisms.

A. (Schwartz) In short, if you shutdown the bioreactor, they go dormant.

A. (Chahbazpour) Yeah.

Q. So are you able to completely control -- to control from time to time during the day if the bioreactor is running or not -- if electricity is needed to --

A. Yes.

Q. -- or not electricity, I'm sorry. What am I try -- if -- if you need to put inputs into the bioreactor, you can control whether or not

you can -- you have to put inputs into the bioreactor --

A. Yes.

Q. -- from time to time?

A. Yes. And they can ramp up and down production based on what's happening on the electric side.

A.L.J. LEARY: Excuse me, Mr. Goodrich. I just want to ask the witness, I was asking you about the name of the laboratory to which you referred.

THE WITNESS: (Chahbazpour) Uh-huh.

A.L.J. LEARY: Did -- is that what was provided in the record the name of the laboratory that you first referred to?

THE WITNESS: Yes.

A.L.J. LEARY: Okay. And that was again?

THE WITNESS: It's Department of Energy's NREL, National Renewable Energy Laboratory.

A.L.J. LEARY: Thank you. And --

THE WITNESS: It's one of the D.O.E. labs.

A.L.J. LEARY: So one of the Department of Energy laboratories?

THE WITNESS: Of energy labs. Correct.

A.L.J. LEARY: The second question I have is, you mentioned that the bioreactor is -- is in demonstration.

THE WITNESS: Uh-huh.

A.L.J. LEARY: Is there an expected date when it will come out of demonstration and be available commercially?

THE WITNESS: I believe it has actually just started operations. It is operational now.

A.L.J. LEARY: Again, is there some timeline that D.O.E. or the lab has to bring it out of a demonstration status?

THE WITNESS: As long as it's sitting at the Department of Energy, it will probably just be a demonstration. So it will be a demonstration for a while. I don't know when that will be.

A.L.J. LEARY: Okay. Thank you.

BY MR. GOODRICH: (Cont'g.)

Q. Is there a length of time -- like a maximum length of time that the bioreactor -- that the organisms can remain dormant and still be able to restart?

A. (Schwartz) To our knowledge, there isn't a maximum length of time that's been determined. They're incredibly resilient organisms, just by virtue of where they naturally habitat which is your deep sea vents.

Q. Turning to Newtown Creek, the R.N.G., Renewable Natural Gas facility at Newtown Creek Water Treatment Plan. As of July 2019, New York City had provided KEDNY with a tax exemption for the first five years of operation of this R.N.G. facility for 14.4 million in utility-owned assets, is that correct?

A. (Chahbazpour) Correct.

Q. And this $14.4 million cap on the property, the assets subject to the exemption was based on a 2012 cost estimate for the Newtown Creek Project, is that correct?

A. Correct.

Q. Is this the current status of the property tax abatement?

A. Yes.

Q. Now KEDNY's current rate plan, which is set forth -- is set forth in the joint proposal adopted by the Commission in its order adopting terms of joint proposal and establishing gas rate plans in cases 16-G-0058 and 16-G-0059 on December 16th, 2016, is that correct?

A. Correct, yes.

Q. In that joint proposal on page 23, it states that New York City will, "Support KEDNY's reasonable efforts to mitigate this -- the effects of the revenue requirement exclusion, including but not limited to supporting KEDNY's applications for property tax abatements, to the extent such efforts are permitted by applicable law." Has KEDNY requested further or updated property tax abatements since December 16th, 2016, for the Newtown Creek project?

A. Yes.

Q. And what are the status of those requests?

A. We, at the time in 2012, we went to the I.D.A. and filed for that property tax exemption and we got it. And the feedback from the

city was, since the project costs, as you mentioned in 2016, is a lot higher than it was in 2012, they could not just extend that property tax abatement to the new project costs.

That offer was issued at that time by the Bloomberg administration and we had to go through an entire application in front of the I.D.A. Board. And typically the city wants a grant, a property tax exemption -- exemption on a project at the time. They don't reopen the cost of the tax exemption if the costs have gone up.

This has been typically a precedent in the city when they do offer property tax abatements. As you can imagine, construction costs do escalate over time, even if they're not projects like Newtown Creek. But that's typically the precedent. So if you wanted another property tax abatement, it would have to be basically an entire new application and going in front of the I.D.A. Board.

Q. Have you submitted -- has KEDNY submitted a new application?

A. At that time, no. The city said they would not be supportive of the new application for the new project costs.

Q. At -- at that time, at what time?

A. Meaning at -- and with 2016. So in other words, the property tax abatement was given to us and approved in whatever the timeline was at 14.4 million in 2012. And they were not in the position to offer that new -- a new tax abatement at the new price with the 2016 project costs.

Q. So you -- you, KEDNY has applied -- had applied for another tax abatement of an additional tax abatement or -- or revised tax abatement in 2016 and that has been denied by the city?

A. I wouldn't say that we applied. Before we even applied, we're in negotiate -- where we were in conversations with the City of New York. And the city was saying before you apply, we are not supportive of that application. In other words, you cannot go in front of this I.D.A. and ask for, without the support of the mayor's office and the city. So in 2016, the city was not supportive of even filing for an application.

Q. And there are no further -- has there been any further talks since then?

A. No.

Q. So the remainder of my tests -- my testimony. My cross examination is simply just to get a -- a number of I.R.s onto the record. And so we'll go through those. The first one is, the response to I.R. D.P.S. 488.

A.L.J. LEARY: Let's assign that 602; date?

MR. GOODRICH: One moment, Your Honor.

A.L.J. LEARY: Sure.

MR. GOODRICH: So I have here, what's been marked as Exhibit 602. It is the response to I.R. D.P.S. 488, provided on June 24th, 2019. And it is eight pages and panel, if you can let me know when you're ready.

THE WITNESS: (Hamlin) Ready.

BY MR. GOODRICH: (Cont'g.)

Q. So was this response prepared by you or under your direction -- supervision?

A. Yes, it was.

Q. And is this a complete and accurate copy of this I.R. response?

A. Yes, it is.

Q. Nothing further on this one. The next one is I.R. -- the response to D.P.S. 850. So I think -- so this would be number 603.

A.L.J. COSTELLO: Yes, 603.

Q. And this is the response to I.R. D.P.S. 850, provided on July 22nd, 2019. And it is three pages in length.

MR. GOODRICH: Let me know when you're ready, panel.

THE WITNESS: (Schwartz) Ready.

BY MR. GOODRICH: (Cont'g.)

Q. Was this prepared by you or under your direct supervision?

A. Yes.

Q. And is this a true and complete copy of this I.R. response?

A. Yes.

MR. GOODRICH: Nothing further on this I.R. response.

A.L.J. COSTELLO: Okay. Before we go forward, just to -- on number 602, just for the record, are there any objections with respect to that?

MR. GAVILONDO: No objection, Your Honor.

A.L.J. COSTELLO: Okay. And as to number six, what's been identified as number 603?

MR. GAVILONDO: No objection.

A.L.J. COSTELLO: Thank you.

MR. GOODRICH: Next is the response to D.P.S. 1014, one zero one four. It was provided on October 11th, 2019. And it's two pages in length and I ask that that's marked as --

A.L.J. COSTELLO: 604.

MR. GOODRICH: 604, thank you.

THE WITNESS: (Schwartz) Ready.

BY MR. GOODRICH: (Cont'g.)

Q. Was this prepared by you or under your direct supervision?

A. Yes.

Q. And is this a true and complete copy of this I.R. response?

A. Yes.

MR. GOODRICH: Nothing further on this I.R.

A.L.J. COSTELLO: Okay. Any objections?

MR. GAVILONDO: No objection, Your Honor.

MR. GOODRICH: Next, I have the response to I.R. D.P.S. 1017, provided on October 11th, 2019. And it is a two-page response.

A.L.J. COSTELLO: That will be marked as Exhibit 605.

A.L.J. LEARY: 604 -- yeah, 605, sorry.

THE WITNESS: (Traczyk) Ready.

BY MR. GOODRICH: (Cont'g.)

Q. Was this prepared by you or under your direct supervision?

A. Yes.

Q. And is this a true and complete copy of this I.R. response?

A. Yes.

MR. GOODRICH: Nothing further on this I.R.

A.L.J. COSTELLO: Okay. Any objections?

MR. GAVILONDO: No objection.

MR. GOODRICH: We're getting there. I promise. Next, we have the response to I.R. D.P.S.

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1019. It was provided on October 15th, 2019 and it is three pages.

A.L.J. COSTELLO: That will be --

MR. GOODRICH: 606.

A.L.J. COSTELLO: -- Exhibit 606.

BY MR. GOODRICH: (Cont'g.)

Q. So I think we may have an issue here. But this -- this is an I.R. that the company, that the staff asked of the company's Future of Heat Panel. And it is responded to by Candice Young. Who Is Candice Young?

A. (Pallet) She is a National Grid employee.

A.L.J. LEARY: And her title?

THE WITNESS: I don't know her title.

A.L.J. LEARY: What section does she work in? Does she work with you?

THE WITNESS: I believe that she is a lead analyst in the National Grid Partners group, no. At the time of responding to this I.R., she was in regulatory and customer strategy within National Grid.

BY MR. GOODRICH: (Cont'g.)

Q. Was this provide -- was this response created under your supervision, panel?

A.L.J. LEARY: Can anybody on the panel testify about the contents of this exhibit?

THE WITNESS: (Traczyk) Yes. So Kenzie Schwartz, Don Chahbazpour and myself, along with Lisa Tallet, all contributed to the generation and response to all of the I.R.s relating to E.A.M.s. Candace was the person who had initially developed the E.A.M. frameworks and so, therefore, was the one that was responded to the IRs. But these responses were all developed under our supervision.

BY MR. GOODRICH: (Cont'g.)

Q. Can you confirm that this is a true and accurate copy of this IR response?

A. I can. And yes, it is.

Q. Is that going to be sufficient to --

A.L.J. COSTELLO: Does the panel adopt this as one of their exhibits?

THE WITNESS: (Schwartz) Yes.

THE WITNESS: (Hamlin) Yes, Your Honor.

A.L.J. COSTELLO: Okay.

MR. GAVILONDO: No objection, Your Honor.

A.L.J. COSTELLO: Thank you.

A.L.J. LEARY: Thank you.

MR. GOODRICH: This is nowhere near as long as some of them. So I have -- I have here the response to D.P.S. 893. This was provided on July 29th, 2019.

A.L.J. LEARY: It should be --

MR. GOODRICH: And thankfully it was provided by one of the people I see on the panel.

A.L.J. LEARY: -- 607.

A.L.J. COSTELLO: Thank you.

A.L.J. LEARY: Mark this as Hearing Exhibit 607.

MR. GOODRICH: Please let me know when you're ready.

THE WITNESS: (Tallet) Ready.

BY MR. GOODRICH: (Cont'g.)

Q. Was this prepared by you or by someone under your direct supervision?

A. Yes.

Q. And is this a complete and accurate copy of this I.R. response?

A. Yes.

MR. GOODRICH: Nothing further on this I.R. response.

A.L.J. LEARY: Do you have a number of pages here?

MR. GOODRICH: Oh, God.

A.L.J. LEARY: Mr. Goodrich?

MR. GOODRICH: Oh, 68, I think.

A.L.J. LEARY: Beautiful.

MR. GOODRICH: Thank you, ma'am.

A.L.J. COSTELLO: And -- any objections?

MR. GAVILONDO: No objection, Your Honor.

A.L.J. COSTELLO: Thank you.

MR. GOODRICH: And with that, I have no further cross at this time.

A.L.J. COSTELLO: We're going to just go off the record.

(Off the record; 10:22 a.m. to 10:36 a.m.)

THE REPORTER: On the record.

A.L.J. COSTELLO: And -- Family Energy has cross-examination next.

MR. BLAKE: Yes, Your Honor. Just to make sure, can you hear me?

A.L.J. LEARY: I can't hear you very well. So you're going to have to --

MR. BLAKE: Can you hear me now?

A.L.J. LEARY: I had someone come up and improve the microphone volume. Are you sure your green light is on?

MR. BLAKE: The green light is on, currently yes. But it doesn't seem to be activated. Can you hear me now?

A.L.J. COSTELLO: We can hear you with that one.

A.L.J. LEARY: Are you okay? Excuse me. Would you please state your identification for the record?

MR. BLAKE: Yes. Kevin Blake, B-L-A-K-E, on behalf of Family Energy. And I'm here also with Jeff Donnelly, D-O-N-N-E-L-L-Y, also with Family Energy.

A.L.J. COSTELLO: Okay. You may begin.

BY MR. BLAKE:

Q. So I want to first discuss the Green Gas Tariff product structure. National Grid is proposing under the Green Gas Tariff, a program by which a residential or commercial customer could voluntarily enroll in a service arrangement whereby National Grid would secure renewable natural gas for some or all of that customer's gas supply. Is that correct?

A. (Schwartz) Yes.

Q. And because the renewable natural gas, which I'll refer to as R.N.G. would be pipeline quality gas that's commingled with traditional gas, customers who choose to enroll in this program would not actually receive R.N.G. molecules, correct?

A. Correct. It would work similar to renewable products on the electric side, where it's a virtual transaction.

Q. Correct. So in this way, just as customers may elect to enroll in a green electricity program or procure renewable energy credits as a means of supporting clean energy development. But are not actually receiving green electrons, a customer may enroll in the Green Gas Tariff to support as a means of decarbonizing their gas supply,

but they're not actually receiving R.N.G. molecules, correct?

A. That is correct.

Q. And to be eligible for the Green Gas Tariff, the customer would need to receive supply from National Grid, correct?

A. That is correct.

Q. And the company's structured the program so that a residential customer -- if a residential customer were to voluntarily enroll, he or she would play -- pay a flat monthly premium on that product that would be locked in for 12 months, correct?

A. That is correct. It was our understanding that residential participants would appreciate a flat monthly premium versus a -- versus a charge associated with their participation for the usage.

Q. And -- and --

A.L.J. LEARY: Excuse me. Whoever has their cell phone on, please turn it off and everybody else please check that your cell phone is off. Judge Costello and I will be collecting the cell phones

that go off again. And you won't get them back until after the hearing. Go ahead. I'm sorry.

MR. BLAKE: Thank you.

BY MR. BLAKE: (Cont'g.)

Q. And similarly with a commercial customer, if the commercial customer were to enroll in the program, the customer would have the option of choosing between a flat rate and a percentage of monthly consumption, and that would require a 24-month commitment, correct?

A. (Schwartz) That is correct.

Q. And so the benefit that the customer is receiving by enrolling in the program is simply the knowledge that the customer is contributing to the decarbonization of their gas supply, correct?

A. That is correct. Similar to how programs work on the electric side.

Q. Correct. And to pay for that benefit, the customer who participates in the Green Gas Tariff firm would -- would pay a premium on their gas supply charges, correct?

A. They would pay an adder, correct.

Q. And that adder would be above and beyond the standard utility rate?

A. Yes, it would.

Q. So is it fair to characterize the Green Gas Tariff product offering as a value added product or service offered by companies?

A. I don't know if there's a formal definition for that. But that sounds like a correct term to use.

Q. Would you characterize the Green Gas Tariff program as a product that is above the standard product that the companies would traditionally offer?

A. Yes.

Q. And with respect to customers who elect not to participate in the Green Gas Tariff program, they do not receive any sort of economic benefit, do they?

A. When it comes to emissions savings, there is a shared economic benefit that comes with those. But when it comes to a program that they're participating in, no. They wouldn't share in those participation -- in that participation.

Q. I'd like to turn your attention to exhibit, what's marked as pre-filed Exhibit 39. Schedule 1, on page 2 of that exhibit, discusses the program costs breakdown. Can you -- can the panel please read that, the last sentence of that paragraph.

A. Could you tell me what, that was located in again?

Q. Yeah. It's -- it's exhibit -- pre-filed Exhibit 39 --

MR. GAVILONDO: Your Honors, it -- it is --

A.L.J. COSTELLO: It is Future of Heat Panel number 1?

MR. GAVILONDO: Yes. It's Future of Heat number 1. Exhibit 2, direct testimony of Future Heat panel.

THE WITNESS: (Chahbazpour) I'm not sure if my other colleagues have it. I don't have it. Would you mind reading it? Whatever that sentence says?

A.L.J. COSTELLO: Sure.

A.L.J. LEARY: No, the witness is --

MR. GAVILONDO: Actually, no. Go ahead, Your Honor. I'm sorry.

A.L.J. LEARY: Stop. Do I have an objection from the company?

MR. GAVILONDO: No, I was going to try to direct the panel to where they could find it within their materials.

A.L.J. LEARY: Please, go ahead.

MR. GAVILONDO: Thank you. It is directly after the testimony of the Future of Heat Panel. It is Future of Heat number 1, schedule 1. So it's the first 2 pages of exhibit -- as counsel indicated pre-filed Exhibit 39, which is Future of Heat-1, schedule 1, page -- pages 1 and 2.

THE WITNESS: (Schwartz) And could you remind me the section you would like me to read?

BY MR. BLAKE: (Cont'g.)

Q. Yes. Could you please read the last sentence of the paragraph section that's titled program cost breakdown?

A. The last sentence, you requested?

Q. Yes.

A. The company further proposes that the cost for the Green Gas Tariff offering would be

borne by customers who elect to participate in the program.

Q. Thank you. And are there any costs of administrating -- administering the Green Gas Tariff program that are directly or indirectly subsidized by non-participating customers?

A. We requested two -- full-time equivalents to support the program.

Q. I'd like to turn your attention to Exhibit 231, which is the labor-related rate year full-time employee costs for the Green Gas Tariff program. Please let me know when you found -- when you found that exhibit.

MR. GAVILONDO: Your Honor.

A.L.J. LEARY: Yes.

A.L.J. COSTELLO: Yes.

MR. GAVILONDO: If I may direct the panel?

A.L.J. COSTELLO: Yes.

A.L.J. LEARY: Please do.

MR. GAVILONDO: Thank you. Panel, this is in the corrections and updates exhibit. And if it's 231, I believe it is corrections and updates Exhibit 13 C.U.

THE WITNESS: (Schwartz) Ready.

BY MR. BLAKE: (Cont'g.)

Q. Is it true that these costs are intended to cover the management of contracts with R.N.G. suppliers, management of the balance between purchases and sales, tracking quantities of Green Gas that's purchased and sold, acquiring environmental attributes, tracking costs and revenues, managing transaction system entries and fulfilling reporting requirements?

A. Yes, that is, as we described the role of the F.T.E.s that would support the program.

Q. And all of those management and administrative responsibilities are also performed regularly by competitive energy service companies in New York State to support the provision of their products, correct?

A. I am -- I'm not aware of how those companies operate.

Q. So isn't it true then that the Green Gas Tariff program is something that a competitive energy service company could also offer if they perform those same functions?

A. Yes, a competitive company could certainly offer a similar program.

Q. I'd like to transition to the -- the company's justification for the program. In particular, did the companies conduct a survey inquiring about, exam -- or examining customer interest in the green gas product?

A. Yes. As a part of a customer survey that we did, we asked a question about customer interest in a green product for their gas consumption.

Q. And the source for that question, do you have -- do you have the specific survey that you can -- you can quote from?

A. I believe we responded to an I.R. to answer that question. But I don't recall the number off the top of my head.

Q. I believe it was a D.P.S. -- interrogatory D.P.S. 841.

A.L.J. COSTELLO: Has that been pre-marked?

MR. BLAKE: I do not believe so.

A.L.J. COSTELLO: Are you -- you're going to be asking questions about that?

MR. BLAKE: Yes.

A.L.J. COSTELLO: Do you have copies for it? Do you want it?

MR. BLAKE: I do, yes.

A.L.J. COSTELLO: Okay.

MR. BLAKE: How many copers -- how many copies of, Your Honors?

A.L.J. COSTELLO: You would need one for each of us as well as the other parties.

MR. BLAKE: Let's see.

A.L.J. COSTELLO: We'll go off the record for a few minutes.

THE REPORTER: Off the record.

(Off the record 10:51 a.m. to 10:55 a.m.)

A.L.J. COSTELLO: And just for the record that's been marked -- the document which you'll identify for the record is being marked as Exhibit 608.

MR. BLAKE: I'd like to direct your attention to Exhibit 608, specifically the -- the survey question.

A.L.J. COSTELLO: Can you just identify -- identify the document?

MR. BLAKE: Yes. I'd like to identify -- this is a interrogatory request from Department of Public Service Staff, Number 841 to National Grid, Future of Heat Panel.

It is -- the date of request is July 10th, 2019 and the date of reply is July 22nd, 2019. It is six pages in length.

A.L.J. COSTELLO: Okay. And you're going to lay a foundation for that?

MR. BLAKE: The --

A.L.J. LEARY: Just ask the witnesses if they've read it.

MR. BLAKE: Right.

A.L.J. LEARY: I think.

MR. BLAKE: Have you -- have you -- have you -- have you read this exhibit? Are you familiar with this exhibit?

THE WITNESS: Yes.

MR. BLAKE: And did you prepare a response --

A.L.J. COSTELLO: What -- do you want to -- okay. Go ahead.

MR. BLAKE: I'm sorry.

A.L.J. COSTELLO: I'm sorry. I didn't mean to interrupt.

MR. BLAKE: It's all right.

BY MR. BLAKE: (Cont'g.)

Q. Did you prepare the response to this interrogatory request?

A. Yes.

Q. And can you please read the question that is listed in the survey on Attachment 1 to this exhibit?

A.L.J. COSTELLO: And let me just add, are there any objections at this point?

MR. GAVILONDO: No objection, but I think the company would stipulate to the language that's actually in the I.R. I don't know if the witness has to read the question.

A.L.J. COSTELLO: Okay. You can proceed.

BY MR. BLAKE: (Cont'g.)

Q. The question reads, "a green option for your natural gas supply with lower environmental impact." Is it true that the companies did not define the words green in that question?

A. No, we did not.

Q. And is it true that there are a number of different ways of interpreting what a green gas product might be?

A. There certainly are although the company favored a green gas product that decarbonized the customer's usage.

Q. And wouldn't a green gas product that's -- that secures carbon offsets or renewable energy credits for customers effectively decarbonize the customers' gas supply as well?

A. It would decarbonize the impact of their energy usage, although, it wouldn't be a way of decarbonizing the energy that they're using, and we think that there is a distinction between those programs similar to how we see distinctions on the electric side.

Q. But is it reasonable to conclude that that question is posed in a generic manner such that it's probing customer interest in green products, generally, not specifically R.N.G.?

A. Yes, this is a general question.

Q. And the companies did not perform any surveys specifically probing customer interest in R.N.G. products specifically?

A. We did not ask any further, more specific survey questions, no.

Q. And the companies did -- also did not perform any surveys probing customer awareness of different types of green gas products offered by energy service companies in New York?

A. No, we did not.

Q. And the panel provided justification for this program based on similarly structured product offerings in other states, is that correct?

A. Yes, we did. I believe we pointed to Vermont Gases Program.

Q. And I believe Owen Brady used to – used to work for Vermont Gas Systems from 2011 to 2017, is that correct?

A. (Traczyk) That is correct.

Q. And to the best of your knowledge, has Vermont ever restructured its power utilities?

In other words, has Vermont adopted retail competition?

A. No, it has not to the best of my knowledge.

Q. Did the panel conduct any analysis of how this green gas program in Vermont would align with or run counter to New York's -- the proposal that's offered in New York?

A. We did not perform an analysis based off of the Vermont Gas Program or the New York market structures.

Q. Given that New York does have retail competition, and ESCO could surely offer a fixed or flat rate product such as this for customers that wish to decarbonize their gas supply, correct?

A. Correct.

Q. And in that situation, isn't it true that an ESCO and the companies would be in competition with one another for customers?

A. To the extent that a customer wished to remain a full-service customer of National Grid, then that is the customer's choice and we believe those customers should have access to an R.N.G. product offering in addition to customers that are served by ESCOs.

So if we take customer choice out of it and simply look at the purchasing or procurement

of energy by those customers, then, yes, there would be competition.

Q. But to be clear, a customer can only either enroll in the Green Gas Tariff supply portion of the company service or receive supply from an ESCO not both, correct?

A. That is correct.

Q. And so in other words, isn't it likely that a customer who is enrolled in the Green Gas Tariff may elect not to switch to an energy service company product offering if one was created that offered the same decarbonization benefits?

A. It is certainly a possibility and to the extent that the customer was under the term outlined in our proposal, they would be unable to switch to an ESCO offering unless they paid the fee that was described in our proposal, effectively allowing them to settle out the contract under the Green Gas Tariff.

Q. So effectively, both the ESCOs and the companies are engaged in -- in competitive energy markets in New York, correct?

A. The companies are seeking to offer an R.N.G. product to their full-service

customers, because as mentioned in the survey before there is an interest in a green offering.

As you described it as a general question and so, therefore, there may need to be additional diligence done as to whether or not an R.N.G. product offering from the company would be the only offering that would satisfy customer's preferences.

However, in this case, both the company and ESCOs could potentially be offering an R.N.G. product.

Q. So in that sense, they would be engaged in competitive energy markets offering a competitive product?

A. They would be offering a similar product, yes. And customers would be making a choice between staying as a full-service customer of the company or transitioning service from an ESCO.

Q. Okay. I'd like to introduce a new exhibit. It's the utilities codes of conduct, and this one will proceed smoother than the last one.

I do have copies here for the judges and the panel and the company.

A.L.J. COSTELLO: Okay. That will be marked as exhibit for identification as Exhibit 609 and we'll go off the record while the copies are being provided.

THE REPORTER: Off the record.

(Off the record 11:02 a.m. to 11:04 a.m.)

BY MR BLAKE: (Cont'g.)

Q. So I'd like to direct your attention to -- to Page 1 of those codes of conduct, in particular, the definition of unregulated competitive energy affiliate.

A.L.J. COSTELLO: I just want to -- any objections?

MR. GAVILONDO: Yes, Your Honor, thank you. And I haven't heard the counsel's question yet. So it's hard to object in advance but to the extent the counsel was asking for a legal interpretation of the application of the codes of conduct, I object to that, that calls for legal conclusion.

A.L.J. Leary: Well, let's see what the question is.

A.L.J. COSTELLO: Is there any objection with respect to the -- this exhibit being offered into evidence?

MR. GAVILONDO: Your Honor, I think this code -- this is the code of conduct that was filed in February of 2019 by the National Grid companies KEDNY and KEDLI. No objection to the substance of the document itself.

A.L.J. COSTELLO: Okay.

MR. WYMAN: I -- I have a question though. If is -- so this is a document that was filed in the Commission's system -- file system under cases 16-G-0058 and 0059?

MR. BLAKE: This was filed in, I believe, it was case 15-M-0501 and 14-M-0101.

A.L.J. LEARY: The document is marked on its face cases 16-G-0058 and 16-G-0059. So --

MR. BLAKE: So that's correct, and I think what I was referring -- I apologize, I was referring to the order, the case where the -- the Commission set the order setting standards of conduct, but the compliance filings were in a separate docket as you -- as you listed.

A.L.J. LEARY: That's very helpful though. Can you give me that 0501? Is that 16?

MR. BLAKE: Yeah, it's 15.

A.L.J. LEARY: 15-M --.

MR. BLAKE: 15-M-0501, and 14-M-0101. And that's pursuant to the order setting standards for codes of conduct that was issued September 15th, 2016.

A.L.J. LEARY: Thank you.

MR. WYMAN: My other question though was if this is a document that is part of the Commission's record in another proceeding and was a filing of the company -- public filing of the company in another proceeding and apparently adopted by the Commission in -- in another proceeding, at least from -- from what counsel is saying, does it need to be an exhibit? Or can we just take reference of it in briefs? And so I'm just like -- I don't want to fully --

A.L.J. LEARY: I think it's very helpful to be an exhibit for this record --

MR. WYMAN: Okay.

A.L.J. LEARY: -- so it's an exhibit to this record. My understanding is this -- this

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particular document was filed as a compliance filing in 16-G-0058 and 0059. It was required under 15-M-0501 and 14-M-0101.

It's a rate proceeding, but it's my understanding it's the compliance filing in the previous rate case, is that correct?

MR. GAVILONDO: Your Honor, I know that the document was filed as part of 15-M-0501 and 14-M-0101. I'm not sure if there was compliance filing in the prior KEDNY-KEDLI rate cases.

A.L.J. LEARY: Counsel, do you know?

MR. BLAKE: I do – I'm not -- I do not know if it was part of a prior KEDLI-KEDNY rate case. I believe it was a compliance -- standard compliance filing in response to that September 15, 2016 order setting standards for codes of conduct.

A.L.J. LEARY: Got it. I'm just trying to figure out why we have rate case numbers associated with this document on its face. So maybe the -- somebody can answer that. I'm just trying to clarify what this document is.

MR. O'BRIEN: Your Honor, the reason why it has the prior rate case number is because this

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is an -- Appendix 12 is an excerpt from the joint proposal --

A.L.J. LEARY: Right.

MR. O'BRIEN: -- in that rate case. So, essentially, what we did is we took the old Appendix 12 and updated it per this filing.

A.L.J. LEARY: Okay. So that piece of information right there if we can go back on the record, if we are still off the record. Are we on the record?

THE REPORTER: We're back on the record.

A.L.J. LEARY: We're on the record. Sorry. Perfect. Thank you.

THE REPORTER: What was your name?

MR. O'BRIEN: Patrick O'Brien.

THE REPORTER: Okay.

A.L.J. COSTELLO: You may continue.

BY MR. BLAKE: (Cont'g.)

Q. Do the companies anticipate administering the Green Gas Tariff program through an unregulated competitive energy affiliate as that term is defined in the code of conduct?

MR. GAVILONDO: Objection, Your Honor, I don't know if the Counsel or I'm sorry, if the panel is prepared to render an interpretation of what the application of the codes of conduct are.

I think the panel -- panel can address the first part of counsel's question in terms of how the company is planning to administer this, but in terms of whether or not the -- the format or the structure for taking that forward is something that meets a particular definition, I don't think this panel is qualified to respond to that.

A.L.J. LEARY: Well, let's -- let them respond if they can. I need the question again. Can you --

BY MR. BLAKE: (Cont'g.)

Q. Yes. The question is do the companies anticipate administering the Green Gas Tariff program through an unregulated competitive energy affiliate as that term is defined in the codes of conduct?

A. (Traczyk) As -- as described in our -- in information requests, Family-1, which was submitted October 29th, 2019, and responded to by the company on November 8th, 2019.

The companies, not an affiliate will offer the Green Gas Tariff program.

Q. And isn't it true that the regulated utilities provision of products and services that are competitive in nature would be inconsistent with the codes of conduct?

MR. GAVILONDO: Objection, Your Honor. Calls for a legal conclusion.

A.L.J. LEARY: I disagree. I mean, you have something in front of you. You can tell whether the two are consistent or not. I mean without rendering some legal conclusion, I would hope.

Overruled.

THE WITNESS: Counsel, could you please repeat the question?

BY MR. BLAKE: (Cont'g.)

Q. Isn't it true that a regulated utilities provision of competitive products and services in the competitive energy markets would be inconsistent with the codes of conduct?

A.L.J. COSTELLO: Can I just -- can you clarify if there's any specific areas of the codes of conduct to which you're referring?

MR. BLAKE: Generally, the distinction between the company's regulated entity and its unregulated competitive energy affiliate.

A.L.J. LEARY: What page?

MR. BLAKE: Page and I -- page one of the codes of conduct, which defines the -- which defines the unregulated competitive energy affiliate.

A.L.J. LEARY: Okay. Helpful. Thank you.

THE WITNESS: The companies as part of their procurement work naturally will procure a gas supply and gas commodity from a variety of sources. To the extent that there is a different product offered as part of the regulated business, we don't view it as being inconsistent with product offering to make that available to our customers.

BY MR. BLAKE: (Cont'g.)

Q. So it's the panel's position that that the -- that the regulated utility company can offer a competitive product and that's consistent with the codes of conduct?

A. I guess I'll offer a non-legal response, what the spirit of the program was. I think the premise of, you know, your questions are,

you know, theoretically if this was offered, what the company's position is, but the spirit of the Green Gas Tariff program was, we see a customer demand and we also talked to suppliers of R.N.G. and they're our supply and they want to be able to offer this product and there isn't a program out there that sort of brings demand and supply.

So the spirit of the program and the intent was we wanted to foster innovation and be the matchmaker between their supply and the demand. Our customers, as Owen referred, they are seeking a product to decarbonize their heat, and suppliers when we talk to them is that they have R.N.G., and they want to be able to sell R.N.G. into the sector, but that mechanism currently doesn't exist.

Q. So I'd like to -- I'd like to turn to the -- the supply that you mentioned. The procurement of R.N.G. for the Green Gas Tariff. Will that be procured through a competitive bidding process?

A. Yes.

Q. And isn't it true that there is only a limited quantity of R.N.G. supply currently available in the market?

A. We don't know. I mean, we will put a solicitation -- market solicitation and we will see what the market comes back with.

Q. Will a likely result of the company's purchasing R.N.G. supply due to its size and purchasing power, it would control most of the R.N.G. supply in the state?

MR. GAVILONDO: Objection, Your Honor. I'm not sure the basis for the objection at this point.

A.L.J. LEARY: Then you can't -- you can't make it. If you --

MR. GAVILONDO: Right.

A.L.J. LEARY: -- don't have a basis, turn your microphone off. So go ahead, proceed. Do you want to repeat the question?

BY MR. BLAKE: (Cont'g.)

Q. So, I should repeat the question. Is it the likely result of the company's purchasing R.N.G. for this program, due to its size and purchasing power, that the company would control most of the R.N.G. supply available in New York State?

A. (Schwartz) I believe that would be a function of the size of the program and the

development of R.N.G. in the state. Two things that the company cannot forecast accurately, because it's not all within our control.

Q. So it's possible.

A. It would be possible if the program was wildly successful.

Q. I want to turn -- shifting gears slightly, turning to the company's anticipated marketing and enrollment of customers into the program. Will the companies actively solicit new customers to enroll in the Green Gas Tariff?

A. (Traczyk) For my information, can you describe what you mean by actively solicit?

Q. In the sense that that you would either conduct a phone call, send a bill insert, generally solicit customers to enter the program.

A. The company would make its customers aware of the Green Gas Tariff offering, yes.

Q. And it's the company's intention to solicit its own distribution customers to provide them with that opportunity to enroll in the program, correct?

A. Correct.

Q. And by soliciting its distribution customers, isn't it true that the companies would also be soliciting a certain percentage of customers that may already be enrolled in a competitive energy service product offering for their supply service?

A. That is certainly a possibility, though the company would undertake all reasonable efforts to ensure that its marketing would reach customers who would be eligible for the product offering similar to many of our other product offerings where we attempt not to send inaccurate or -- or messages to customers that would be ineligible to take action based upon them.

Q. So it's possible that a customer who may currently be receiving a decarbonized gas product from an ESCO may be convinced to terminate his or her service and elect to receive service from the companies under the Green Gas Tariff?

A. That is a possible result, yes.

A. (Schwartz) However, it is not the target of the program.

Q. I'd like to turn your attention to the panel's direct testimony. It's page 108 of

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120. Specifically, lines 11 to 17 and I'll read those lines.

"The proposed carbon reduction E.A.M. is composed of two metrics. One, metric tons of CO2 equivalent reduced from all full-service customers, and two, metric tons of CO2 reduced from transportation customers.

This metric will focus on the adoption rate of three specific offerings, oil to gas geothermal conversions, sales of R.N.G. via the Green Gas Tariff, and use of compressed natural gas and R.N.G. and N.G.V.s to offset the use of gasoline and diesel.

A.L.J. LEARY: Excuse me a second. I have oil to geothermal conver -- conversions. And I just thought I heard you say oil to gas.

MR. BLAKE: Apologies, oils to geothermal is correct.

A.L.J. LEARY: Thank you.

BY MR. BLAKE: (Cont'g.)

Q. My question for the panel is, is it true that the companies would be incentivized to enroll customers into the Green Gas Tariff in order

to achieve the company's earning adjustment mechanisms for CO2 reduction?

A. (Traczyk) You are correct that sales of R.N.G. via the Green Gas Tariff is one of the inputs into the carbon reduction E.A.M., yes.

Q. And is it true that the companies would be further incentivized to enroll customers into the Green Gas Tariff program in order to achieve the company's E.A.M. for supply diversification?

A. (Schwartz) That is incorrect. The supply diversification E.A.M. is targeted at how much R.N.G. we connect into our system rather than how much R.N.G. we deliver to customers.

Q. But those two numbers would be related, would they not?

A. They would -- they would not be related, no.

Q. And is it true that a competitive retail supplier of a green gas product such as an energy service company would not be entitled to receive those E.A.M.s?

A. (Traczyk) I believe that is true, yes.

Q. Is it the panel's position that the companies thereby have a -- an advantage in the market over competitive suppliers offering a similar product?

A. To the extent that the E.A.M. is accepted as proposed by the company and that there is a financial incentive relating to the sale of R.N.G. and to the extent that that R.N.G. is sold and generates a financial incentive for the company, yes, there would be a beneficial positioning from the company.

Q. And have the companies developed a marketing strategy for the Green Gas Tariff program?

A. We have not as of yet, no.

Q. And have the companies developed any sales agreements for customers that are solicited to enroll into the program?

A. We have not as of yet, no.

Q. If a customer enrolls into the Green Gas Tariff program and terminates before the expiration of the agreement, would the companies charge an early termination fee?

A. We are working through the process of customer both enrollment and departure from the program. The fact that there is a term limit offered in our proposal would generally lend itself to an early termination such that the cost of the R.N.G. that were procured on behalf of that customer do not become borne by non-participants of the Green Gas Tariff program.

Q. And assuming the company -- the companies did charge an early termination fee for an early exit from the green gas program, would the companies abide by the early termination fee limits that are imposed on mass market customers as set forth in the uniform business practices for energy service companies?

A. Our intention would be to abide by all regulations and policies outlined for sale of gas to our full-service customers.

Q. And would that include the uniform business practices imposed on energy service companies?

A. I don't have knowledge of those specific practices, but I -- our intention, of

course, would be to comply with all practices that were relevant for the offering.

Q. And have the companies developed any disclosure statements, bill inserts, marketing scripts, verification scripts, customer notices or other material that will be used in the marketing of the program?

A. No.

Q. Did the companies anticipate complying with all public service commission orders imposing restrictions on the sales and marketing of value-added supply products to customers?

A. Yes.

Q. So the company's intend to comply with the December 16th, 2016 order adopting a prohibition on service to low income customers which is applicable to ESCOs, the companies plan on complying with that as well?

A. I believe that topic was covered in I.R. I'm simply looking for the reference. I apologize for the delay. I'm unable to find the I.R. I believe there was one on this topic and the company's position of course is that it would comply

with all relevant policies and -- and practices for the product offered.

Q. So is it the company's position that the policies relevant to the green -- to the company's provision of the Green Gas Tariff are the same as those policies, rules, and regulations applicable to energy service companies offering a compet -- a competing product?

A. The company -- the company's position is that we would want to offer a product that was similar to the extent that it eliminated customer confusion and to the extent that it allowed for fair competition within the marketplace, given that it is a competitive market.

Our belief is that this product offering focused on full-service customers would allow those customers to have access to an R.N.G. product offering should they choose to remain full-service customers of the company.

Q. So is that a yes or a no to that question?

MR. GAVILONDO: Objection, Your Honor. The counsel is asking the witness to determine whether or to state whether or not the company is --

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A.L.J. LEARY: Do you want to rephrase the question? Because I'm going – I'm going to sustain that. I mean, that's my sense that there are two sets of rules here. The company's made clear in testimony on the panel that goes without saying almost that they expect and intend to comply with applicable orders, et cetera, from the Commission, policies, et cetera, prior orders. So do you want to rephrase?

MR. BLAKE: I under -- I understand that the -- the company is willing to comply with applicable rules and regulations and orders. My question more specifically was, does the company believe that the rules and regulations and orders applicable to the competitive retail energy markets in New York State are applicable to its provision of this program?

MR. GAVILONDO: And I would restate my objection, Your Honor.

A.L.J. LEARY: I think -- I think it is a legal conclusion because the rules for ESCOs,

energy service companies, my understanding are fairly consistent with regulated utilities in our state, but that's not necessarily an issue for this panel or us to decide today.

Are you asking the company if they will comply with what's applicable to ESCOs?

MR. BLAKE: Correct.

A.L.J. LEARY: They're not an ESCO.

MR. BLAKE: Right.

A.L.J. COSTELLO: And that's also something that you can --you can raise as a legal argument. Your position that the companies because of their offerings would be subject to those rules. That's a legal argument. It's not a factual argument.

I think they've answered the question with respect to they intend to adhere to all applicable rules. I don't believe they've indicated that they also are subject to those ESCO rules.

If you want to ask just their question, do they believe they're subject to the rules applicable to ESCOs? That's fine, but that's something that you can argue as a legal matter in your brief.

MR. BLAKE: Well, then, I'd like to ask that question then. Do they -- do they believe that they're applicable to the rules subject to energy service companies?

A.L.J. LEARY: You can ask that question. The panel can answer.

A.L.J. COSTELLO: That's if you know. If you know it.

THE WITNESS: (Traczyk) Sure. So I'm not -- I'm not a lawyer, but my -- my belief is that as we are not an energy services company, we are not directly subject to the same policies.

MR. BLAKE: Thank you. And turning briefly to the -- to the company -- to pre-filed Exhibit 39. Schedule one, page two.

MR. BLAKE: And this is the -- the section.

A.L.J. LEARY: And I just need to know, Hearing Exhibit 39, is that correct?

MR. BLAKE: Yes, correct.

A.L.J. LEARY: And on our exhibit list it would be PMN-5, is that correct?

MR. BLAKE: I -- I have it as -- I have it listed as -- as pre-filed Exhibit 39. This is -- this is the same exhibit that --

A.L.J. COSTELLO: This is the Future of Heat Panel 1?

MR. BLAKE: Correct, correct.

A.L.J. LEARY: Okay. That's -- I don't think that's Hearing Exhibit 39, is it? Do you have that?

A.L.J. COSTELLO: Yeah.

FEMALE: I have the wrong goddam list --

MR. BLAKE: This is the same document that was the first exhibit.

FEMALE: -- of work papers. This is for depreciation. This is to -- this is from -- okay.

A.L.J. LEARY: Go ahead.

BY MR. BLAKE: (Cont'g.)

Q. Is it true that the panel's analysis of potential alternatives is limited only to the two short sentences at the end of the alternative section?

A. (Traczyk) No, that is not true and in -- was it supplemental -- rebuttal. In a -- in an additional testimony filed by the panel I believe it was the rebuttal testimony additional detail was provided about both of these alternatives.

Q. And did the company consider proposals to create or encourage a renewable energy credit type market for R.N.G. that would recognize the environmental attributes of green gas?

A. Can you explain what you mean by renewable energy credit market?

Q. Did -- did -- in exploring alternatives to this product, did the company consider any proposals to create a program similar to renewable energy credits on the electric side for renewable natural gas to recognize those environmental attributes?

A. The company did not. I don't believe that would be within our purview necessarily.

Q. And to the best -- to the best of your knowledge, has the Commission ever authorized a regulated utility company to offer a competitive product such as this in the past?

A. I don't have knowledge of -- of all offerings that have been made available so I cannot speak to that question.

Q. But the panel -- the panel is not able to name -- to name one?

A. (Chahbazpour) The only other similar product that we are aware of is the Vermont Gas, Green Gas Tariff.

Q. And that product is in Vermont, not New York, correct?

A. Correct.

A. (Traczyk) The only other piece of information -- forgive me. The only other piece of information as included in D.P.S. 841, information request D.P.S. 841 was that Consolidated Edison Company of New York proposed a green gas tariff in it's rate case, cases number 19-E-0065 and 19-G-0066.

MR. BLAKE: Thank you. No further questions.

A.L.J. LEARY: Thank you.

A.L.J. COSTELLO: Okay. Up next is Paul.

MR. BERKLEY: Thank you, Your Honor. I was going to say good afternoon panel, but we still have a few minutes of morning left.

THE REPORTER: What is your name?

MR. BERKLEY: My name is Richard Berkley, and I'm representing the Public Utility Law Project of New York.

A.L.J. LEARY: I can hardly hear you and I'm going to bet the people in the back can't hear you, so you really need to speak into the microphone.

MR. BERKLEY: I'll try to be louder, Your Honor.

A.L.J. LEARY: Thank you.

BY MR. BERKLEY:

Q. On page 7 of the -- your direct testimony lines 3 and 4 you complete a paragraph on the purpose of your testimony wherein in the last two lines you state, "… the state, the commission and the companies recognize more is needed to meaningfully change the current climate trajectory."

Are you familiar with those words and were they created under your direct -- under your direction or by you yourselves, panel?

A. Yes.

A.L.J. LEARY: Can I -- excuse me. Can I just ask every lawyer who's doing cross-examination to read the complete sentence that you are citing? Don't start your citation in the middle of the sentence.

So can we have that again, Mr. Berkley, for the record?

MR. BERKLEY: Of course, Your Honor.

A.L.J. LEARY: I think it starts yet comma.

BY MR. BERKLEY: (Cont'g.)

Q. "Yet, with the challenges presented by climate change, the state, the commission and the companies recognize that more is needed to meaningfully change the current climate trajectory." Are you familiar with those words?

A. Yes.

Q. Are you also familiar with the company's position as stated in the future of HEAT testimony that it will promote non-pipe alternatives and demand reduction?

A. (Chahbazpour) Yes.

Q. Are you familiar with the company's 80x2050 Conference and commitment to decarbonize and otherwise address climate change?

A. 8050 Conference?

Q. Yes. It is also referred to in the Future of Heat testimony. If you prefer, the 80x50 Pathway?

A. The companies, yes.

Q. No, not the states, thank you.

A. The Northeast Pathway, yeah.

Q. How long has the company been aware that it was approaching a need for a moratorium? I refer to the natural gas moratorium in Long Island and portions of New York City.

MR. DECICCO: Your Honor, I'm going to object to the question. It's beyond the scope of the panel's testimony. I don't believe that they addressed the moratorium in their filing or that these folks are qualified to discuss that issue.

A.L.J. LEARY: Well, let's see if they are. Can anybody on the panel answer the question?

THE WITNESS: (Chahbazpour) I cannot.

THE WITNESS: (Schwartz) No.

MR. BERKLEY: Thank you, Your Honor. I'll continue.

BY MR. BERKLEY: (Cont'g.)

Q. How long have you and the panel been aware of the state's goals for decarbonizing?

A. I don't know the exact date, but as long as the administration and the previous administrations have been talking about climate change.

Q. Is it fair to suggest that the company's actions as proposed by the Future of Heat Panel are aimed at de-carbonization among other things?

A. Yes.

Q. Would the company's institution of more non-pipe alternatives and demand reduction reduce the actual usage of natural gas in its service territories in New York City and in Long Island?

A. Yes.

Q. Would imposing the company's planned demand reduction and non-pipe alternatives and other measures as put forward in the Future of Heat have an effect on the revenue requirement sought by the company?

A. Just so I'm clear, are you talking about energy efficiency programs or are you talking about Green Gas Tariff or just broadly?

Q. Broadly.

A. I think the answer -- I think -- I believe the answer is yes, but I'll let my other colleagues chime in.

A. (Traczyk) I'll amend that answer only to say to the extent that there were non-pipeline alternatives, N.P.A's., that were proposed and adopted by the company, the N.P.A. incentive mechanism is based around a net reduction in plant investment.

And so therefore, if there was a reduced capital investment over the course of the case, it could potentially result in a reduced revenue requirement. Though I'm not -- I'm not an expert and -- and can't speak on the exact accounting as to how and when that would influence the -- the company's overall costs.

Q. Thank you. Would the company's institution of non-pipe alternatives demand reduction or energy efficiency cause an increase in customer's bills?

A. (Tallet) From an energy efficiency perspective, the increased collection costs to fund the energy efficiency programs would initially increase the customer's bill with an expectation of driving energy efficiency savings.

Q. Would that savings be felt within a year?

A. Potentially.

A. (Traczyk) And -- and counsel, the only other thing I'll say is a distinction might be helpful between customer with a lower case C, so to speak, an individual customer as opposed to customers at large to the extent that there are additional expenditures in energy efficiency demand response specifically those two which tend to be more outlays of -- of funds into those programs that would likely reduce -- sorry, likely would increase costs.

Non-pipes alternatives to the extent that they are accepted by their definition have a lower plant investment. So unless and project was accepted where it was a project that would not have been pursued by the company, there would be a net reduction in cost based off of the N.P.A. mechanism.

Q. Thank you. Are you aware of the increase to KEDNY delivery rates that was proposed by the company in this rate case?

A. I am not specifically aware of those, no.

A.L.J. LEARY: Anyone else on the panel?

THE WITNESS: (Schwartz) No.

THE WITNESS: (Tallet) No.

THE WITNESS: (Chahbazpour) I'm not familiar either, no.

BY MR. BERKLEY: (Cont'g.)

Q. Would you accept, subject to check, that in the company's filing letter on page one, the letter dated April 30, 2019 that it stated that KEDNY would increase revenues by 19.3% in the proposed rate plan and KEDLI 6%.

MR. TRACZYK: Your Honor, I'm not sure if the panel is familiar with that, but if counsel is asking if that's what the letter says, I think the letter is filed in the -- in the proceeding and I think we'll stipulate that that's what the letter says.

A.L.J. LEARY: Does anyone in the panel have an answer to the question? Do you know what the increase in revenue percentage is?

MR. TRACZYK: I do not have specific knowledge of the proposed increase percentages.

A.L.J. LEARY: Anyone else on the panel?

THE WITNESS: (Schwartz) No.

THE WITNESS: (Pallet) No.

THE WITNESS: (Chahbazpour) No.

THE WITNESS: (Hamlin) No.

A.L.J. LEARY: Okay.

MR. BERKLEY: Thank you, Your Honor. Does the panel know whether the E.A.M. set forward in the Future of Heat testimony would have an impact upon delivery rates?

THE WITNESS: (Schwartz) I am not familiar with how the specifics of how the E.A.M.s will be recovered from.

A.L.J. LEARY: Anyone else?

THE WITNESS: (Tallet) I'm not aware of what the impact would be of the E.A.M.s to the delivery rate.

MR. BERKLEY: Are E.A.M.s additional sums of money that the company would receive if it met certain goals that it proposed in its Future of Heat testimony that have not yet been approved by the Public Service Commission?

THE WITNESS: It would be.

MR. BERKLEY: Is it unreasonable to ask if the addition of more money to the company's revenue would have an effect on bills and delivery rates?

THE WITNESS: It would, as far as the recovery of the E.A.M.s, I'm not -- I'm not aware myself of exactly how it would impact all of the -- how all of the E.A. -- E.A.M.s proposed would impact the delivery rates.

MR. BERKLEY: Thank you. Is the panel able to discuss projected bill impacts to residential customers subject to the proposals put forward in the Future of Heat testimony?

MR. GAVILONDO: Your Honor, we do have a panel that will be testifying in this proceeding that will address rate impacts. I think that's beyond the qualifications and the matters that this panel is to testify on.

A.L.J. LEARY: I think Mr. Berkeley is aware of that. And, again, the panel can say what you just said, but what I would prefer is if you don't do that so that they say that. You let them answer the question unless you have an objection, like, calls for a legal conclusion is fine. Let the panel tell Mr. Berkley no idea or whatever they're going to say, or this panel will talk about it.

So let's -- let's just let this panel answer the question.

You want to repeat the question, Mr. Berkley?

MR. BERKLEY: Yes, Your Honor. Would the panel be able to speak to the impact upon customer bills by the proposed E.A.M.s set forth in the testimony?

THE WITNESS: (Traczyk) This panel was involved in the development of the E.A.M. proposals. But the rate design panel will be better positioned to speak about potential impacts from their achievements should they be approved.

MR. BERKLEY: Thank you. How much is a basis point in the case that the company has filed?

THE WITNESS: This was addressed in information requests D.P.S. 361 which was submitted to the company on June 4th, 2019 and responded June 14th, 2019. In page 2 of that response, the value of 10 basis points on a rate year basis were described and it is outlined in the table within that specifically for KEDNY.

Should I read the table? Okay.

A.L.J. COSTELLO: You can, if you want to. You -- you don't have to, you can wait for a question.

THE WITNESS: Okay.

MR. BERKLEY: Read the table.

THE WITNESS: Sure, absolutely. For -- for KEDNY and again, this is the value of 10 basis points. In 2020, the value was 3.301 million. In 2021, it was 3.847 million, 2022, 4.338 million and 2023, 4.769 million.

For KEDLI, over the same four rate years, the values are 2.132 million, 2.497 million, 2.788 million and 3.054 million respectively.

A.L.J. LEARY: Same years, correct?

THE WITNESS: Correct.

A.L.J. LEARY: Thank you.

BY MR. BERKLEY: (Cont'g.)

Q. Thank you. And in 2020, in a table in your testimony on page 96 that is entitled KEDNI E.A.M. maximum incentive basis point. And tell me when you've gotten to that page.

A. (Traczyk) I'm on the page.

Q. Great. At the bottom of the column for 2020 you list 78 basis points. Is 78 basis points the number of basis points the company could qualify for if it met all of its proposed goals in 2020?

A. As outlined in our initial direct-filed proposal, yes.

Q. And since this is KEDNY is that 78 basis points each valued at 3 million of basis point?

A. No. So the value of the basis point was one tenth of the numbers that I provided, that was the value for 10 basis points. So it would be 78 multiplied by one tenth of the values provided by four -- provided before.

Q. I would say that like most attorneys, I don't do math very much. But --

A.L.J. LEARY: I want to make sure the record's clear. So it would be 7.8 --

THE WITNESS: Correct, 7.8 --

A.L.J. LEARY: -- times --

THE WITNESS: -- multiplied by 3.3 million.

A.L.J. LEARY: -- 3.301 million for 2020, the same true for each year?

THE WITNESS: Correct, Your Honor.

A.L.J. LEARY: Thank you.

BY MR. BERKLEY: (Cont'g.)

Q. Is it fair to say that the impact of the basis points would be several million dollars?

A. That is fair to say.

Q. 2020. Would it also -- and forgive me, on page 97 there is an identical table for KEDLI which would result in 77 basis points if the company qualified for or met all of its E.A.M. goals in that year. And for 10 basis points you asserted that would be 2.13 million dollars in KEDLI?

A. I believe -- I believe that is correct. I don't have the page open in front of me.

Q. So this also would add -- would this also add several million dollars to the company's revenue if it met all of its requirements?

A. Yes.

Q. Would the ratepayers be the ones to pay for that additional revenue?

A. Yes.

Q. Would they be true also for KEDNY?

A. Yes.

Q. So in the end the company's achievement of its maximum E.A.M.s would be paid for by ratepayers on their bills?

A. That is correct. And to your earlier question about the impact on delivery rates, our responses -- our response was not to say that it would not have an impact on delivery rates, simply that we are not aware how the direct accounting mechanisms would impact the delivery rates or what those rate changes would be.

And again, I would -- I would direct you to the rate design panel who would be better positioned to answer those questions.

A.L.J. LEARY: But your response is that it will impact delivery rates. Is that your testimony?

THE WITNESS: Yes.

A.L.J. LEARY: Thank you.

MR. BERKLEY: Thank you. And -- and we will ask similar questions of that panel.

BY MR. BERKLEY: (Cont'g.)

Q. Is the panel aware of the allocation of -- the impact of these E.A.M.s between residential and commercial and industrial customers in its service territory?

A. By impact, are you referring to the financial impact of their achievement on delivery rates or on the -- on -- on a different aspect of the E.A.M.s?

Q. Yes, I am referring to the impact upon delivery rate.

A. I don't have knowledge of the -- the allocation.

Q. Thank you. Is the panel aware of the return on equity sought by the company in this case?

A. I don't have that specific information in front of me. So I would say I'm not aware at this time.

Q. Okay. I'll wait for revenue. Would the panel agree, subject to check, that the company's R.O.E. after receipt of all of the E.A.M.s, the maximum E.A.M. set forward in the Future of Heat testimony could exceed the R.O.E. sought by the company in the first place? And sorry for the court reporter return on equity.

A. I suppose that is -- is theoretically possible. Again, I'm -- I'm not -- I'm not aware of the mechanisms by which the -- the money would be recovered, nor how it would impact the overall finances of the case.

Q. Thank you. Bear with me for a moment please. The panel puts forward in the Future of Heat testimony, again, I'm still working on the direct, it describes energy efficiency programs for low- and moderate-income customers. Is that correct?

A. (Tallet) Yes.

Q. How does the company or in this case, this panel and the testimonies define what a low-income customer is?

A. A low-income customer for the company is defined as a customer who is a recipient of a HEAP grant or HEAP eligible. They may be enrolled in our -- our energy affordability program, low-income bill discount.

Q. Would the company's definition exclude income eligible ratepayers who are not enrolled in the company's assistance plan?

A. That would fall under the category of a customer who might be HEAP eligible as opposed to a HEAP recipient.

Q. How does the panel define a moderate income household?

A. The moderate-income customer might be a customer with an income limit between 60 to 80% of the State Median Income. Little bit higher than what the HEAP recipient is. So too -- income level is too high to receive HEAP.

Q. Thank you. Is the panel able to describe what the expected allocation would be for E.E. programs between low income and moderate-income households?

A. We have not made that determination yet with the -- the New Efficiency New

York Order that was recently approved in December that requires the company to dedicate at least 20% of our incremental energy efficiency funding to the low- and moderate-income sector.

So the company is engaged with NYSERDA and the joint utilities on, on what that -- what that effort might be, would ultimately be described in our energy efficiency plans.

Q. Does the company have a potential date when it will know that allocation between low- and moderate-income households?

A. For the effort that the company is working on with the New Efficiency New York, the details of that will be described in an implementation plan that would be prepared by NYSERDA and the joint utilities and I believe the filing date is March 16th of this year.

Q. Thank you. Does the panel have a definition of what it would consider to be a high usage residential customer?

A. I -- I do not have a specific definition of that. I don't know if anyone else.

Q. Does the panel have a definition of what it would consider to be an average use residential customer?

A. (Traczyk) Just for information, counsel, is -- is there a place in the testimony where these terms are -- are used that we could be referencing to ensure we're -- we're looking at the same information you are or -- or is this outside of the testimony?

Q. It's outside of the testimony.

A. Thank you.

Q. It may be in your -- in the affordability area. And forgive me while I dig that up. So page 106, line 10, you describe an average low-income customer bill. It has no definition, which is why I asked.

A. (Tallet) Right. I don't have it in front of me, but the company does calculate an average low-income bill for our customers that are on the energy affordability program. It's part of the annual calculation to determine the bill discount.

Q. Thank you. No further questions for this panel, Your Honors.

A.L.J. LEARY: Thank you.

A.L.J. COSTELLO: Yeah. Utility Intervention Unit will be next.

MS. O'HARE: Your Honors, we have no questions at this time.

A.L.J. COSTELLO: Okay. Then we go to the City of New York.

THE REPORTER: What was your name?

MS. O'HARE: Kathleen O'Hare.

A.L.J. LEARY: So U.I.U. has no questions.

MR. FUNG: Your Honors, Justin Fung for the City of New York, just give me a second to get situated and can you hear me fine?

A.L.J. LEARY: Just make sure the microphone is close to your mouth.

MR. FUNG: Morning, panel. I'm going to mostly focus on your rebuttal. So if you could just turn to --

A.L.J. LEARY: You need to be speaking into the microphone.

MR. FUNG: If you could turn to your rebuttal page 53. Actually, sorry, 55.

On page 55 of your rebuttal, on line -- starting on line 2 the panel stated, "Although the

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companies are open to collaborating with the City, including the Retrofit Accelerator, to determine how to improve information sharing and communications consistent with applicable guidelines, we do not agree that new reporting requirements should be implemented in the context of these rate cases."

I just want to focus on the portion regarding the Retrofit Accelerator. Could the panel clarify for the record whether the company's willingness to collaborate with the Retrofit Accelerator is within the context of this rate case or whether the panel's testimony is that such collaboration would occur through some other means?

And what I'm trying to get at is earlier in your rebuttal testimony the panel opposed some of the city's proposed energy efficiency program improvements as outside of the context of this rate case and that those matters should be better addressed as part of a generic proceeding.

So I just want the panel to clarify whether that also applies to collaboration with the Retrofit Accelerator.

THE WITNESS: (Tallet) So as far as the collaboration and the attempt of that, the

19-G-0309/19-G-0310 - 2-10-20 - Brooklyn Union Gas company is very interested in collaborating with any stakeholder as this is a -- it's a -- it's a huge undertaking to address the energy needs of the city.

With respect to the reporting, I don't think the company is opposed to additional reporting for maybe new efforts coming into it. I think in the context of the -- the testimony, the topic of the energy efficiency reporting was -- was really what the -- what we were interpreting there.

The energy efficiency reporting is -- is very comprehensive. It's very complex and with the New Efficiency New York, the utilities are waiting on some additional guidance that may change some of that. So that directing it to the energy efficiency proceeding might be that opportunity for all of us to discuss collaborative activities and how they should be reported in support of the -- the statewide targets.

MR. FUNG: Thank you. And with respect to collaborating with the Retrofit Accelerator, has the panel -- is the panel proposing any specific types of collaboration or is there any -- any sort of collaboration that you've envisioned, you know, for example, monthly meetings with a

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specified energy efficiency representative from the companies with the Retrofit Accelerator?

Has -- has the panel, or have the companies proposed -- are the companies proposing any such collaboration or is -- is anything -- anything being envisioned at this time?

THE WITNESS: I'm not aware of anything specific at this time.

MR. FUNG: Okay.

THE WITNESS: But with that said, with the New Efficiency New York order, it's giving us some opportunity to rethink and -- and consider other proposals.

MR. FUNG: Okay. Thank you. If I could turn to page 59 of the rebuttal.

BY MR. FUNG: (Cont'g.)

Q. Starting on line 16 and spilling over into the next page. The panel stated the companies acknowledged the importance of electrifying transportation in order to achieve the state's emissions objectives.

However, the current commercially available electric vehicle or E.V. market is focused primarily on passenger vehicles, whereas the vast

majority of the company's fleet is comprised of other vehicle types.

The companies know that E.V. market will continue to evolve and that they will continue to pursue appropriate opportunities to introduce E.V.s into their fleets. However, in the absence of commercially available E.V. options to meet many of the company's vehicle needs, the companies do not agree with the city's proposal.

Would you agree that electrifying the company's fleet of passenger vehicles is one of the various steps that the companies could take towards meeting state emissions objectives?

A. (Tallet) I -- I would agree. However, I'm not familiar with -- with the work that we've done in this space with respect to E.V.s. I don't know if there's anyone else on the panel, who might be able to address that.

A. (Schwartz) And I -- I believe we responded to some questions along the same lines with a -- for request number 98 from the City of New York, as well.

A.L.J. LEARY: Excuse me. Please refer again. Are you referring to information request 98?

MS. SCHWARTZ: Yes. Sorry, Your Honor, information request.

A.L.J. LEARY: And who served that information request? Can you give the full numerical value?

MR. FUNG: Your Honor, --

A.L.J. LEARY: And --

MR. FUNG: -- this was City of New York Number 98, which was included as one of the exhibits to N.Y.C.E.E.P.-1 on page 28 out of 57 and I think that's Exhibit 548.

A.L.J. LEARY: Thank you. So if you're going to refer to an exhibit, you're going to have to give more information than just 98 or I.R.9 -- you know -- do you follow me? So we can follow it along in the record.

THE WITNESS: Yes.

A.L.J. LEARY: Thank you.

THE WITNESS: Thank you, Your Honor.

BY MR. FUNG: (Cont'g.)

Q. So -- so the basis of my questions on electric vehicles was that the panel did respond to some of the -- the city's I.R.s on that topic. And I'm basing my understanding -- I am -- my understanding is that this panel is qualified to talk about E.V. procurement and E.V. fleet issues on behalf of the companies. So if -- if that is not the case.

A. (Schwartz) I don't believe anyone on this panel works on E.V. procurement for the company, although I am somewhat familiar with our responses to the information requests.

Q. Okay. Then on that, if you look at the top of page 60 of the rebuttal, where the companies had stated that they'll continue to pursue appropriate opportunities to introduce E.V.s into their fleets.

Could you elaborate on what the companies considering appropriate opportunities?

A. I believe in our responses to the information request just mentioned, the company has set targets about E.V. procurement.

Q. Are you referring to your response to City Number 98 where the response stated

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that National Grid has a commitment to spending 5% of annual acquisition dollars on plug-in electric vehicles and that is across all National Grid companies, not just -- not just KEDNY-KEDLI.

A. Yes, that's correct.

Q. And are you aware of any internal policies within KEDNY & KEDLI related to that 5% target that is, do the companies have any internal targets for procuring E.V.s out of KEDNY & KEDLI's fleet budgets?

A. I'm not aware of any jurisdictionally specific targets. But I believe the response to the information request Number 99 from the City of New York elaborates further on our plans for E.V. procurement within KEDNY-KEDLI.

Q. Could I turn to that response? I believe it's City 99? It is. It's marked -- it's -- it's marked as Exhibit 548. It's part of N.Y.C.E.E.P.1 and it's on page 29, I believe.

A.L.J. LEARY: Do you have a proposed hearing exhibit number on our list?

MR. FUNG: Yeah. It's -- it's pre-filed 548.

A.L.J. LEARY: Thank you.

MR. FUNG: It's already -- it's already in the record.

BY MR. FUNG: (Cont'g.)

Q. So as part of that response, the panels stated that National Grid continues to convert its passenger car fleet to all electric vehicles and intends to accelerate that process in fiscal year '19 and fiscal year '20.

Is the panel -- can the panel provide any specifics on how many electric vehicles KEDNY-KEDLI have procured -- passenger vehicles as part of that effort or is that just a general statement?

A. I'm not familiar with any specific plans or numbers of vehicles expected to be procured.

Q. And would it be fair to say that, if there are commercially available passenger E.V.s in the market today that would be an appropriate opportunity for the companies as you stated in your rebuttal to introduce E.V.s into their fleets?

A. Sorry. Could you -- could you restate that question?

Q. Sorry. So, in your rebuttal, you stated that the companies will pursue appropriate

opportunities to introduce E.V.s into their -- their fleets.

Is it fair to say that if there are commercially available passenger E.V.s in the market today that would constitute an appropriate opportunity?

A. I'm not familiar with our procurement practices when it comes to our fleet. However, I do know that we are moving towards procuring electric vehicles as a company and so if that aligns with our expectations within our procurement practice that theoretically could be a realistic trust.

Q. Do you -- do you have any specifics on what the company's expectations are for fleet procurement? When -- when does a specific procurement opportunity get looked at in the context of whether it should be an E.V. or not an E.V.? Is there a specif -- specific threshold that the companies have on -- in terms of dollars spent?

A. I'm not aware of our procurement practices for our fleet.

Q. Right.

A.L.J. LEARY: Excuse me a second. Okay. So the testimony I'm reading says, "Appropriate opportunities". And I'm missing the answer to that question. What are you talking about in your testimony, all the panel members, when you say appropriate opportunities to introduce E.V.s into their fleets, what does that mean?

THE WITNESS: So I am not familiar with how we make specific decisions when it comes to exactly what we procure. I believe --

A.L.J. LEARY: I'm not asking you about -- I'm not asking you about procurement. I'm trying to clarify the record on this. I'm not asking about procurement because you've made clear you can't talk about that.

What did you mean when you said appropriate opportunities to introduce E.V.s into their fleets? What's that mean?

THE WITNESS: (Traczyk) Your Honor, in this case, appropriate refers to commercially available vehicles that meet the operational requirements of their purpose effectively.

So to the extent that there are commercially available passengers E.V.s that also

meet any operational needs for the company in terms of range, in terms of resiliency, in terms of their use in a storm response or something like that then I believe they would be considered by the -- the company's fleet procurement practices.

A.L.J. LEARY: Okay. Great. Thank you. Anybody have anything to add on that? Okay. Thanks. Sorry to interrupt.

MR. FUNG: Thank you, Your Honors.

BY MR. FUNG: (Cont'g.)

Q. Building off of that, you have no -- is it fair to say that that you have no specific knowledge on whether there are specific criterion or criteria in order to ascertain whether such commercially available E.V.s should or should not be procured as part of that overall procurement plan?

A. (Chahbazpour) I do not have knowledge of any specific criteria, no.

Q. The rest of the panel, same thing?

A. (Tallet) No.

Q. Thank you. I'd like to switch gears a little bit. Is the -- the panel’s generally familiar with the company's settlement in case 19-G-

0678, which is the proceeding on motion of the commission to investigate denials of service requests by National Grid specifically the portion of the settlement that calls for the companies to pursue incremental energy efficiency and demand response programs?

A. Yes.

Q. And are those efficiency and demand response initiatives separate and apart from the efficiency and demand response initiatives that you are proposing as part of your -- your testimony?

A. Believe there's an I.R. that describes them. So some -- some would be enhancements to existing energy efficiency efforts and the demand -- the demand response efforts.

I would have to ask Owen to speak to those.

Q. I think the response that you're referring to is a City of New York 158. Is that correct?

A.L.J. LEARY: And do you have a hearing exhibit number for that?

MR. FUNG: That is not in the record and I -- I'd be happy to pass it out.

A.L.J. LEARY: Great.

MR. FUNG: We only have about 20 copies, so if anyone is okay with doubling up, that'd be great.

A.L.J. COSTELLO: Thank you.

A.L.J. LEARY: 610.

A.L.J. COSTELLO: Yeah.

MR. FUNG: Did you say 610, Your Honor?

A.L.J. COSTELLO: That's marked, yeah, as Exhibit 610. And just identify it for the record, please.

MR. FUNG: Yes. So 610 would be the company's response to City of New York 158, which was issued on January 20th, 2020 and it's four pages.

BY MR. FUNG: (Cont'g.)

Q. And this I.R. was prepared under your direct or indirect supervision?

A. (Tallet) Yes.

A. (Traczyk) Yes.

Q. And it is a -- is a -- is it correct to say that it's a true and accurate representation of the response that you submitted?

A. (Tallet) Yes.

Q. I don't have any specific questions on this I.R. itself. I just -- I just put it into the record because you're referencing it.

Is it the panel's understanding that these incremental energy efficiency and demand response initiatives, that were agreed to as part of the settlement are to be shareholder funded? Is that correct?

A. Yes.

Q. And to the panel's knowledge, will the costs to implement these incremental settlement initiatives be tracked separately and apart from how the companies track its spending in its other energy efficiency programs?

A. Yes.

Q. And in the panel’s understanding, will performance of these incremental initiatives also be separately tracked?

A. Yes.

Q. And will the -- will any savings that are achieved as a result of these incremental initiatives be counted towards the company's proposed E.A.M.s, specifically the energy efficiency E.A.M.?

A. I don't think the company has developed a position on that.

Q. Okay. So it is possible that the companies will track these incremental efficiency initiatives and any savings that are achieved through those initiatives could be considered for the energy efficiency E.A.M. it's possible?

A. Yeah. So -- so as this is -- is evolving and we have set up our -- our separate reporting and tracking for these particular initiatives. I think with respect to the savings that come from those initiatives and how those savings could be contributing to the statewide goals. I think those are still discussions that the company has to work out with respect to the settlement and our New Efficiency, New York.

As I mentioned previously, there is reporting guidance that will be coming out from -- from the commission with respect to how all of that is tracked.

Q. Okay. So as of right now, nothing is set in stone.

A. Correct.

MR. FUNG: No more questions, Your Honors.

A.L.J. COSTELLO: Do you want to go off?

A.L.J. LEARY: Yeah.

A.L.J. COSTELLO: We're going to -- thank you. We're going to go off the record for a second.

THE REPORTER: Off the record.

(Off the record 12:17 p.m. to 1:06 p.m.)

A.L.J. COSTELLO: And Sane Energy Project.

MS. ZIESCHE: All right. Yeah. Can you hear -- everybody hear me? Is this working? Okay. My name is Lee Ziesche, for the record. On page six of the direct testimony filed in April, the companies say as the energy provider to nearly two million customers in Downstate, New York, the companies are committed to advancing clean energy solutions that further National Grid's commitment to environmental stewardship.

The company then goes on to use clean energy many times throughout that testimony. How does the company define clean energy?

A.L.J. LEARY: Excuse me a second. Would you please again, state your name and spell your last name for the record.

MS. ZIESCHE: Sure. It's Ziesche, Z-I-E-S-C-H-E.

A.L.J. LEARY: Thank you.

MS. ZIESCHE: Yeah, no problem.

THE WITNESS: (Chahbazpour) So this is Donald Chahbazpour. I -- we did not have a technical definition for what the term clean energy means. The intent here was to what -- we are moving towards to achieve the policy objectives in all of our jurisdictions, including, New York. So it's basically wanting to get to an 8050.

MS. ZIESCHE: So you're referring to greenhouse gas emissions?

THE WITNESS: Yes.

MS. ZIESCHE: Thank you.

THE WITNESS: Generally speaking.

A.L.J. LEARY: And just for the record, please identify 8050. What does that mean?

THE WITNESS: Reduction of 80% of greenhouse gases by 2050, off a 1990 baseline. That is generally a term when policy makers talk about 8050.

MS. ZIESCHE: Thank you.

A.L.J. LEARY: Thank you.

BY MS. ZIESCHE: (Cont'g.)

Q. What percentage of the gas that you're currently delivering has been extracted through hydraulic fracturing?

A. (Chahbazpour) I don't know the answer to that question. Our procurement team will be able to answer that question.

Q. Would it be fair to -- would you at least it be -- would it be accurate to say at least a growing percentage of the gas, if you don't know an exact percentage?

A. Yes. That's fair, yeah.

Q. Okay. Thank you. And yeah, so you -- you defined the 80 by 50 as your emissions reduction goals. How does the company -- how do the com -- how does the company calculate total network methane emissions?

A. We calculate methane emissions under -- currently when we report methane emissions is under the E.P.A.s or known as the G.A.G.R.P. greenhouse gas reporting program. That's a regulatory filing, but in addition, we disclose under a voluntary program C.D.P. Carbon Disclosure Project.

So we -- we report our methane emissions under C.D.P. as well. In addition, we report methane emissions under other voluntary programs. One of them falls under the methane challenge that's under, again in the E.P.A. program. And the methane challenge has a program under -- there is two programs under the E.P.A.s methane challenge.

One of them is, One Future and we report our emissions to One Future as well.

Q. Yeah. I wasn't asking about how it's reported. How has that total methane emissions calculated, like, how do you find out that number?

A. Understood. So it's -- it's the current methodology for calculation, it's under the E.P.A.'s regulations. It's the G.A.G.R.P. emission factors. So it's really a function of your inventory, the assets that you have.

Types of mains, you look at your -- every asset class and you have an emission factor and you multiply that -- by that emission factor. That's how it's done.

Q. Okay.

A.L.J. LEARY: Excuse me. You just referred to an E.P.A. document and you used an acronym for it. Would you please repeat that for the record?

THE WITNESS: I -- I said G.A.G.R.P. It's Greenhouse Gas Reporting Program.

A.L.J. LEARY: Thank you.

BY MS. ZIESCHE: (Cont'g.)

Q. We would like you to look at a response to Sane R-1 which is not already a marked exhibit. So we are handing it out now.

A.L.J. LEARY: Is this the attachment that you need?

MS. ZIESCHE: No, this -- no.

A.L.J. LEARY: Okay.

MS. ZIESCHE: That will be coming up.

A.L.J. LEARY: We meant to see you at the break, so.

A.L.J. COSTELLO: And this will be marked as Exhibit 611 for identification.

BY MS. ZIESCHE: (Cont'g.)

Q. Did you prepare this response?

A. Yes. Yes.

A.L.J. COSTELLO: Yes.

MS. ZIESCHE: Sure. No problem.

A.L.J. COSTELLO: Yeah. I --

MS. ZIESCHE: No problem.

A.L.J. COSTELLO: And also just identify the document for the record --

MS. ZIESCHE: Okay. Got you.

A.L.J. COSTELLO: -- just basically. But wait until everybody has it, okay.

BY MS. ZIESCHE: (Cont'g.)

Q. So this is -- the document is the response to Sane Energy, I.R.-1. The company responded on October 28th. Did you prepare this response?

A. Yes.

Q. And is it a fair and accurate -- sorry, I don't know what the -- the question was supposed -- exactly to be. The truth -- is this a true and correct copy of that response?

A. Yes.

Q. Thank you. Sorry, I'm not a lawyer. All right. So in that response, the company said that in 2018, the methane emissions for their total network was, you said was 18,853, was that figure reported? Did you calculate it using that method that you just described?

A. I don't remember the exact number that we filed with E.P.A., but I'm pretty comfortable speculating that this number, it would matched what we filed with E.P.A.

Q. Okay. And when you're looking at the 1990 total emissions that you were supposed to reduce by 60% by 2035, what is that final number you're looking to achieve as far as methane emissions?

A. I'm not sure if I understood the question as --

Q. Or I guess maybe, --

A. -- you're --

Q. -- do you know what the methane total emissions were in 1990?

A. Yes, we do know what they are in 1990.

Q. What -- what are they? What were they?

A. I don't recall, and I don't have that number in front of me. But we did -- we calculate it in 1990 emissions basically.

Q. Okay. Or do -- are you -- do you know what the number is you are aiming for in 2035?

A. It is a reduction off that baseline.

Q. Okay. And we can get that information if you don't have it right now?

A. Yes.

Q. So how would the --

A.L.J. COSTELLO: Can I just say -- is that information already in the record? To -- are you aware, is it in the record or not? If you know.

THE WITNESS: I don't recall. It might be as part of our -- some of the I.R.s from E.D.F. that went to our engineering, my colleague in engineering. but I can't remember off the top of my head.

A.L.J. COSTELLO: Okay.

A.L.J. LEARY: What document would the total methane emissions in 1990 be? If the company

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calculated it where -- what document are you talking about? What did it calculate?

THE WITNESS: That --

A.L.J. LEARY: -- total emissions in 1990 to be?

THE WITNESS: And my colleague can keep me honest. It would not -- if I understand the question, it was not a number that was reported to E.P.A. at the time. If that's what you asked?

A.L.J. LEARY: It's not what I'm ask -- that's not what I'm asking.

THE WITNESS: Okay.

A.L.J. LEARY: You indicated that the company calculated a number, a baseline number for 1990. You're now saying that that was not reported to E.P.A. but there is a number within the company that was calculated. Is that your testimony?

THE WITNESS: Yes.

A.L.J. LEARY: Okay. Where would we be able to find that number if it's not in the record?

THE WITNESS: We would have it -- my-- our group, the regulatory strategy would have that

number and my colleagues in the gas engineering would have that number as well.

So either -- either -- either our group or gas engineering would be the source of the information.

A.L.J. LEARY: Okay. Not this panel. This panel does not know what the 1990 baseline is. Is that right?

THE WITNESS: We do know. We have the spreadsheet that has that number, but we don't have that spreadsheet with us. Does that make sense?

A.L.J. LEARY: The only thing that makes sense to me is, if you know the number or you don't know the number.

THE WITNESS: I do not know the number.

A.L.J. LEARY: So it's helpful that there you can get the number, but right now you don't know the number.

THE WITNESS: I do not know the number right now.

A.L.J. LEARY: Okay. So how do you want to deal with this Judge Costello? If we want the number and I'm assuming Sane wants this 1990

number that's internally and Mr. Wyman is shaking his head.

Anyone else interested in this? Okay. So I'm going to ask -- I don't know, for the company, counsel, how you can do this, and I don't know if the spreadsheet is some confidential information, but can you get through gas?

Did you say gas opera -- infrastructure and operations or gas safety? What -- what panels were you -- or what groups were you just referring to that have this number?

THE WITNESS: I was referring to my colleagues in gas engineering. I'm also referring to actually our group in the gas utility, the future of the regulatory strategy. We also have that spreadsheet, but we don't have it with us.

A.L.J. LEARY: Okay. Can we ask counsel for the company to solve this problem by getting us the number that was calculated on the 1990 baseline levels from which, as I understand your testimony, everything else is driven on the 80 by 50 or the 60 by 35.

MR. GAVILONDO: Yes, Your Honor. We can take a record request to produce that information and --

A.L.J. LEARY: No, I'm not -- I'm not suggesting that there be any --

MR. GAVILONDO: Okay.

A.L.J. LEARY: -- I'm asking you to do this to expedite this --

MR. GAVILONDO: Certainly.

A.L.J. LEARY: -- and if you really need 10 days, tell me why --

MR. GAVILONDO: No, no. That -- that wasn't -- I guess I was just trying to formalize -- to bring it into the record. We can certainly produce it in whatever manner that the Judges would like.

A.L.J. LEARY: Well, the panel has seen the spreadsheet, I'm assuming or somebody on the panel has seen the spreadsheet. So --

MR. GAVILONDO: They just don't have it at their fingertips, right.

A.L.J. LEARY: Exactly. So I'm assuming we can somehow stipulate to get this in if it doesn't come in today so.

MR. GAVILONDO: Yes. The company would stipulate to that.

A.L.J. LEARY: Thank you.

BY MS. ZIESCHE: (Cont'g.)

Q. Okay. So in addition to the 80 by 50 goal, on page nine of your direct testimony, beginning line two, building on the 80 by 50 pathway, National Grid proposes an aggressive goal of reducing total network methane emissions, 60 -- 60% by 2035. That is correct?

A. Yes.

Q. How do -- how will the company achieve that 60% reduction goal by 2035?

A. A majority of the methane emission reduction will come from leak-prone pipe replacement -- replacing casts iron and bare steel.

Q. Are there any other programs that will help achieve that goal?

A. So the way I would -- our methane emission reduction strategy has two components, a long-term strategy and a short term strategy. A long-term strategy is to replace leak-prone pipe, as I mentioned.

Our short term strategy is to -- to assess the latest technology in advanced leak detection tech -- advanced leak detection space and that will give us the ability to prioritize the biggest leaks.

We are assessing that technology and my colleagues in the other panels and particularly the infrastructure and operations panel would talk about that. But over time, in addition to leak -- leak-prone pipe replacement, we'll probably be utilizing that technology as well.

Q. And so the number that we have that -- that in 2018 that there was 18,853 metric tons of methane emissions. What percentage of that came from leak-prone pipe?

A. So you can -- just the way you calculate your emissions is you take your inventory, okay, and you take your mains and services by asset infrastructure.

So plastic, bare steel and you multiply an emission factor for each one of those and you multiply by the length of miles that you have. That's how you would be able to show your methane emissions.

Q. How is that emissions factor calculated? What is that number from?

A. The emission factor that's -- that's E.P.A. is using went through a process. It was as, you know, a whole, legal making process and it was based on a study that was peer reviewed and those emission factors were adopted by E.P.A. on their --

Q. So these are estimates based on E.P.A. factors, not actual calculations from say, like, a leak-prone pipe in person?

A. No, they are based on -- the emission factors are based on surveys action in the field. So, you know, I mean, you can go down this path here, but the first study was done by Washington State University by a number of professors.

So they went out and actually leaked the -- they surveyed actual leaks in different materials. And those are the emission factors that E.P.A. is currently using. Since then, there have been some additional studies that E.P.A. has not yet adopted, but those emission factors are based on actual leak surveys --

Q. I guess what I'm --

A. -- and studies.

Q. -- what I meant was these are based on estimates nationally, not specifically towards National Grid's system?

A. It is a national number, but it is reflective of the entire country. So when they -- when these surveys are done, they are -- they do -- they try to do their best to capture every region of the country.

Q. And so when you're look -- you're saying you're looking at these emission factors. The main one for National Grid is the leak-prone pipeline, the largest percentage?

A. Yes.

Q. Okay. Do you know about what percentage came from L.N.G. facilities?

A. I do not know them.

Q. How about Compressed Natural Gas facilities?

A. I don't have that number.

A.L.J. LEARY: Does anybody else on the panel have anything to offer in this regard?

THE WITNESS: (Schwartz) No.

A.L.J. LEARY: Speak up, okay, if you do.

BY MS. ZIESCHE: (Cont'g.)

Q. All right. And so in your direct testimony from April on page 14, line 21, it says as shown in figure one, between 2014 and 2018, the companies replaced approximately 733 miles of leak-prone pipeline, preventing approximately 3,388 metric tons of methane emissions each year. Is that correct?

A. Yes.

Q. And how did you calculate that 3,388 metric tons of methane emissions each year figure?

A. It is a difference between the emission factor of plastic and whatever that was replaced. So you -- we file, you know, whatever number of miles that you have in plastic that gets replaced by bare steel, I mean, or a cast iron with plastic and then you can calculate the difference.

Q. And so with that figure though 3,388 metric tons a year, that's clearly not cumulative that you're reducing every year, because if you look at that graph, in those four years, it

looks about like there is not an exact number where it starts but it has only gone down 3,147 in over those four year times, which is less than what you're claiming is reducing every year.

Can you describe, like, describe that discrepancy?

A. (Chahbazpour) Yeah, I'm not sure there is a discrepancy you have. It is basically looking at a snapshot. You have your emissions in a year and then you look at your emissions in the following year. The decline is obviously by the rate of the leak-prone pipe replacement.

Q. So even though you reported this number that the leak-prone pipe replacement is preventing about 3,388 metric tons. The actual total reduction though is only about 800 a year?

A.L.J. LEARY: 800 metric tons.

MS. ZUSCHE: Metric tons of methane.

BY MS. ZUSCHE: (Cont'g.)

Q. And I mean I -- I can have a follow-up question to that. That could mean that other sources are growing as you're reducing methane through --

A.L.J. COSTELLO: Well, let -- let him --

A.L.J. LEARY: Let the witnesses --

A.L.J. COSTELLO: Yeah, let him answer his question and this is -- are you referring to Figure 1?

MS. ZUSCHE: Yes, Figure 1.

A.L.J. COSTELLO: Okay.

A.L.J. LEARY: Page 15 of the direct testimony of the Future in Heat Panel.

THE WITNESS: Could you repeat the question again just --

BY MS. ZUSCHE: (Cont'g.)

Q. So what I'm trying to get at is actually what is the total methane reduction that you're seeing cumulative -- cumulatively from leak-prone pipe --

A. (Chahbazpour) Uh-huh.

Q. -- because it's clearly not 3,388 metric tons a year because in that four year period, your total methane only went down 3000 for four years and you're claiming 3000 something for each year.

A. (Schwartz) So I think that it's -- Is this on? I think the distinction here is that

it's cumulative. So in the last four years we've replaced enough leak-prone pipes so that each year moving forward, we're emitting 3000 -- sorry, I don't have the number in front of me. 3,388 million metric tons of methane emissions each year.

And so yes, it is a line graph which may not be the best representation of what's occurring, but over the total of four years, we've replaced that amount of leak-prone pipe to achieve on an annual basis that much emission savings because methane emissions happen constantly.

So you see essentially methane emissions savings in perpetuity when you make these types of changes.

Q. Okay. And are there sources of methane that are growing within your system? Are there sources of methane emissions that are growing?

A. (Chahbazpour) Specify somewhat that you mean you're growing.

Q. Well --

A. You're talking about this size of our -- so since if you look at the main growth right and would replace that would mean -- I don't know if that, you mean, is our system getting larger? Are

you talking about certain sources if their emissions are increasing?

Q. I'm asking if there are --

A. Is it a distinction?

Q. -- certain sources, yes, where methane is increasing currently?

A. I would say no. What is -- has the highest emission is our leak-prone pipe.

Q. Got you. On page 17 line 15, the company states to that end, the companies propose to continue their aggressive proactive L.P.P. replacement efforts, increase the replacement miles to 225 miles per year for a total of 930 miles over four years.

Altogether the companies anticipate these efforts once complete will result in greenhouse gas reductions of approximately 3,627 metric tons of methane each year. Is that still correct?

A.L.J. LEARY: And you are reading page 17 --

MS. ZUSCHE: Yes.

A.L.J. LEARY: -- of the direct testimony for the --

MS. ZUSCHE: Starting at line 15.

A.L.J. LEARY: -- Future of Heat Panel. Just make sure the record reflects what you're up to.

MS. ZUSCHE: Sorry, I thought I said that, but I will be more clear.

THE WITNESS: Yes.

BY MS. ZUSCHE: (Cont'g.)

Q. And so if you look at the numbers of metric tons per mile that the company is replacing, in the past four years, the number that you gave, it averages out to about 4.6 -- 6 metric tons of methane per mile.

In the next four years you're projecting about 3.9. Why is the amount of methane per mile going down as you are replacing leak-prone pipe?

A. (Chahbazpour) It has to do with the overall system having less leak-prone pipe.

Q. Okay.

A.L.J. LEARY: So in other words, each year as you replace more leak-prone pipes, the total overall emissions are going to be less.

THE WITNESS: Correct. Because you are having now a plastic system and it has less emissions.

A.L.J. LEARY: And I'm assuming the company is going to -- well, I should -- I'll ask you this. Is the company in its replacement of leak-prone pipes prioritizing to get the bang for the buck on the parts of the pipeline that may have the greatest emissions or is this just – how -- how was it approached?

THE WITNESS: So I'll give you my answer and the experts will be my colleagues in the gas infrastructure and operations panel. They adopt an algorithm that looks at lots of factor. They look at corrosion and safety.

A.L.J. LEARY: Age and so forth.

THE WITNESS: All of that, exactly. So all of those factors go into it. One of the factors is emissions and leakiest, you know, the leakiest segments. So they try to -- balancing all of those factors, they try to identify the leakiest sections. So they tried to do that.

A.L.J. LEARY: So as time goes on you're going to have, I think, where you're going with this is less emissions.

THE WITNESS: Yes, absolutely.

A.L.J. LEARY: Okay. Thanks.

BY MR. ZUSCHE: (Cont'g.)

Q. Yes. But -- so there would be less emissions total, but the actual amount that you're reducing emissions per year with the investments per mile will be less moving forward.

A. Yes.

A. (Schwartz) That -- and that may be a function of what is being replaced because cast iron and bare steel do have different emission factors and we are targeting the weakest according to our algorithm portions of our system to replace first.

And so you are correct, as we move through our -- the leakiest and oldest portions of our system, we then move on to portions of our system that are marginally less so.

A. (Chahbazpour) And they also tend to be the oldest, right.

Q. Did the company do an analysis comparing greenhouse gas reductions for retiring infrastructure versus replacing it with new pipeline?

A. It is a combination of those. Again, a panel that can address this in more detail is the gas infrastructure and operations panel. When you are replacing the leak-prone pipe there is part of that that's also being retired as well.

Q. So the company --

A. Explain it in -- in much more better detail than I can.

Q. So the company does have an analysis of retiring pipeline, a greenhouse gas analysis versus replacing it?

AI believe so. But that is sort of beyond the scope of my testimony.

Q. Yeah. We will be -- we'll be talking to the gas operations too. So we can save it for then. Did the company do an analysis of greenhouse gas emissions for new infrastructure that went into use during that same time period between 2014 and 2018?

A. Just so I'm clear. When you say, did we do an environmental emissions analysis of new

infrastructure, we know what the emission factor of the new infrastructure is. So we know what the emission factor of a new plastic is, and we know what we are replacing, so.

Q. Okay. Because in a response to a Sane I.R.-4 which is a pre-filed Exhibit number 581. The company said they do not -- they have not done a greenhouse gas emissions evaluation of the metropolitan reliability infrastructure project. Is that still true?

A. That is beyond the scope of our testimony about infrastructure.

A.L.J. LEARY: So just let me just make sure that our expectations are very clear to the panel whether or not it is within the scope of your testimony, in our view it has some level of relevance, but if you know the answer to the question, answer the question.

If you have knowledge that is responsive to that question, unless counsel objects, answer the question, somebody on the panel. This concept of what's in and what’s outside the scope, we decide that, not you.

So make sure that that's not the reason you not -- you don't answer a question. Okay. If you know, answer it, if you don't know, that's okay too. Just say it on the record.

THE WITNESS: Yeah. I do not know any emission analysis that was done with the M.R.I. projects.

BY MS. ZUSCHE: (Cont'g.)

Q. But you said you do have emission factors that you use while -- for evaluating new projects.

A. We know what the emission factors are of the plastic that we insert versus the old ones.

Q. But you don't look at a greenhouse gas emissions of the gas itself, that kind of thing.

A. I am not sure I'm following your question. I know what the emission factor of cast iron is, bare steel and what plastic is. So when you put a new pipeline that's plastic and we knew what we took out was cast iron or bare steel, we can calculate the difference.

Q. Does the company do analysis of the full life cycle of greenhouse gas emissions including upstream emissions?

A. We do not do it as part of the E.P.A.'s filings, that's not acquired, however, we do it in a voluntary programs under One Future, which is an E.P.A. methane program.

Q. So the company wants to add two new liquefied natural gas vaporizers at your Greenpoint Facility. Was any greenhouse gas emissions assessments done for that infrastructure?

A. I don't know.

Q. Trucking --

A.L.J. LEARY: Anyone else on the panel know whether there was something like that done.

THE WITNESS: (TALLET) No.

A.L.J. LEARY: Thank you.

BY MS. ZUSCHE: (Cont'g.)

Q. The company is also proposing to add trucking abilities at the Greenpoint Facility. Was any greenhouse gas emissions assessment done for that project?

A. (Chahbazpour) I don't know that.

Q. You're also proposing to double the capacity of a compressed natural gas facility in Glenwood, was any greenhouse gas emissions reduction or greenhouse gas emissions done for that -- a study done?

A. I don't know.

Q. The company also wants to add a compressed natural gas filling station in Staten Island, was a greenhouse gas assessment done of that project?

A. I do not know.

Q. The company is also proposing the Riverhead Transmission and the Suffolk -- South East Suffolk Phase 2 projects. Was a greenhouse gas emission assessment done of those projects?

MR. O'BRIEN: Your Honor, I'm just going to object. I mean this is beyond the scope of the panel's testimony. I mean it's clearly within the scope of the G.I.O.P. panel's testimony. They testify to it, they supported it.

I mean they're the appropriate witnesses for the subject matter.

A.L.J. LEARY: There's another panel that you can ask these questions to. The witnesses

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should respond and direct counsel inquiring that there is another panel or individual witness who can respond to that question. That is what will be helpful to this process.

And I appreciate that you have directed us to gas infrastructure and operations panel, which is on I think later this week. So do you want to just ask sort of an umbrella question?

MR. ZUSCHE: I mean, I think they answered it that they're not doing greenhouse gas assessments of their new projects.

A.L.J. COSTELLO: I think his answer was, he didn't know.

THE WITNESS: Yeah.

A.L.J. LEARY: Yeah.

MS. ZUSCHE: Well, earlier they said that -- okay. That there is a factor for pipelines.

A.L.J. COSTELLO: Well, the -- the record will --

MS. ZUSCHE: Sure. Well, I'll ask, yeah.

A.L.J. COSTELLO: -- will reflect that.

MS. ZUSCHE: I'll ask, yeah.

A.L.J. LEARY: So is there -- is there any assessment -- greenhouse gas emissions assessment that you know of that's done for any -- that has been done or is planned to be done for any of the company's new projects?

THE WITNESS: I do not know. The only project that I was involved in where we did upstream -- upstream methane emissions analysis that was the NESE Projects.

MS. ZUSCHE: Yeah.

THE WITNESS: That was the proposed project but with a --

A.L.J. LEARY: The North --

THE WITNESS: -- a full life cycle analysis of that one phase.

A.L.J. LEARY: And would you spell out the acronym for N.E.S.E. because it's slipping my mind. Northeast --

THE WITNESS: Northeast Supply Enhancement.

A.L.J. LEARY: Northeast Supply Enhancement Project.

THE WITNESS: Right.

A.L.J. LEARY: N.E.S.E. Project. Thank you.

BY MS. ZUSCHE: (Cont'g.)

Q. So if you're not aware of the assessments of a greenhouse gas emissions, how can you be confident that you're actually going to be lowering emissions overall by 2035?

A. The question you're asking me is -- I think you're asking me about specific questions like Glenwood, Riverhead Transmission. I don't know if there wasn't -- that assessment has been done. And again, gas infrastructure and operating panel -- operations panel can answer that.

The reason that I can tell you I know what -- that emissions are declining over every year. When we look at the E.P.A. filings, we'll look at the entire system. So gas engineering and infrastructure, you know, they're -- they're obviously doing their infrastructure and investments.

When we look at it from our perspective and we file, we know what the system looks like every -- at the end of the year. So we know at the beginning of every year what our

infrastructure look like and we know what it looks like at the end of the year.

So even though I can't talk about specific projects, we know at the end of the year and every year as we file with E.P.A. and we file with D.O.T., we know what the overall system looks like. And when you know your overall system, you can break that inventory down by different asset classes and that's how you calculate their methane emissions.

Q. And so there are emission factors for things like liquefied natural gas and compressed natural gas as well as pipelines?

A. Yes, I -- the -- the L.N.G. has its own reporting under E.P.A. sub-part W.

Q. And that's all taken into account in that total methane that you're talking about, the whole network?

A. Yes. What -- whatever is the regulation under E.P.A. is greenhouse gas reporting program.

Q. And once again though, those are all national averages and not specific to the company's infrastructure?

A. Correct.

Q. And would the gas operations panel know what that factor is so that there is an analysis of what these facilities are emitting in these new projects?

A. The emission factors is a public record under E.P.A.'s greenhouse gas supporting program. Those emissions --

Q. But you don't know if that was looked at for these projects?

A. No, I do not know.

Q. Okay. Has the company done an -- so all this analysis that we've talked about so far is about methane that's leaking from the system. Has the company done an analysis of greenhouse gas emissions released when the product you are delivering is combusted or consumed?

A. As the company that -- no, I believe that falls under the overall E.P.A.'s greenhouse gas emission inventory.

Q. But that's not included in this figure that you've been talking about as far as reducing greenhouse gas emissions. That's just methane --

A. Correct, yes.

Q. Just methane.

A. Right. Just methane.

Q. Okay. So in your direct testimony Future of Heat filed in April on page 16, line 15.

A.L.J. COSTELLO: Okay. One -- just one second.

MS. ZUSCHE: Sorry.

A.L.J. COSTELLO: Let them get to the page before you -- and once they get to the page then you can read it. So we're all on the same page.

THE WITNESS: You said the rebuttal testimony or the --

BY MS. ZUSCHE: (Cont'g.)

Q. No, this is the direct testimony.

A. Direct test, okay, go ahead. I'm there.

Q. So second, as part of the E.P.A.'s methane challenge program, One Future emissions intensity commitment option. The companies are working to reduce the methane leakage rate from the production processing transmission and distribution of natural gas to 1% or less by 2025.

A. Uh-huh.

Q. How is this program calculating methane emissions?

A. The emission in sense -- intensity protocols that One Future came up with were all reviewed and approved by E.P.A. and they are on One Future's website and also E.P.A.'s methane challenge.

Q. And what is the difference between an intensity leakage and then just the pure leakage rate?

A. So a complicated technical question, I'll try to, uh, simplify it. It is basically taking your emission and dividing it by a through point and you come up with an intensity.

The difficulty is when we were developing One Future and it took us several years to develop these protocols, every sector calculates their emissions differently. So what we try to do is to come up with a single protocol that is able to cut through all of these sectors.

So when you look at the emission protocols and when we say we have a goal of 20, 25, 1% or less, it is from wellhead to burner tips. So looking at from all the way upstream to the

distribution sector and all of those protocols and what -- how intensities are calculated are in great detail in the One Future protocols.

Q. I'd like to introduce Sane Energy -- your response to Sane Energy request number 2. We're passing that out now. And then the attachment is what you have.

A.L.J. LEARY: Do you need --

MS. ZUSCHE: We have the -- the answer. We just don't have all the attachments.

MS. FRACZEK: The attachment was the study. You have --

A.L.J. LEARY: Okay. So let me just make sure I have what you have.

MS. FRACZEK: Yeah, you have the first two pages.

A.L.J. LEARY: Okay.

MS. FRACZEK: And then the study was --

A.L.J. LEARY: This lifecycle --

MS. FRACZEK: Yeah, exactly.

A.L.J. LEARY: Let me just say for the record, the attachment to Sane information request Sane-2 dated October 28th, 2019 is a 30 page

19-G-0309/19-G-0310 - 2-10-20 - Brooklyn Union Gas attachment called Lifecycle Analysis of the Northeast Supply Enhancement Project.

For this record, I have that attachment. I'm assuming other parties have been served with this. You do not -- as I understand it, Sane does not have this attachment available for you.

If anybody would like it and that does not have it -- I'm looking at the company or D.P.S. staff. Let me know.

MS. FRACZEK: I'm sorry. We would have printed it, it was just prohibitive --

A.L.J. COSTELLO: Yeah, can -- can we go off the record one second.

(Off the record 1:45 p.m. to 1:47 p.m.)

A.L.J. COSTELLO: This -- this is an exhibit that you've just offered is going to be marked as Exhibit 612.

MS. ZUSCHE: Yes. And this is the company's response to Sane Energy's second I.R. which was submitted -- the response was submitted on October 28th.

BY MS. ZUSCHE: (Cont'g.)

Q. Did you prepare this response?

A. (Chahbazpour) Yes.

Q. Is this a complete and accurate copy?

A. Yes.

Q. Okay. So in that response -- oops, let me bring it back up -- to question 1B, the company said the emission intensity level is the total methane emissions divided by gross natural gas production.

So that number that we're looking at is -- is smaller than the actual total methane leakage, correct?

A.L.J. LEARY: Sorry. The number you are looking at, what are you referring to?

MS. ZUSCHE: Sorry. The 1% leakage rate. They're talking about intensity level and he just explained this entire process of how that's calculated.

THE WITNESS: Uh-huh.

A.L.J. LEARY: So did you intend to read the preceding sentence as well?

MS. ZUSCHE: I can if -- sorry, this is my first time doing this --

A.L.J. LEARY: Well, again --

MS. ZUSCHE: -- but I'm happy too.

A.L.J. LEARY: It's okay. I just want the record to be clear. So set it up with the previous sentence.

MS. ZUSCHE: Sure.

BY MS. ZUSCHE: (Cont'g.)

Q. So in 1 -- in 2014, One Future established a goal of achieving 1% or less methane emissions intensity level by 2025. The emission intensity level is the total methane emissions divided by gross natural gas production.

So when the company talks about the intensity level that's what you're talking -- the intensity level is what you're talking about in your Future of Heat testimony, correct?

A. Yes.

THE WITNESS: (Schwartz) A clarifying question.

THE WITNESS: (Chahbazpour) Yeah.

THE WITNESS: (Schwartz)are you referring to our 60% target or other portions of our testimony?

MS. ZUSCHE: I'm referring to the other part of the testimony. When you began talking

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about the upstream emit -- emissions in your Future --

A.L.J. LEARY: Can you give a page number in the direct --

MS. ZUSCHE: Yeah, it's what I --

A.L.J. LEARY: -- testimony.

MS. ZUSCHE: Sorry. Yes. Which is what I -- sorry, I said right before this. It's page 16, line 15, which I had read before, but I can read again.

THE WITNESS: Sorry. Could you repeat your question one more time to that?

MS. ZUSCHE: Yeah.

BY MS. ZUSCHE: (Cont'g.)

Q. So in your testimony, when you're looking at that methane reduction, you're not looking at a total percentage, you're looking at this intensity level, right? Which is the total emissions divided by gas production -- total gas production.

A. (Schwartz) Correct. For the upstream emissions discussion contained on this page, we're talking about emissions intensity. For the company's 60% target we're talking about total emissions reductions.

Q. So the 60% reduction includes upstream emissions?

A. No, just from the company's assets.

Q. Okay. But for the upstream emissions, do you have a stated reduction goal?

A. We have sectorial goals as a part of One Future. I -- let's see. They're not included I don't believe in our testimony or in the information response to Sane-2.

Q. But would it be accurate to say that the company's upstream emissions goals are all based on intensity emissions?

A. (Chahbazpour) Yes.

Q. Okay. Is the panel familiar with the reduction goals set out by the Climate Leadership and Community Protection Act?

A. Yes. I'm assuming you're referring to the 85%, the broad policy objectives.

Q. The broad project goals, but also how the C.L.C.P.A. defines emissions.

A. I'm not familiar with the details how C.L.C.P.A. defines emissions, I don't know.

A. (Schwartz) Is that something you'd be willing to share?

Q. Sure. I don't have a copy though. I forget --.

A.L.J. LEARY: What are we -- what are we talking about?

MS. ZUSCHE: I'm asking if they were familiar how the C.L.C.P.A. defines emission reduction goals.

A.L.J. LEARY: You don't have a copy of the --

MS. ZUSCHE: C.L.C.P.A. on me.

A.L.J. LEARY: C.L.C.P.A.

MS. ZUSCHE: It just -- it was referred to a lot in other people's testimony. So I made the assumption because they -- they talk about the C.L.C.P.A. in their own testimony, but --

A.L.J. COSTELLO: If you have specific questions from their testimony, you can point to that and ask those questions.

MS. ZUSCHE: Okay. It's okay. I can move on. No, I have it, but it's not -- it's not --

MR. GOODRICH: Your Honors, may we --

A.L.J. COSTELLO: Yes.

MR. GOODRICH: -- clarify something about Exhibit 612.

A.L.J. LEARY: 612?

A.L.J. COSTELLO: Yes.

A.L.J. LEARY: Yes.

MR. GOODRICH: So what I understood was we have the -- the two page response itself and that you had -- there was specific reference to attachment 1 to that response, the 30 page report. But then there's also attachment 2, a 16 page report.

A.L.J. LEARY: Yes.

MR. GOODRICH: Was that addressed? I'm sorry if I missed that.

A.L.J. LEARY: I don't think counsel has inquired about either attachment 1 or attachment 2 yet. Is that right?

MR. GOODRICH: No, I mean -- I mean what is included in the exhibit?

A.L.J. LEARY: Yes, those two things are the only two things I have included in the exhibit. It's attachment 1 and I gave you the title Lifecycle Analysis of the Northeast Supply Enhancement Project. Attachment 2 is One Future 2017 methane emission intensities --

MR. GOODRICH: So that is included.

A.L.J. LEARY: -- initial progress report, November 15th, 2018. This would be attachment 2 to Sane-2. The first document is -- attachment 1 is 30 pages long. The second is 16 pages long.

MR. GOODRICH: Thank you. I just wasn't clear on -- on what we had -- if we had addressed that attachment, thank you.

A.L.J. COSTELLO: Thank you.

A.L.J. LEARY: For our purposes, if this exhibit and I'm assuming since we've given it an exhibit number. I just don't want to hear any objection to somebody not having these if you all have it. So let me just make that clear.

This was a logistical glitch that we are willing for this party to be able to ask the panel about it. They have it in front of them. I think the company is okay with this.

MR. O'BRIEN: No objection from the company.

A.L.J. LEARY: Okay. Are you okay with it, Mr. Goodrich?

MR. GOODRICH: Yes. I just wanted to clarify that we have the full response that was all.

A.L.J. LEARY: Yeah, I -- I agree. Thank you for doing that. You want to go ahead?

MS. ZUSCHE: Thank you.

BY MS. ZUSCHE: (Cont'g.)

Q. So in response to Sane I.R.-2 the company said the companies are exploring methods to evaluate upstream emissions when determining the over -- overall greenhouse gas emissions of a project. Has any progress been made on exploring those methods since this response was given in October 20 -- 28th?

A. (Chahbazpour) I think, you know, we work with upstream companies through collaborations like One Future and under initiatives. But we don't directly engage upstream companies to develop their emission protocols.

Q. So when you said the companies are exploring methods, what are those methods?

A. I think those methods refer to really to more upstream companies being engaged in up -- in One Future. And we have also in our procurement and my colleagues from the energy procurement team when they are here, they can speak.

We have developed under an initiative called N.G.S.C., Natural Gas Supply Collaborative. We have developed finance -- environmental and social performance indicators and we urge the upstream companies to begin to reporting those metrics.

So it's really a collaboration in engaging those upstream companies to begin to report their emissions.

Q. So this is dependent on other companies reporting.

A. Correct.

Q. Has the company looked at research that evaluates upstream emissions conducted by Robert Howarth of Cornell who is recently appointed to the Climate Action Council for the Climate Leadership and Community Protection Act?

A. I have not.

Q. In response to Sane I.R.-2, the company referenced the lifecycle analysis of the Northeast Supply Enhancement Project conducted by M.J. Bradley & Associates, which everybody has now in front of them.

In that report, M.J. Bradley & Associates use a 100 year global warming potential to

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estimate the compare -- estimate and compare life cycle greenhouse gas emissions from different energy options.

I'm sorry, let me look at the page real quick. I thought I had it written down. So this is page 7 of 30 of that attachment. So this is using the 100 year global warming potential. The Climate Leadership and Community Protection Act now requires using a 20 year global warming potential.

Has the company done an analysis looking at the 20 year -- using a 20 year global warming potential?

A. Just to be clear, the reason that we are using a 100 year global warming potential -- that is consistent with how E.P.A.s uses that emission factors for all greenhouse gas emissions. We know what the emission factors are on a 25 year or a 20 year global warming potential for methane.

But that is, we do know what they are, but that is not what goes into the reporting.

Q. So all of your reporting on this is based off the E.P.A. of a 100 year timeframe even though state law now requires a 20 year timeframe.

A. Yeah, all of our reporting to E.P.A. is based on a 100 year emission factors.

Q. So all of the emission factors you just talked about, that's all a 100 year?

A. Yes.

A. (Schwartz) Correct. Our reporting has historically used E.P.A.'s framework because that's who requires us to report.

Q. But you said the company does know what it looks like on a 20 year timeframe.

A. (Chahbazpour) Yeah, I mean, by the global warming potential on methane or any other gasses, whether it's 20 years, that's just public information. Yes. But you're -- you're talking about how we filed our emissions, which is consistent using E.P.A. protocols.

Q. Well, I'm asking how you calculate your emissions and look at emission reductions and if it matches what the state is requiring because there is very different calculations when looking at methane over a 20 year timeframe then a 100 year timeframe.

A. We currently file it under E.P.A. as far as I know, there is no New York State Methane

Emissions Reporting that's their filing compliance requirement.

Q. But does -- but since the C.L.C.P.A. requires a 20 year timeframe, do you plan on starting to?

A. Yeah. When -- when that becomes, you know, codified and there is regulation we would. But currently all of our reporting again to E.P.A. is not -- it's based on a 100 year global warming potential.

A.L.J. LEARY: Excuse me.

THE WITNESS: I misunderstood the question.

A.L.J. LEARY: You indicated that the company knows what the 20 year figure is. Is that correct? Did you -- I thought I heard that.

THE WITNESS: Yes, the global warming potential for methane, yes.

A.L.J. LEARY: For a 20 year period.

THE WITNESS: Yes.

A.L.J. LEARY: So let's put aside the E.P.A. filing for a moment. What is the number for that 20 year period?

THE WITNESS: Right now, the global warming potential for methane over a 100 year period is 25 times and --

A.L.J. LEARY: I said 20.

THE WITNESS: And under 20 years I believe is 80 -- 80 --

THE WITNESS: (Schwartz) 86.

THE WITNESS: 86 times global warming potential of CO2.

A.L.J. LEARY: Okay. Thank you.

BY MS. ZUSCHE: (Cont'g.)

Q. How long does the company estimate the useful life of new pipeline replacing leak-prone pipe will be?

A. (Chahbazpour) I do not know the answer to that. And my colleagues in the gas infrastructure and operations will be able to give you a definitive answer.

Q. I'd like to direct your attention to Sane Exhibit-2 which is pre-marked as Exhibit 582.

MS. ZUSCHE: When -- so on page 27, the final page --.

THE WITNESS: (Chahbazpour) I'm sorry, I don't have that. Was that -- was the same one, was it a different --

MS. ZUSCHE: The -- yeah, it's Sane Exhibit 2 and it was pre-filed as Exhibit 582.

MR. GAVILONDO: Your Honor, I'm not sure if the panel has a copy of that. If counsel could identify what that is and maybe we'll be able to hunt it down.

MS. ZUSCHE: It is a transcript of a community board meeting where National Grid was presenting on the Metropolitan Reliability Infrastructure Project.

A.L.J. COSTELLO: Thank you. Let's go off the -- go off the record for a second.

(Off the record at 2:02 p.m. to 2:02 p.m.)

MS. ZUSCHE: That is the -- that's the end of our cross.

A.L.J. LEARY: Thank you.

A.L.J. COSTELLO: Okay. Thank you. The next party up for cross-examination is A.A.R.P.

CROSS EXAMINATION

BY MR. RIGBERG:

Q. Good afternoon. I'm Saul Rigberg with A.A.R.P. I'm going to focus on your direct testimony. I'm picking up on the issue methane emissions.

Does the panel know if -- if methane is a greater contributor to climate change than carbon dioxide?

A.L.J. LEARY: Excuse me, Mr. Rigberg. I'm sorry, the court reporter cannot hear you.

MR. RIGBERG: Yeah.

A.L.J. LEARY: So --

MR. RIGBERG: Well, we'd like to have another microphone then.

A.L.J. LEARY: Is there a small microphone somewhere over there?

A.L.J. COSTELLO: This one seemed to be working well.

MR. RIGBERG: Okay.

A.L.J. LEARY: Okay. I actually need the court reporter microphone, unless you really speak up.

MR. RIGBERG: One of these?

A.L.J. LEARY: Yeah. Can you move that over?

MR. RIGBERG: Yeah.

THE REPORTER: You need this.

A.L.J. LEARY: Thank you. Just be careful you don't trip over anything.

MR. RIGBERG: Can you hear me?

THE REPORTER: Yes.

MR. RIGBERG: Okay. Good. All right. I'm Saul Rigberg with A.A.R.P. I'm going to focus on the panel's direct testimony.

BY MR. RIGBERG: (Cont'g.)

Q. If you look at page 14 of your testimony where you -- again, the discussion of reducing methane emissions and my -- my first question is does the panel know if methane is a greater contributor to greenhouse gas emissions and climate change than carbon dioxide?

A. (Chahbazpour) So I believe the biggest contributor when you look at the overall emissions is the combustion of all fossil fuels that lead to CO2. So if I understand your question correctly, the biggest source of GHG emissions is what is known as scope 3 emissions which just means combustion of fossil fuels that lead to CO2.

Q. Okay. Thank you. And in -- on lines 19 and 20, the -- the company says your -- your panel testimony is that the most cost effective means of reducing methane emissions is to eliminate system leaks through the replacement of aging infrastructure. Do you see that testimony?

A. Yes.

Q. Okay. Now --

A.L.J. LEARY: Mr. Rigberg, you need to speak in both microphones because I can barely hear you and I -- I'm going to bet that the room does the -- yeah, the bigger one is more important.

MR. RIGBERG: Yes. Okay.

BY MR. RIGBERG: (Cont'g.)

Q. All right. Are you referring to both distribution and transmission pipes in that sentence?

A. This is referring only to the Utility National Grid, our assets.

Q. Right. Okay. Well, the reason I'm asking is on page 16, on line 14, you talk about gas distribution system and on line 18 you refer to transmission.

So is there a difference in -- in the impact of leakage of a distribution pipe compared to a transmission pipe?

A. So let me just make clear --on the previous page we were talking about company's effort to reduce emissions. On page 16, we're trying -- we're talking about the overall natural gas industry from wellhead to burner tip.

So establishing those protocols that we talked about previously One Future, in these protocols we are looking at the emissions by sector. Was that helpful? I'm not sure if I understood your question.

Q. Well -- okay. So on Line 17 --

A. Which page?

Q. On page 16, line 17 through 19 you're saying the companies are working to reduce the methane leakage rate from the production processing transmission and distribution of natural gas --

A. Uh-huh.

Q. -- to 1% or less by 2025. So what steps are the company -- companies doing to achieve that goal?

A. We can talk about what step -- when you say what companies, you mean KEDNY-KEDLI, am I correct? Are you talking about --

Q. Well --

A. -- upstream company just --

Q. -- when you use the word companies on lines 17 with a capital C, what -- whatever -- what do you mean by those, that word?

A. Here, we are talking about through the protocols of establishing One Future to urge the other sectors of the natural gas to reduce their emissions to achieve this emission intensity target by 2025, so.

A. (Schwartz) Right. We actively take part in a collaborative that includes distribution companies like ourselves, transmission, production and gathering and boosting companies to come up with protocols for better measuring and reporting methane emissions and then to encourage other companies to join suit in pursuing methane emission reductions and that's what we do with -- and through One Future.

Q. Okay. So those efforts have or have nothing to do with the replacement of methane of -- of leak-prone pipe. That's a separate initiative.

A. (Chahbazpour) Well, those efforts reduce our emissions in the distribution sector -- the leak-prone pipes.

Q. The leak-prone pipes. Okay.

A. So when we replaced leak-prone pipe --

Q. Okay.

A. So -- yeah, just saying when we reduced leak-prone pipe that reduces National-Grid's emissions which is a distribution sector, the collabora -- the collaboration that my colleague was talking about, we are working with the other sectors of the industry to -- they reduce their emissions.

How they reduce their emissions is -- is their own strategy.

Q. Okay. Now, how do you go by -- go about identifying which pipes to replace?

A. My colleagues in G.I.O.P. gas infrastructure will take you through a protocol, they have a protocol, an algorithm that looks at lots of factors and I do know that age is one of those

factors and age tends to be also a correlation to emissions. Older pipe, that's more --

Q. Okay.

A. -- you know, cast iron and barrel -- bare steel and risk, but I do know that they also consider the leak surveys and the data from those surveys to identify the leakiest sector -- segments and replace those as well.

Q. Okay. So does the companies possess technology to actually measure whether methane is leaking from a particular pipe?

A. Yes, we conduct leak surveys. My colleagues in the gas infrastructure can again provide you a lot more detail how they do it and how often.

Q. Okay. Thank you. Okay. Could you please turn to page 21 of your testimony?

A.L.J. LEARY: Twenty-one?

MR. RIGBERG: Yes, the section beginning on line 14 titled Energy Efficiency.

BY MR. RIGBERG: (Cont'g.)

Q. Is the panel aware of any of the company's energy efficiency programs that provide

weatherization or insulation of buildings, whether residential or apartment buildings?

A. (Tallett) Yes. I believe there was an I.R. that -- that described the company's energy efficiency programs and the sectors that they impact.

Q. Okay. Does the panel know how many buildings are weatherized each year by the -- by the company's efforts?

A. I can't say the exact number of buildings within our energy efficiency reporting that the company submits on a quarterly basis. There would be -- there would be information there about multi-family buildings, however, the company may not directly provide weatherization services to all sectors.

There may be weatherization assistance that's provided by the local community agencies or NYSERDA in the instance of -- of KEDNY.

Q. Okay. And do you know if -- if another panel would -- would have that information be -- be able to talk about that -- that issue?

A. It -- it would not be another panel, that would be a number that the energy

efficiency reporting could derive. I just don't have it right now.

Q. Okay. Thank you. Okay. Turning to page 29 of your testimony, on -- on lines 14 and 15, you say in addition KEDLI proposes to increase the budget from $1.9 million to $2.5 million annually.

KEDLI will use the additional funding to expand the offerings to moderate income customers to find of those between 60% and 80% of state or area medium income whichever is higher. And has the company identified, what it will do with that extra $600,000.00?

A. In the company's supplemental testimony filed on January 29th, we described that the KEDLI empower replacement program known as HEAP would be transitioned into the energy efficiency portfolio.

So that funding would be used to provide additional energy assistance to low income customers as well as expand to moderate income customers.

Q. Okay. What do you mean by the term energy assistance?

A. So it -- it could be whatever types of services would be provided to that individual customer who may be applying. The customers -- it would vary what is installed in each customer's home based on their energy assessment.

Q. Okay. So that could -- that could include furnaces for -- for instance?

A. It -- it could --

Q. Okay.

A. It could include a variety of energy efficiency, direct install measures.

Q. And -- and does the panel know how many people might be served by the extra $600,000.00?

A. I don't recall if we calculated the -- the number of customers. I would offer that by transitioning this effort into our energy efficiency programs, we would look at this as we would any other energy efficiency program.

So the company would have the flexibility to spend in addition to that -- that $2.5 million if -- if it were -- if it were appropriate.

Q. Okay. Thank you. Please turn to page 30 of your testimony. On line 7, you state the

companies are committed to empowering and enabling customers to take more control over their energy usage, reduce consumption and proactively embrace products and services that align with the state's clean energy goals.

What are some of the ways that the panel believes a custo -- customers can take more control over their energy usage?

A. (Traczyk) So within the scope of the proposal and the direct testimony, we were referring specifically to the Green Gas Tariff, to the expanded demand response proposal, and to the fuels switching calculator.

All of which would provide options either for customers to become more informed about total cost of ownership for various energy choices or to select new products to procure a green gas solution for instance or to participate in a demand response program that would allow them to effectively leverage their own flexibility to achieve a financial incentive.

Q. All right. Is the panel aware of any rate design elements that can be -- help a customer achieve control over -- over their usage?

A. I'm not sure what you mean by rate design elements.

Q. Well, for instance with -- with the size of a customer charge, you know, a fixed charge have any impact on the customer's ability or -- or send signals to a customer about controlling usage.

A. So certainly I think the -- the rate design panel will have more insight into the various interplay of the factors. I think the idea behind the proposal such as the demand response proposal is that you can create financial signals for customers that would allow them to make choices regarding their energy consumption.

We don't currently have in place time varying rates or -- or any analogous product offering for -- for gas customers and so, therefore, the -- the signals are flatter than they are in -- in some other systems and so we are trying to start to introduce these sorts of products that would be available to customers.

Q. Okay. So the panel would not be -- would not have an opinion about whether placing more of a customer's bill on volumetric -- on use

would have an impact on a customer's ability to control usage.

A. I think if -- if rates were redesigned and to the extent that they were differently structured than they are today in terms of volumetric versus fixed charges that would have some impact on customer usage. I do not have an official opinion as to your question.

Q. Okay. thank you. If you could please turn to page 31. Let's see, hold on. Let's see. Actually, I'm sorry. Thirty -- page 32.

I had a question about R.N.G. and I think -- actually, is there -- is R.N.G. in the sys -- the -- do either of the companies inject R.N.G. into the system now?

A. (Chahbazpour) The only R.N.G. that's going into our system directly today is the -- from the Staten Island Landfill in New York.

Q. And -- and then -- and that's distributed -- the molecules just mix with the other molecules.

A. That's correct.

Q. Okay.

A. It goes into our transmission system --

Q. Okay.

A. -- in Staten Island.

Q. Okay. Now, on page 32 on the top of the page you -- you have a chart where you proposed four tiers for -- for use of different pricing structures for your new products and services.

Are -- are these -- for the column -- it's the third column over, percentage of monthly consumption. Is -- is that based on historic average, that percent or is that -- is that going forward, that --

A. (Traczyk) So that would be --

Q. -- that percentage?

A. -- that would be retroactively billed to the commercial customer.

Q. Okay.

A. So they -- they would be signing up effectively to offset 5% of their consumption. So once their consumption was known, there would be a -- a retroactive true-up for those customers.

Q. Okay. All right. So now looking -- moving to the first column, which I think is for residential customers, correct? How -- how is -- how are those categories defined in terms of actual, like, therms?

A. And this information was provided as an attachment to an I.R., I don't have the number in front of me.

Q. Okay.

A. But basically we took the average residential consumption for a customer in Downstate, New York and we calculated what the cost would be using and assumed $15.00 per million B.T.U. or -- or per dekatherm for renewable natural gas which was provided based off of existing market prices that we are aware of today.

Using that, we then backed into these estimates which again, it is provided in an attachment and I -- I can certainly find that and -- and provide that to you.

Q. Okay. If you could just identify it for me.

A. Of course.

Q. Do you -- do you know offhand what the average usage is that you -- number that -- that you use?

A. (Schwartz) The -- the information request, I believe he is describing is DPS 365.

Q. Okay. And does that response indicate in terms of therms what -- what the categories of low, low middle, high middle and high are?

A. I believe the estimated average annual consumption was a 100 dekatherms for a residential customer and let's see --

A. (Traczyk) That's a heating customer?

A. (Schwartz) A heating customer, yes. Low was equivalent to roughly 5%, low to middle roughly 20%, high to middle roughly 25% and high $50.00 per month. And I believe we also made a correction to this testimony to the table and this testimony for the high to middle residential customer.

Q. Okay. All that information is in the response to DPS 365? All those numbers you just --

A.L.J. LEARY: Do we -- excuse me. Mr. -- pardon me. Do we have that I.R. in the record already as a -- as a hearing exhibit on our exhibit list? D.P.S. staff, do you -- did you --

MR. GOODRICH: I'm -- I'm looking.

A.L.J. LEARY: Okay. You can just wait. Hold on a second because I'd like to identify for the record if, in fact, it is a hearing exhibit.

MR. GOODRICH: Your Honor, this is Brandon Goodrich for staff. I don't believe it is, I just looked at our -- our pre-filed exhibits and I didn't see it listed. It's possible somewhere, but I don't believe it is.

A.L.J. COSTELLO: There is the company's response to DPS 365 looks like it's in Appendix A to the corrections and updates testimony by the Future of Heat Panel.

MR. GAVILONDO: Thank you, Your Honor. Your Honor, that should be hearing Exhibit 229, it's look like.

A.L.J. LEARY: Thank you. Does the panel have a copy of this, the corrected testimony so you can proceed?

MR. RIGBERG: Okay. I -- I was just -- I don't have that exhibit before me. I'm wondering if -- does that exhibit contain all those numbers that the panel responded regarding the number of therms assigned to each category of -- on the table -- first column of the table?

THE WITNESS: (Schwartz) So the exhibit includes the estimated average annual consumption and the percentages, so the therms can be inferred from those two pieces of information.

MR. RIGBERG: Okay. Thank you.

BY MR. RIGBERG: (Cont'g.)

Q. Turning to page 32 of your testimony, on line 14, there is a sentence beginning, to the extent that companies monetized the environmental attributes of excess cash, such amounts would reduce R.N.G. cost for participating Green Gas Tariff customers.

Could -- could you explain the process involved in -- in -- in monetizing environmental attributes in the context of this sentence?

A. (Traczyk) Yeah, I'll -- I'll explain high level of what people mean when they talk

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-- talk about monetizing the environmental attributes of R.N.G.

Currently, today, R.N.G. is mostly being utilized in the transportation sector under E.P.A's R.F.S. program, renewable fuel standards, and it generates something known as R.I.N.S., Renewable Identification Numbers.

So if you use R.N.G. in the transportation sector, you generate R.I.N.S. and they have market value. So this really refers to selling R.N.G. into the transportation sector and specifically within the scope of the Green Gas Tariff proposal.

The idea here would be to allow some mechanism such that if the company had excess gas as referred to in line 15 on page 32 that the company would be able to -- to do something with that volume and to the extent that there was a positive financial benefit from that, a revenue stream effectively from monetizing the environmental attributes of the R.N.G.

That cost would then go back into the -- the program cost at large to reduce the R.N.G. costs for participating Green Gas Tariff customers.

Q. So let me understand this. So the company would -- would be selling what you called them R -- R-I-N-S?

A. R.I.N.S. are --

Q. R-I-N-S.

A. R.I.N. Yeah. Renewable Identification Numbers.

Q. And then the company would put that into an account that would benefit the participants of this program and --

A. Correct.

Q. -- reduce their cost.

A. Yeah.

Q. Okay.

A.L.J. COSTELLO: Could you just -- what does R.I.N. stand for again?

THE WITNESS: Renewable Identification Number.

A.L.J. COSTELLO: Okay, thank you.

THE WITNESS: Of course.

BY MR. RIGBERG: (Cont'g.)

Q. On -- on the bottom of page 33 and the top of 34, the panel indicates that R.N.G. --

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using R.N.G. would not cause -- require a customer to change any -- any, you know, furnace or anything.

So -- so wouldn't it make sense for the company to do everything possible to inject as much R.N.G. as -- as possible into the system?

A. Yes.

Q. Okay. So -- okay. If we could turn to page 39 of your testimony on page fifth -- line 15 where the discussion of natural gas vehicles begins.

Does the panel know if -- if N.G.V. vehicles or natural gas vehicles are considered more or less a contributor to greenhouse gases than electric vehicles?

A. (Schwartz) I think that would depend on the carbon intensity of the electric grid, which I am not aware of today.

Q. Okay.

A.L.J. LEARY: Sorry. I thought the question was and I may have misunderstood. If you're talking about an E.V., an electric vehicle versus a vehicle that uses renewable natural gas, which of those in and of themselves would have more greenhouse gas emissions?

THE WITNESS: Right. And -- and depending on the feedstock used to produce C.R.N.G. we can calculate the emissions impact of that vehicle's R.N.G. usage, however, the emissions associated with an E.V. depends on where that electricity is coming from.

So whether it's coming from coal, natural gas or solar and wind and that's not information that I have in front of me for -- of KEDNY and KEDLI.

A.L.J. LEARY: Okay. Thank you.

BY MR. RIGBERG: (Cont'g.)

Q. Okay. If we could turn to page 59 of your testimony. The -- I'm interested in the section beginning on line 8, the L.M.I. gas conversion program.

I think earlier you were asked a question about how -- how you -- how you defined a low income customer and I -- I think part of -- I remember part of the response was if -- if a person is HEAP eligible that -- that's one way of identifying a low income customer. Is that correct?

A. (Tallet) Yes.

Q. Okay. How do you determine if somebody is HEAP eligible, but who does not receive HEAP?

A. So if -- if a customer is HEAP eligible, this customer may also qualify for another benefit program that might give the company insight into their income level, you know, such as, you know, public assistance or the SNAP program.

The company might rely on other human service agencies to help identify those customers.

Q. Well, for KEDLI, what -- what type of interaction does the comp -- does KEDLI have with the county social services that would allow that interchange of information about whether somebody is HEAP elg -- eligible but doesn't receive HEAP.

A. So the company has consumer advocates on our staff who they may interact with the individual department of social services. They may either, you know, be within their offices or have some discussions and referral mechanisms in place.

Q. But you say -- you use the word may. Do you know if -- if KEDLI actually speaks with county social service agencies on Long Island to --

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to ask for a list of people who are HEAP eligible but don't receive HEAP?

A. We do. With respect to our low income bill discount program we do have a -- a file match type program in place to help identify those customers.

Q. Okay. Thank you. Does the panel know how many low income people have converted from oil or propane to gas in the last year?

A. (Tallet) I do not.

Q. Okay. Thank you. And would that be the same answer for the number of moderate income people?

A. Yes.

Q. Well, in -- on page -- please turn to page 64 of your testimony. Do you see the discussion there about the reasons why the company underspent economic development grant revenues?

MR. O'BRIEN: So Your Honor, at this point I'm going to object. I'm looking at the topic list, you know, the company gave Mr. Rigberg some leeway, nowhere on the topic list or anything, discussion about a Green Gas Tariff, N.G.V.'s, R.N.G., methane reduction, economic development.

And my understanding was to keep the parties fully informed that we were to give all the parties advanced notice of the topics we wanted to discuss. And I do look at the time and it looks like we're over the estimated time and I want to make sure everybody here has a chance to participate today.

A.L.J. LEARY: So you need to let us worry about that. Thank you. How much more do you have, Mr. Rigberg?

MR. RIGBERG: Not too much.

A.L.J. LEARY: Okay. So proceed.

MR. RIGBERG: Okay. Thank you.

BY MR. RIGBERG: (Cont'g.)

Q. My question regarding there is a discussion, the factors that are the reasons for the underspend starting on lines 11 through 21 on page 64 and going on to line -- at 15 on page 66. My question is what efforts the companies intend to take to ensure that there is not an underspend in the future?

A. (Hamlin) I think the -- the main thing will be for us to ensure that customers are aware of the programs. These are relatively new programs for the KEDNY, KEDLI territories. So we are

still building awareness. And with the existing programs and in addition what we're proposing for the future, it just puts more of a impetus on the company to get the information out there through company personnel, community facing, customer facing personnel, largely.

Q. Okay. Well, these programs have been in place for three years, have they not?

A. That's correct.

Q. So what type of outreach has the companies undertaken so far?

A. We have one full-time employee who is responsible for meeting with customers directly. Also with meeting with the economic development network. And by that I mean, New York State Economic Development Agencies, regional economic development organizations, local economic development organizations to make them aware of the programs.

Q. Now, do you -- does the panel know if -- if these -- if the cost of these programs are recovered in rates currently?

A. They are.

Q. Okay. So they're recovered in rates but, not fully use -- used?

A. That's correct. And there is a mechanism for tracking and recovering the grant program expenses through a deferral. So there is a rate allowance provided each year, actual grant program costs are tracked against that rate allowance. And that balance is carried forward throughout the term of the rate agreement for either refund potentially to customers and potentially, if our costs exceed that, we can petition a commission for additional funds.

Q. Okay. Thank you. Turning to your page 68, the --

A.L.J. LEARY: Can I ask a question to clarify the record? So you have an underspend of 4.7 million for KEDNY and 5.2 million for KEDLI?

THE WITNESS: That's correct.

A.L.J. LEARY: And in the rate year for which this filing is made, what are you requesting in addition to that?

THE WITNESS: We are request --

A.L.J. LEARY: What is the company requesting? I apologize.

THE WITNESS: The companies are requesting a reduction in that annual rate allowance from its current $2 million per year for each company to $500,000 per year in that year for each company and we did a forecast of expenditures and the -- the delta. So we are forecasting actual expenditures to be higher than that new rate allowance of $500,000. That difference would be taken from that accumulated deferral balance that I was just referring to.

A.L.J. LEARY: So you've got one full-time employee. Is the company seeking any additional full-time employees for this particular program to deal with getting the word out, I think is sort of what you meant in your testimony?

THE WITNESS: No, we're not requesting any additional labor. We do expect some of the proposed program changes to also drive an increase in spending.

A.L.J. LEARY: Like what?

THE WITNESS: We're creating a -- an entirely new grant program called economic development in the Future of Heat. So it's not in the previously existing portfolio of programs. And it's really designed to support the larger Future of

Heat initiative that the companies are putting forth. And it allows for funding. It will encourage investments by customers in things like non-pipeline alternatives as well as investments in new efficient gas technologies.

So we do expect significant new demand for the programs to come. From that some other minor proposed changes to the programs as well in addition to maintaining and increasing the level of outreach to make customers aware.

A.L.J. LEARY: With that one full-time employee?

THE WITNESS: One full-time employee directly assigned to the economic development function. However, that employee relies heavily on utilizing other customer community facing employees. So folks in, what we call, the jurisdiction organization who have assigned accounts, who may be positioned to take advantage of the grant programs, our energy efficiency folks because there energy efficiency and economic development tend to be complementary, in terms of helping to reduce the customer's costs of doing business.

There's a lot of, I would call it, cross outreach going on. So our economic development representative informs customers of energy efficiency and our energy efficiency team is out there as they interact with customers to inform them about our economic development programs.

A.L.J. LEARY: So is there any plan here? Because I hear you citing the economic development and Future of Heat, which it sounds like it's a plan of some kind. Is there any plan here to increase public outreach or have this under -- under expenditure addressed in the immediate term?

In other words, you got to plan that, you know, somebody is meeting with somebody else and saying okay, this is the strategy to address this issue. Is there anything in the record about that?

THE WITNESS: I don't believe so in terms of any specific measures to increase the level of outreach.

A.L.J. LEARY: So this is really sort of the informal, we know it's a problem, somebody might be assigned to it to work with a bunch of other people I think. Is that right?

THE WITNESS: Yes, it's in that it hasn't been specified in a plan, but there is a general acknowledgement that with -- particularly with the changes that we are proposing for the programs, there will be a need to increase the level of outreach even to revisit with folks who are already informed of the programs to make them aware of the new and revised program offerings.

A.L.J. LEARY: Thank you. Sorry about that.

MR. RIGBERG: No -- no problem.

BY MR. RIGBERG: (Cont'g.)

Q. Please turn to page 68 of your testimony. I had a question about the -- the business incentive rate that you described starting on line 15.

A. Yes.

Q. Yes. My question is doesn't offering a 50% discount on incremental gas usage encourage the use of gas?

A. Yes, it does. And it seeks to, for customers who are needing to use more gas, in order to expand their operations, it provides them cost savings to do that.

Q. And does the program or -- also involve a component that encourages efficient use of gas through changing out equipment or, you know, other measures?

A. It does. And I don't have the tariff in front of me, but I believe there is a requirement that customers explore energy efficient opportunities as they're expanding their gas usage.

Q. Okay. Thank you. All right. Turning to page 100 of your testimony. I wanted to ask you a question about the supply diversification metric, that starts on line -- line 6 of a page 100. Do ratepayers pay for the costs of developing R.N.G.?

A. (Schwartz) Today, they do not. The cost of developing is solely borne by the R.N.G. developers.

Q. Okay. So what -- what part of the development process do the companies participate in?

A. We participate in evaluating the interconnection of R.N.G. facilities on approving the interconnection of R.N.G. facilities into the gas network.

Q. Is that something that the companies are obligated to do as -- as part of the normal course of business?

A. That's not something that we're obligated to do, but it is something that we're interested in encouraging, in part, because it's a local supply of gas and also to encourage low carbon sources of renewable natural gas.

Q. Okay. All right. And turning to the energy efficiency, E.A.M. -- E.A.M. on -- starting on page 103 of your testimony. On the top of -- and going to the top of page 104, there's a -- there is a description of the metric and how it's measured. I'm wondering as part of the measurement of -- of determining whether there's a decrease in use, do you -- do you account for exogenous factors like weather or the robustness of the economy?

A. (Tallet) Those are not factored in.

Q. Okay. Thank you. All right. Please turn to page 106 of your testimony. The affordability metric that's described starting on line 9, the companies say that -- that the -- the

metric would measure the annual percentage reduction in average low income customer bills.

Did the company also look at the median bills or -- or mode or other ways of looking at distribution of bills and determining to use average as -- as the, you know, the standard?

A. We did not look at -- at the other variables. We looked at the average bill because that is information that we have. So for customers that participate in our E.A.P. or affordable -- Energy Affordability Program, we calculate the bill discount based on the average bills. So we use that as a -- as a metric because it's -- it's information that we had.

Q. Now does the panel know how many low income customers have deferred payment agreements?

A. I do not know that. That might be something that our shared services panel could respond to.

Q. Okay. And would that panel also know whether the differ -- the -- how many of the low-income customers with deferred payment agreements are considered high usage?

A. I don't know if they would -- if they would have that information on considering them high usage.

Q. Okay. Did -- did the panel in -- in setting forth and developing this metric look at the impact of terminating -- if high usage customers, low income customers were terminated due to non-payment, how that would -- how that would affect the -- the average that would be used in this metric?

A. So just to clarify your question is if -- if a customer had been cut out for non-payment if that impact was factored in --

Q. Well, if it was a high usage customer.

A. So a high usage customer who was cut out for non-payment. If that -- that activity of cutting them out impacted the average bill?

Q. Well, if -- yeah, if removing high usage customers from the total number of low income customers, what would be the impact on the ability of the company to achieve this E.A.M.?

A. So that particular scenario is not one that we modeled.

Q. Okay. Thank you.

MR. RIGBERG: That's all the questions I have, Your Honor.

A.L.J. LEARY: Thank you.

A.L.J. COSTELLO: Thank you. I'm just going to -- do you want to take a break? Yeah. We're going to take like a 10 minute break, but before we do, I just want to -- something I want to address what Mr. O'Brien said earlier. We do have estimates of cross examination times and that we are using and we're basing their days and the amount of time that we're going to be spending here based on those estimates.

We understand they're just estimates and today we've had -- we have a little leeway because of some people have waived their cross examination but going forward we would like you to adhere as close as possible to your estimates or provide revise estimates in advance if you're going to be going over significantly. So thank you. We're going to take a 10 minute break. Thank you.

(Off the record; 2:54 p.m. to 3:05 p.m.)

THE REPORTER: On the record.

A.L.J. LEARY: We have E.D.F., Environmental Defense Fund.

MS. MURPHY: Yes, this is Erin MURPHY for E.D.F. Is this working?

A.L.J. LEARY: Got to stay real close to it, sorry. I'm trying to fix this logistical problem, which may not be fixed till tomorrow, but I've sent an email on it. So the real problem is we only have four microphones that the court reporter can pick up that microphone. So that microphone has to be really good for the record to be as clear as possible, but we're working on that as well so.

CROSS EXAMINATION

BY MS. MURPHY:

Q. Good afternoon, to the panel. Thanks everyone for staying awake. I think some of these questions may sound familiar from Sane Energy. I've tried to condense them to avoid duplication. Methane is the primary component of natural gas, is that correct?

A. (Chahbazpour) Yes.

Q. And methane is a greenhouse gas that contributes to climate change, is that correct?

A. Yes.

Q. And is it correct that the companies have established their own greenhouse gas reduction goal and have expressed commitment to taking actions that reduce methane emissions from the gas network to comply with those goals?

A. Yes.

Q. So your testimony indicates that you are aware of the Climate Leadership and Community Protection Act, which I'll refer to as the C.L.C.P.A., which took effect on January 1st, 2020. Is that correct?

A. Yes.

Q. I'm going to read briefly, your testimony states in the rebuttal testimony at page 2, there are two significant changes that impact the panel's proposals and then at rebuttal page 3, your testimony states "second New York adopted the Climate Leadership and Community Protection Act of 2019 codifying greenhouse gas emission reduction and renewable power generation requirements while also creating a climate action council to drive policy goals for achieving the clean energy requirements". Is that correct?

A. Correct.

A.L.J. LEARY: What page you're on?

MS. MURPHY: That was the rebuttal -- the Future of Heat rebuttal testimony pages 2 to 3.

A.L.J. LEARY: Thank you.

BY MS. MURPHY: (Cont'g.)

Q. And at the Future of Heat supplemental testimony page 11, your testimony states the companies developed a suite of proposals in the Future of Heat filing that align with the – N.E.N.W. orders, clean energy focus as well as the broader C.L.C.P.A. clean energy goals, is that correct?

A. Yes.

Q. So is it correct to say that your testimony indicates the companies are planning and implementing projects with the emissions reduction goals of the C.L.C.P.A. in mind?

A. Yeah. Broadly, we are trying to be consistent with the broad objectives of the C.L.C.P.A.

Q. And what are the company's primary sources of methane emissions?

A. The primary source of methane emissions is a leak-prone pipe, cast iron and bare steel mains.

Q. And I think we covered this to some -- Sane Energy asked some questions on this topic, but I'd just like to verify. What methods are the companies using to quantify those emissions?

A. We use E.P.A.'s greenhouse gas reporting program to quantify methane emissions from our system.

Q. You mentioned earlier, a couple of other tools that you use, I think to also quantify emissions such as the carbon disclosure project and One Future, can you talk more about those?

A. So carbon disclosure program is -- they're both voluntary. I believe that's when you do carbon disclosure program and one of my other colleagues may have more insight, but that's something you do for part of the shareholder initiatives. You provide your entire emissions and I think this shows not only our U.S. emissions but it's U.S. and U.K.

So if you go to our carbon disclosure pro -- C.D.P. reporting is, it will show National Grid U.S. and U.K. One Future -- this started under the Obama's administration and it was a voluntary program developed by E.P.A. and it's still there

today as a voluntary program to reduce methane emissions.

Q. And so --

A. And one thing I would just add, the one thing -- the unique thing about One Future was to engage the industry and to cut across sectors. So this was a collaborative that engaged the entire value chain of the natural gas supply.

Q. And so I think we've talked about three different reporting tools then, the E.P.A. emissions factors and then these other two voluntary programs you're describing. I mean, did those overlap, did they describe the same emissions or are they each describing distinct sets of emissions from your system?

A. The scope of the reporting is different and also the greenhouse gas reporting program is using emission factors that are quite old. For One you -- One Future, there are emissions that are -- we are using that are newer, that E.P.A. is already adopting under G.H.G.I., Greenhouse Gas Inventory, helped develop by number of studies by -- by environmental defense fund.

E.P.A. has expressed an interest to adopt those emission factors from G.H.G.I. to G.H.G.R.P., however, the reason that they have not, this is a regulatory proceeding. So the emission factors are different and there are also emission criteria under G.I. -- G.H.G.R.P., which is the E.P.A. program and under the One Future we go beyond those emissions sources.

So we go above what's require -- what's required to be reported in the One Future. So it's more and I believe also the carbon disclosure program is the same in a sense that it goes beyond what's required under E.P.A.'s. G.H.G.R.P.

Q. Would it be fair to say then that maybe the -- the carbon disclosure program and One Future characterize, you know, the difference if I'm missing it, but would it be fair to say that those are more precise than -- than the E.P.A. emissions factors if they're using more updated methods?

A. They are in a sense that the new emission factors actually are more reflective of the infrastructure today.

Q. Sure. And you mentioned sort of at the beginning of your previous answer that the

current E.P.A. emissions factors, which I believe you're referring to sub-part W that you mentioned earlier with Sane Energy. So you mentioned that those are quite old. Do you recall what -- when those were first established?

A. I think about 2010, I'm not sure, but it's somewhere around -- they may even be older.

Q. Sure. Thanks. And so with regard to the -- the E.P.A. emissions factors, how confident are you in that quantification method?

A. The method I'm confident, it's just that the emission factors are old because those emission factors represent the gas infrastructure and I actually remembered, because people refer to them, they're 20 years old. So that's older than 2010. The emission factors today that you would be adopting from G.H.G.I. would be -- are more reflective of what the infrastructure is out there today. I'll give you an example.

The emission factors -- this gets technical for M. and R. stations, metering and regulating stations, are much more accurate today than the 19, you know, the emission factors that are 20 years old or over 20 years old because some of

that was -- some of those assets are no longer in the field, like, you know, blow, you know, they call them low bleed valves, no bleed valves, they've just been -- they no longer exist. Everything is no bleed valve.

And that's why if you look at the recent emission factors that have adopt -- adopted under G.H.G.I., they're much lower than the emission factors for the same material in the G.H.G.R.P. So the -- you know, to answer your questions, they reflect a different time period of what the assets look like.

Q. Thank you. And have the companies undertaken any studies that compare or correlate your system wide emission rate to the E.P.A. emissions factors?

A. We have been involved in a number of studies, most prominently the one with Brian Lamb and Environmental Defense Fund.

Q. Again, this may have been covered with Sane, but I want to just clarify again. To the extent that the companies are quantifying greenhouse gas emissions, do you include emissions that result

from use of the gas downstream of the company's meters?

A. Just to be clear, you're talking about scope 3 emissions, right, as part of combustion of natural gas?

Q. Correct.

A. National Grid does not report those emissions, but that's part of the U.S. G.H.G.I. emission inventory.

Q. Right. So the numbers that you are reporting under the E.P.A. emissions factors do not include those?

A. That's correct.

Q. Any past the meter?

A. That's our operational emissions.

Q. And so we talked about a number of programs, maybe to try to pull some of that together. Do you currently have a measure or estimate of the baseline greenhouse gas emissions that are emitted across the company's operations?

A. Yes. You're referring back to the 1990 baseline, right? Am I -- the answer is yes.

Q. I'm talking about the present -- present baseline?

A. Yes, we know what the present baseline is.

Q. And -- and you're saying that based on the E.P.A. reporting only?

A. Correct.

Q. The companies are -- and I'm now referencing page 91 of the Future of Heat testimony though I'm not quoting it. The companies are --

A.L.J. LEARY: Direct testimony, direct?

MS. MURPHY: Direct, yes.

BY MS. MURPHY: (Cont'g.)

Q. The companies are proposing a new Future of Heat engineering group, is that correct?

A. Yes.

Q. And when assessing the viability of new projects, would that group consider the level of greenhouse gas reductions that a project under consideration is projected to achieve?

A. So this group has not been formed yet. And they would sit in the group that my colleagues will testify later. They would sit actually under the gas infrastructure and operations. So I would -- I would say when that Future of Heat

engineering group does exist, we would probably -- to begin integrate what you just said.

Q. So to -- to clarify, yes, your expectations --

A. Yes.

Q. -- if that --

A. Yes. Currently no, but yes, my expectation is that we will.

Q. So referencing the Future of Heat rebuttal testimony at page 60. Your testimony states that the companies agree that understand --

A. I'm sorry, which line on page 60?

Q. I will have to check, one second. Line 19. Your testimony states that the companies agree that understanding the nature and extent of greenhouse gas sources is vital to developing effective strategies to address emissions, is that correct?

A. Correct.

Q. And can you explain why understanding the extent of greenhouse gas emissions is vital?

A. It addresses how do you get to the ultimate goal of, you know, low carbon future of 8050.

Q. And when you referenced that ultimate goal, my thought is that that describes both the company's own goal as well as the goals of -- of New York State Law under the C.L.C.P.A., do you agree?

A. Yes, agree. An addition context into this comment was I remember writing this that what you were asking E.D.F. and what we're saying we're supportive of, the policy framework does not exist to get to what you are asking for.

Q. And thank you for that response. I -- additional question would be in light of the emissions reductions targets of both the C.L.C.P.A. as well as the company's own greenhouse gas emissions reduction goals, would it be helpful for the companies to have an up-to-date calculated emissions baseline?

A. Yes.

Q. MS. MURPHY: Thank you. That concludes my cross.

A.L.J. COSTELLO: Okay. The cross examination will now be conducted by Bob Wyman.

MR. WYMAN: May I first say this is the first time I've done this. And I'm not a lawyer. So please accept in advance my apologies for the inevitable mistakes I will make. I will do my best.

CROSS EXAMINATION

BY MR. WYMAN:

Q. Let's see, to start off --

A.L.J. LEARY: You need to put the microphone a little bit closer, Mr. Wyman.

MR. WYMAN: Yes, is -- how is that?

A.L.J. LEARY: Just speak into it.

MR. WYMAN: The other one too?

BY MR. WYMAN:

Q. To start off, I was a little confused when you were talking with -- saying on several occasions, you mentioned an 80 by 50 goal. Now, I know 80 by 50 was the goal under the old executive order and it's -- it's -- it's the way the goal is expressed in Local Law 97 of 2019 for New York City, but the state has a goal of 85 by 50, not 80 by 50.

And I'm wondering if you can tell me, are you aware that the C.L.C.P.A. has an 85 by 50 goal and by saying that your goal was 80 by 50, was that simply a mistake or are you somehow rejecting the state's goal of 85 by 50?

A. (Chahbazpour) When we provided initial testimony C.L.C.P.A. was not final, when we provided the -- the testimony in September, which was the rebuttal. And the first question on Page 2 talks about some of the changes that have happened, we actually highlight C.L.C.P.A.

Q. Okay. So today --

A. So we acknowledge basically that --

Q. Is --

A. -- when we filed it was 80 percent and in September we made our rebuttal, that target has been updated. And we are consistent as a company policy, whether it's 8050 or not a C.L.C.P.A.

Q. So is it -- is it correct to say that today, your goal is to -- is to meet the requirements of 85 by 50, not 80 by 50?

A. Correct.

Q. Okay. Thank you. Thank you. Now also, there was some discussion of the -- the CO2 emissions baseline, the 1990 and by implication, do you have data for other -- other years as well? I think you said you did. And I'm curious, it would be wonderful if we could have that data as well.

But I'm wondering, as I mentioned in my testimony, which I believe you all are familiar with. New York City has an emission -- greenhouse gas emissions inventory, which reports on the emissions from a variety of sectors, but it includes reports on emissions from natural gas.

Is there -- can you tell us -- I assume you're familiar with that, as any -- as any responsible organization to providing gas in New York City would be. Can you tell me, do your estimates of emissions generally agree with the estimates that are reflected in New York City's official greenhouse gas emissions inventory?

A. That's the first question I want to make you -- you said that we have our 1990 CO2 emissions baseline, we have our 1990 methane emissions baseline.

Q. Only methane emissions?

A. Methane emissions, CO2 emissions means that's a Scope 3 emissions. So we're talking about methane emissions from our system at from a 1990. I haven't -- I don't remember when was the last time I looked at the emission factors from the New York City reporting, but I'm pretty confident or pretty comfortable speculating that they're pretty consistent and close to what we report to E.P.A., but they use their own methodology.

So you know, New York City has their own compliance reporting mechanism. Is that fair, I think.

Q. So would it be fair to say --

A. So in other words --

A.L.J. LEARY: Excuse me.

MR. WYMAN: Sorry.

A.L.J. COSTELLO: Yeah.

MR. WYMAN: I'm sorry.

A.L.J. COSTELLO: They haven't finished.

MR. WYMAN: I'm sorry.

THE WITNESS: (Schwartz) No worries. Right, we do not report to New York City, they have their own compliance reporting that they do. And I

am not familiar with where the discrepancies are between --

MR. WYMAN: Yeah.

THE WITNESS: -- E.P.A. requirements versus the City's requirements.

A.L.J. LEARY: So I need the record clarified Mr. Wyman, this New York City document pursuant to Local Law of 97 something. Is there a name on this document? Is there a name that you can put on the record?

MR. WYMAN: Well, there is --

A.L.J. LEARY: For this document?

MR. WYMAN: Well there is the -- the law itself, Local Law 97 of 2019.

A.L.J. LEARY: 2019, but I'm asking you a different question. I thought that there was a compliance --

MR. WYMAN: There is a --

A.L.J. LEARY: -- document --

MR. WYMAN: Sorry.

A.L.J. LEARY: -- that you are referring to, you know, she's not going to be able to pick us up both at the same time. So just wait till I finish. I thought you were referring to the New

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York City greenhouse gas emissions inventory calculations. I assumed that that was in a particular document. But I'm not hearing on the record what -- how to identify that document. I hear that the panel is familiar with it. That's what I imply from your response. But what is the name of that document?

MR. WYMAN: First, I'd like to distinguish the discussion of Local Law 97 of 2019, which is New York City Local Law, it's part of the Climate Mobilization Act, is that -- yeah, Climate Mobilization Act. That is the law that establishes in New York City the 80 by 50 goal for New York City.

The New York City greenhouse gas emissions inventory is actually created by an earlier as -- as a result of earlier legislation. Frankly, I can't remember its number. It is -- it is -- I have always accessed it as a website, not as a document, per se. It is -- it is a website that's maintained by the City of New York and it's entitled their greenhouse gas emissions inventory.

A.L.J. LEARY: I think that's --

MR. WYMAN: The earl of the --

A.L.J. LEARY: -- sufficient for the record.

MR. WYMAN: Right. The earl or the -- the reference to it can be found in my testimony, my -- my direct testimony.

A.L.J. LEARY: Okay.

MR. WYMAN: Not the rebuttal.

A.L.J. LEARY: You can go ahead.

MR. WYMAN: Okay.

BY MR. WYMAN: (Cont'g.)

Q. Okay. So but -- but what we did establish at least was that to the best of your knowledge, the discrepancies between your own estimates of emissions from natural gas and the estimates made by the City of New York and the several years that they've been maintaining their inventory have never risen to the point where you have -- have -- where you personally know that it has become an issue that that needed to be worked.

Can -- is that safe to say?

A. (Chahbazpour) I think the distinction is that New York City does its own greenhouse gas inventory reporting and we are not an input into that process. In other words, National

Grid does not file it with the City of New York and the city puts out their own. The differences when we do to E.P.A., we are an input. So we have a regulatory compliance that would be filed with E.P.A. on the Greenhouse Gas Reporting Program.

Now, how -- at National Grid as far as I know, we haven't looked at exactly what the city does, what is their methodology, and to try to reconcile what did they do and how different is an E.P.A.. If that's your question, we haven't done that.

Q. Okay. You haven't done it, but also you never found the need to do it. Okay. That's fine. Okay. So Future of Heat, this is -- this is a great subject. Of course, the fact that we mentioned the Future of Heat implies that somehow the future is going to be different from what we have currently obviously.

I wonder, could you -- could you say a bit, just briefly summarize why it is necessary or appropriate for us to be envisioning a Future of Heat, which is different from what we have today?

A. Sure, I think when it comes to, I think we all know addressing climate change is

extremely challenging. And you have to decarbonize every sector of the economy. And within that, one of the most difficult sectors to decarbonize, it's heat. So going to your point, we -- we view the vision, the Future of Heat, the way we decarbonize heat will be very different than it is today, and it will utilize a portfolio of technologies that we're looking at including geothermal and decarbonizing of the gas network. So those will be two of the tools in the kit, you know, in the toolkit.

Q. Okay. Would it be fair to suggest that one of the reasons we need to move forward is not only that it's -- it's sort of right or meet just right for us to decarbonize and clean things up, but also legislation and regulations such as Local Law 97 of 2019, the Mobilization Acts, the C.L.C.P.A., et cetera, are -- are requiring that we do this?

A. Yes, I think, you know, right now the C.L.C.P.A. is very high level has it -- has just a 85 percent emission reduction targets. It doesn't talk about sector specific, but you will pop -- you do need to decarbonize every sector of the economy

and heat would be included to achieve those kind of emission reduction targets.

Q. Can you -- can you let me know, other than sort of the two legislative pieces that are driving us in this directive that I just mentioned, can you quickly summarize any others that you may know have -- know of and perhaps considered while -- while putting together the Future of Heat study?

A. Any legislations?

Q. Any legislation or regulations that that compel us to move towards a Future of Heat?

A. So I don't think it was -- the impetus wasn't the C.L.C.P.A. I think National Grid has been advocating for a low carbon future for a very long time. We publicly supported the Paris Agreement, we publicly even urged the Trump administration to stay within the Paris Agreement. So we have been supportive of that -- it used to be what it is 8050. Now -- now it is more aggressive, it is C.L.C.P.A. So it goes further back in time.

And all of our efforts over that last few years have been moving the company to be part of that low carbonate future. So in other words, just

to answer your questions, we did not come -- sort of with the Future of Heat program as a response of the C.L.C.P.A. We're already thinking about these initiatives well in advance.

Q. But whether or not it was your motivation, it will certainly compel us to take action in the future, right? So I'm wondering, are you familiar with the New York City Mayor's Executive Order 52 issued actually just a couple days ago, that states the city's position as being opposed to the expansion of fossil fuel infrastructure, and doing things like requiring those who speak for, represent the city, to oppose fossil fuel infrastructure expansion in all forms. Are -- are you familiar with this executive order?

A. I've read what's been in the newspapers over the weekend. The one that you're talking about, the one that just came out last Thursday, correct?

Q. Yes. Okay. And -- and it -- is it fair to say that this is yet again another piece of legislation, policy making or whatever which will compel us to move towards the Future of Heat?

A. Agreed.

Q. Okay. But emission reductions themselves are – are a pretty big reason to move forward. Now, one of the reasons that I referred to the, you know, when we look at the Future of Heat, when we look at the plans, we have to sort of ask ourselves, you know, are they going to be adequate to meet our -- our purposes. And those purposes are not only, I think, the ones we feel because we're responsible people, but also the ones that are expressed from the various bits of legislation that we're compelled to respond to.

And one of the things I laid out in my testimony was that according to New York City, the emissions from natural gas burned in buildings today, okay, in New York City is at a rate which is equivalent to about 144 percent of the city's goal for all CO2 equivalent emissions in the entire city from all sources in 2050 -- about a 104 percent. I'm assuming you're familiar with that because you guys have read that testimony.

So what this tells us is that we've got to have --

A.L.J. COSTELLO: Are you asking a question?

MR. WYMAN: Yes, yes, yeah.

A.L.J. COSTELLO: Okay.

MR. WYMAN: I -- my apologies, I try hard to be terse, but I'm not good at it.

BY MR. WYMAN: (Cont'g.)

Q. So the question is, this would seem to imply to us that we've got to get the emissions from natural gas below a 100 percent of the total goal by 2050. So that'd be a -- over a 30 percent reduction in emissions from natural gas by 2050. Are the plans that you have documented in the Future of Heat, are they sufficient to get us a reduction at that level by 2050?

A. Whether or not I think in my opinion, they're sufficient or not, we will know over time and what we'll really depend on is the policy support to implement those programs. I'll give you a sort of quick anecdotal when National Grid released a white paper on R.N.G. in 2010, there were only a handful of projects in the United States that were injecting R.N.G. That was always the initial question I would get. Ten years later in 2020, there are now over 120 projects in the U.S. injecting renewable natural gas into the gas network. And

what's been the catalyst has been E.P.A.'s renewable fuel standards program. So what we are doing is, you know, we obviously don't have a crystal ball to what 2050 looks like. And we know that we need a lot of technology solutions. We are proposing, I would say, most of them in our Future of Heat, whether or not how successful, we will find out over time. But I'm -- one thing I would say I'm confident is that without the appropriate policy framework, we probably would not be able to decarbonize the heat sufficiently to the numbers again that he has mentioned.

Q. But do you think the things which are identified in your testimony, did they encompass sub – a substantial portion of the things that we will have to have -- of the tools and such that we'll be using between now and 2050 to accomplish those goals?

A. I believe so. I mean, I -- in -- in 10 years from now, we are able to bring a lot of R.N.G. from biomass and geothermal programs. And we talked a lot about hydrogen. We talked about it today -- the power to gas, what we're doing. Right

now the space of hydrogen is getting a lot of attention -- a lot of attention.

It has the potential to be a genuine game changer. It could change the energy industry landscape significantly, not only to decarbonize heat, but they also decarbonize in other sectors of the economy that are very difficult to decarbonize, and particularly, industrial and transportation for heavy duty.

Q. Of the various things that you proposed in the Future of Heat testimony, given our understanding of technology today, and in -- in the very near future, say just looking out three to four or five years or so. Which of the various approaches that you laid out there has the greatest potential to -- to allow us to -- to decarbonize the -- the things that use natural gas today, between now and 2050?

A. I think if you ask 10 people on National Grid, you'll get 10 different answers because you're basically asking which technology, you know, it's, you know, which technology do you think will take off on a bigger scale than the others. If you probably asked me, I'll probably say R.N.G. from

my -- biomass. If you probably asked my other colleagues, they might say geothermal.

So we don't know, right, we don't know the answer which one of them has the biggest promise. But what we're trying to do is to put some of these programs as pilots and to begin fund them and to again, give them some policy support. So in other words, you know, we are looking -- we -- you know our hypothesis is that you need all of these technologies. We are looking at all of them too and you will probably need all of them to decarbonize the heat. I would say there's probably no silver bullet to decarbonize the heat sector. You need all of these solutions.

Q. But you said your personal feeling is that R.N.G. is the leader of the group?

A. From biomass today, yes. And you said the next three to five years, I would say, yes.

Q. And the other panel members, do you -- you want to -- can we do a vote here? What do you --

A.L.J. COSTELLO: Well this is a -- it's the testimony by the panel, it's not what individuals --

MR. WYMAN: So should I take that answer that R.N.G. is the number one technology --

A.L.J. LEARY: I think his testimony -- I think his testimony is there's no silver bullet, and that you know, the -- you need to use whatever is feasible out there. And I don't mean to mischaracterize your testimony, but I think Mr. Wyman, that's probably, you know, your best answer. And if you could sort of move on from that.

MR. WYMAN: Okay. I'll move on. Okay.

A.L.J. LEARY: And by all technologies, I would assume if there's a technology that comes up tomorrow that the company -- that -- that the company thinks has some -- some merit, some feasibility, that would be considered as well. Is that correct?

THE WITNESS: Yes, absolutely.

THE WITNESS: (Schwartz) That is correct. And I would like to clarify, because we're speaking about an 8050 target, but what we're doing over the next four years, and those are two very different things. This is us trying to do everything we can think of to move the conversation forward

about how we can decarbonize heat and looking at all the technologies we know of today that we think are feasible to do so. And if another technology comes across our desks, we would -- would of course, evaluate that.

BY MR. WYMAN: (Cont'g.)

Q. Could I just clarify? Did you say, old here. Was that 8050 or was it 8550?

A. Sorry, that was 2050. Too many numbers going around. 85 percent by 2050.

Q. Okay. Thank you.

A. Yeah the 2050 goal.

Q. Thank you. Let's see. So renewable natural gas, powered gas, hydrogen, a whole bunch of technologies here essentially putting something else in the pipes other than natural gas. Essentially, we're -- we're talking about replacing natural gas with synthetic gas.

Is it -- is it fair to say this is essentially going to full circle we started off with a gas system that was all coal gas and town gas. We went to natural gas. Are we now talking about going back to -- of course, it's a lot cleaner than town gas or coal gas, but -- but are we now talking

basically going back to distributing synthetic gas as opposed to natural gas? Is that the long term future to get to an 85 percent reduction?

A. (Chahbazpour) I wouldn't say we're going back, I would say there's a distinction here between the gas network and what's flowing in it.

Q. Yeah.

A. And what we're saying is what is flowing in it over time will change. And the carbon footprint of the gas network is declining over time -- will decline over time by integrating these sources.

Q. But are you saying that what will flow through those pipes is something other than natural gas?

A. Yes. I mean, to what percentage will be natural gas over time, I don't know. But what's flowing through the pipe will be different in the future than it is today.

Q. And the primary gases that flow, other than those which are natural gas will be synthetic of some form. They will have been produced

or constructed not simply harvested the way natural gas is.

A. Yeah, they will not be geological.

Q. Okay. Now, on page 75 of your testimony line 16 to 17. You refer to a 2010 R.N.G. report that has said – that -- that predicts a significant contribution to clean energy for R.N.G. for -- for renewable natural gas. Is that report the report which is known as the renewable gas vision for a sustainable gas network, which the company did -- what in July 2010 and was -- was actually filed as part of the staff policy -- filed as part of an I.R. and attached as an exhibit to the staff policy panel in August?

A. Yes.

Q. Okay. Has the company done any study of the potential for renewable natural gas -- renewable gas since 2010?

A. Yes, we realized that we know that's really, when you talk about studies 2010 feels like centuries ago. We and here we I mean the National Grid and the collective natural gas industry wanted to do -- update the study in early 2019. We

19-G-0309/19-G-0310 - 2-10-20 - Brooklyn Union Gas hired I.C.F. on behalf of A.G.A. the American Gas Association, to conduct a national study. That study was released right before Christmas.

So we update -- that study is on A.G.A.'s website and we can definitely share it. It's not part of record yet because it was just released before Christmas. And I'll sort of give you the key takeaways from this study compared to the 2010. It essentially concluded that by 2040, the entire U.S. residential sector could realistically come from renewable natural gas.

And it includes nine feedstocks, eight biomass and it's the first study that actually quantifies power to gas and hydrogen. And there is no upper limit to power to gas and hydrogen, so we were fairly conservative. But that study answers few questions in the R.N.G. industry, which is how much is out there and how much would that cost. And again, that study -- something we will share with you that just came out before Christmas.

Q. Okay. Can I -- can I state my disappointment here because we have been promised many times by representatives of National Grid that

if that study was ever completed, that it would be provided to the parties of this rate case and I am --

A.L.J. LEARY: Okay. So --

MR. WYMAN: Okay. So --

A.L.J. LEARY: -- we're going to get it --

MR. WYMAN: --let me just say that, okay.

A.L.J. LEARY: -- and I'm looking at. Excuse me, Mr. Wyman.

MR. WYMAN: Sorry.

A.L.J. LEARY: Because we can move on quickly from this.

MR. WYMAN: Anyway, no.

A.L.J. LEARY: Is this something that we can expect from the company? I know I already have one thing on your list, but --

MR. GAVILONDO: Yes, Your Honor.

A.L.J. LEARY: -- would you mind adding this? Does anybody have a problem with that from the company? Great. Okay, Mr. Wyman.

MR. WYMAN: Okay. Now --

A.L.J. LEARY: No more complaining.

BY MR. WYMAN: (Cont'g.)

Q. I'm very surprised by what you said, because the 2010 studies said that the potential for renewable gas in New York State was 17 percent of overall demand or 25 percent of non-power generation demand. So something dramatic has changed between 2010 and now in our understanding, and apparently it's been discussed or reported in this report that is missing.

Can you say what has caused the potential to grow four times, which is essentially you said, we can handle our full requirement with -- with renewable natural gas. But the 2010 study said, you know, at best 25 percent and only if it's non generation. So what has changed since 2010 that has allowed the estimate to go up four times?

A. So the overall potential from 2010 is about doubled. So not four times. What I referred that -- what the study -- 2019 study showing that you can decarbonize the entire residential sector in the U.S. The biggest difference between the recent study and the last study is the amount of biomass. So the amount of biomass and that really comes from the federal government data.

So we referenced a billion tons study in our study, and if you compare the amount of biomass feedstocks over the last 10 years, that has grown significantly. And at the same time, there is like all technologies, lower costs of technologies has made an advancement in the various technologies that can be utilized to produce R.N.G. and particularly anaerobic digestion gasification. But that study I do want to indicate you will see it break down potential by region. So it's not just a national study it will show potential by region.

Q. By region or does it include data by state?

A. It does not include state, it is by region.

Q. So it will actually provide us data at a less useful aggregation than the 2010 study, is that correct?

A. That is correct. National Grid is taking efforts, so we are -- last time was the other way around in 2010. We did the national -- we did our National Grid study and in 2011 a national study came out. This time around there is a national study that came out and we are in the next few months

hoping to release our own white paper using this report to show potential specifically in our states.

Q. In your states?

A. In our jurisdiction.

Q. Now the 2010 study as well said that the potential carbon reduction from renewable natural gas in New York State was 11.3 million tons. I guess I'm assuming you're going to tell me that the new study will say that the savings would be larger than 11.3 million tons. Is that correct?

A. That's probably true. I mean, I don't have the numbers in front of me, but it's yes.

Q. Okay.

A. That will be a fair assumption.

Q. Might it be double that?

A. I don't know. And one -- one thing I can tell you is that this study is far more complicated in the sense that in the 2010 study, we use R.N.G.'s carbon neutral. And if you use carbon neutral, you can just assign a value of zero, but there has been a lot of litter -- literature and studies that says depending which policy paradigm or regulatory structure you adopt, you may not be able to use a carbon neutrality for R.N.G. therefore, you

need a carbon intensity similar to what California does under low carbon fuel standards.

So in this latest study, we have also show emission reduction two different ways. One, if you assume R.N.G. is carbon neutral, two, if you apply a carbon intensity, and if you do we have actually used a separate carbon intensity for each feedstock. So every biomass feedstock has its own carbon intensity, and it's shown so you'll see the emission reductions by feedstock as well.

Q. Okay. But even if it was doubled, it's still not a lot, right because that'd be like 22 million tons. New York City alone, according to the inventory, produces about what, 15 million tons a year just in that one city. So even if it was double the 2010 study, it's still only to about half of what one city produces. So is it -- is -- is that -- do you think a fair statement?

A. I would say that is not a fair statement.

Q. Okay.

A. You are comparing New York City, the largest city in the country and you're comparing one city that is very unique than any other city in

the United States. Almost every city in the United States has the highest source of emissions transportation, unique -- New York City's unique that their number one source is buildings.

So comparing the largest building in the country, and the largest emission source, which is buildings it is a very significant challenge to decarbonize the entire heating sector in New York City. If you look at it that way.

Q. Okay.

A. So New York City, I mentioned how difficult it would be to decarbonize heat. New York City will even be more difficult than any other city. Given just the fact what -- what I just stated.

Q. Right. Right. Okay. Let's see. Let's talk about hydrogen a bit if we could. I think on page 87 of your -- your testimony line 10 it says hydrogen blending holds promise with future of the heating --

A.L.J. COSTELLO: Okay.

MR. WYMAN: -- sector for --

A.L.J. COSTELLO: Mr. Wyman, Mr. Wyman, just let them get to the page --

Mr. Wyman: Yeah.

A.L.J. COSTELLO: -- first and then --

MR. WYMAN: Okay. Sorry.

A.L.J. LEARY: And you are referring to the direct testimony --

MR. WYMAN: Yes, yes.

A.L.J. LEARY: -- is that --

MR. WYMAN: All my -- all my references here are --

A.L.J. LEARY: Everybody that does cross please remember there's a lot of direct, rebuttal, supplemental, et cetera. You need to clarify what you're talking about not just saying testimony, but direct testimony, supplemental dated whatever, third supplemental, dated whatever.

Page number again, Mr. Wyman?

MR. WYMAN: Page number 87. Line --

A.L.J. COSTELLO: Yeah.

MR. WYMAN: -- 10.

BY MR. WYMAN: (Cont'g.)

Q. And then actually if once you're there, if we could skip down to lines 12 through 16, where it gives one of the reasons of using, okay. I was -- I was seeing you there. Using hydrogen blending as a low carbon supply side source may prove

to be an efficient way to rapidly lower the carbon intensity of and uses fueled by natural gas et cetera, et cetera.

I was intrigued by the word rapidly because I have been under the impression that the hydrogen sources and such are -- are not developed well enough to produce large quantities at the moment and are some years out. So can you tell me what -- what is your -- what is meant by rapidly in that context?

A. So when you look at hydrogen just a few years ago, there were not a lot of demonstration projects around the world. Looking today, there are multiple hydrogen blending projects in Europe, including U.K. So it feels like the hydrogen sector has hit a tipping point where, in fact that even though there are a lot of demos, there are some cities and there are demos in Germany and Denmark and France, where they're starting to look at certain towns and campuses that will begin to blend hydrogen.

So that -- that was sort of from the lens we were looking at this. But we don't see actually beginning to blend hydrogen in the near term

here in the U.S. in our footprint. We see mostly decarbonization coming from biomass sources. And part of our testimony when we talk about the future of gas, Future of Heat engineering and the demos we're doing, we're basically doing our due diligence to lay the foundation over the next few years that we can begin to scale hydrogen over time.

Q. So I think what I just heard you say was that that you would no longer stand behind this statement, that hydrogen blending will prove to be an efficient way to rapidly lower the carbon intensity of in use cases. Did I misunderstood -- stand you?

A. Yeah, yes, I'm saying hydrogen has great promise to reduce emissions. And -- but what I'm saying it is not here and now. So if your definition of rapidly are we able to deploy hydrogen blending in the next two years, the answer is no. And again, we're talking about the context of 2050.

So in a few years, you're able to blend hydrogen, we don't know what that number is, you know, there's a lot of literature out there that says it's somewhere between 5 to 20 percent. And if you're able to begin to introduce that hydrogen into

your gas network, that would be a significant decrease of emissions.

BY MR. WYMAN: (Cont'g.)

Q. Do -- do you know of anybody in the U.S. doing any substantial hydrogen blending in natural gas systems?

A. (Chahbazpour) No, the only one I am aware of is currently is in California, SoCalGas is blending hydrogen into a campus into a col -- U.C.R., but I believe a college campus.

Q. Now, on that same page, there's a reference to the fact that to -- to excess renewable energy, which will be the source of energy needed to produce this hydrogen, okay.

I think we heard you earlier in the day sort of express the opinion that, that most of the thoughts about excess renewable energy seem to be reliant on things like California and places out west that have a lot of solar, but we don't have a lot of excess renewable energy here, at least in the New York area, the Brooklyn and Long Island areas. Is that -- isn't that correct?

A. What I was referencing from this morning was where is that happening today and today

where you do have excess renewable energy that's being curtailed is California and Colorado.

Q. Do -- do you anticipate there being significant quantities of excess renewable energy anywhere near New York City -- Brooklyn or Long Island anytime soon?

A. I would say that's probably another one of those speculative personal question in the sense that, do you believe that the 9 gigawatts will materialize. I think it's in the C.L.P.A. -- C.L.C.P.A. by 2035 and I think it's 2035. If there is 9 gigawatts of offshore wind by New York, wherever that is, could be downstate by New York City or Long Island, I would probably say, yes. And it also depends of how much solar is built over that time.

So it's one of those questions as, you know, what's your perspective of how much renewables do you think will be coming on over the next few years.

Q. Okay. Now, the company is required by the N.E.N.Y. process to -- okay. Between now and 2025 we're going install lot of geothermal heat pumps. You all are proposing geothermal projects. Also there's a lot of effort being put in

to convincing people to drive electric vehicles. Okay. There is -- and all sorts of other kinds of beneficial electrification.

One of the intriguing things about the geothermal is of course the consumption load profiles of geothermal patterns happen to match rather nicely with the production profiles of things like that 9 gigawatts of wind power.

Is -- is it possible that as we dramatically ramp up the amount of geothermal and heat pumps in general that we're using, as well as ramping up electric vehicles and all the other forms of beneficial electrification as appears to be what's going to happen over the next couple of years that essentially beneficial electrification in its various forms will consume any excess renewable energy in this area?

Why would we believe that there will be enough additional access to -- to allow us to do something as inefficient as produce biofuels based on electrolysis of water and such?

A. So Your Honor I'm -- I'll doing my best not to say this is beyond the scope of this testimony, but I'll just say that you're asking me to

speculate and how I think the future will look like and how much geothermal will be ramping up, how much electrification will happen, and how much decarbonization hydrogen will replace.

It is possible that the scenario that you laid out may happen, but it's just -- sitting here it's impossible to say whether it's 2030 or 2040, how much each technology would be contributing to decarbonize heat.

Q. Your document is about the Future of Heat, it is -- it is a visionary document.

A. Uh-huh.

Q. I'm asking you to be no more speculative than you already have been, but --

A.L.J. LEARY: So is there something in the --

BY MR. WYMAN: (Cont'g.)

Q. -- anyway let's talk about thermal --

A.L.J. LEARY: Excuse me, Mr. Wyman. Is there something in the testimony that you can link for the benefit of the panel your -- your question to? Is there something in the testimony itself any of?

MR. WYMAN: Yeah, the testimony speaks of excess renewable energy. I'd like to know where it's coming from.

A.L.J. LEARY: Okay. Is that a question that is -- is also too speculative?

THE WITNESS: (Chahbazpour) No, I mean I -- I kind of answered it. If you think there is 9 gigawatts of offshore wind, right. Let me look at New York City today. New York City today peaks at 11.5 gigawatts on July. So if you put 9 gigawatts of offshore wind, I don't know where it will go, but I am assuming most of that will be dedicated to New York City.

So let's if -- let's just say 9 gigawatts is connected to the New York City system and if that was operating today, I would say you would definitely have excess -- times of excess renewable electricity, just given the fact that the city peaks at 11.5 gigawatts today. Which means the city is in a base load as about 4 or 5 gigawatts.

So if that 9 gigawatts is spinning at times where the city demand and -- and could be a spring evening if it's 4 gigawatts and if you have 9

gigawatts of turbine spinning, you could definitely say a condition occurred.

A.L.J. LEARY: Excess.

THE WITNESS: Excess.

A.L.J. LEARY: Okay.

THE WITNESS: And I think that's one of the reasons that the New York -- NYSERDA recently announced a New York offshore consortium. And I believe that NYSERDA is starting to also look at some of these production technologies in particularly hydrogen because they're starting to think about these conditions occurring.

BY MR. WYMAN: (Cont'g.)

Q. Okay. On page 93 line 1, you refer to the thermal lift heat pump.

A.L.J. COSTELLO: I'm sorry, what line?

MR. WYMAN: Line 1.

BY MR. WYMAN: (Cont'g.)

Q. You refer to the thermal lift heat pump, which is a gas powered heat pump and you say that it has exceeded its testing goals. So I couldn't find anywhere in the exhibits or testimony, any statement of what those goals were. Could you

say anything about what those goals were that it exceeded?

A. (Traczyk) I don't have the -- the testing goals in front of me at this time and I don't believe they've been entered into the record, but I think to the extent that they are non-confidential within the -- the testing arrangements that exist, we would be -- we'd be happy to provide them.

Q. Okay. I think -- do you know for instance, off the top of your head what -- what you believe their C.L.P. goal is?

A. I don't know the number and wouldn't wish to speculate.

Q. They've often made public statements of their goals and most of them are below the C.L.P.s that we would anticipate for -- for geothermal heat pumps. Do you believe that is the case?

A. Yes. We believe geothermal is the -- the most efficient heat pump technology that's available.

Q. Okay. Great. Thank you. Now, I am -- I am curious about how you see these -- these

less efficient, but nonetheless gas powered heat pump -- pumps fitting into the Future of Heat. And one of the things that strikes me is that if they did get nice C.L.P -- say C.L.P. -- let's just be really exotic 6 -- sorry, 3 which is -- which is higher than what they're shooting for. Then essentially anybody who is using one of these heat pumps to heat their house would only need one third of the amount of gas as they do today.

Could you talk perhaps for a moment about what implications that might have on -- on rates, on delivery rates specifically on how we would accomplish the company's revenue requirement? If suddenly everybody was doing what they're doing today, but only doing it with one third the amount of gas that they use today.

A. I think that's sort of a wholesale change is a theoretical scenario that could go along many -- alongside many other theoretical scenarios including increased company investment in geothermal ground loops for instance, which might make for a more complicated analysis of the impact on overall customer rates.

So I would not wish to speculate what the direct adoption of one technology would mean in the grand scheme of -- of customer rates.

Q. Now when you select these technologies to support to a tune here of 175,000 a year, is any thought given in advanced or -- or in parallel with the projects as to what the actual impacts and -- and what the imp -- actual impact will be if the technology program is successful?

A. Certainly to the extent possible, we -- we consider the impacts of supporting a technology both in terms of sort of each individual customer that might adopt it as well as our -- our broader customer base. And all of that is why we are proposing a portfolio of solutions that we believe explore a number of technologies and a number of ways that we can invest in the Future of Heat.

And we will continue to modify and adjust course as -- as we learn more about what the future looks like.

Q. Okay. But I'm -- I'm hearing you say that at least in this particular context you haven't thought about the impact that this might have on the business itself. This is -- it sounds like

this is just a technology thing you -- you haven't really looked at what it would mean to things like revenue recovery, bill impacts, whatever it might be.

A. If you're asking if we have considered what would happen if all of a sudden nearly 2 million customers adopted a brand new technology that's still in -- in early stages of development, no, we have not considered that.

Q. Darn, that was going to be my next question. What if 2 million people -- sorry, I can't do that, my apologies. Anyway, on geothermal, it says on page 5 of Freedom of -- sorry, Future of Heat testimony Exhibit 4, direct testimony Exhibit 4 on line 2.

It indicates that -- I hope you're there. That basically it indicates that what you're trying to test for in that program is whether or not geothermal is cost effective. I'm sorry if I missed it, but I couldn't find anything in the testimony that indicated. Do you think it's cost effective or did the result -- did the test indicate to you that it either was or is likely to be cost effective?

A. Forgive me Mr. Wyman, just to make sure I'm referring to the correct thing. Page 5 of 13 in the Exhibit 4, Future of Heat 4?

Q. It's page 5 of what was Exhibit 4 attached to the Future of Heat direct testimony.

A.L.J. LEARY: Okay. Just a second. You need to state the name of the document. So as I see this, this is the National Grid Geothermal Gas R.E.V. demonstration project, Long Island, New York Quarter 4 2018 report, January 31, 2019. Is that the document to which you're referring? That is F.O.H 4.

MR. WYMAN: F.O.H. 4, yes, that's the one.

A.L.J. LEARY: Okay. So if you're going to ask the panel about a particular document, please state the name of the document and if -- if you have it, the date so that we can match the record with the exhibit list.

MR. WYMAN: I'm -- I'm sorry, I thought the number was good enough. I'll keep that in mind.

A.L.J. LEARY: No, it's not good enough.

MR. WYMAN: Okay. Thank you.

BY MR. WYMAN: (Cont'g.)

Q. Are we there?

A. (Traczyk) Yes, we are there.

Q. Okay. Right.

A. Could you restate the question please?

Q. So you say there that what you're trying to determine is whether or not the geothermal systems will be cost effective. I -- I didn't see sort of an explicit conclusion, I may have missed it. But did you decide that they -- did the test indicate that they are cost effective or that they're likely to be or -- or what did you discover about cost effectiveness?

A. If you give me one second, Mr. Wyman, I wish to look into the one I.R. because I believe some information was provided about this, so if you'll let me. So I'm unable to locate the I.R. at this point in time. I certainly can check on that and get back to you.

But I can say that the Q4 2018 report that was included, Exhibit F.O.H.-4 to which you are referencing. At that time there had not been a

19-G-0309/19-G-0310 - 2-10-20 - Brooklyn Union Gas economic analysis of the geothermal system installed as part of the -- the Riverhead demonstration.

Subsequent to that, there have been additional quarterly reports filed by the companies or by -- by the company KEDLI on behalf of the Riverhead Project that have shown that all of the participants of the project have recognized a savings based upon their participation as compared to their prior fuel consumption. That does not -- since they were part of a demonstration project and therefore not paying the market rate for the system that does not necessarily conclude that all participants of a geothermal project would by definition receive a savings.

However, it is a positive indication that there is a savings potential to the technology. As we seek to expand our geothermal demonstration project within our proposal, we would hope to understand what conditions will allow customers to achieve savings. And in what ways we can interact with and support the geothermal industry such that the system costs can be brought down, adoption can be increased, and the savings can be increased for customers as well.

Q. Okay. Now, -- well, I think you know, I supported your -- your proposals. Staff suggested that it -- it may be inappropriate for you all to get into the geothermal business. They also said that they would brief on this subject sometime later in the case.

Are you -- are you looking forward to the -- to that brief?

A. Looking forward, meaning eagerly anticipating or looking --

Q. You continue to anticipate that it will happen.

A. Certainly we are -- we are hopeful for our continued dialogue with -- with all parties in the case on -- on these topics and specifically geothermal, we -- we look forward to -- to hearing opinions upon that topic.

Q. Do you continue to believe that it's a -- that it would be a good thing for ratepayers and for shareholders if the company were allowed to advent -- invest substantially more in geothermal systems, particularly geothermal district heating systems in one form or another?

A. It's a lot to unpack in that question. We continue to believe that investing in geothermal and being able to consider geothermal investments alongside gas pipe investments is a positive thing that is consistent with the company's vision for the Future of Heat.

The impacts on ratepayers and shareholders would of course be dependent upon the positioning in the marketplace, the eventual billing model and fee structure, et cetera, et cetera.

Q. Is it fair to say that because the geothermal assets are essentially plastic pipe stuck in holes in the ground that -- that the -- that they present much the same investment challenge that your exists -- that you're totally familiar with -- with the gas pipes of basically investing in pipes in the ground for 60, 80, 90 years or so that they're basically -- they're very similar these two businesses aren't they? Or are they not?

A. I would say I see it as an investment opportunity, not an investment challenge, but yes, we believe that they are analogous investment models.

Q. Now, I wonder if we could just briefly go back to hydrogen for a moment. My apologies for having forgotten something before. And that is to reflect on the hydrogen atom itself, which is a very tiny atom. And we know that one of the difficulties that gets tremendous amount of discussion whenever we're talking about natural gas is -- is leakages and lost and unaccounted for gas.

Given that hydrogen is a smaller molecule than methane, it is likely is it not that, if hydrogen is injected into pipes that currently are leak prone or leaky that the leakage rate will increase or essentially allow for it to go up?

A. (Chahbazpour) As part of our hydrogen blending study that we put in the Future of Heat that's one of the things we want to look at. So we want to understand the impact on infrastructure, particularly our infrastructure. And one of the answers that our engineering team, the infrastructure and operations would like to know is some of the questions where you're going.

In other words, can we begin to blend hydrogen into the system if you have any cast iron or does it need to be plastic? So part of these efforts

is to answer a lot of questions. Some of those include what you just stated.

Q. But -- but you are at least concerned that there is a risk that a lot of would increase dramatically by introducing these tiny atoms -- tiny molecules into pipes that are leaking bigger ones?

A. Yeah. Hydrogen leak, you know, looking at the hydrogen molecule and -- and that's one of those study -- issues you want to study as well a lot indicate that there's another demo that we're trying to in the power to gas, we're actually looking at hydrogen and then finding a steel to stream and putting -- putting them back together to get back to renewable natural gas or synthetic methane as well.

So in other words, hydrogen could either be blended or you could find hydrogen and a CO2 stream and put them back together and get to which shows the bioreactor I was discussing earlier this morning to get back to methane.

Q. Right. Yeah, my question here was just about --

A. Just about a leak. It's one of the issues.

Q. Just about hydrogen.

A. It's one of the issues that we'll be studying.

Q. Do you have any gut feel at the moment what the situation is? Do -- will hydrogen leak more than methane?

A. From what I've read is molecules are smaller, but it is much lighter, right, than the air, so disperses immediately. But from everything I've read is that your system does need to be tight in order for them not to leak.

Q. Right.

A. Have you read anything differently?

Q. Have you all in the past been able to make a system which is tight enough so that it could be used for hydrogen?

A. Well, I think we're not introducing hydrogen to our system yet. So I'm not going to speculate what we'll do. But again, going back to our hydrogen blending study, there's a lot of technical questions and that's why, you know, I was

mentioning that to your question as hydrogen is a promising technology. We're taking a really close look at it and all of the due diligence that we are doing as the next few years to build that foundation to ask a lot of those questions, which is going back to our previous conversation.

I don't view hydrogen as something that we are ready to do over the next few years because we have to address all of these technical and engineering questions.

Q. Right. So that would tend to support the idea that perhaps blending hydrogen will not rapidly reduce our carbon issues. Is that correct? We talked about this before.

A. (Traczyk) When referring to the word rapidly in the testimony, one could interpret that to mean the solution could be rapidly deployed at scale or it could mean that the emissions would rapidly decrease if it were deployed at scale.

And our vision for how we would introduce hydrogen into the system, which is consistent with SoCalGas and with what we're seeing with our colleagues in the U.K. as well, is that it makes sense to strategically identify portions of the

system that are tight, that are ready for hydrogen investment and that would be prime sites for exploring how we would blend this.

So we don't see this as being something that you would just wholesale inject into the system and assume that the system is entirely ready for it. You would make strategic investments in order to support operations in key parts of the system, which would allow for a tighter system as you were referencing before.

Q. So of the two meetings that you suggested, which was the meeting that was intended when you wrote the report --

A. Since --

q.-- when you wrote the testimony?

A. Since the report was also referring to without needing to modify end use equipment, we were referring to that it would rapidly decrease emissions as opposed to that it would be a technology that would be ready for rapid scale deployment. Which is why our proposal in this case was merely a study of hydrogen blending and its impacts on the system.

A.L.J. COSTELLO: Mr. -- Mr. Wyman, let me just -- do you have many more questions? We've -- we've --

MR. WYMAN: No, I don't.

A.L.J. COSTELLO: Okay. I'm just trying to see if you had a sense of when you were going to wrap it up.

A.L.J. LEARY: We have you down for 30 minutes. No topics identified. You're now at an hour. So -- yes, you can believe me.

MR. WYMAN: My apologies. I'm -- I'm actually -- that was -- that was the end of it so.

A.L.J. LEARY: Great.

A.L.J. COSTELLO: Thank you.

A.L.J. LEARY: I'd like to thank the panel for your --

A.L.J. COSTELLO: Wait, wait.

A.L.J. LEARY: Sorry.

A.L.J. COSTELLO: Do you have any re-cross?

A.L.J. LEARY: Do you have any redirect?

A.L.J. COSTELLO: I mean redirect, I'm sorry?

MR. GAVILONDO: Yes, Your Honors. If the -- if we could take a break for about 10 minutes and I think we'll probably only have about 10 or 15 minutes of redirect.

A.L.J. LEARY: Perfect.

MR. GAVILONDO: Thank you.

A.L.J. LEARY: Let's take a --

A.L.J. COSTELLO: Off the record.

A.L.J. LEARY: -- less than 10 minute break, less.

(Off the record 4:21 p.m. to 4:31 p.m.)

THE REPORTER: We're back on the record.

A.L.J. COSTELLO: Okay.

MR. GAVILONDO: Thank you, Your Honor. Carlos Gavilondo for the companies. I would like to conduct some brief redirect of the panel.

REDIRECT EXAMINATION

BY MR. GAVILONDO:

Q. Earlier today the staff asked some questions regarding energy efficiency, full time equivalents, F.T.E.s and specifically Mr. Goodrich

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referred to what's been pre-marked as Exhibit 599, which was the company's response to D.P.S. I.R. 266.

`And I asked you Ms. Tallet, I believe you testified to that. Is that correct?

A. (Tallet) Yes.

Q. And could you please clarify your answer with respect to that response of the company?

A. Sure. So within the -- within the I.R. response I had previously mentioned that there were 163 F.T.E.s charging to KEDNY and the same for KEDLI. In that attachment we also provided the hours that were charged to the energy efficiency companies.

What happens within that, the energy efficiency space and I'll use myself as an example. I'm one employee, however, I charge my time to the different companies that I -- that I work with, with the energy efficiency portfolio. So it could be for myself, KEDNY, KEDLI, NIMO Electric and NIMO Gas.

So in the attachment 1 to D.P.S. 266 where it actually shows the -- the departments and it might say like employee 114. So when we look at -- at all of those they may be one employee who, like me, will charge their time accordingly. So with

respect to this I.R., the actual number of F.T.E.s would be found by looking at the attachment, on attachment 2 and looking at the total number of hours divided by the F.T.E.s.

So for KEDLY for 2017, it would come out to be 9.11 F.T.E.s. For 2018, it would be 7.82 F.T.E.s. For KEDNY, for 2017, it would be 8.29 F.T.E.s, and for 2018, 8.03 F.T.E.s.

Q. Thank --

MR. GOODRICH: I'm sorry, I'm very slow at writing now.

A.L.J. LEARY: I think you also said KEDNY, if I -- if I heard that incorrectly, I'm sorry. But I think you did the KEDNY numbers 9.1 F.T. --

THE WITNESS: (Tallet) Okay. I'll read them again.

A.L.J. LEARY: Great.

THE WITNESS: So for KEDLI, for 2017 it would be 9.11. For 2018 for KEDLI it would be 7.82. And for KEDNY for 2017, it would be 8.29 and for 2018 for KEDNY it would be 8.03.

MR. GAVILONDO: Thank you.

BY MR. GAVILONDO: (Cont'g.)

Q. Directing the panel to the cross examination by Family Energy earlier today, those discussion of the company's proposed Green Gas Tariff offering. Would the pricing and terms for the company's proposed Green Gas Tariff product be defined in the companies and their respective company's regulated tariffs?

A. (Traczyk) Yes.

Q. Thank you. And would the company oppose or object to any other entity including an E.S.C.O. offering a product which included a renewable gas offering?

A. No, we would not.

Q. Thank you.

MR. GAVILONDO: Almost done, Your Honors.

A.L.J. LEARY: I do want to ask you if you can provide for the record where in -- where is that pricing defined in the tariff?

MR. GAVILONDO: So --

A.L.J. LEARY: So the -- the -- the question you asked is would the pricing be defined in the current tariff. I believe that's what you asked, and the answer was, yes. So where is it defined?

MR. GAVILONDO: I'm sorry, if I -- Your Honor, I'm sorry if I said the current tariff. I -- I thought I intended to say the company's regulated tariffs.

A.L.J. LEARY: Okay.

MR. GAVILONDO: If I said the current tariffs, I apologize.

A.L.J. LEARY: I am not sure you did. So I don't mean to say that you did. I'm slow at writing things down as well. So it is defined in the company's regulated tariffs.

MR. GAVILONDO: Actually, Your Honor, the -- the -- the product is a proposed product. So there is no tariff at this time for that product offer.

A.L.J. LEARY: But the pricing itself is, I believe, what you were referring to for that product?

MR. GAVILONDO: No, Your Honor. The -- the thrust of the question was to point out that the product would be offered on a regulated basis, not --

A.L.J. LEARY: I see.

MR. GAVILONDO: -- as an ESCO offering a product on a competitive basis.

A.L.J. LEARY: Got it, thanks.

MR. GAVILONDO: Should I clarify?

A.L.J. LEARY: You can proceed. No, I think --

MR. GAVILONDO: Okay.

A.L.J. LEARY: Well, unless someone else, this was a clarification for me, apologies, but that was helpful.

MR. GAVILONDO: Your Honor, there is no more important clarification than for the bench, so.

BY MR. GAVILONDO: (Cont'g.)

Q. The next question relates to a question actually that Judge Leary had regarding the 1990 baseline that was spoken about in the context of questioning by Sane Energy. Does the panel recall that?

A. (Schwartz) Yes.

Q. Okay. Thank you. And Ms. Schwartz, do you have any information that you can provide for the record that might shed a little bit more light on that?

A. I believe I have an accurate estimate of our 1990 baseline pending confirmation. For KEDNY 85 kilotons of carbon dioxide equivalent, so that's converting the methane into its carbon dioxide equivalency. And for KEDLI 329 kilotons of carbon dioxide equivalent.

Q. Thank you.

A.L.J. LEARY: Very helpful, thanks.

MR. GAVILONDO: Thank you.

BY MR. GAVILONDO: (Cont'g.)

Q. Panel, earlier today there was some discussion of the tax abatement relative to the Newtown Creek Project and Mr. Chahbazpour, you spoke to that. Could you please provide some additional information regarding the status of the tax abatement from New York City?

A. (Chahbazpour) Sure. I had indicated that we had not filed a new property tax abatement, but we have worked with the City of New York and they've been very cooperative and particularly the New York City Economic Development Corporation to retain the existing 5 year property tax abatement.

In other words, when we originally signed a property tax abatement it had a date that our plants that had to be operational by that date has already passed. So the City of New York has moved that 5 year window. So from the time to -- so again, to retain the 5 year property tax abatement.

A.L.J. LEARY: Just so I understand that you're in negotiations with the City or have they agreed to retain the existing tax abatement?

THE WITNESS: (Chahbazpour) No, we have agreed, there's -- there's an agreement.

A.L.J. LEARY: There's an agreement.

THE WITNESS: Yeah.

A.L.J. LEARY: Thank you.

MR. GAVILONDO: I have no further questions for the panel, Your Honors.

MR. GOODRICH: I just have one question based on that that last one.

RECROSS EXAMINATION

BY MR. GOODRICH:

Q. You said that they moved the 5 year window. What -- when did they move it to, what -- what is the window?

A. We've moved it I believe twice. Now the latest window is what's being -- it's about -- it's July. So in other words, our plant needs to be operational by July.

Q. Do you expect your plant to be operational by July?

A. Yes. Right now we expect the plant to be operational hopefully, fingers crossed, in April, May.

Q. And if it is not operational by July, what happens?

A. We -- we will go back to the Economic Development Corporation, I think they'll be very cooperative to again move that 5 year window.

Q. And to be clear what that moving the window means, I'm -- I'm taking that it means that from the day that the plant is operational is the day that it would be subject to tax. And so it's from that day --

A. Exactly.

Q. -- 5 years that it's exempt.

A. Exactly.

Q. Okay.

A. Correct.

A.L.J. LEARY: And just so I can relate back, this is the 4.4 million dollars that the panel testified about earlier today. 4.4 million, is that right as a cap?

THE WITNESS: (Chahbazpour) It was a five year property tax abatement on a 14.4 million dollar project.

A.L.J. LEARY: 14 --

THE WITNESS: Yes.

A.L.J. LEARY: -- sorry.

THE WITNESS: Yeah.

MR. GOODRICH: I don't have anything further.

A.L.J. LEARY: Anybody else have re-cross based on redirect?

MS. MURPHY: Yes. This is Erin MURPHY with Environmental Defense Fund. Just one question. The 1990 emissions baseline numbers that were just provided, and my apologies if I missed this in the answer. But what was the method of calculation for that 1990 baseline?

THE WITNESS: (Schwartz) That was using the E.P.A. sub-part W method.

MS. MURPHY: Thank you.

A.L.J. COSTELLO: Anything further? No. Not hearing any, we want to thank the Future of Heat panel, the witnesses. Thank you very much for your testimony today. You're excused.

A.L.J. LEARY: The panel is, but everybody else has to stay for a very short time and you can stay too, if you'd like.

A.L.J. COSTELLO: But we'll --

A.L.J. LEARY: Let's go off the record.

A.L.J. COSTELLO: We'll go off the record, yeah.

A.L.J. LEARY: We're off the record.

(Off the record; 4:42 p.m.)

STATE OF NEW YORK

I, KAYLA ALLEN, do hereby certify that the foregoing was reported by me, in the cause, at the time and place, as stated in the caption hereto, at Page 1 hereof; that the foregoing typewritten transcription consisting of pages 1 through 522, is a true record of all proceedings had at the hearing.

IN WITNESS WHEREOF, I have hereunto subscribed my name, this the 14th day of February, 2020.

Kayla Allen, Reporter